



Interplay® | Production Best Practices Guide

Version 3.8

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Contents

	Using This Guide	9
	Symbols and Conventions	9
	If You Need Help.	10
	Viewing Help and Documentation on the Interplay Production Portal	10
	Interplay Documentation.	11
	Avid Training Services	12
Chapter 1	Interplay Production Components	13
	Using MultiRez and Dynamic Relink.	13
	Interplay Engine	14
	Interplay Access	14
	Interplay Administrator	15
	Interplay Window	16
	Interplay Transfer	18
	Avid Instinct.	19
	Interplay Production Services Providers	19
	Interplay Assist	20
	iNews and Avid Instinct.	21
	Archive and Restore Services.	21
	MediaManager to Interplay Migration Tool	22
	Interplay Media Indexer	23
	Service Framework Services	23
	Interplay Capture	23
	Access Control and Media Deletion Model.	24
	Interplay Application Icons	24
Chapter 2	Setting Up a Folder Structure and a User Database	26
	Permissions and Folder Structures.	26
	Setting Up a Folder Structure in Interplay Access	27
	Basic Setup: Assigning Access to User Groups.	28
	Assigning Database Access on a Project Basis	31
	Defining the Editor Database Settings	34
	Managing the Instinct/Assist User Settings.	37
	Creating Folders for Capturing Media.	37

Chapter 3	Developing an Interplay Delete Strategy	38
	Introduction to Interplay Deletion	38
	Deletion Guidelines	39
	Interplay Database Folder Structure	43
	Creating Media Composer Projects in an Interplay Environment	48
	Interplay Production and MediaCentral UX	51
	Project Workflow Samples	52
	Example: Deleting Metadata and Media	54
	Example: Deleting Metadata Only	64
	Determining How Much Space a Delete Operation Will Free Up	68
	Searching for Media	70
Chapter 4	Refining Your Search in Interplay Production	71
	Using the Search Root and Attributes to Speed Up Interplay Production Search	71
	Basic Interplay Production Search Attributes	72
	Searching in Interplay Assist, Avid Instinct, and the Interplay Window	73
	Searching in Interplay Access	74
	Example: Finding Assets By Using a Unique Identifier	77
Chapter 5	Using AMA Material in Interplay Production	82
	Overview of Working with AMA Material	82
	AMA on Interplay Production Workflow	82
	How Interplay Media Indexer Works with AMA Material	85
	Configuring Interplay Administrator and Interplay Media Indexer to Manage AMA Media	87
	Using Third-Party Storage for AMA Media	93
	Viewing AMA Clips in Interplay Access	97
	Deleting AMA Clips Using Interplay Access	101
	Resyncing AMA Media Files	102
Chapter 6	Long GOP End-to-End Workflows	106
	Long GOP Overview	106
	Long GOP Media Wrappers	108
	Long GOP Workflows	108
Chapter 7	720p MultiRez Workflow	119
	Overview of the 720p MultiRez Workflow	119
	Editor MultiRez Workflow for 720p	120
	720p MultiRez Workflow Components	124
	Setting Up the Production Services Profiles for a 720p MultiRez Workflow	134
Chapter 8	Determining Interplay Database Scalability	139
	Definition of Terms	139
	Total Used Database Pages and Estimated Database Page Limit	140
	Comparing Database Page Count and Cache Size	142

	Tips for Improving Database Performance	144
	Establish a Stable Ratio Between Database Objects and Database Pages.	145
Chapter 9	H.264 End-to-End MultiRez Workflow	149
	H.264 Resolutions Supported by Interplay Production	149
	Installation and Configuration for an H.264 Workflow.	151
	Acquiring Media for an H.264 Workflow	162
	Editing with H.264 and High-Res	166
	Archiving, Restoring, and Deleting H.264 and High-Res Media	166
Chapter 10	HD 23.976p and HD 24p Support	167
	Interplay Production Support for Native HD 23.976p and HD 24p Formats	167
	Interplay Production Support for HD 23.976p and HD 24p Proxy Resolutions.	170
Chapter 11	Apple ProRes Resolutions Supported in Interplay Production.	171
Chapter 12	JPEG 2000 Resolutions Supported in Interplay Production	174
Chapter 13	Support for Stereoscopic 3D Workflows	175
	Overview of Stereoscopic Support	175
	Stereoscopic Media and Stereoscopic Clips.	176
	Stereoscopic 3D Support in Interplay Access.	178
	Stereoscopic 3D Support in Interplay Transcode	180
	Stereoscopic 3D Support in the Interplay Window, Interplay Assist, and Avid Instinct.	181
	Properties for Stereoscopic Clips	181
Chapter 14	DNxHD Resolutions Supported in Interplay Production	184
Chapter 15	DNxHR Resolutions Supported in Interplay Production	187
Chapter 16	XDCAM HD and XDCAM EX Operating Points Supported in Interplay Production 189	
	Operating Points Qualified for Omneon Servers.	191
	Operating Points Qualified for AirSpeed Multi Stream	191
Chapter 17	XAVC-Intra HD Resolutions Supported in Interplay Production	192
Chapter 18	AVC-LongG Formats Supported in Interplay Production	193
Chapter 19	XAVC Long GOP Formats Supported in Interplay Production	195
Chapter 20	Using Frame Chase Editing	198
	Frame Chase Editing and Frame Chase Capture.	198
	Understanding Frame Chase Editing	198
	Workflow for Frame Chase Editing	200
	Limitations When Working With In-Progress Clips	203
Chapter 21	Additional Interplay Production Topics	204
	Changing Source Information for a Clip in Interplay Production.	204
	Ingest and Playout of Ancillary Data.	205
	Support for AFD Ancillary Data.	206

	Dynamic Relink: Working with a Single Resolution	207
	Disconnecting an Avid Editing Application from the Interplay Environment	209
Chapter 22	Working with Interplay Media Indexers v3.5 and Later	210
	MongoDB Database	210
	MI Memory and Disk Space Requirements	212
	Support for AVC-LongG Media	215
	Media Indexer Independence from ASF	215
	Interplay Media Indexer Overview	218
	Web-Based User Interface	218
	Web Interface Components	219
	Configuring Media Indexer Servers and Clients	235
	Managing Interplay Media Indexers in an Interplay Environment	240
	Troubleshooting Interplay Media Indexer	245
	Estimating the Indexing Time	257
	Prebuilding a Media Indexer Cache During an Upgrade	259
Appendix A	Interplay Production Maintenance Recommendations	261
	Interplay Weekly Maintenance Check List	262
	Interplay Monthly Maintenance Check List	263
	Avid ISIS Recommended Maintenance	265
	Complete Server Room Shutdown	265
Appendix B	Archive Migration (Workgroup 4)	266
	Searching for Migrated Workgroup 4 Archive Items	266
	Restoring Migrated Assets	267
	Restoring Different Versions of a MediaManager Archive Clip	268
	Working with OMF Media Files After an Archive Migration	272
	Retained Workgroup 4 Clip Attributes	274
	Index	275

Using This Guide

Revision History

Date Revised	Changes Made
6/28/2017	Updated for Interplay v3.8
12/22/2016	Updated for Interplay v3.7
6/30/2016	Updated for Interplay v3.6

Congratulations on your purchase of Interplay® Production, a powerful system for managing media in a shared storage environment. This guide contains an overview of Avid Interplay and also contains task-oriented instructions, conceptual information, and reference material you need to use your Avid Interplay system. This guide is intended for all Interplay users, from beginning to advanced

Symbols and Conventions

Avid documentation uses the following symbols and conventions:

Symbol or Convention	Meaning or Action
	A note provides important related information, reminders, recommendations, and strong suggestions.
	A caution means that a specific action you take could cause harm to your computer or cause you to lose data.
	A warning describes an action that could cause you physical harm. Follow the guidelines in this document or on the unit itself when handling electrical equipment.
>	This symbol indicates menu commands (and subcommands) in the order you select them. For example, File > Import means to open the File menu and then select the Import command.
►	This symbol indicates a single-step procedure. Multiple arrows in a list indicate that you perform one of the actions listed.
(Windows), (Windows only), (Macintosh), or (Macintosh only)	This text indicates that the information applies only to the specified operating system, either Windows or Macintosh OS X.
Bold font	Bold font is primarily used in task instructions to identify user interface items and keyboard sequences.
<i>Italic font</i>	Italic font is used to emphasize certain words and to indicate variables.
Courier Bold font	Courier Bold font identifies text that you type.
Ctrl+key or mouse action	Press and hold the first key while you press the last key or perform the mouse action. For example, Command+Option+C or Ctrl+drag.

Symbol or Convention Meaning or Action

(pipe character)	The pipe character is used in some Avid product names, such as Interplay Production. In this document, the pipe is used in product names when they are in headings or at their first use in text.
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If You Need Help

If you are having trouble using your Avid product:

1. Retry the action, carefully following the instructions given for that task in this guide. It is especially important to check each step of your workflow.
2. Check the latest information that might have become available after the documentation was published. You should always check online for the most up-to-date release notes or ReadMe because the online version is updated whenever new information becomes available. To view these online versions, select ReadMe from the Help menu, or visit the Knowledge Base at www.avid.com/support.
3. Check the documentation that came with your Avid application or your hardware for maintenance or hardware-related issues.
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Viewing Help and Documentation on the Interplay Production Portal

You can quickly access the Interplay Production Help, links to the PDF versions of the Interplay Production guides, and other useful links by viewing the Interplay Production User Information Center on the Interplay Production Portal. The Interplay Production Portal is a Web site that runs on the Interplay Production Engine.

You can access the Interplay Production User Information Center through a browser from any system in the Interplay Production environment. You can also access it through the Help menu in Interplay | Access and the Interplay | Administrator.

The Interplay Production Help combines information from all Interplay Production guides in one Help system. It includes a combined index and a full-featured search. From the Interplay Production Portal, you can run the Help in a browser or download a compiled (.chm) version for use on other systems, such as a laptop.

To open the Interplay Production User Information Center through a browser:

1. Type the following line in a Web browser:
`http://Interplay_Production_Engine_name`

For *Interplay_Production_Engine_name* substitute the name of the computer running the Interplay Production Engine software. For example, the following line opens the portal Web page on a system named docwg:

`http://docwg`

2. Click the “Interplay Production User Information Center” link to access the Interplay Production User Information Center Web page.

To open the Interplay Production User Information Center from Interplay Access or the Interplay Administrator:

- ▶ Select Help > Documentation Website on Server.

Interplay Documentation

The following documents describe how to use Interplay Production:

- *Interplay / Production Best Practices* — provides an overview of the major Interplay components, shows sample configuration diagrams, and describes several of the key features, such as setting up a user database and using the Interplay Archive Engine.
- *Interplay / Production Software Installation and Configuration Guide* — describes how to use the Interplay Production installation programs to install and configure software on the various systems that make up a Interplay environment.
- *Interplay / Engine Failover Guide* — describes how to set up a cluster configuration for the Interplay Engine and the Interplay Archive engine. There are two versions of this guide: one for SR2500 systems and one for AS3000 systems.
- *Interplay / Engine and Interplay / Archive Engine Administration Guide* — describes how to administer your Interplay Engine or Interplay Archive Engine and Avid Interplay database.
- *Interplay / Access User's Guide* — describes how to use Interplay Access to browse, search, and work with assets in the Avid Interplay database.
- *Interplay / Assist User's Guide* — describes how to use Interplay Assist for logging, archiving, and sending to Playback material in an Interplay environment.
- *Avid Instinct User's Guide* — describes how to use the Avid Instinct application to write iNEWS stories and put together simple audio and video sequences.
- *Interplay / Production Services Setup and User's Guide* — Interplay Media Services lets you control and automate Transcode, Archive, and other Interplay Media Services.
- *Interplay / Transfer Setup and User's Guide* — describes how to use Interplay Transfer to transfer media to and from another workgroup, send finished sequences to a configured playback device, ingest media from a configured ingest device, and perform standalone transfers between workstations.
- *Avid Service Framework User's Guide* — describes how to use the logging, monitoring, and management features of the Avid Service Framework applications.
- *Interplay / Capture User's Guide* — describes how to use Interplay Capture, a newsroom ingest tool that enables automated recordings.
- *Interplay / Capture Administrator's Guide* — describes administrative concepts, tasks, and reference material for Avid Interplay Capture.

All documents are available in PDF form on the Avid Knowledge Base at www.avid.com/onlinesupport.

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1 Interplay | Production Components

This chapter provides a brief overview of the main components that make up an Interplay Production environment.

- [Using MultiRez and Dynamic Relink](#)
- [Interplay | Engine](#)
- [Interplay | Access](#)
- [Interplay | Administrator](#)
- [Interplay Window](#)
- [Interplay | Transfer](#)
- [Avid Instinct](#)
- [Interplay | Production Services Providers](#)
- [Interplay | Assist](#)
- [iNews and Avid Instinct](#)
- [Archive and Restore Services](#)
- [MediaManager to Interplay Migration Tool](#)
- [Interplay | Media Indexer](#)
- [Service Framework Services](#)
- [Interplay | Capture](#)
- [Access Control and Media Deletion Model](#)
- [Interplay Application Icons](#)

Using MultiRez and Dynamic Relink

MultiRez (a term derived from “multiple resolutions”) allows you to associate a single master clip with multiple media files of different resolutions. For example, you can associate the same master clip with low-resolution MPEG-2, JFIF 15:1s, and uncompressed media files. MultiRez also works with audio files: you can associate an audio clip with more than one sample rate.

In an Interplay Production environment, you can capture a master clip in multiple resolutions or transcode a clip to another resolution, and you can easily switch between the different resolutions. This switching is referred to as *dynamic relink*. Dynamic relink is useful when you want to work with low-resolution media and create your final master in high-resolution media. For more information, see “Using MultiRez and Dynamic Relink” in the *Avid Media Composer Editing Guide*.

This information is also included in the Help for these products or the Interplay Production Help. For additional information, see [“Dynamic Relink: Working with a Single Resolution” on page 207](#).

Interplay | Engine

The Interplay Engine forms the backbone of the Interplay environment. The Interplay Engine is a server that combines an asset database with workflow management software, both of which are integrated with Avid shared storage and Avid archive solutions. This topic provides basic information about the functions and components of the Interplay Engine.

The Interplay Engine provides the following features:

- Management of *Avid assets* (for example, master clips and sequences) as well as *file assets* (for example, Adobe Photoshop® and Microsoft® Office files)
- Version control for sequences and file assets
- Integration with Avid ISIS™
- Support for MultiRez (multiple resolution) workflows
- Access control through groups as follows:
 - Five fixed user roles: Administrator, Read/Write/Delete, Read/Write, Read, and No Access
 - A user can be part of several groups
 - Permissions can be modified at the folder level
- Import on the desktop level to Interplay Access
- Use the Interplay Window to check in and check out to the database from the Avid editing application
- Support for MXF media files and AAF metadata. Interplay v3.0 adds support for AMA media.
- Failover capability (using Windows® 2003 Enterprise Server), including Auto-Failover to hot spare (Interplay Engine and Interplay Archive Engine only)
- Simplified workgroup delete rules including the following:
 - The administrator can specify the delete permissions that users and groups have on particular folders. You use Reservations to protect the media files related to Avid assets.
 - Media can be protected by reservations that have time and date expirations
 - Reservations protect assets and media placed in a protected folder
 - You can use the Interplay Access software to place personal reservations on folders

For information on the Interplay Engine, see *Interplay Engine and Interplay Archive Engine Administration Guide* and the *Interplay Access User's Guide*.

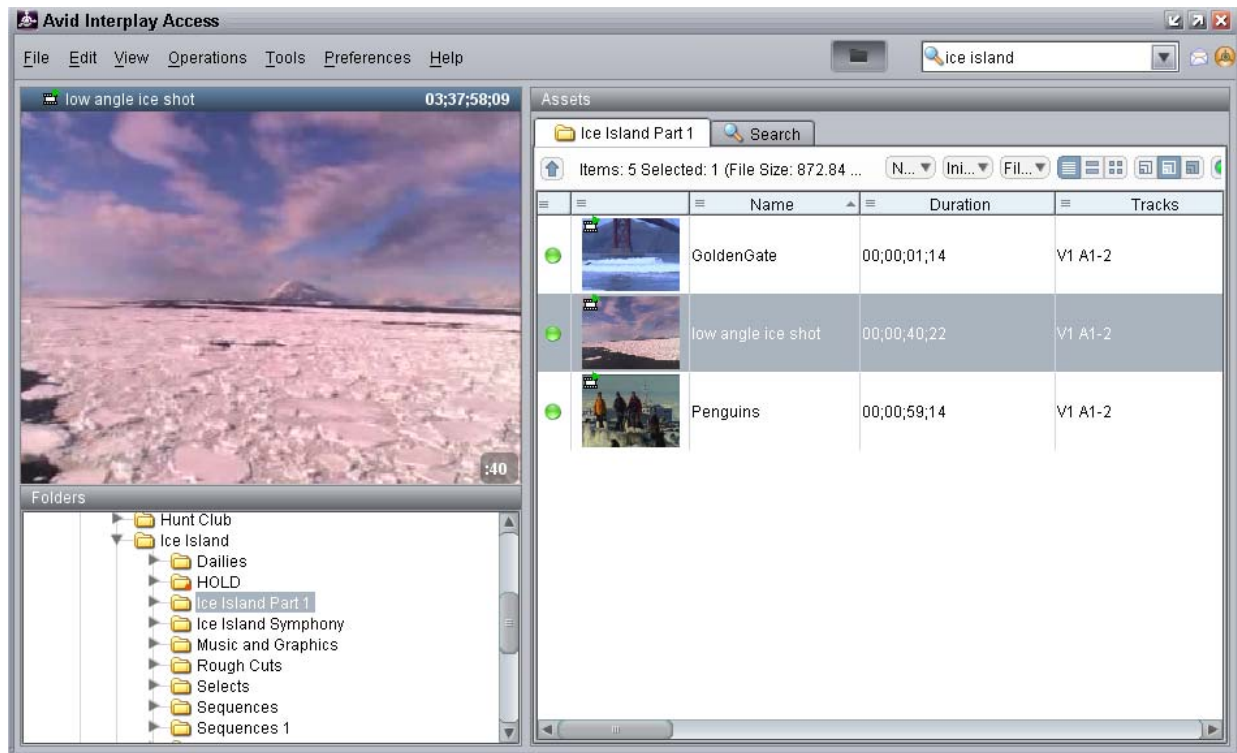
Interplay | Access

The Interplay Access software is a dedicated application that can be used by system administrators and other designated users such as editors, assistants, and producers to interact with the Interplay Engine.

The Interplay Access user interface allows you to quickly search or browse its contents for assets of interest. Drag-and-drop support allows you to obtain Avid assets by dragging them to the Avid editing applications Interplay Window.

Interplay Access provides search tools and an Object Inspector that allow anyone from artists to editors to easily find shots, clips, sequences, or other elements defined for the project without even knowing file names, and to quickly see the dependencies and relatives involved. You can also use Interplay Access to manage sequences and other Avid assets whose media is no longer online. You can then search for these assets, view information about them such as the source tape name, obtain the source footage, and recapture it.

The following illustration shows Interplay Access.



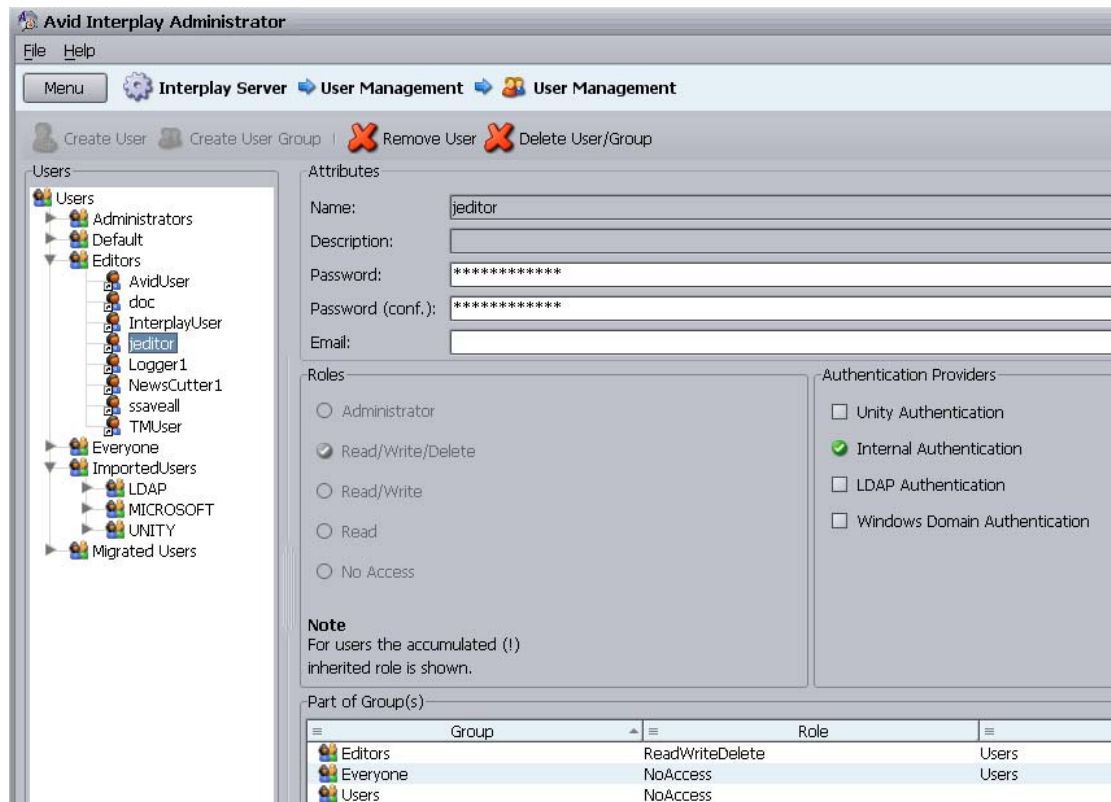
For more information about Interplay Access, see the *Interplay Access User's Guide*.

Interplay | Administrator

The Interplay Administrator is a cross-platform application used for basic administration of the Interplay Engine and the Interplay Archive Engine. Supported capabilities include the following:

- Database activation, creation, and security
- Scheduling backups
- Server licensing configuration
- Central configuration server setup of users

The following illustration shows the User Management window of the Interplay Administrator application.



For information on using the Interplay Administrator, see the *Interplay Engine and Interplay Archive Engine Administration Guide*. For information on setting up a user database, see [“Setting Up a Folder Structure in Interplay / Access” on page 27](#).

Interplay Window

The Interplay window allow you to share Avid assets with other users in the Interplay Production environment. The Interplay window provide the following capabilities:

- Avid editing applications such as Avid Media Composer® can work with Avid assets through the Interplay window. The Interplay window displays the same folder structure you see in the Interplay Access interface. You can capture material to an Interplay window and edit those clips into a sequence in a bin. You can also check sequences into the Interplay window for asset management and for sharing with other editors.
- Avid Instinct and Interplay Assist allow you to access remote assets directly. The interface is similar to the Interplay window. For example, an Instinct journalist can work with remote assets without copying the files locally and without having to check the files in and out of the Interplay Engine. The interface provides quick access to story information. You can work on and finish your stories without having to copy files to your local system.

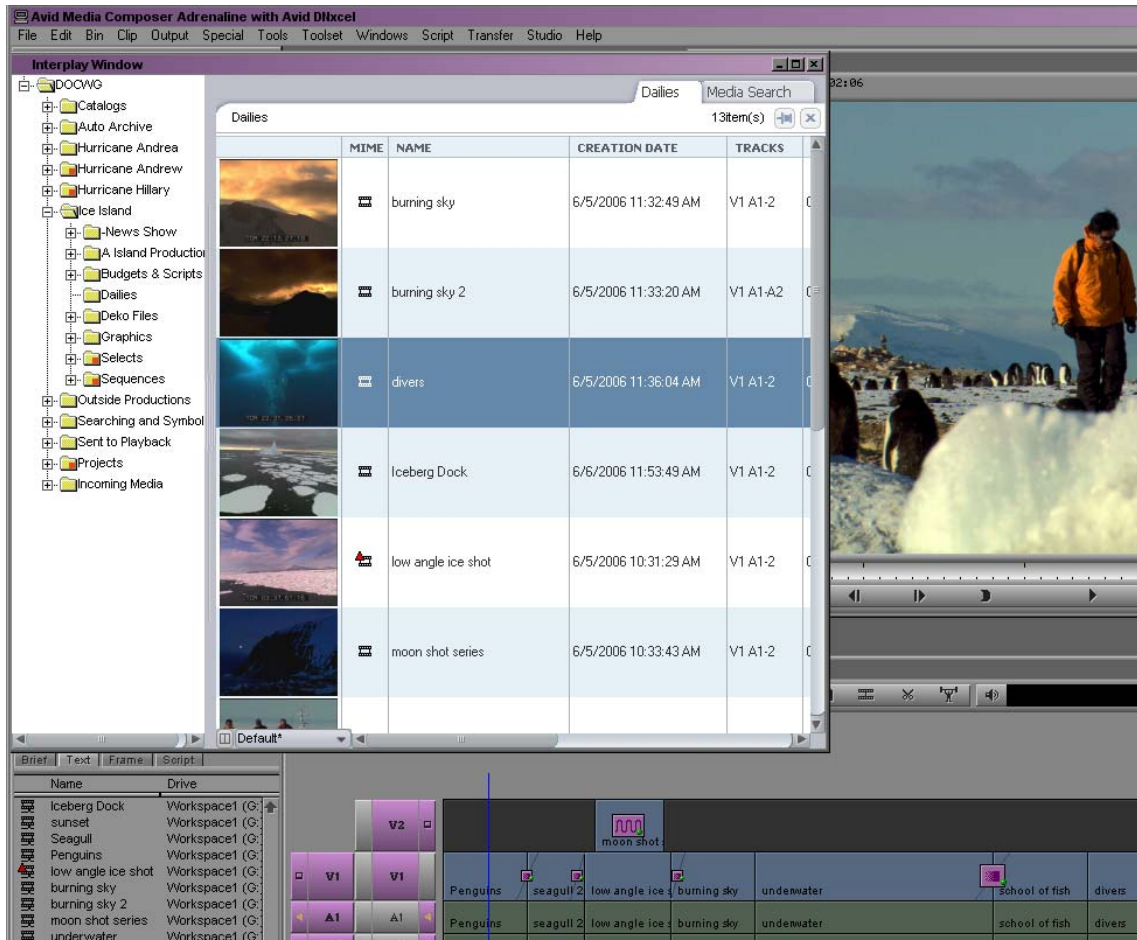
Key features of the Interplay Window include the following:

- Capture (digitize) directly into the Interplay Window
- Save current state and restore on relaunch
- Drag from editing application bins to the Interplay window and vice versa

- Edit remote clips directly into a sequence
- Search the Interplay Production database
- Create shortcuts

Avid editors must edit sequences, render, and consolidate in bins.

The following illustration shows the Interplay Window within an Avid editing application.



For more information, see “Working with Interplay Production from an Avid Editing System” in the Interplay Production Help, the Help for your Avid editing application or the *Avid Media Composer Editing Guide*.

Interplay | Transfer

Interplay Transfer lets you transfer Avid assets to and from another workgroup, send finished sequences to a configured playback device, and capture media from a configured ingest device. You can also use Interplay Transfer in a standalone environment (an environment other than Avid shared storage) to move Avid assets between workstations.

The Interplay Transfer Engine and client software operates as TransferManager does in a MediaManager environment. In addition, it supports the following features:

- You can perform MediaManager-to-Interplay Production transfers from a MediaManager browser to an Interplay Transfer Status window or to a bin in an Avid editing application within your Interplay Production environment.
- Standalone Interplay Transfer support is provided in the Avid editing applications.
- Support for a Transfer Cache Engine that can monitor transfers for a large number of users. This removes the task of reporting transfer status from the Interplay Transfer Engine.

The following illustration shows the Interplay Transfer Status window viewed from the Interplay Administrator.

Status	ID	Name	Type	Source	Destin...	% co...	User	Start ...	Finish ...	Tape ID	Transf...T
	16	Wg5 ...	Work...	172.2...	bsiwg...	91%	Remote	02:43...			bsiwg...
✓	1	imx c...	Playb...	BSIX...	bsiwg...	100%	bsiwg...	02:25...	02:25...	rimx50	bsiwg...
✓	2	read ...	Playb...	BSIX...	bsiwg...	100%	bsiwg...	02:28...	02:28...	rwm	bsiwg...
✓	3	Wg5 ...	Work...	172.2...	bsiwg...	100%	Remote	02:31...	02:31...		bsiwg...
✓	5	Wg5 ...	Work...	172.2...	bsiwg...	100%	Remote	02:31...	02:32...		bsiwg...
✓	4	read ...	Playb...	BSIX...	bsiwg...	100%	bsiwg...	02:31...	02:31...	rwm2	bsiwg...
✓	7	Wg5 ...	Work...	172.2...	bsiwg...	100%	Remote	02:32...	02:32...		bsiwg...
✓	6	nc2_s...	Playb...	DEF10	bsiwg...	100%	bsiwg...	02:32...	02:32...	ss2	bsiwg...
✓	8	Wg5 ...	Work...	172.2...	bsiwg...	100%	Remote	02:32...	02:32...		bsiwg...
✓	9	Wg5 ...	Work...	172.2...	bsiwg...	100%	Remote	02:32...	02:33...		bsiwg...
✓	10	Wg5 ...	Work...	172.2...	bsiwg...	100%	Remote	02:33...	02:33...		bsiwg...
✓	11	Wg5 ...	Work...	172.2...	bsiwg...	100%	Remote	02:33...	02:34...		bsiwg...
✓	12	Wg5 ...	Work...	172.2...	bsiwg...	100%	Remote	02:34...	02:34...		bsiwg...
✓	13	Wg5 ...	Work...	172.2...	bsiwg...	100%	Remote	02:34...	02:36...		bsiwg...
✓	14	Wg5 ...	Work...	172.2...	bsiwg...	100%	Remote	02:36...	02:36...		bsiwg...
✓	15	Wg5 ...	Work...	172.2...	bsiwg...	100%	Remote	02:36...	02:37...		bsiwg...

For information on Interplay Transfer, see the *Interplay Transfer Setup and User's Guide*.



Interplay-to-OMF transfers are not supported.

Avid Instinct

Avid Instinct™ is a story creation tool that introduces a story line designed to support the way that journalists create a story. Avid Instinct, an iNEWS client, is a script-based application that is tightly integrated with iNEWS. It uses iNEWS metadata for the story that is being written.

Instinct supports a subset of the Avid editing features, including the following:

- Recording voice-over to the story line
- Simple searches
- Edit while capture
- Send to Playback

Instinct also incorporates the following new features that support the Interplay Production environment:

- Support for MultiRez workflow
- Access to iNEWS stories and Interplay Window assets eliminates the need to copy media and story information to a local system
- Integrated asset management, providing search, query, and organization capabilities with the Interplay Window.

For more information, see the *Avid Instinct User's Guide*.

Interplay | Production Services Providers

The Interplay Production Services application supports the following services:

- Interplay Transcode allows you to transcode clips from one Avid-supported resolution to another. For example, you can create a low-resolution version of a sequence or master clip. You can perform transcode operations from within the Avid editing application or from within Interplay Access.
- The Copy service allows you to copy media and metadata between different Interplay Workgroups.
- The Move service allows you to move media and metadata between different workspaces on the same Avid ISIS system.
- The STP Encode service allows you off load time-consuming processing involved in exporting and transferring of Long GOP OP1a media, during a send-to-playback request from the Avid editing application or Avid Interplay Assist. See [“STP Encode Workflow: Background Processing and Transferring of Long GOP Media”](#) on page 116.
- The Delivery service lets you perform workgroup-to-workgroup transfers of shotlists, subclips, master clips, cuts-only sequences, and their media files. Interplay Delivery Receiver is also a service that is part of Interplay Production Services. However, it runs as a Windows service.
- The Interplay Archive service and the Interplay Restore service. See [“Archive and Restore Services”](#) on page 21.

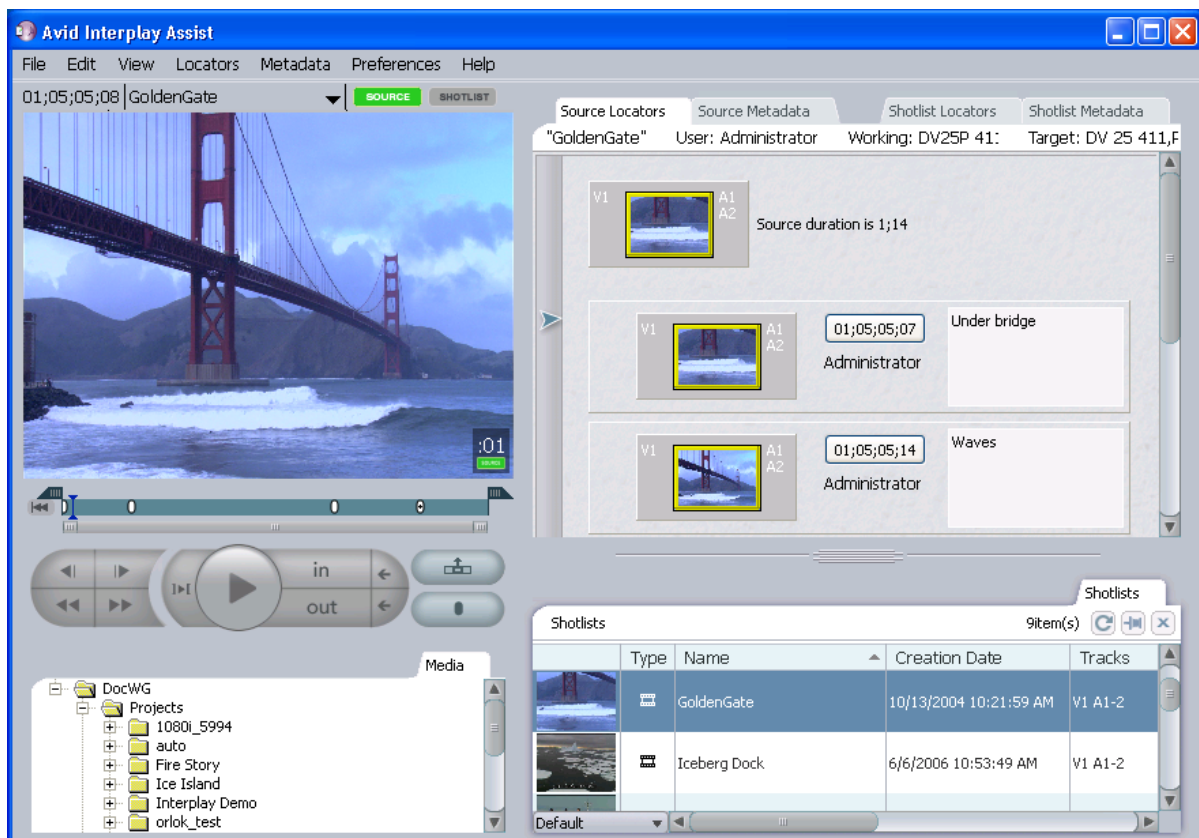
For more information, see the *Interplay Production Services Setup and User's Guide*. For hardware configuration information, see the *Interplay Software Installation and Configuration Guide*.

Interplay | Assist

Interplay Assist allows you to perform the following tasks:

- Search the Interplay Engine database
- Play media
- Add locators and locator information
- Select, view, and create shotlists
- Send to playback
- Archive

The following illustration shows Interplay Assist.



You can also use Interplay Assist to quickly preview clips. All resolutions supported by the Avid editing applications are supported.

Interplay Assist also provides the following additional features:

- Restrictions — Support for a restriction marker indicating that a range of media being used is somehow restricted for use. These markers do not prevent usage, but alert the user that a restriction is in place. This feature is also referred to as Digital Rights Management (DRM).
- Annotations — labels you can attach to locators. Annotations support special markers such as the following:
 - Restricted flag
 - Best Picture
 - Best Sound
 - User-created custom annotation markers
- Metadata — textual data you can use to identify and describe the creation, contents, and disposition of a clip or shotlist
- Searching across multiple logs
- Log filtering

For more information, see the *Interplay Assist User's Guide*.

iNews and Avid Instinct

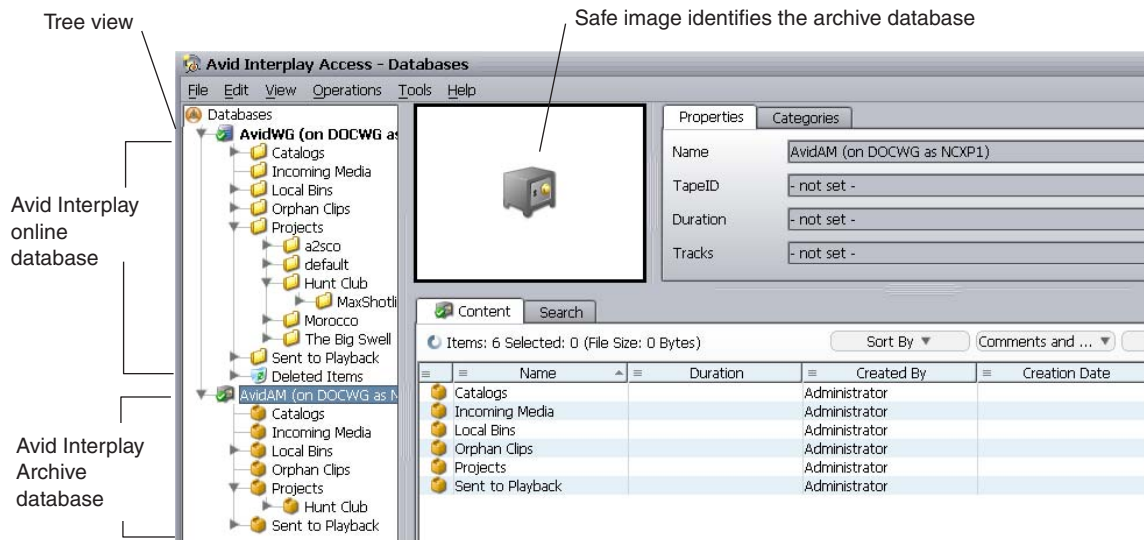
The Avid iNEWS newsroom computer system (NRCS) is a powerful tool for developing news, rundowns, and stories. Avid Instinct allows producers, writers, and journalists to write Avid iNEWS stories and combine them with simple audio and video sequences that are available through the Interplay Engine.

For additional information, see the *Avid Instinct User's Guide*.

Archive and Restore Services

An Interplay Production environment supports a separate asset manager known as the Interplay Archive Engine to track archived assets. The basic features and functionality that are used to track assets in the online storage system, including MultiRez support, are also available for the archived assets. The Interplay Archive Engine also supports partial file restore of Avid MXF media in the archive. A *partial restore* refers to a situation where there might be a large master clip in the archive but your sequence only uses a small portion of the master clip. A partial restore restores only the portion of the file that you need.

The following illustration shows an archive database displayed in Interplay Access.



The Interplay Production Services application supports Archive and Restore services that manage the process of moving data. You can access the Archive and Restore operations from within the Avid editing applications and from within Interplay Access.

You can send high-resolution media to archive and keep a low-resolution version of the asset in the online storage system. This allows you to edit with the low-res asset and conform a “partial” file of the high-res asset from the archive.

The Auto Archive feature includes the following:

- An auto-archive folder that triggers an archive of any material placed in the folder
- Duplication of archive items automatically prevented (default option)
- Ability to append metadata and logging information to archived material

For additional information, see the *Interplay Production Services Setup and User's Guide*.

MediaManager Archive Solutions

Starting at Interplay v1.1.4, you can migrate MediaManager v4.5.x (also known as Workgroup 4) archived assets to an Interplay Archive environment. See your Avid representative for details on performing the migration. See [“Archive Migration \(Workgroup 4\)” on page 266](#).

MediaManager to Interplay Migration Tool

You can migrate much of your MediaManager environment metadata to an Interplay Production workgroup with the Migration tool. You can migrate clips, sequences, and effects from projects, catalogs, workspaces, and users. You can also migrate MediaManager Archive metadata to an Interplay Archive system. For more information, see your Avid representative.

Interplay | Media Indexer

The Interplay Media Indexer is a background service that keeps track of the media files in storage locations that you identify. In an Interplay Production environment, each Avid editing application has its own local Interplay Media Indexer that manages the local storage areas on that system. Each Interplay Production environment also has at least one Interplay Media Indexer that manages media on shared storage. When the Avid editing application needs to access media on shared storage, it communicates with its local Interplay Media Indexer, which then communicates with the shared storage Interplay Media Indexer. For more information, see [“Working with Interplay | Media Indexers v3.5 and Later” on page 210](#).

Service Framework Services

Avid Service Framework is a common service distribution framework, which enables the configuration and monitoring of Avid services and applications in an Interplay environment.

Avid Service Framework consists of applications and services that provide dynamic discovery and lookup mechanisms, a scalable notification system, a centralized configuration tool, diagnostic logging for workgroup-enabled applications and services, system health monitoring, and so on. Some of the services are known collectively as Support services. The Support services are viewed, monitored, and configured using the Service Framework applications.

The Avid Service Framework provides the following tools for Interplay Production:

- Scalable notification mechanism
- Configuration tool for centralizing configuration tasks
- Diagnostic Logging application for consistent logging across all Interplay applications
- Health and Monitoring application

Interplay | Capture

Interplay Capture is intended for customers who require a management system for video capture. This application is a key part of the MultiRez workflow. Interplay Capture stores its schedule and other metadata in the Interplay database, which allows for easy access to the captured material within the Interplay work environment.

For information on Interplay Capture, see the *Interplay Capture Administration Guide*, the *Interplay Capture User's Guide*, and the *Interplay Capture Install Guide*.

Access Control and Media Deletion Model

One of the most essential aspects of asset management is developing a system by which assets and their associated media files (for example, video and audio files) can be automatically and accurately protected from normal deletion. The administrator can specify the delete permissions that users and groups have on particular folders. Administrators and users can also use folder reservations to protect media files.

- A reserved folder protects any media files referred to by the assets in that folder.
- Folder reservations can have an expiration date so they don't needlessly protect media files after they are inactive.
- Administrators and file owners can override folder reservations.

For more information, see the *Interplay Access User's Guide* and [“Developing an Interplay Delete Strategy” on page 38.](#)

Interplay Application Icons

This section describes the Interplay Production icons.

Interplay | Access, Administrator, Instinct, and Assist

The Interplay Access icon has a purple background to signify that it is the main entry point that most people will use to work with the Interplay Production database. The Interplay Administrator has a similar look but with a key to denote the administrator. The Assist icon represents adding metadata to a locator track in Assist. The Avid Instinct icon is based on the iNEWS icon that has the same script “i” with a circle around the character.



Interplay Access



Interplay Administrator



Avid Instinct



Interplay Assist

Avid Service Framework

The Avid Service Framework icons use a tools-based hex nut design in all four icons.



Avid Health Monitor



Service Configuration



Avid Diagnostics



Workgroup Properties

Interplay | Production Services

The Interplay Production Services Engine icon represents a controller or Broker controlling the Production Services jobs. The image of the curved line and connecting dots is repeated in several of the icons. In the Stream Publish icon the image represents the fact that the service continually watches for files to publish. Delivery represents the files being delivered across the world to different Interplay environments. The Copy and Move icons also use the curved line to represent the fact that Copy works on one ISIS system (between workspaces) while Move works between two ISIS systems.

The Interplay Transcode and STP Encode icons use broken lines and arrows to indicate that the application processes the media. Interplay Archive and Restore use the metaphor of moving media to and from an archive.



Interplay Production Services Engine



Interplay Archive



Interplay Transcode



Interplay Delivery



Interplay Copy



Interplay Move



Interplay STP Encode



Interplay Restore

Interplay | Transfer

The Interplay Transfer icons share a double arrow that represents transfers. In the case of FTP Clips List and FTP Media Browse, the arrows show that the application only transfers in one direction.



Interplay Transfer



Interplay FTP Clip List



Interplay Production Services and Transfer Status



Interplay FTP Media Browse



Interplay Transfer Configuration

2 Setting Up a Folder Structure and a User Database

This chapter describes the process of setting up a user database for your Interplay Production environment. This chapter includes the following topics:

- [Permissions and Folder Structures](#)
- [Setting Up a Folder Structure in Interplay | Access](#)
- [Basic Setup: Assigning Access to User Groups](#)
- [Assigning Database Access on a Project Basis](#)
- [Defining the Editor Database Settings](#)
- [Managing the Instinct/Assist User Settings](#)
- [Creating Folders for Capturing Media](#)

Permissions and Folder Structures

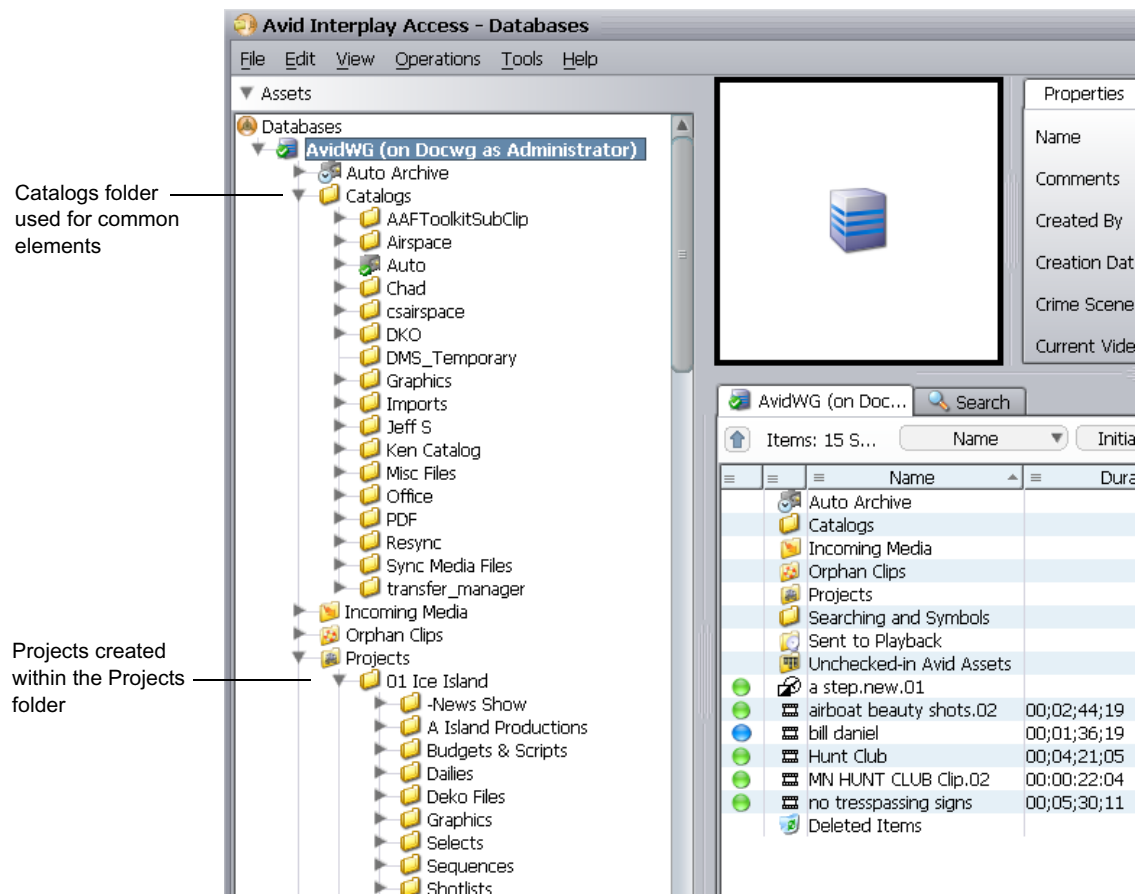
The permissions that you assign to users in your workgroup define which folders users have access to. For example:

- At a small site, you might have a small group of editors and graphic artists that should always have Read/Write/Delete privileges on all files. You might also have a group of assistants that should have Read/Write privileges. In this case you can set up two user groups that have different sets of privileges. The access privileges you assign to those groups apply across the entire database. You can also change access for particular folders when necessary.
- At a larger site, you might have editors and graphic artists that work on different projects. Depending on the workload, they might be assigned to different projects at different times. In this case you might want to avoid confusion and not allow editors on one project to have Write/Delete privileges on projects they are not working on. This is easy to accomplish by creating project groups that have privileges that are inherited by any user that is added to the group. In this scenario, an editor can work on more than one project at the same time and they inherit permissions from both projects.

This chapter describes how to set up both scenarios.

Setting Up a Folder Structure in Interplay | Access

Every project that you work on must have a folder structure. You can create folders within the default folders or create new ones. You can set up a folder structure that complements your workflow and allows people in your Interplay workgroup to easily navigate to their working folders. The following illustration shows folders used by the DocWG database. The administrator uses the default Catalogs folder for common elements that might be used by different projects. A new folder was created in the Projects folder for a documentary named 01 Ice Island. For additional information, see [“Creating Media Composer Projects in an Interplay Environment”](#) on page 48.

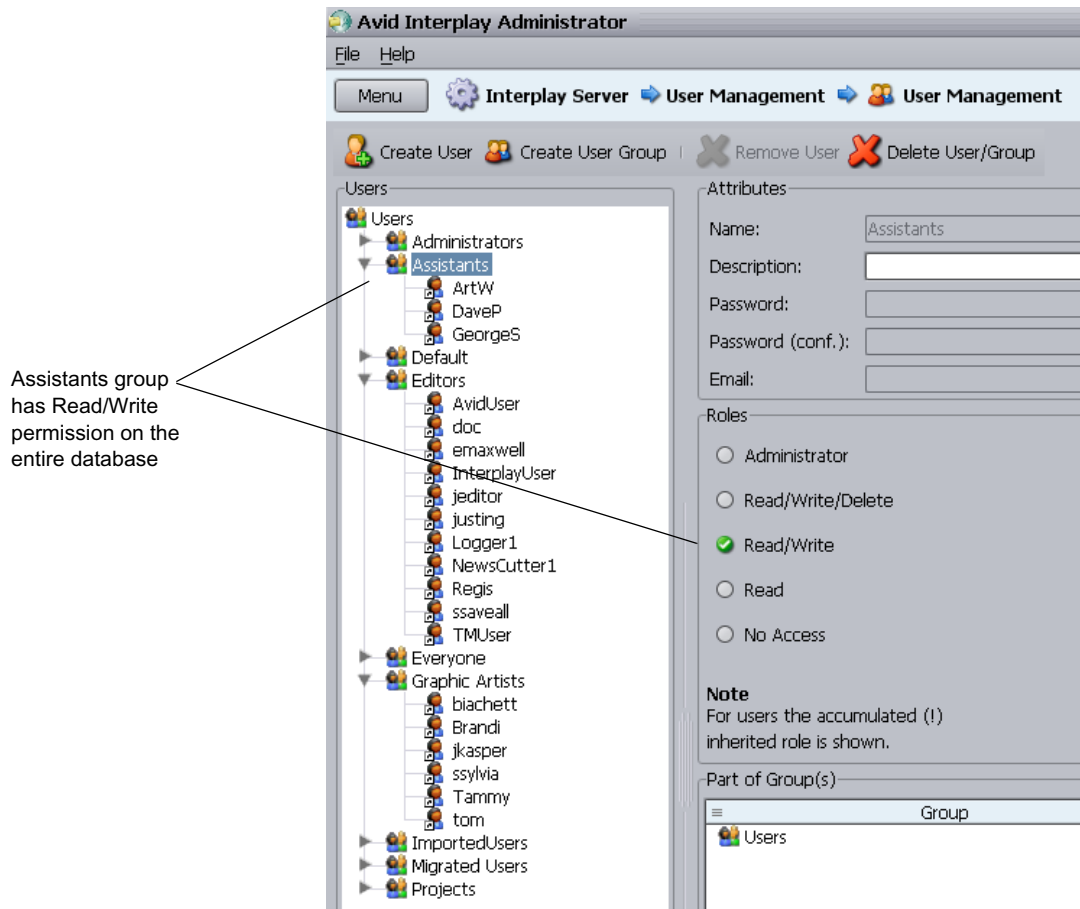


Maximum Number of Characters for Clip Names, Folders, and Files

Avid recommends adhering to a best practice of a 255 character limit for clip names. While it is technically possible for clip names to be longer, folders and files are hard set at this 255 maximum and using it as a guideline may be easier to enforce over time across staff or teams.

Basic Setup: Assigning Access to User Groups

If you work at a relatively small site where all of the editors have access to all of the current projects, then you can set user permissions by user groups. For example, in the following illustration, the Editors group and the Graphics Artists groups have Read/Write/Delete privileges for the entire database. The Assistants group has Read/Write access to the database. You set up the permissions for each group in the User Management View of the Interplay Administrator window.



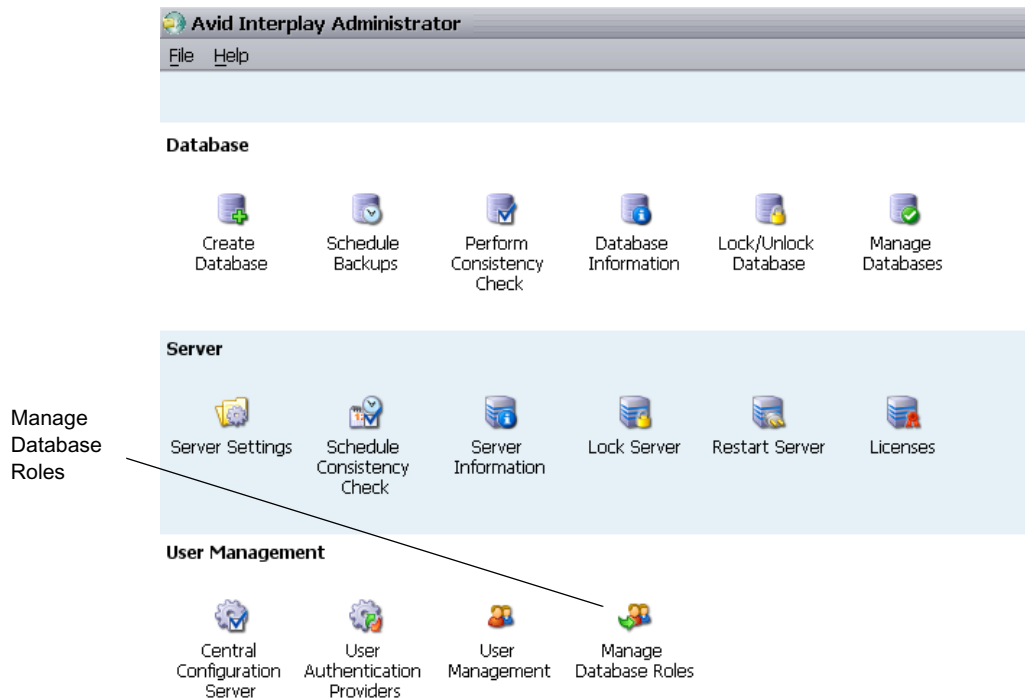
For more information, see the *Interplay Engine and Interplay Archive Engine Administration Guide*.

Assigning Access Rights to User Groups on a Folder Basis

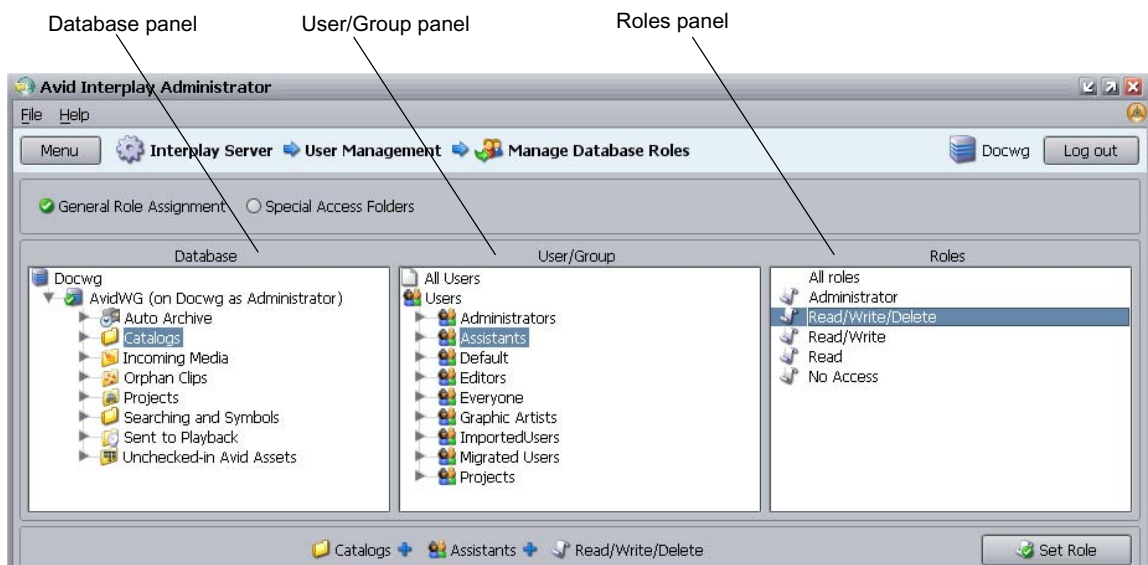
You can use the Manage Database Roles view of the Interplay Administrator window to set permissions on specific folders for user groups. In this example, the Assistants group has Read/Write access to the entire database but you would like to give them Read/Write/Delete permission to the Catalogs folder.

To use the Manage Database Roles window to assign permissions for a group:

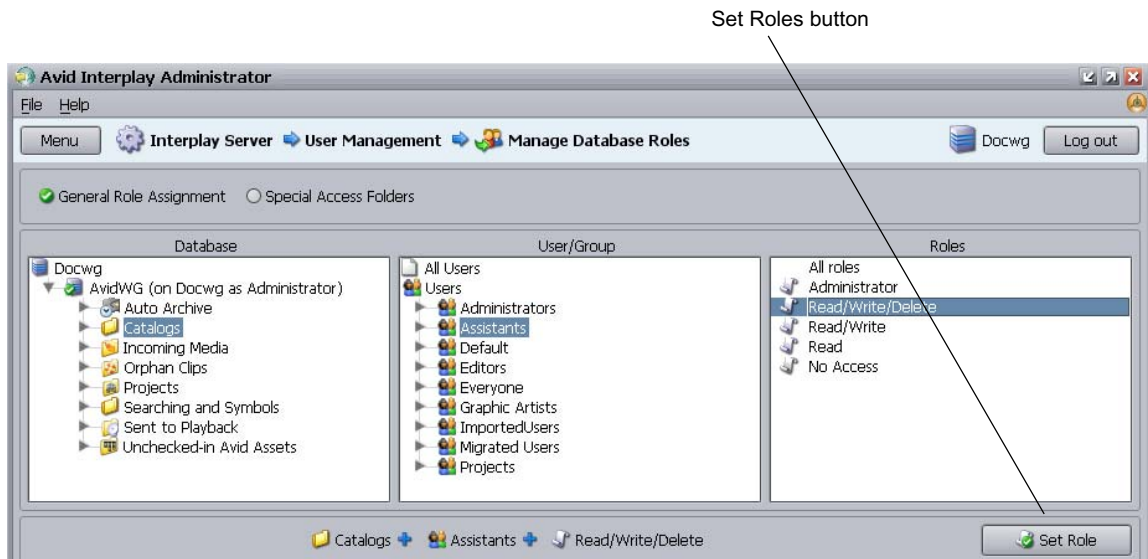
1. Click Manage Database Roles in the Interplay Administrator tool.



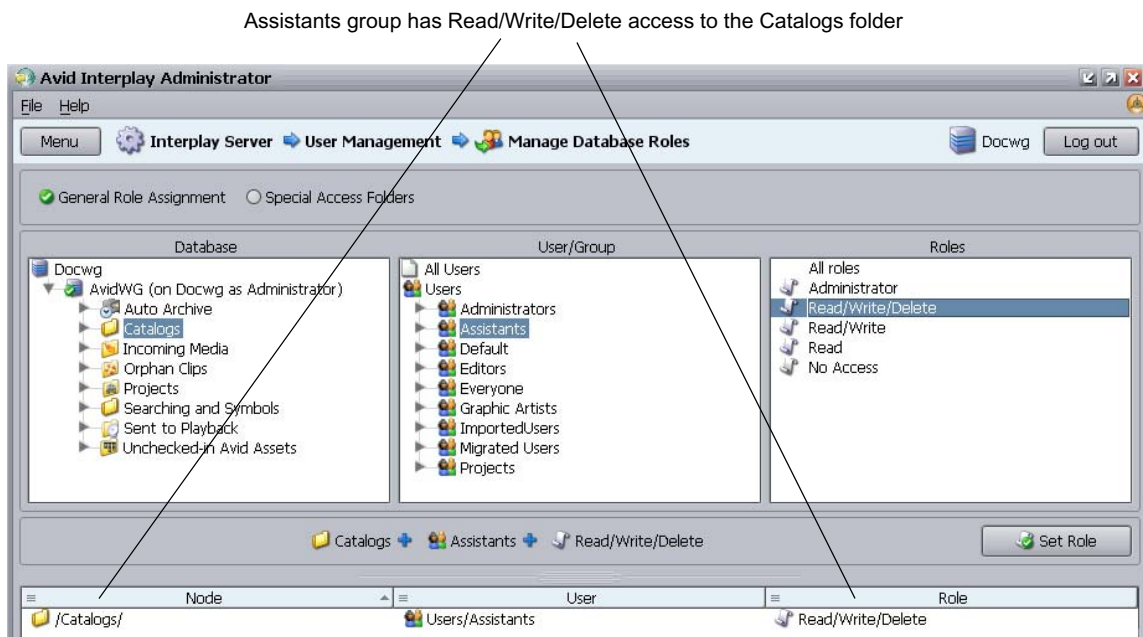
The Manage Database Roles window opens.



2. Select the folder that you want to assign permission to in the Database panel.
3. Select the folder containing the group in the User/Group panel and select the permissions in the Roles panel.
4. Click the Set Roles button.



The permissions for that user group appear in the bottom panel as shown in the following illustration.



Assigning Database Access on a Project Basis

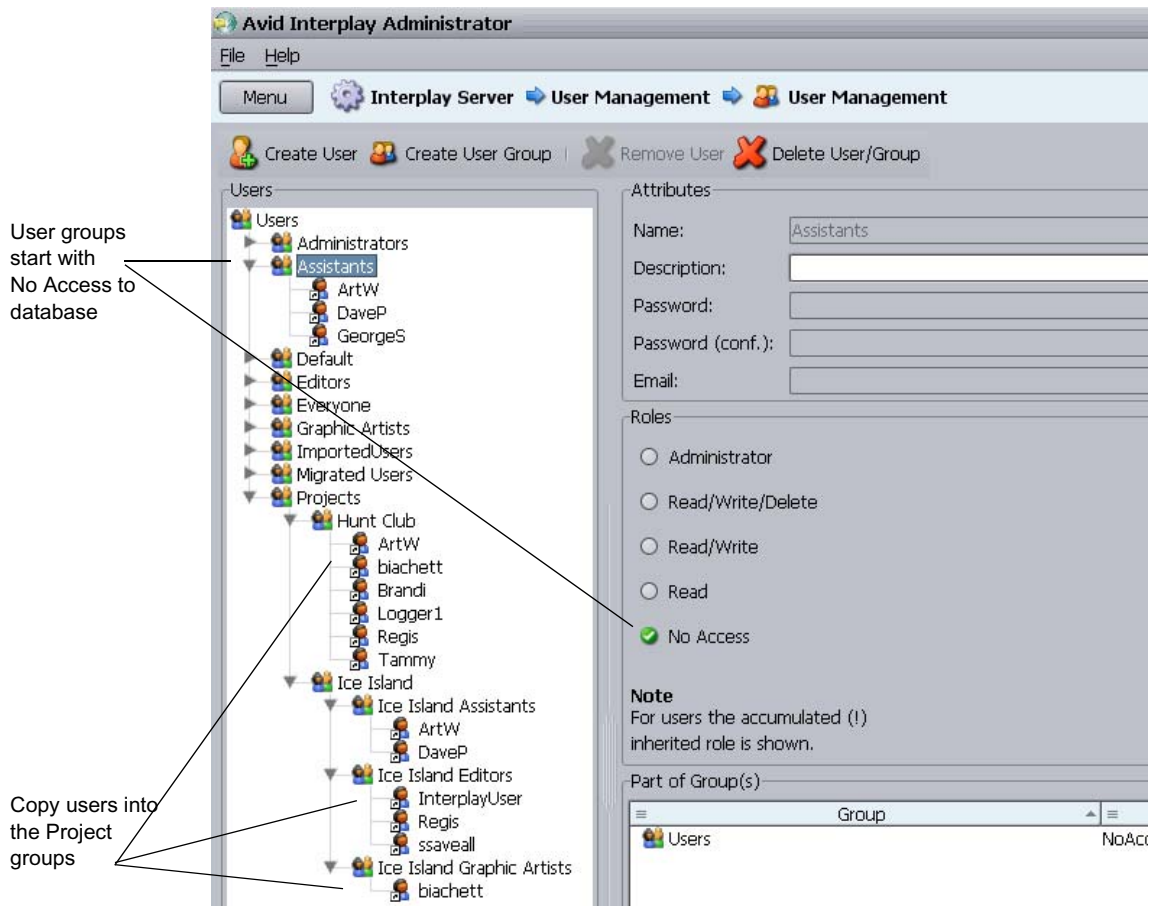
It is often useful to prevent users on one project from modifying or deleting material used by other projects. There are several ways to achieve this type of structure and the following guidelines show one way to accomplish this:

- Create user groups (for example, Editors, Graphic Artists, and Assistants) and give each user group “No Access” (or perhaps just Read-only access) to the database. Then add your users to the groups.
- Create another user group for Projects and assign either No Access or Read only access to that group. Then create folders within that group for each project.
- Assign permissions to the folders within the Projects folder.
- Copy your users into the folders in the Projects folder. A single user can be copied to more than one Project folder.

This section covers each of these topics.

The following illustration shows the user structure for two sample projects:

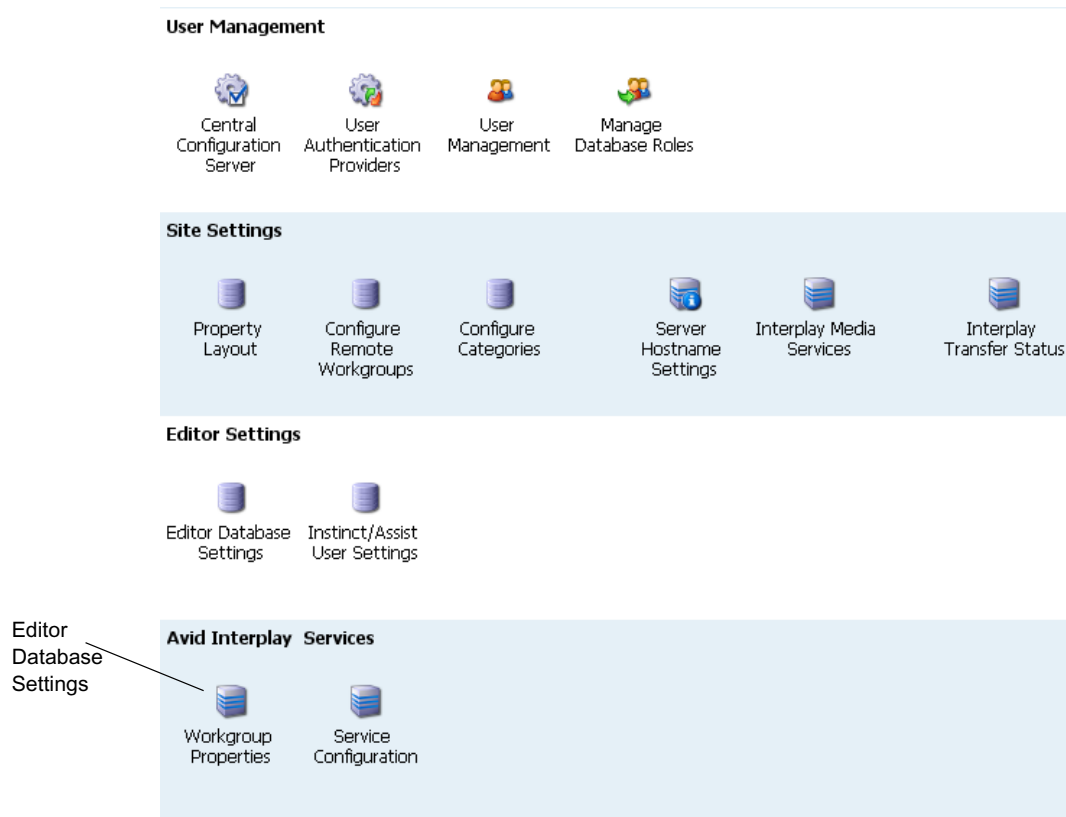
- The Hunt Club project group uses a one-level hierarchy. All users in the Hunt Club group will be given Read/Write/Delete access to all the files in the Hunt Club folder.
- The Ice Island project group shows a more complex structure where users are copied to different folders within the Project Folder. In this model you can assign different permissions to the different user groups within the Ice Island project.



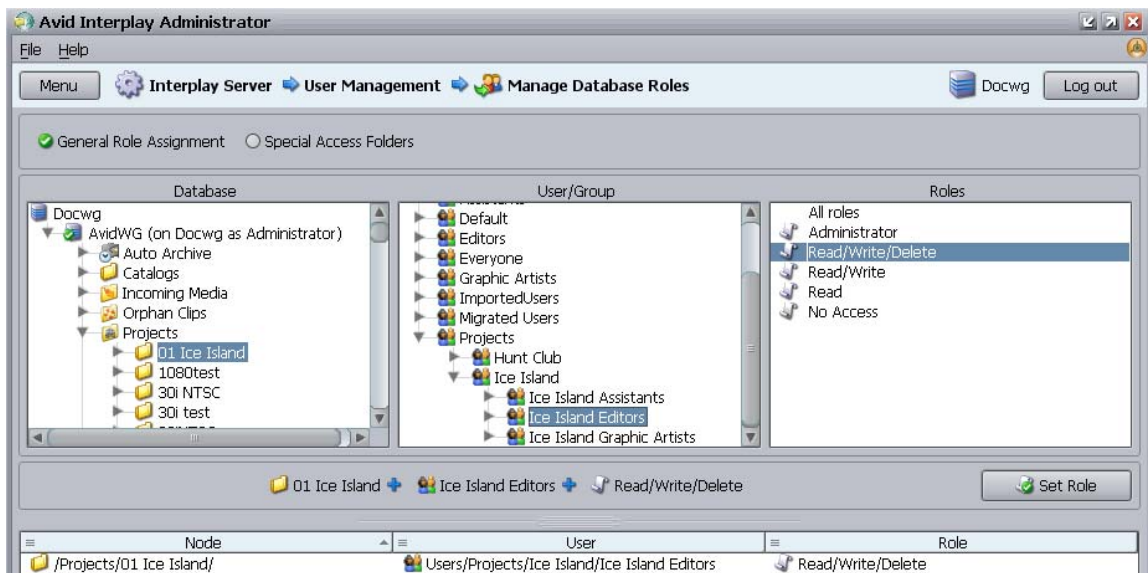
Users can belong to multiple groups. The user inherits the privileges from both groups. In this example we initially give no access to Project groups. Use the Manage Database Roles window to assign the permissions for each group.

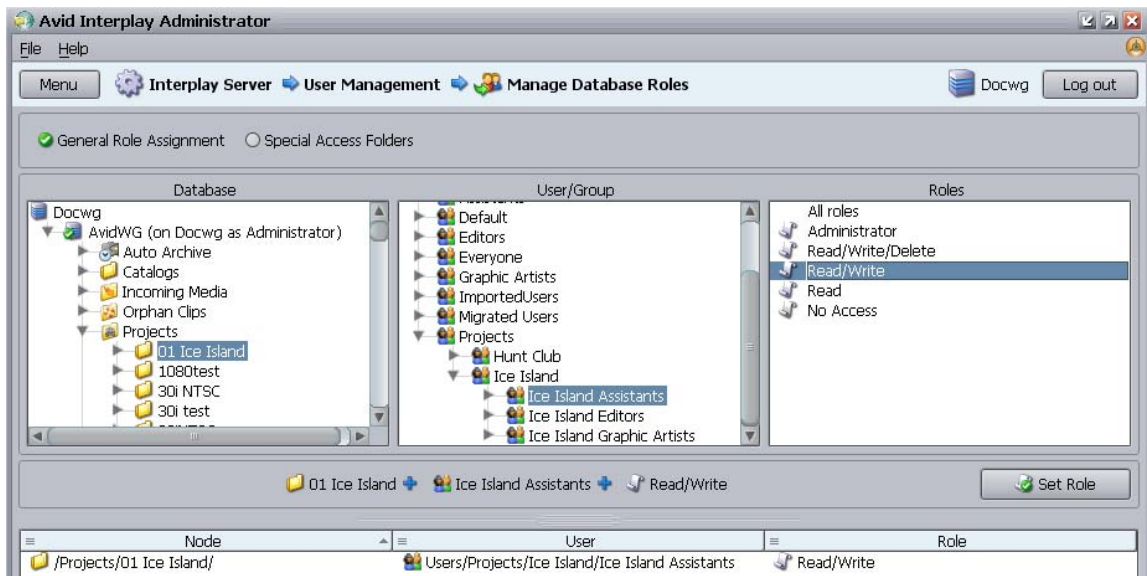
Managing Database Roles

In this example, users and groups initially inherit their Read/Write/Delete permissions from the folder they are working in. You use the Manage Database Roles window to assign these permissions. The following illustration shows the Manage Database Roles button in the Interplay Administrator window.



The following illustrations shows the permissions for the Ice Island Editors and Ice Island Assistants groups. Any user copied to these folders inherits these permissions when he or she is working in the 01 Ice Island folder.





For details on assigning permissions, see [“Assigning Access Rights to User Groups on a Folder Basis” on page 29](#).

Defining the Editor Database Settings

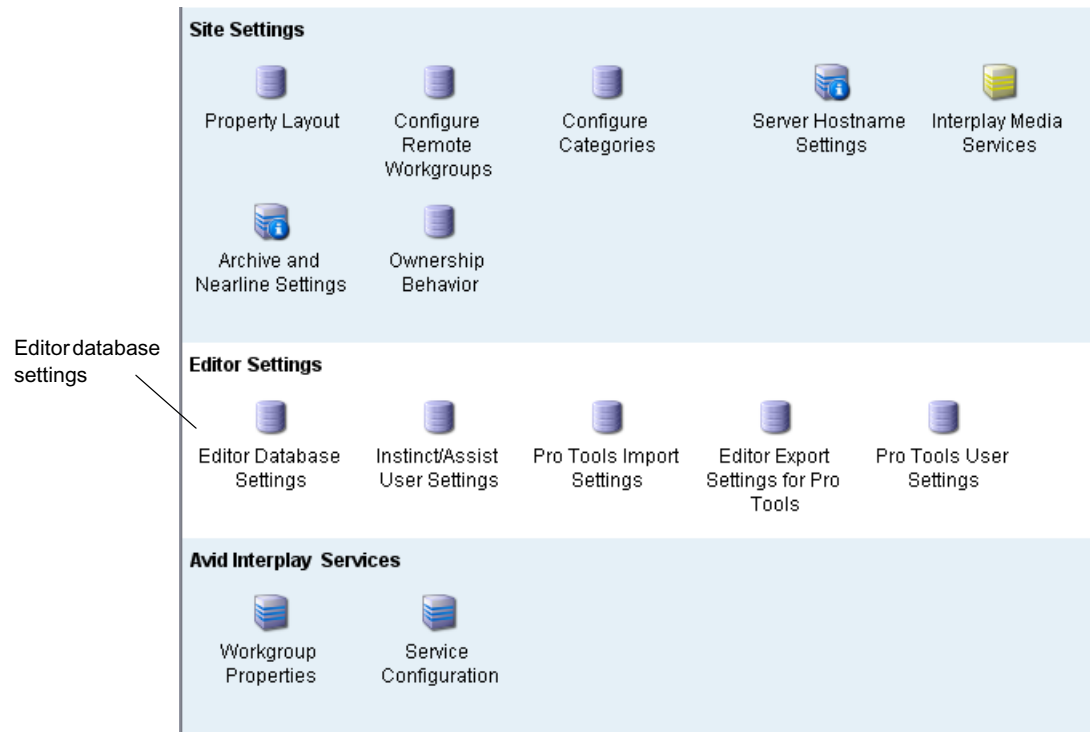
After you set up your folder and user structure, you use the Editor Database Settings window to define the following settings:

- Automatic Checkin and Auto Reservation settings. These settings apply to all editing applications, including Media Composer, Symphony, and NewsCutter Adrenaline, and NewsCutter XP.
- Project-related settings for Avid Instinct and Interplay Assist such as target resolution and audio settings.

For additional information on the Editor Database Settings, see the *Interplay Engine and Interplay Archive Engine Administration Guide*.

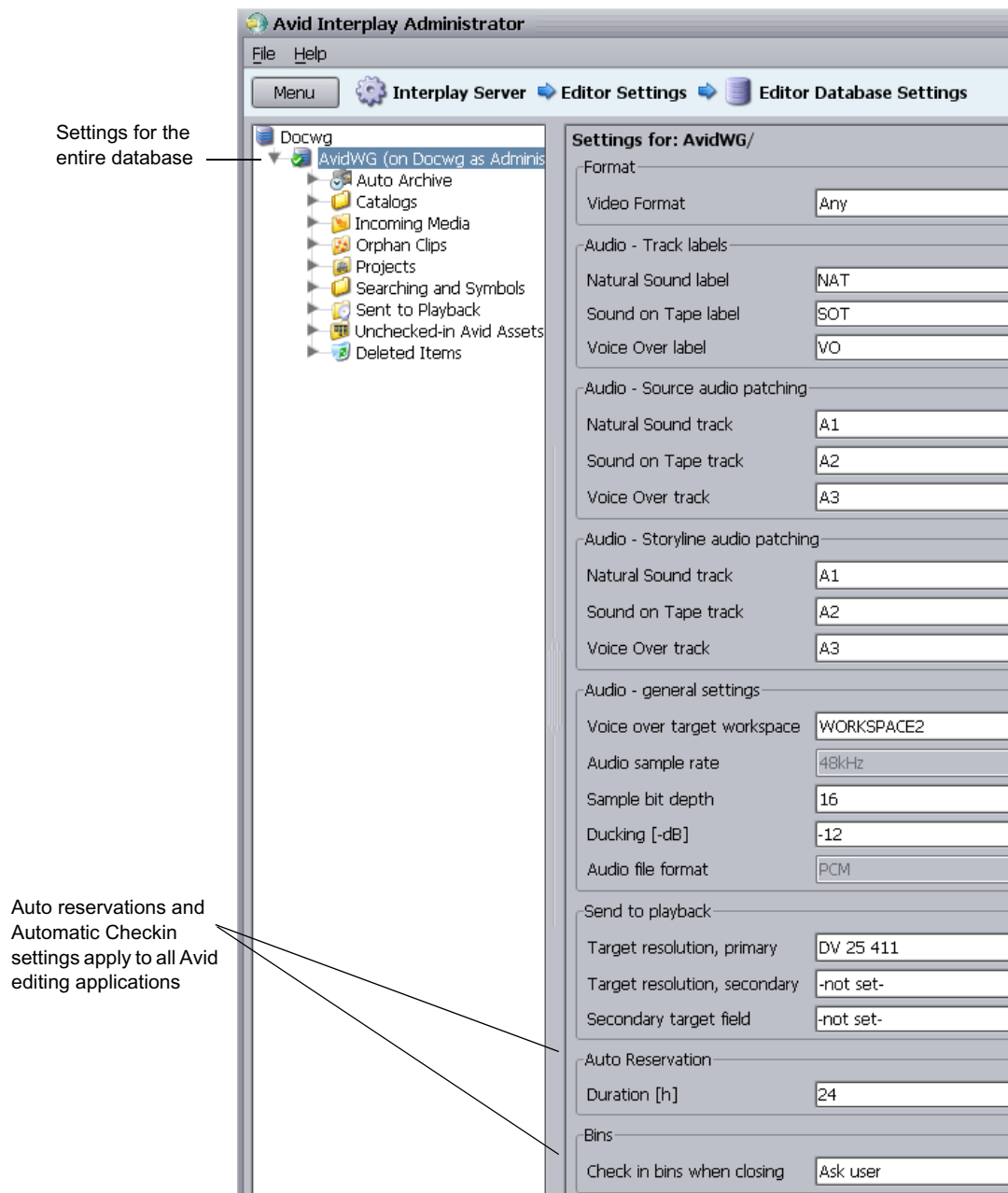
To set the Editor Database settings:

1. Double-click Editor Database Settings in the Interplay Administrator window.



The Editor Database Settings window opens.

2. Click the root node for the database and define the default settings for the entire database.



3. Click the Apply button in the lower right of the window. The settings apply to all folders in the hierarchy.
4. (Option) Select a folder and define the settings for the folder.

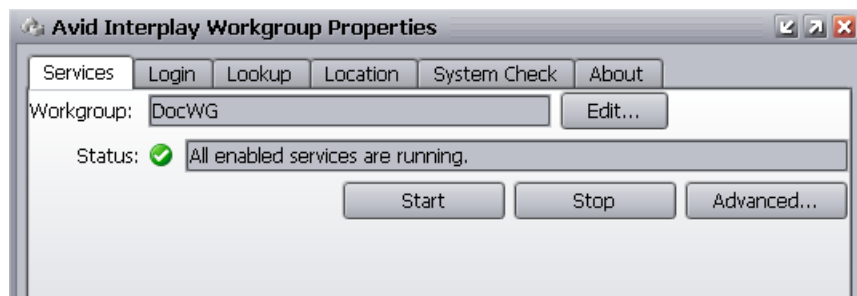
Managing the Instinct/Assist User Settings

If you have Avid Instinct or Interplay Assist users in your Interplay environment, an administrator must select options in the Interplay Administrator window:

- Set the following in the Instinct/Assist User Settings window:
 - The working resolution for each Avid Instinct or Assist user. For this release, this value must be for each user individually.
 - Access settings that allow users to perform operations such as adding locators and restrictions, creating shotlists, and sending to playback. These can be set on a user basis or on a group basis.
- Set the project-related settings, such as target resolution, and audio settings, in the Editor Database Settings window. See [“Defining the Editor Database Settings” on page 34](#).
- Set the workgroup name in the Server Hostname Settings window. The Avid Instinct and Interplay Assist clients use this setting to locate the Interplay Media Indexer that monitors shared storage.



The workgroup name you enter in the Server Hostname Settings window must match the Workgroup name used by the Service Framework. The name must match exactly and it is case sensitive. You can locate and set the name in the Interplay Workgroup Properties window. The following illustration shows the Interplay Workgroup Properties window.



For more information, see the *Interplay Engine and Interplay Archive Engine Administration Guide*.

Creating Folders for Capturing Media

When you are setting up your folder structure, you can determine how you want to set up the folders that you will use for temporary storage of captured material. For additional information, see [“Interplay Database Folder Structure” on page 43](#).

To ensure the best performance with the Interplay Engine, you should not store more than 20,000 files in a single folder. For example, if you are going to use the Incoming Media folder for captures, perform periodic deletions of files you no longer need in that folder. If you know that you will have recurring captures, you can set up folders for those captures. For example, you could name folders for the days of the week or for specific types of captures.

3 Developing an Interplay Delete Strategy

The following topics describe how to develop a workflow for deleting material from your Interplay system:

- [Introduction to Interplay Deletion](#)
- [Deletion Guidelines](#)
- [Interplay Database Folder Structure](#)
- [Creating Media Composer Projects in an Interplay Environment](#)
- [Interplay | Production and MediaCentral | UX](#)
- [Project Workflow Samples](#)
- [Example: Deleting Metadata and Media](#)
- [Example: Deleting Metadata Only](#)
- [Determining How Much Space a Delete Operation Will Free Up](#)
- [Searching for Media](#)

The strategies are guidelines and may not be suitable for all workflows. Each site needs to develop its own deletion strategy, according to its own workflows, media throughput, storage capacity, archiving, and media protection policies.

Introduction to Interplay Deletion

Developing a successful strategy for deletion involves the following:

- Become familiar with the default folder structure of the Interplay database.
- Develop a consistent folder structure that complements your site's workflow.
- Develop a consistent way to create (and subsequently delete) projects among the Media Composer systems and among the users that capture, transcode, or otherwise create media stored in the Interplay database.
- Become familiar with the Interplay Production documentation that describes deletion features and settings. In particular, refer to the following:
 - "Deleting Assets" in the *Interplay | Access User's Guide*.
 - "Setting Options for Deletion" in the *Interplay | Engine and Interplay | Archive Engine Administration Guide*.
- Become familiar with the Interplay Production scheduled deletion feature and its options.
- Determine a regular schedule for deleting material from the following locations:
 - Interplay Production database
 - Media Composer systems

- Develop a consistent way of preserving assets and media from unwanted deletion
- Become familiar with the Interplay Access Extended Search feature to assist you in locating material that should be deleted.

Use Interplay Access when deleting assets from the Interplay database and deleting the associated files from shared storage. Avid strongly recommends that you do not delete files directly from the file system because the relationships between the clip metadata and the specific files that make up the clip are not obvious from the file system. Those relationships are managed by the Media Composer systems and the Interplay Engine. Interplay Access is the best tool for managing the files in the Interplay Production database.



For information on deleting the material associated with AMA media, see “Deleting AMA Clips Using Interplay | Access” on page 101.

Deletion Guidelines

The following sections provide general information you should keep in mind when developing an Interplay Production deletion strategy:

- [Do Not Perform Large Deletions During Prime Time](#)
- [Create a Custom Deletion Schedule](#)
- [Do Not Delete Items from the Orphan Clips Folder During a Scheduled Deletion](#)
- [Use Reservations](#)
- [Other Strategies to Protect Assets and Media](#)
- [Delete Sequences When Possible](#)
- [Actions to Avoid on Shared Storage](#)
- [Do Not Delete More Than 2,000 Assets at a Time](#)
- [Folder Limitations](#)
- [Deleting By Workspace](#)
- [Perform Deletions from One Interplay | Access Client](#)
- [Guideline for Number of Deletions Per Hour](#)
- [Priorities with Deletion](#)
- [Amount of Space Freed Up by Deletion](#)

Do Not Perform Large Deletions During Prime Time

Perform deletions during times when the system is not heavily used. A large delete operation can slow down the response time on the Interplay Engine. This also slows down the response times on Media Composer applications and other Interplay clients.

Deletion consists of three parts: deletion of links, deletion of media, and deletion of database assets.

- In Interplay v2.3 and earlier, deletion was *synchronous*: links, media, and database assets were all deleted within the same interactive delete operation. Disadvantages to synchronous deletion were the impact on the engine (particularly caused by the deletion of database assets) and the length of time required for the deletion to take place. Starting with Interplay Engine v2.4,

deletion is *asynchronous*: the deletion of database assets is disassociated from deletion of links and media and can be scheduled for off hours (see the next section, “Create a Custom Deletion Schedule.”) Scheduling reduces the performance impact of the initial deletion.

- Deletion of media impacts Interplay Media Indexer performance. For each file that the Interplay Engine deletes on the share storage system, a notification is sent to Interplay Media Indexer. These notifications cause Interplay Media Indexer to update its internal tables.

Create a Custom Deletion Schedule

By default, a 24-hour time span (00:00 to 24:00) is specified for deletion. Avid recommends that you create a custom deletion schedule during times of low activity. This chapter includes an example of scheduled deletion. See [“Example: Deleting Metadata and Media” on page 54](#).

There are other default settings in the Delete Behavior dialog box in the Interplay Administrator that you need to review. For example, you can delete both assets and media by selecting the option “Permanently delete assets and associated online media from the Orphan Clips folder.” This chapter describes workflows when this option is selected and when it is not. By default, this option is not selected. See [“Example: Deleting Metadata and Media” on page 54](#) and [“Example: Deleting Metadata Only” on page 64](#).

Do Not Delete Items from the Orphan Clips Folder During a Scheduled Deletion

The Orphan Clips folder plays an important role in the Interplay Production deletion process. (see [“Interplay Database Folder Structure” on page 43](#)). The top level of the Orphan Clips folder holds assets whose links have been deleted from all other folders, but whose database assets have not yet been deleted. Assets remain visible in the Orphan Clips folder until they are deleted through a scheduled deletion.

If necessary, you can manually delete assets outside of a scheduled deletion time span. For example, if deletion is scheduled for 1 a.m. to 3 a.m., you can delete assets at any other time. Deleting assets during a scheduled deletion time span can cause various problems.



By default, a 24-hour time span (00:00 to 24:00) is specified for deletion. If you have not set a custom deletion schedule, do not manually delete assets in the top level of the Orphan Clips folder.

The Kept Media subfolder has a different use. It holds assets whose media has not been deleted because the option “Permanently delete assets and associated online media from the Orphan Clips folder” is not selected. These assets are not deleted through the schedule deletion process, so you must review and manually delete these assets at a time that suits your schedule.

Use Reservations

Reservations are an excellent method for protecting media and metadata from deletion. Reservations work on folders, not on individual assets. So, to protect an asset from deletion it must be in a reserved folder.

A special case is that sequences and subclips in reserved folders protect the media files of referenced master clips, even if the master clips are in a non-reserved folder. For example, Sequence A includes Clip 1. Sequence A is in a reserved folder, and Clip 1 is in a different, non-reserved folder. If you try to delete Clip 1 and its associated media, you can delete the master clip, but the media remains protected, and the media files are shown as reserved in the Delete Status dialog box.

This protection occurs because a folder that contains a complex asset like a sequence (in this example, Sequence A) also contains referenced assets, which are links for the assets used by the complex asset. (You can view referenced assets in Interplay Access, depending on the Filter Settings). In the case of master clips (Clip 1 in the example) or rendered effects, reserving referenced assets also reserves their media.

Use the following guidelines when using reservations:

- Do not reserve the top level folders with an expiration date of “never.” This protects all of the assets in the top level folder and all subfolders. This effectively prevents you from deleting any assets in those folders.
- Reserve special items such as templates, frequently used graphics, and station logos that are used on a daily basis with “never.”
- Reserve only folders that need to be reserved, especially when auto-reservation is on. By default, auto-reservation extends the reservation for an additional 24 hours from the time that a new or modified Avid asset is checked in from a Media Composer application.

For details on using reservations to protect media from deletion, see “Setting Reservations” in the *Interplay Access User’s Guide*.

Other Strategies to Protect Assets and Media

In addition to reservations, Interplay Production offers other strategies to prevent unwanted deletion of assets and media:

- User roles that limit deletion to particular groups and users. See “Assigning User Roles” in the *Interplay / Access User’s Guide*.
- Freezing a folder. See “Freezing Folders and Content” in the *Interplay / Access User’s Guide*.
- Starting with Interplay v2.3, you cannot delete a referenced asset unless you also delete the asset that references it. See “Restricted Deletion of Referenced Assets” in the *Interplay / Access User’s Guide*.
- Delay the scheduled deletion

The Delete Behavior view in the Interplay Administrator includes an option to delay the scheduled deletion by a specified number of days. This option gives you the opportunity to review the contents of the Orphan Clips folder before the scheduled deletion. This option protects assets (metadata) but not media, unless a “metadata only” strategy is used (see [“Example: Deleting Metadata Only” on page 64](#)). See “Scheduling the Deletion of Database Assets” in the *Interplay / Engine and Interplay / Archive Engine Administration Guide*.

Delete Sequences When Possible

If possible workflows should include deletion of sequences and their referenced assets. By default you cannot delete referenced assets unless you delete the asset that references them. However, you can overwrite this behavior. See “Activating the Option to Delete Referenced Assets” in the *Interplay / Engine and Interplay / Archive Engine Administration Guide*.

Actions to Avoid on Shared Storage

- Do not delete material directly from the Avid MediaFiles folders in the shared-storage workspace.
- Do not use the Media Tool on shared-storage workspaces for searching or deleting.
- Do not delete workspaces without first deleting Interplay Production assets and media files from the workspace. Avid ISIS will not let you delete a workspace with any files on it (whether the files are media or any other type of file, even hidden files). Use the following procedure to delete a workspace:
 - Perform an extended search in Interplay Access, using the workspace name as a search criteria. For search examples, see [“Searching for Media” on page 70](#).



Interplay Access v3.2 and earlier clients contained a bug which sometimes led to missing workspaces in assets, resulting in incomplete workspace search results. This was fixed in v3.3 and later, with the latest patches.

- Use Interplay Access to delete the files on the workspace.
- Use the ISIS Administration tool to delete the workspace.

Do Not Delete More Than 2,000 Assets at a Time

Earlier versions of Interplay Production documentation recommended deleting up to 2,000 assets at a time. With the introduction of scheduled deletion, this recommendation is less critical **unless you are deleting media**. If you delete media (with or without its corresponding assets), wait at least 5 minutes in between deletions to give the Interplay Media Indexer time to update its structure.

Folder Limitations

It is important that you do not store more than 20,000 assets in Interplay database folders. In general, you can keep up to 5,000 assets in a folder without affecting system performance. One of the problems with storing many assets in a folder is the amount of time it takes to list the items in the folder when you open the folder for browsing.

Periodically check that folders do not contain more than 5,000 assets, for example, the Kept Media subfolder in the Orphan Clips folder. Delete or move assets to other folders. You can use the delete_kept_media.exe tool to help clean up the Kept Media folder. For more information, see the *Interplay / Engine and Interplay / Archive Engine Administration Guide*.

If a folder must contain a large number of assets it is more efficient to add multiple subfolders for storing the assets. An exception is the Orphan Clips folder.

This limitation also applies to capture folders. If you know that you will have recurring captures, you could set up folders for those captures. For example, you could name folders for the days of the week or for specific types of capture.

Deleting By Workspace

The Delete dialog box includes an option to select which workspace you want to delete media from. If you select media files and resolutions for deletion, all workspaces that contain the selected media are listed and are selected by default. This can help in management of shared-storage capacity.

Perform Deletions from One Interplay | Access Client

Media management and large deletions should only be executed from one Interplay Access client. If you start several clients on different machines and perform large deletions on each, it is actually slower compared with doing all the deletions on one client in a serialized order. Overall performance of the Interplay Engine is also affected.

Guideline for Number of Deletions Per Hour

Under normal load conditions, a scheduled deletion on the Interplay Engine can delete approximately 25,000 assets per hour, and possibly more. This guideline can be useful when determining the length of a scheduled deletion window.

Priorities with Deletion

- By default, a scheduled deletion stops when an Interplay Production backup begins, and resumes when the backup ends or at the next scheduled deletion. You can select an option in the Delete Behavior dialog box in the Interplay Administrator to allow both processes to operate at the same time, but this is not recommended unless deletion slots and backups cannot be separated.
- Ingest operations are not impacted by scheduled deletions because scheduled deletion has a lower priority.

Amount of Space Freed Up by Deletion

The Delete dialog in Interplay Access provides an estimate of the space that will be freed up by a Delete operation. Performing a search in Interplay Access provides a less reliable estimation. For more information, see [“Determining How Much Space a Delete Operation Will Free Up” on page 68](#).

Interplay Database Folder Structure

Once you become familiar with how Interplay uses the various Interplay folders, you can organize and maintain your media more efficiently and start developing your own delete strategy. The following table lists the default folders.

Folder	Function
Catalogs	You can create subfolders in the Catalogs folder that are specific to your overall workflows. For example, you can create folders with site-specific names such as Highlight Reels, and create folders for Auto-archive or Auto-transcode operations. In addition, two types of subfolders might be automatically created: • When ingesting media through an AirSpeed 5000/5500 server, recorded clips can be automatically stored in a Catalogs folder. See “Folders Used by AirSpeed 5000 Software and Interplay Capture” on page 45. • When you resynchronize the Interplay database with media stored on Avid shared-storage workspaces through the Resync command in Interplay Access, the system automatically places a Resync folder in the Catalogs folder. Any assets checked in while the system is performing a Resync are placed in this folder. You can move them to an appropriate location after the Resync is completed. For more information, see “Resyncing Media Files with Avid Shared Storage Workspaces” in the <i>Interplay Access User’s Guide</i> or Help.

Folder	Function
Incoming Media	This folder is the destination for clips recorded by AirSpeed 5000 software or transferred by Interplay Transfer. The folder structure is organized by workspace name. The system does not create the workspace folder until you perform the first transfer or ingest to that workspace. You can then move assets from the Incoming Media folder to another folder. See “Folders Used by AirSpeed 5000 Software and Interplay Capture” on page 45 and “Folders Used by Interplay Transfer Checkins” on page 47.
Orphan Clips	This folder and its subfolders holds several types of assets: - Top level: Links for assets that have been deleted from all other Interplay folders and assets that are orphaned through other means, usually rendered effects replaced by a new version of a sequence. These links are stored until they are completely deleted through a scheduled deletion, moved to Kept Media or Failed to Delete by a scheduled deletion, or deleted manually.  See “Do Not Delete Items from the Orphan Clips Folder During a Scheduled Deletion” on page 40. - Kept Media folder: Links for master clips and rendered effects that have been deleted from Interplay folders but still point to media files on shared storage. Links are moved to this folder only if the option “Permanently delete assets and associated online media from the Orphan Clips folder” is not selected (Delete Behavior view in the Interplay Administrator). These links are not deleted through a scheduled deletion. An administrator needs to manually delete the links and media when the media is no longer needed. This subfolder is created when needed. You can use the delete_kept_media.exe tool to help clean up the Kept Media folder. For more information, see the <i>Interplay Engine and Interplay Archive Engine Administration Guide</i>. - Failed to Delete folder: Links for assets that could not be deleted by the scheduled deletion process (for example, if a database asset is corrupted). An administrator needs to check these links for system problems. This subfolder is created when necessary. - Resync folder: If the Resync process identifies a clip with a missing audio or video track, it creates a special master clip and stores it in this folder. An administrator needs to check these links for system problems. Such problems are not database problems, but rather result from files manually deleted from ISIS shared storage. This subfolder is created when necessary. By default, only administrators can view this folder. An administrator can assign read/write/delete permissions for this folder to users or groups through the Managing Database Roles view in the Interplay Administrator. For more information, see “Orphan Clips (Weekly)” on page 61.
Project	You can create subfolders in the Projects folder that are specific to your overall workflows. Additional subfolders are created automatically: - This is the default folder for Media Composer projects. See “Project and Bin Folders Created by Media Composer” on page 48. - Interplay Capture creates subfolders here. See “Folders Used by AirSpeed 5000 Software and Interplay Capture” on page 45. - AirSpeed 5000 software creates subfolders if configured to create them. See “Folders Used by AirSpeed 5000 Software and Interplay Capture” on page 45.

Folder	Function
Sent to Playback	This folder contains the sequences created when you perform a Send to Playback operation. See “Folders Used by Send to Playback Operations” on page 47.
Un-Checked-In Avid Assets	This folder serves as a temporary folder for assets associated with newly created media. Media Composer automatically checks newly created media into this folder. When you check in the bin or drag the assets to the Interplay Window, the Interplay Engine moves the assets to the corresponding folders. For additional information, see “Un-Checked In Avid Assets (Weekly)” on page 62.
Deleted Items	Serves as a recycle bin for non-Avid assets such as graphics files and Microsoft Word files. You can restore deleted items from this folder. The folder is created the first time you delete non-Avid assets from the Interplay database. Users only see what they deleted. An administrator sees all items deleted by all users. When an administrator deletes material from this folder, the files are permanently deleted.

Folders Used by AirSpeed 5000 Software and Interplay | Capture

A recording device such as an AirSpeed 5000/5500 server works with Interplay Capture software to store captured media in an ISIS workspace and create master clips in the Interplay Production database.



“AirSpeed 5000/5500 server” refers to both the AirSpeed 5000 and AirSpeed 5500 servers. “AirSpeed 5000 software” refers to the software used by these servers.



In Interplay Access, a single media file can be represented by multiple master clips, sometimes referred to as “links.” This is important to keep in mind when developing an Interplay delete strategy.

Device Clips and Master Clips

Interplay Capture creates both device clips and master clips (sometimes referred to as Capture clips). A *device clip* is a clip created by a recording device, such as an AirSpeed 5000/5500 server. A *master clip* created by Interplay Capture refers to one or more device clips and is used for editing.

In most cases, you should not use device clips for editing. This is especially important in a multi-res workflow. For example, AirSpeed 5000 software can simultaneously create a high-resolution device clip and a low-resolution (proxy) device clip. The master clip created by Interplay Capture is associated with both device clips. Avid recommends that folders that hold device clips be hidden from all users except administrators. See the following section for more information about these folders.

Folders Used by AirSpeed 5000

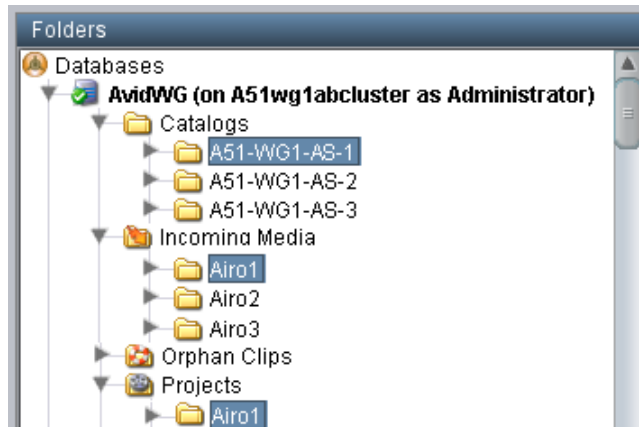
AirSpeed 5000 software can check clips in to one, two, or three locations in the Interplay Access folder hierarchy, depending on your settings:

- Catalogs*server_name*, where *server_name* is the network name of the AirSpeed 5000/5500 server. A setting on the Inventory page lets you enable or disable checkins to this folder.
- Incoming Media*workspace_name*, where *workspace_name* refers to the shared-storage workspace you set in the Destination template. You must specify a workspace in this template to perform an ingest operation, and the name of this workspace is used to name the folder.
- Projects*name*, where *name* is the folder name you set in the Destination template. This field is not needed if you manage ingest through Interplay Capture, which disables this function.

The clips contained in these folders are considered device clips.

The following illustration shows these three subfolders selected.

- The Catalogs subfolder uses the network name of the AirSpeed 5000 server (A51-WG1-AS-1).
- The Incoming Media subfolder uses the name of the shared-storage workspace (Airo1).
- The Projects subfolder also uses the name of the shared-storage workspace (Airo1).



These folders should be accessible only by administrators and hidden from other users. See [“Assigning Access Rights to User Groups on a Folder Basis” on page 29](#).

Folders Used by Interplay | Capture

Interplay Capture checks clips in to two locations in the Interplay Access folder hierarchy:

- **Projects\Device:** This folder holds device clips that are created by a recording device, such as by AirSpeed 5000 software. You set this location in the Capture Settings view in the Interplay Administrator. By default this file is named Projects\Device. This setting disables the Projects folder setting in the AirSpeed 5000 Management Console.
- **Master clips folder:** Interplay Capture periodically searches the device clips folder. When it finds a new device clip, Interplay Capture creates a master clip in the folder specified for the recording. This folder is set in a scheduled recording template or manually before recording.

In a multi-res workflow, Interplay Capture identifies the high-res device clip and the low-res device clip and associates both with the master clip.

This is the clip that should be used for editing.

The following illustration shows these two subfolders selected, and also shows a subfolder created by AirSpeed 5000 software.

- The Incoming Media subfolder is created by AirSpeed 5000 software and uses the name of the shared-storage workspace (Airo1). Note that there is no Catalogs folder created by AirSpeed 5000 software because the folder is disabled in the Inventory page of the AirSpeed 5000 Management Console.
- The Projects subfolders are created by Interplay Capture. The Devices folder holds device clips. The Big Swell holds the master clips for editing folder.



 A setting in the *Capture Settings* view, named “Delete device clips after —” specifies how long the device clips stay in the device clips folder. Capture automatically deletes clips from this folder after the specified period of time. The associated media is not deleted. It remains associated with the master clip, which is created and checked in by Interplay Capture.

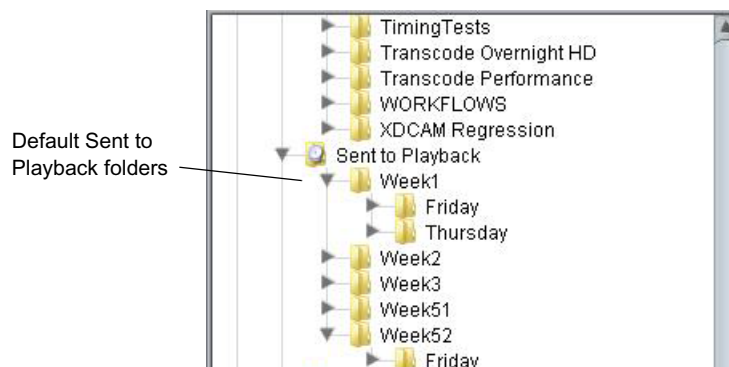
Folders Used by Interplay | Transfer Checkins

Assets checked in by a workgroup-to-workgroup transfer or other Interplay Transfer operations are placed in the Incoming Media folder. Assets are organized by the name of the workspace that the media is transferred to.

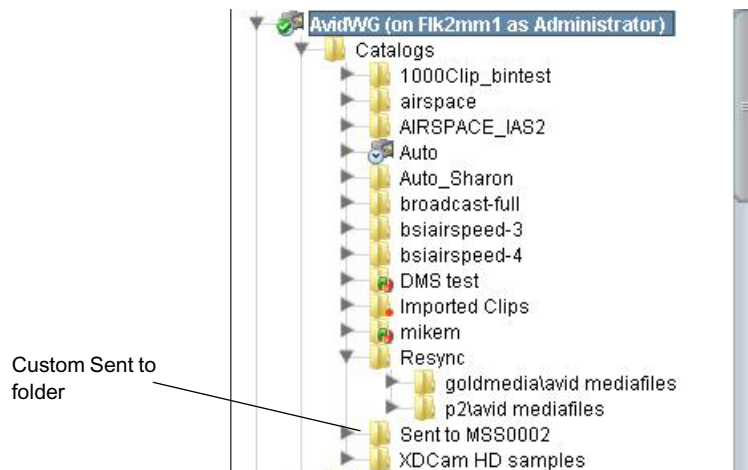
 For additional information on files checked into the Incoming Media folder, see “*Folders Used by AirSpeed 5000 Software and Interplay | Capture*” on page 45.

Folders Used by Send to Playback Operations

When you use Interplay Transfer to send a sequence to playback, the system automatically checks the sequence into the Sent to Playback folder. The default organization is by “week of the year” and “day of the week” as shown in the following illustration.



You can also instruct Interplay Transfer to create an additional folder in the Catalogs folder. The following illustration shows a Sent to MSS002 folder created in the Catalogs folder. In this case an asset is also checked in to the Sent to Playback folder.



For more information on creating an additional folder, see “Configuring an Ingest Device Catalog” in the *Interplay Transfer Setup and User’s Guide*.

Project and Bin Folders Created by Media Composer

After logging in to Media Composer, an editor selects or creates a project to work in. This project and its associated bins are checked in to the Interplay database when the editor uses the Check in Bin to Interplay command. An Interplay administrator can set an option to automatically check in bins when the editor closes Media Composer.

For additional information, see “[Creating Media Composer Projects in an Interplay Environment](#)” on [page 48](#).

Creating Media Composer Projects in an Interplay Environment

An important decision you need to make about your workflow is the location of the projects and bins created by Media Composer. A predictable folder structure allows for the following:

- Creates an easily maintained folder structure in the Interplay database
- Makes it easy for editors and other users in the organization to locate assets that are used by different projects
- Makes it easier for the administrator to delete older projects on the Media Composer systems

The following topics describe the basics of creating a project structure that is easy to maintain in an Interplay environment:

- [Creating Projects](#)
- [Defining the Interplay Folder Setting](#)
- [Defining the Media Creation Settings](#)

Creating Projects

In a large Interplay environment it is important to locate Media Composer projects in areas that are easily accessible by the administrator that has to delete the projects. This section describes the options for choosing a location for newly created projects.

When you start the editing application, the system displays the project log in window. From this window, you can create a project that is either Private, Shared, or External. Starting with Media Composer v, you can also select Synced Project.

- A Private project is stored locally. It can be accessed only by the user currently logged on to the editing machine. On a Windows system, a Private project is always stored under the user's Documents directory. An administrator can access this project if he or she has read/write permission to the user's directory.
- A Shared Project is stored locally. It can be accessed by any user that can log on the editing machine. On a Windows system, a Shared Project is always stored in the machine's Shared Documents directory.
- An External Project can be saved to any directory, either local or remote. Depending on the file-system permissions set on the selected folder, other users or an administrator may or may not be able to access the project. The external option is particularly useful when you want to save the project on shared network drive to make project maintenance easier.
- A Synced Project is stored locally and in your Interplay database. It can be accessed by any user who has permissions for the project. On a Windows system, a Synced Project is always stored in the machine's Shared Documents directory and in a corresponding project folder on Interplay.



Interplay Synced Projects were introduced in Media Composer v7.0. They provide you with an efficient way to link your Media Composer projects and bins with the folder structure in the Interplay Production database. When you make changes in your project, the changes are automatically saved to the database. In Interplay, synced projects are created in a root-level folder called Synced Projects. For more information, see "Interplay Synced Projects" in the Interplay Help or the Avid Media Composer Editing Guide.

The best choices for a shared structure are either Shared or External. These types of projects are created in a common location, making it easier for the administrator to locate and delete old projects. However, in an Interplay environment, using local projects guarantees that only one user will be in a project at any given time. Other editors must access the material through the Interplay folder structure.



Any user can create or delete a project through Media Composer. Keep your project structure consistent to make the Interplay database easy to maintain.

If you are working with projects created on local drives, then note the following:

- A local project resides on the C: drive of the Media Composer system.
- If the connection between the Media Composer system and shared storage is lost, the project and its contents are unlikely to become corrupt as a by-product.
- You cannot log in to the project from another workstation. You (or other users) must go to the Interplay Projects folder to locate the contents created in this project.
- From an administration standpoint, you must clean these projects up or archive them from each editing system. At a large site this can become time consuming and needs to become part of the administrator's workflow.

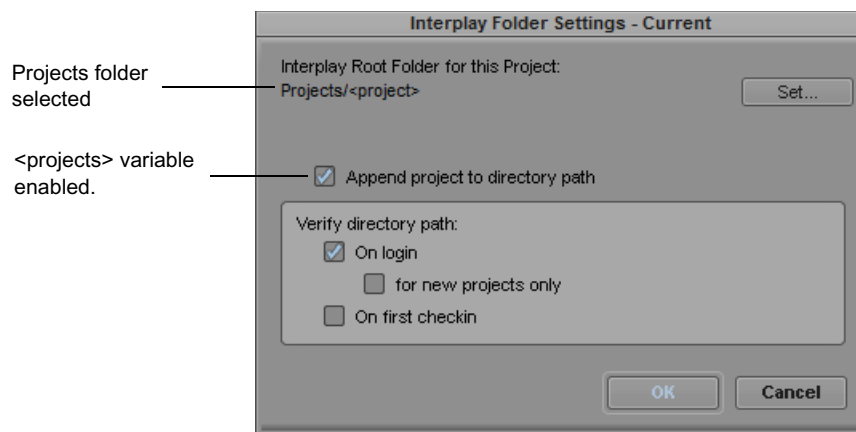


Avid does not support sharing bins in an Interplay environment. If you create projects on shared storage, do not allow editors to work in the same bin at the same time. This can result in data loss or media corruption. Instead of sharing bins, editors should use the Interplay Window to collaborate and share material.

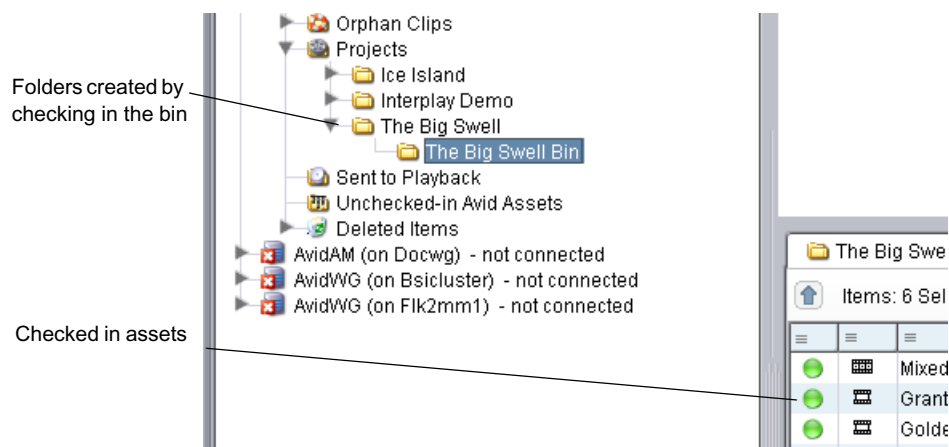
Defining the Interplay Folder Setting

The Interplay Folder setting in Media Composer defines where assets will be checked into the Interplay Production database. You need to define the location for each project. The examples in this topic assume that you are using the Projects folder in the Interplay Production database, which is the default.

The following illustration shows Interplay Folder Settings window, with a recommended folder path displayed. By default the folder setting is blank. In the following illustration, Projects/<project> represents the name of the folder automatically created the first time you check in bins from a project. Click the Set button to select the Projects folder. Then select “Append project to directory path” to add the <project> variable.



The following illustration shows a folder for a project named “The Big Swell” created in the Projects folder. The project folder was created automatically when the editor checked the first bin into Interplay Production using the Check in Bin to Interplay command. Interplay Access automatically created both folders and checked the assets in to the folder.



For a day-of-the-week workflow, use the Projects\<project> setting in the Interplay Folder Settings window. Then, when multiple editors work in the 01Monday project, there will be one 01Monday folder populated by the bins from each system. This keeps the organization easier to maintain.

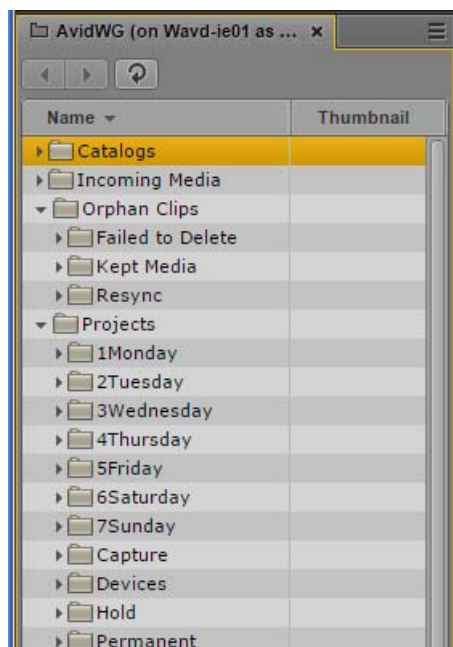
To instruct the system to automatically check in bins when the editor closes the application, select the Bins option in the Editor Database Settings window in the Interplay Administration tool. The default is to ask the editor before checking in the bin.

Defining the Media Creation Settings

Use the Media Creation dialog box to define where you want the Media Composer application to store newly created media. This is usually a workspace on shared storage that is monitored by Interplay Media Indexer.

Interplay | Production and MediaCentral | UX

MediaCentral | UX is a web and mobile front end to the Avid MediaCentral Platform. It provides tools for journalists, editors, and producers in a single user interface. Users can work with iNEWS, Interplay Production, and Interplay MAM in any combination that supports their environment. MediaCentral | UX provides a direct window to the Interplay Production database through the same hierarchy of folders as displayed in Interplay Access.



You can create and manage Interplay Production folders through MediaCentral UX similarly to the way you create and manage folders in Interplay Access. However, you need to use Interplay Access for your deletion workflow, as described in this chapter.

Project Workflow Samples

This section describes sample folder structures for Broadcast and Post environments.

Whatever the structure, the more simplified the Project and the Interplay Folder structure, the easier it is to maintain the Interplay environment.

Local Projects

This document assumes that you are using Local projects (projects created on local drives of the Media Composer system). For more information, see [“Project and Bin Folders Created by Media Composer” on page 48](#).

Local projects guarantee that only one user will be in a project at any given time. Other editors must access the material through the Interplay Production folder structure.

The following topics give examples of project workflows:

- [Broadcast Project Examples](#)
- [Post Environment Project Example](#)
- [Using Project Templates](#)

Broadcast Project Examples

Many broadcast sites use a day-of-the-week structure for organization and ease of administration. Broadcast projects are time-sensitive and items are often removed from the system within days. This convention makes it easy for users to locate material, and it makes it easy for administrators to maintain the system.

For example, if a site decides to use a 1Monday through 7Sunday project structure, this allows them to regularly delete the daily projects. This reduces the likelihood of the project becoming corrupted from reuse and also cleans out old tape names.

The following table shows some sample project names.

Weekly Structure	Monthly Structure (September)
1Monday	0903 Monday
2Tuesday	0904 Tuesday
3Wednesday	0905 Wednesday
4Thursday	0906 Thursday
5Friday	0907 Friday
6Saturday	0908 Saturday
7Sunday	0909 Sunday
Hold	Hold
Permanent	Permanent

In these examples the day of the week is the project and the story is the bin. The Hold folder is for continuing stories and the Permanent folder is used for items such as frequently used graphics, music, bumps, and show openers. The following illustration shows this file structure in Interplay Access. Red dots on the Hold and Permanent folders indicate reservations.



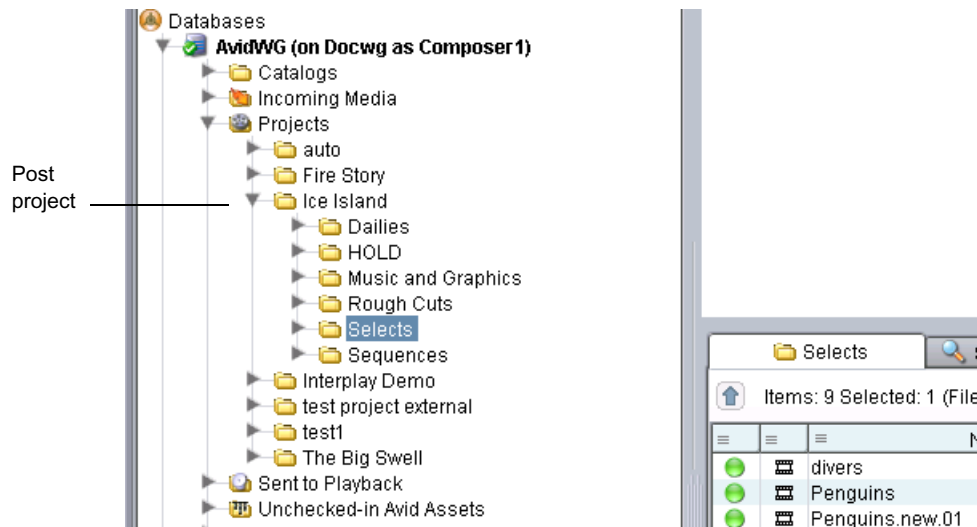
The day of the month monthly structure provides the ability to work further than a week ahead but requires more work creating projects. When using this model you have to create 28-31 days of the month projects manually. To simplify the operation on Media Composer systems, you can copy from a template and rename the project and the .avs and .avp files within the project. See [“Using Project Templates” on page 54](#).

Before deleting these folders, check whether editors want to archive finished sequences or source clips. Editors can move the assets to the Hold folder. In addition, you can use Interplay Archive to send the assets to an archive system. For more information, see “Working with the Archive and Restore Services” in the *Interplay / Production Services Setup and User’s Guide*.

Post Environment Project Example

The day-of-the-week structure is usually not appropriate for Post environments because a Post site may work on projects for several weeks or months at a time. You should develop your own structure based on your particular workflow. Keep in mind that when projects and bins are clearly named (for example by program name) it is much easier to maintain the folder structure.

The following illustration shows a sample Post structure for a project named Ice Island. Most of the subfolders were created when bins were checked in by the editors. The HOLD folder is created by an administrator in Interplay Access and is used for master clips, frequently used graphics, and other elements that should not be deleted. In this example when editors want to make sure certain assets are not deleted, they either copy the assets to the HOLD folder themselves or request that the assets be copied to the folder.



In this example the HOLD folder is on a per-project level because there may be some elements that you do not want to share between different shows.

Using this model, the administrator could delete the entire project folder after production has finished. Before deleting the folder, check whether editors want to archive finished sequences or some of the dailies. The following are some archiving methods:

- Move the assets to a reserved HOLD folder elsewhere in the Interplay folder structure
- Perform a digital cut to tape
- Use Interplay Archive to perform a Send to Archive operation

Using Project Templates

A facility can create a folder called Project Templates (either locally or on a shared storage workspace) for use on an editor's local machine. In the broadcast day-of-the-week structure, a media manager deletes the current "1Monday" project from the editor's local machine and copies and pastes the template to provide a new starting point.

To copy a project template to a local system:

1. Navigate to the location of the Project Templates.
2. Right-click on the 1Monday folder and select Copy.
3. Navigate to the current 1Monday folder on the local machine and delete the folder.
4. Paste the template 1Monday folder into the same location.

Example: Deleting Metadata and Media

The following example of a deletion workflow is based on the day-of-the-week structure described in ["Broadcast Project Examples" on page 52](#). This deletion strategy involves deleting both metadata (master clips, subclips, sequences and so on) and media as part of the same operation.

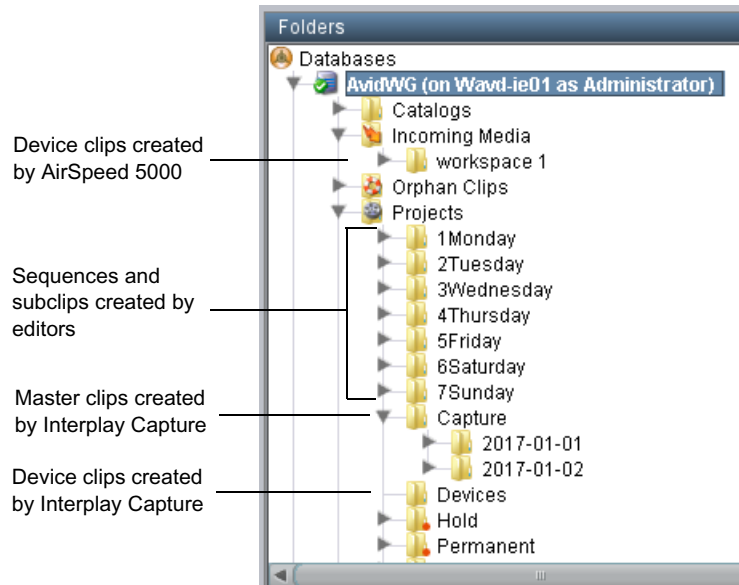
Folder Structure

For this example, ingested media is created in the following folders:

- Incoming Media\workspace 1 (holds device clips created by AirSpeed 5000 software)
- Projects\Devices (holds device clips created by Interplay Transfer)
- Projects\Capture (holds master clips created by Interplay Capture)

Editors work in the day-of-the-week projects: 1Monday, 2Tuesday, and so on.

The following illustration shows these folders.



Protecting Media

In this workflow, media that should not be deleted is kept in folders labeled Hold and Permanent. These folders use the Reservations feature to prevent media from being deleted. Reservations can be applied to any folder. For more information, see [“Use Reservations” on page 40](#).

Archiving Assets and Media

This workflow does not describe how assets and media are archived for storage and future use. Interplay Production offers a full-featured archiving solution. For more information, see the *Interplay / Engine and Interplay / Archive Engine Administration Guide*.

Manual and Automatic Deletion Schedules

In this workflow, manual and automatic deletion is scheduled for midday, which can be a slow time at a 24-hour broadcast facility. A media manager begins deletion at approximately 11 am, and the complete deletion process takes place between 1 pm and 3 pm, seven days a week. The following illustration shows this custom schedule, as configured in the Delete Behavior view in the Interplay Administrator.

Scheduled Deletion Options

☒ Enable custom schedule

☐ Allow scheduled deletions during database backups

☒ Permanently delete assets and associated online media from the Orphan Clips folder

☐ Delay scheduled deletion by at least days from the asset deletion date

Deletion Time Spans

From : to :

☒ Mon ☒ Tues ☒ Wed ☒ Thurs ☒ Fri ☒ Sat ☒ Sun

You can create more than one deletion time span, and set different time spans for different days of the week. For more information, see “Setting Options for Deletion” in the *Interplay / Engine and Interplay / Archive Engine Administration Guide*.

Deletion Activities

The following table lists deletion activities based on the day-of-the-week workflow and points to the sections that explain the operations.

Interplay Production Component	Activity
Interplay Access	Delete daily project and those other projects ready for deletion. See “Project Folders – Editor Projects (Daily)” on page 57. Delete material created by AirSpeed 5000 software. Sift the results by creation date and delete. “Projects Folder – Folders Created By AirSpeed 5000 and Interplay Capture (Daily)” on page 59. Delete material created by Interplay Capture under the Projects folder. See “Incoming Media (Daily)” on page 60. Delete Sent to Playback for the day of the week that you deleted above-located under the week number. See “Folders Used by Send to Playback Operations” on page 47. Manage items in the Orphan Clips folder and subfolders. See “Orphan Clips (Weekly)” on page 61. Delete material from the Un-Checked in Avid Assets folder. See “Un-Checked In Avid Assets (Weekly)” on page 62. Delete from the Deleted Items folder. See “Deleted Items (Weekly)” on page 62.
Media Composer applications	Delete daily project and those other projects ready for deletion. You must do this on each editing system. See “Media Composer Maintenance” on page 63. Delete the project or projects deleted in the step above out of the Attic. You must also do this on each editing system.

Project Folders – Editor Projects (Daily)

The fastest way to clean up space on the system is to delete entire folders from the Interplay database. You can safely delete an entire folder and all the media referenced by the assets in that folder if reservations are correctly used (see [“Use Reservations” on page 40](#)). The Interplay Engine checks to make sure assets are deleted only if links are not used in other folders.

After using the system for a few weeks, determine how long it takes to reach the maximum storage capacity of your shared storage workspaces. Then you can determine how often you will need to delete material.



On Avid ISIS, attempt to keep the used space under 80% of the capacity of the Storage Group for each Storage Group on the ISIS. Use the ISIS Administration Tool/Storage Groups tab to view this information.

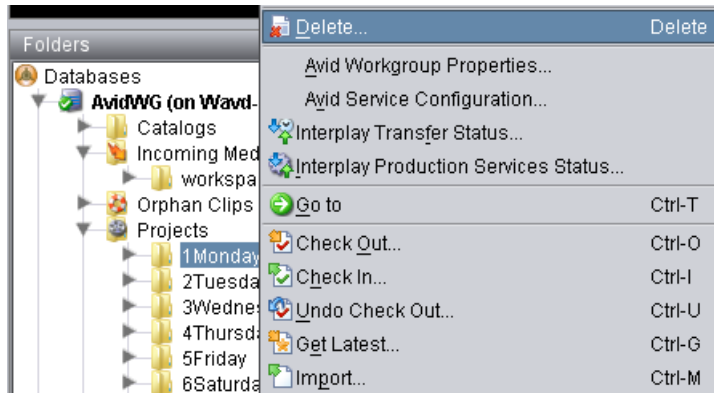
For the day-of-the-week workflow, after you determine how long it takes to reach your maximum capacity, set up a schedule to delete the day of the week project a specific number of days after the project day. For example, always delete the 1Monday project on Friday. Note that you will also have to delete the corresponding local projects on the Media Composer systems as well.

To delete Project folders created by editing projects and associated media:

1. Open the Projects folder in Interplay Access.
2. Locate the project to delete, for example, 1Monday.

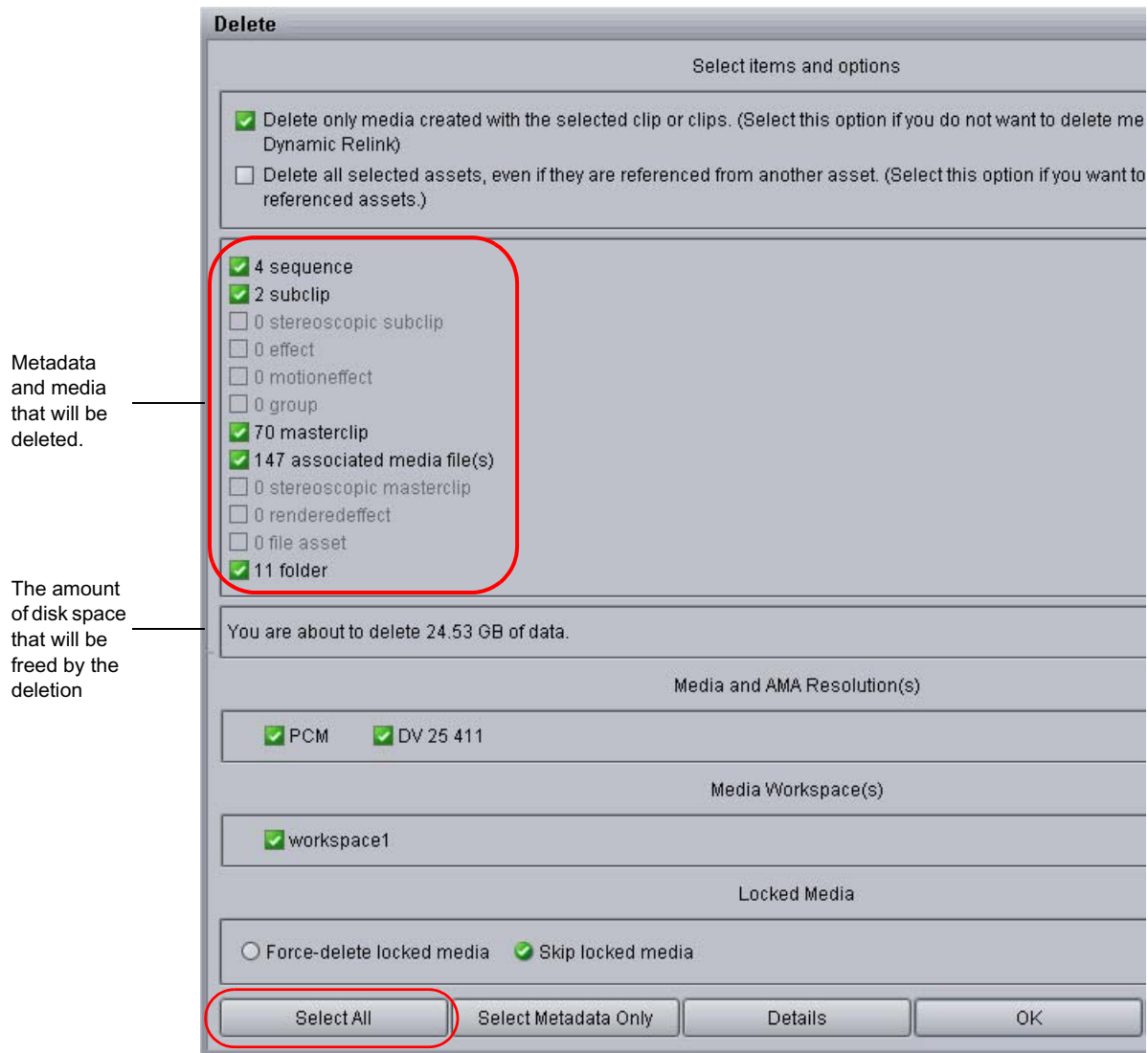
After the deletion process is complete, you can recreate the 1Monday folder

3. Right-click the folder and select Delete.



The Delete window opens and displays the contents of the folder.

4. Click Select All to select all the assets, files, and resolutions.



Note that all assets, along with associated media files, are selected if you click “Select All.”

The center portion of the dialog box shows how much free space you will gain from the Delete operation.



The number of media files and the amount of data that will be deleted are affected by the option “Delete only media created with the selected clip or clips.” This information is not affected by your selections in the Media Resolutions or Media Workspace sections of the dialog box.

For more information about options in the Delete dialog box, see “Deleting Avid Assets” in the Interplay Access User’s Guide.

5. Click OK.

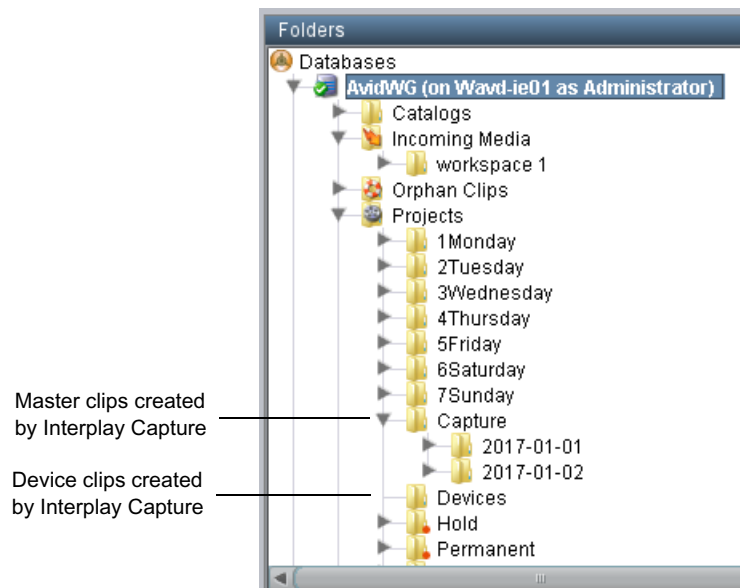
The system displays the Delete in Progress window and indicates whether any files were reserved. The system does not delete any media files associated with reserved items and does not delete the folder containing the reserved assets. You can move reserved items to a Hold folder and then delete the folder that contained the reserved assets.

- After the folder is emptied, the system deletes the folder. If there are any non-Avid assets in the folder (for example, graphics files or text files) the system asks if you want to delete the files and the folder from the Interplay database.

Projects Folder – Folders Created By AirSpeed 5000 and Interplay | Capture (Daily)

The Projects folder contains folders that contain assets for media created by AirSpeed 5000 software and clips created Interplay Capture. The folders that you delete depend on how you have configured these folders. See [“Folders Used by AirSpeed 5000 Software and Interplay | Capture” on page 45](#).

In this example, Projects folders for ingested material are named Capture and Devices, as shown in the following illustration:



In this case, do not delete an entire folder because it contains material from all seven days of the week. This section describes how to delete selected assets.

To delete assets from the Airspeed 5000 and Interplay Capture folders:

- In Interplay Access, click on the Projects folder.
- Open the folder that contains the device clips, for example, the folder named Devices.
- Select Creation Date from the Sort By list. This will show the oldest material.

Sort By list

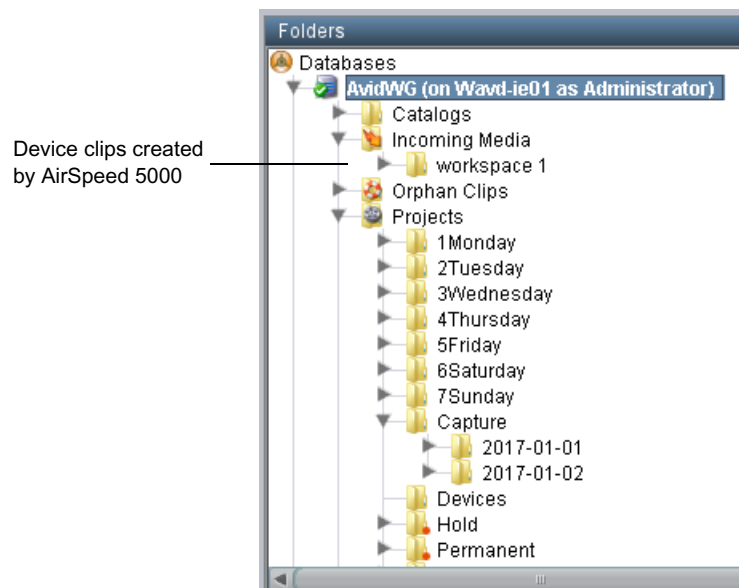
Selects					
Simple					
Items: 9 Selected: 0 (File Size: 0 Bytes)					
			Creation Date	Initial Preset*	
		Name	Duration	Tracks	Creation Date
		Penguins.new.01	00:00:24;19	V1 A1-2	6/13/06 2:37 PM
		school of fish.new.01	00:00:07;18	V1 A1-2	8/27/06 2:26 PM
		Episode 1 opening	00:01:27:16	V1 A1-2 TC1	8/27/06 2:46 PM

- Select all the clips from the day whose project folder was deleted.

5. Do one of the following:
 - ▶ Right-click one of the clips and select Delete.
 - ▶ Press the Delete key.
6. Select the metadata and media that you want to delete.
7. Click OK.
8. Open the folder that contains master clips, for example, the folder labeled Capture. Select the subfolder for the day whose project folder was deleted and delete the clips and media as described in the previous steps.

Incoming Media (Daily)

Media that is transferred from an Interplay Transfer server or captured by AirSpeed 5000 software creates a device clip under the Incoming Media Files. The folder is sub-divided by the name of the workspace that the media was transferred to or captured to. See [“Folders Used by AirSpeed 5000 Software and Interplay | Capture” on page 45](#) and [“Folders Used by Interplay | Transfer Checkins” on page 47](#). In this example, the folder is named workspace 1.



Filter each workspace as described in the previous procedure, and delete the material for the day whose project folder was deleted.

Sent to Playback (Daily)

The Sent to Playback folder is the area where sequences get created when you send to playback devices. It is structured by weeks and then by the days within the week. See [“Folders Used by Send to Playback Operations” on page 47](#).

Delete the day corresponding to the 1Monday project deleted previously.



You can also use a search in Interplay Access to identify the files for deletion. The following procedure searches the Send to Playback folder for assets whose media status is offline and that were created before a particular date. In this case the files are offline because you deleted the media when you deleted assets in the Projects folder.

To delete using search results:

1. In Interplay Access, open an extended search and select the Send to Playback folder.
2. Add the criteria “Media Status is offline” and “Creation date before” a specified date, and click Search.



3. Delete the search results.

Orphan Clips (Weekly)

In this workflow, deletion of assets and media is scheduled and automatic (see [“Manual and Automatic Deletion Schedules” on page 55](#)) so there is no need to delete items from the top level of the Orphan Clips folder. Check the following subfolders:

- **Kept Media:** Contains links for master clips and rendered effects that have been deleted from Interplay folders but still point to media files on shared storage. These links are not deleted through a scheduled deletion. An administrator needs to manually delete the links and media when the media is no longer needed. This subfolder is created when needed.
- **Failed to Delete:** Contains links for assets that could not be deleted by the deletion process (for example, if a database asset is corrupted). An administrator needs to check these links for system problems. This subfolder is created when necessary.
- **Resync:** If the Resync process identifies a clip with a missing audio or video track, it creates a special master clip and stores it in this folder. An administrator needs to check these links for system problems. This subfolder is created when necessary.

Some workflows may involve deleting assets when they are no longer needed and later deleting the media from the Orphan Clips folder. See [“Example: Deleting Metadata Only” on page 64](#).

Un-Checked In Avid Assets (Weekly)

This folder serves as a temporary folder for assets associated with newly created media on Media Composer systems. For example, when you capture video, consolidate, transcode, or render effects in your bin, Media Composer automatically checks the asset into the Un-Checked In Avid Assets folder. When you check in the bin or drag the assets to the Interplay Window, the Interplay Engine moves the assets to the corresponding folders.

Administrators see a folder for each user; users see only their own folder. If editors always check in bins when they close them, this folder should be easy to maintain. Be careful of deleting assets and associated media from this folder because if an editor forgot to check in a bin, you might delete media such as rendered effects that the editor recently created.

To delete using search results:

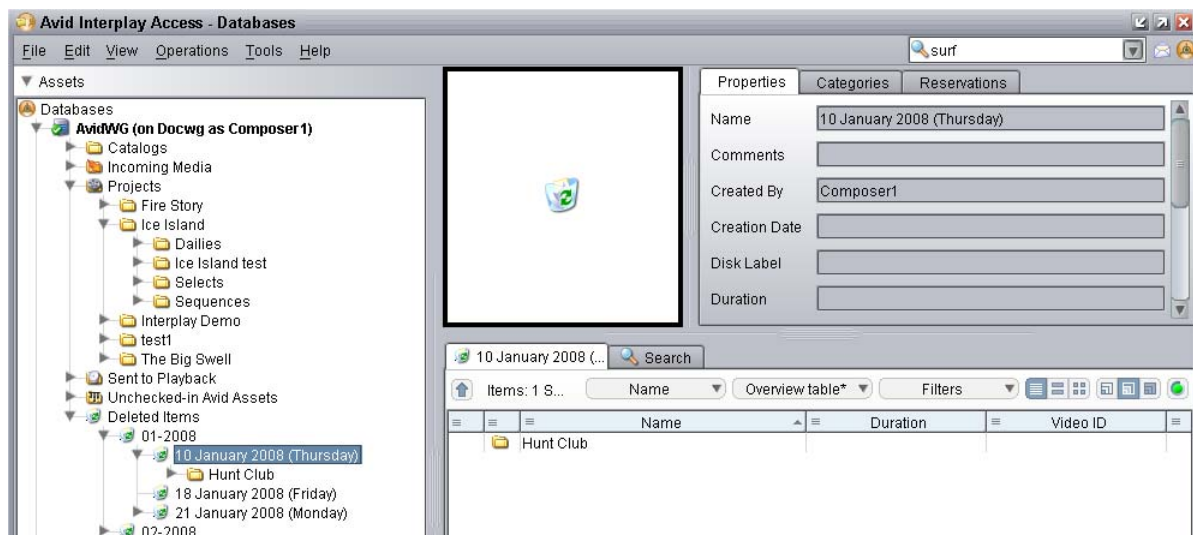
1. In Interplay Access, open an extended search and select the Unchecked-in Avid Assets folder.
2. Add the criteria Creation date before” a specified date, and click Search.
3. Delete the search results.

For additional information on extended searches, see [“Searching for Media” on page 70](#).

Deleted Items (Weekly)

The Deleted Items folder serves as a recycle bin for non-Avid assets such as graphics files and Microsoft Word documents. You can restore deleted items from this folder. When an administrator deletes files from this folder, the files are permanently deleted. The folder is created the first time you delete non-Avid assets from the Interplay Production database.

The Deleted Items folder for a non-administrator shows only the items deleted by the user who is logged in, while the administrator sees all items deleted by all users. The following illustration shows the Deleted items folder.



Media Composer Maintenance

After you clean up the Interplay Production folder structure, you should also delete the corresponding projects and bins used by Media Composer. These projects and bins reside locally on the Media Composer system and you have to clean them up on each editor. The files exist independently of the Interplay Production database.

This procedure ensures the following:

- Prevents the editor from opening projects and bins that contain clips and sequences that have been deleted from Interplay Production. If they are re-opened, and the metadata gets checked in again, you will have to repeat the deletion procedures through Interplay Production.
- Prevents projects from becoming corrupted by regularly replacing the project folders with fresh folders.
- Clears out the old tape names that build up in the project every time a tape is put in to a deck on the editing system.



Depending on your workflow you may find it more convenient to delete the projects on the editor systems before you delete the corresponding assets and media from Interplay Access. This avoids the possibility of editors re-opening the projects and checking in material after it has been deleted from Interplay Production.

Deleting the Project Files

This section describes how to replace a project file folder with a clean template. If you aren't using templates, you can just delete the project file and create a new project.

To delete projects when using the 01Monday – 07Sunday workflow:

1. Open My Computer on the Media Composer system.
2. In the Network Drives section, locate the workspace that contains the Project Templates folder and open the folder.
3. Locate the day of the week you are going to delete and replace, right-click the folder and select Copy.
4. Navigate to the Project folder that you want to replace.
5. Select the folder that you want to delete and press the Delete key
6. Perform a right click-Paste and paste in the new project template.

Cleaning Up the Attic

This section describes how to delete files from the Avid Attic on the local Media Composer system.

To clean up the Attic:

1. Navigate to the following location on the Media Composer system:
C:\Program Files\Avid\<application name>\Avid Attic
2. Sort by creation date.
3. Delete projects no longer needed.

Example: Deleting Metadata Only

This deletion strategy involves deleting only metadata (master clips, subclips, sequences and so on) from Interplay Access folders.

It is most useful at sites where users regularly copy assets between Interplay folders. It avoids the problems of deleting the media associated with a clip that someone else might be using. There are several practices that can also prevent unwanted deletion of media, but depending on your site's workflow, you might find that this provides an additional security step before media is fully deleted.



For information about protecting media from deletion, see “Use Reservations” on page 40 and “Other Strategies to Protect Assets and Media” on page 41.

In this workflow, use reservations for assets which will be needed in the future.

There are two options for this strategy:

- Delete metadata only and let the scheduled deletion delete the media of the assets that were deleted. This speeds up the time needed for manual deletion of the assets.
- Delete metadata only and manually delete online masterclips, rendered effects, and their associated media from the Kept Media folder. This gives you more control over which media is deleted, but requires more work.

To delete metadata only and delete media automatically:

1. Review the settings for scheduled deletion in the Delete Behavior view of the Interplay Administrator. Make sure the setting “Permanently delete assets and associated online media from the Orphan Clips folder” is selected, as shown in the following illustration.

Scheduled Deletion Options

☒ Enable custom schedule

☐ Allow scheduled deletions during database backups

☒ Permanently delete assets and associated online media from the Orphan Clips folder

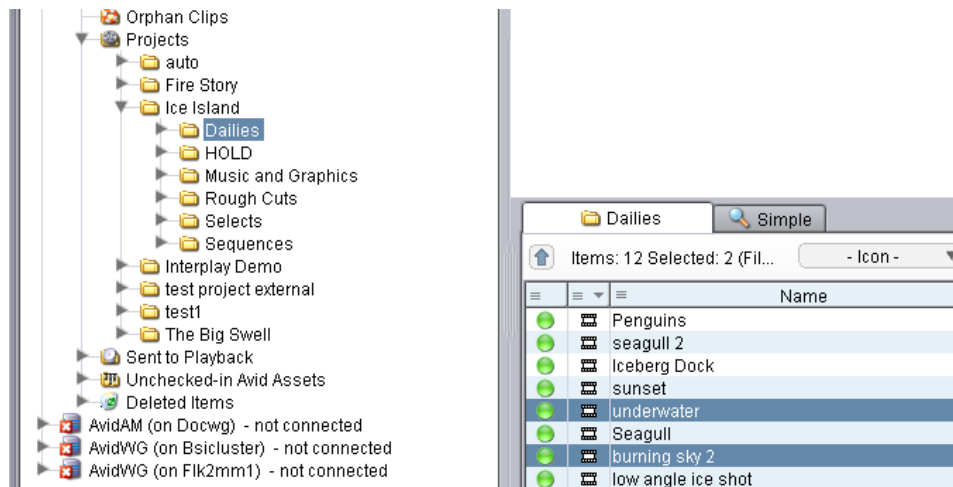
☐ Delay scheduled deletion by at least 1 days from the asset deletion date

Deletion Time Spans

From 13:00 to 15:00 ☒ Mon ☒ Tues ☒ Wed ☒ Thurs ☒ Fri ☒ Sat ☒ Sun

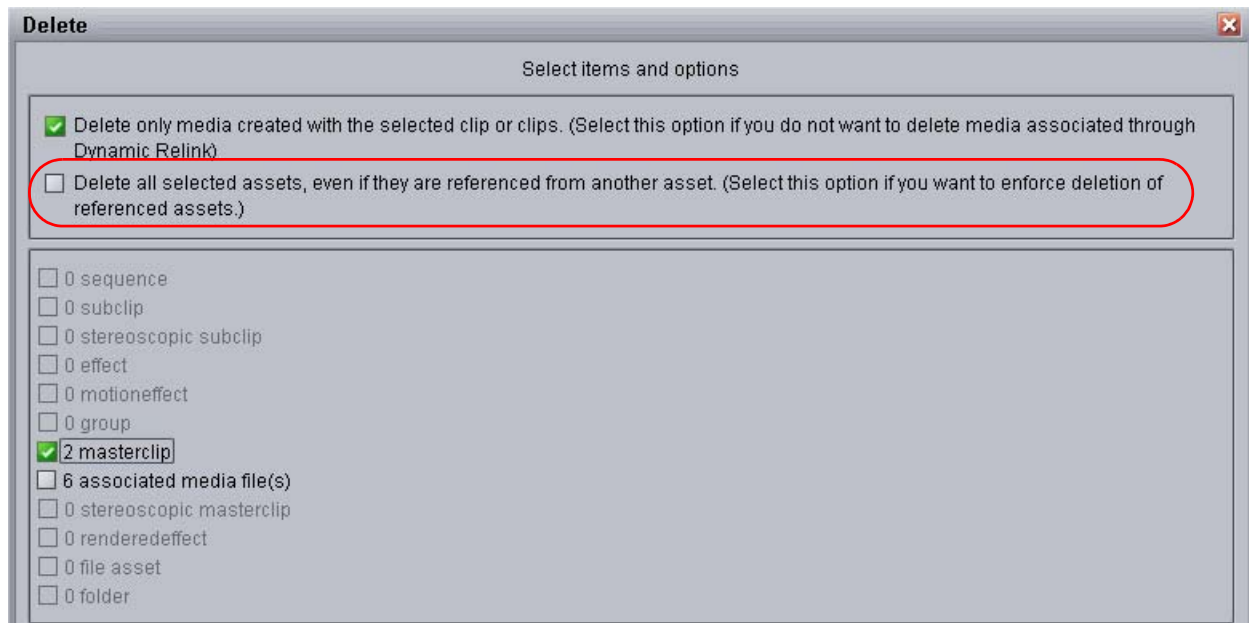
2. Delete assets, but not media, from the various Interplay Access folders as your workflow requires. For example, delete from the folders as described in [“Example: Deleting Metadata and Media” on page 54](#).

The following illustration shows two master clips in the Dailies folder. These are the last links for these assets, and neither asset is used in a sequence.



- a. Right-click the assets and select Delete.
- b. In the Delete dialog box, select the metadata but do not select the media.

The following illustration shows the Delete dialog box for this operation. Note that the option “Delete all selected assets, even if they are referenced from another asset” is not selected. Keep this deselected if you want to be sure the clips are not being used.



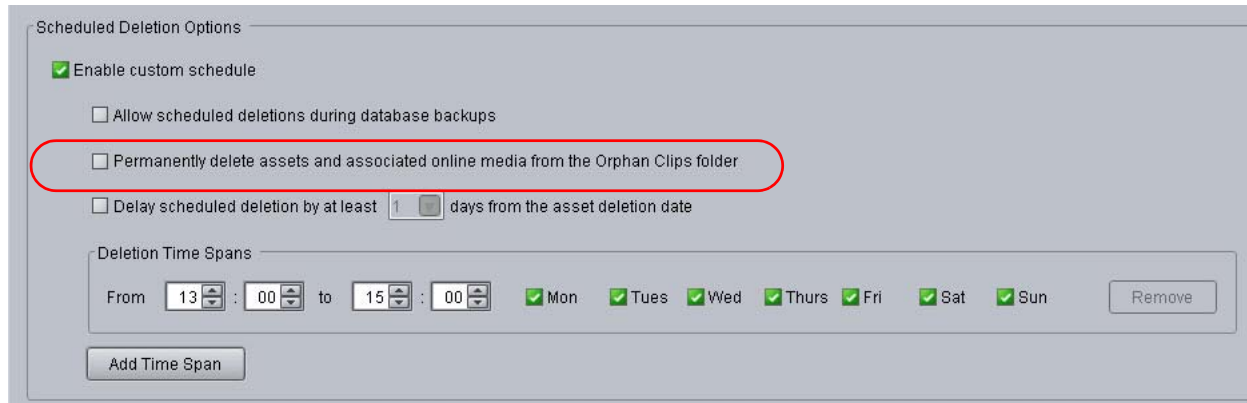
For more information about options in the Delete dialog box, see “Deleting Avid Assets” in the Interplay Access User’s Guide.

- c. Click OK.

Database assets and media will be deleted during the scheduled time slot. Assets scheduled for deletion are listed in the Orphan Clips folder until they are deleted.

To delete metadata only and media manually:

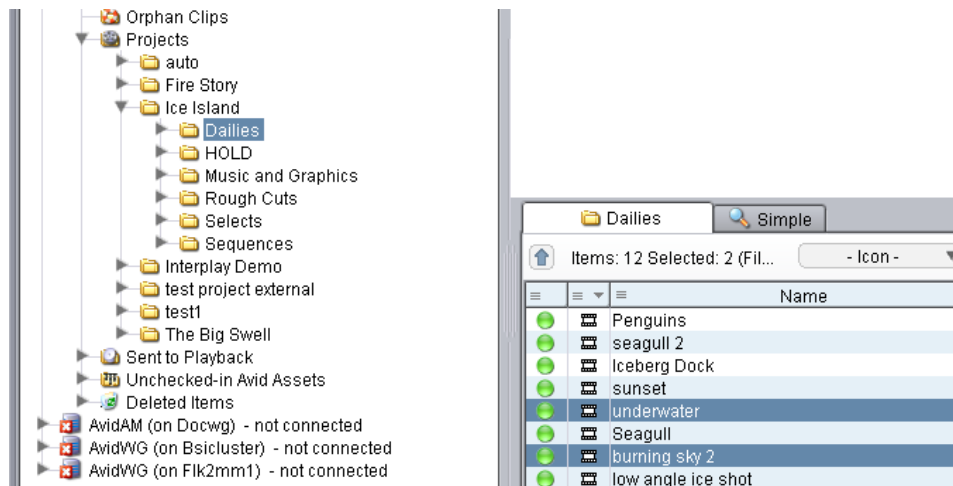
1. Review the settings for scheduled deletion in the Delete Behavior view of the Interplay Administrator. Make sure the setting “Permanently delete assets and associated online media from the Orphan Clips folder” is **not** selected, as shown in the following illustration.



Note that a deletion time span is scheduled. Database assets without media will be deleted during this time span.

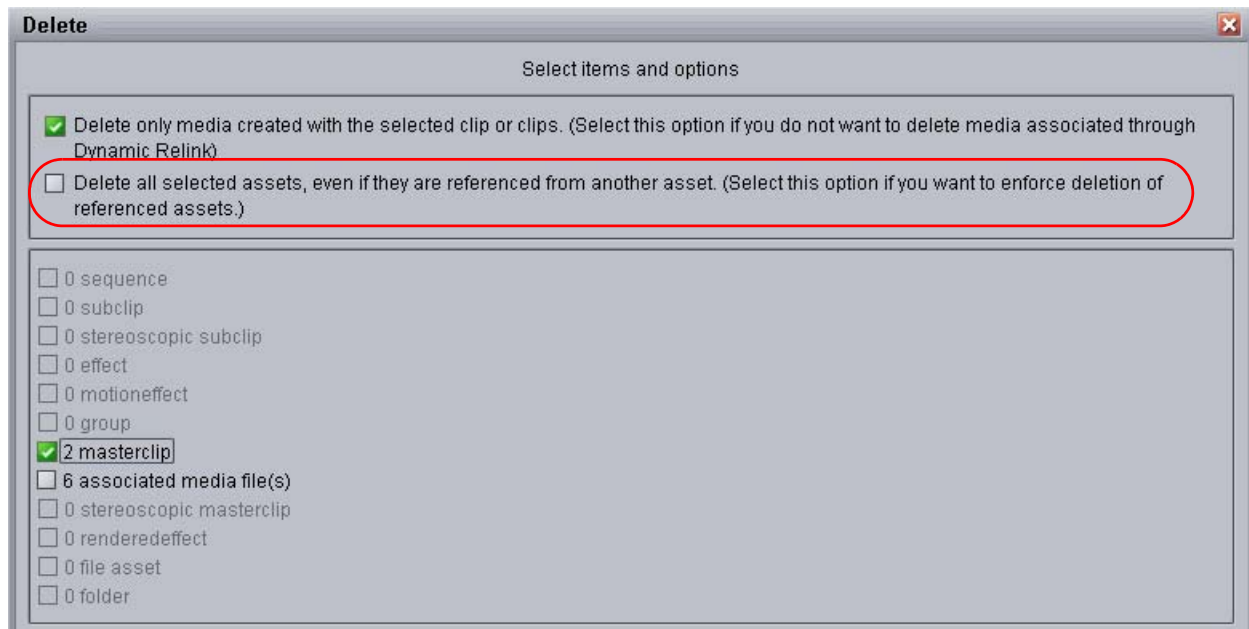
2. Delete assets, but not media, from the various Interplay Access folders as your workflow requires. For example, delete from the folders as described in [“Example: Deleting Metadata and Media” on page 54](#).

The following illustration shows two master clips in the Dailies folder. These are the last links for these assets, and neither asset is used in a sequence.



- a. Right-click the assets and select Delete.
- b. In the Delete dialog box, select the metadata but do not select the media.

The following illustration shows the Delete dialog box for this operation. Note that the option “Delete all selected assets, even if they are referenced from another asset” is not selected. Keep this deselected if you want to be sure the clips are not being used.



For more information about options in the Delete dialog box, see “Deleting Avid Assets” in the Interplay Access User’s Guide.

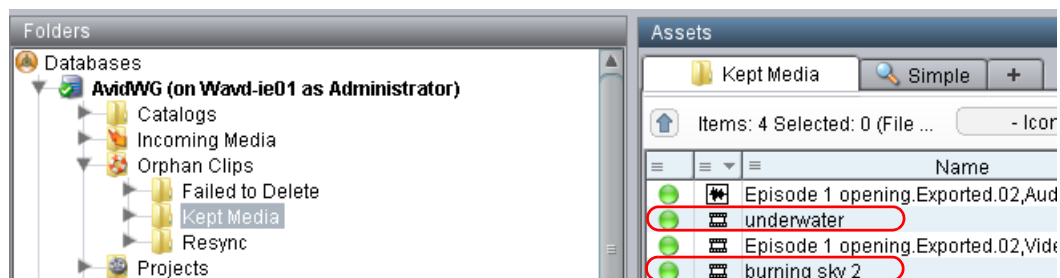
c. Click OK.

- When you are ready to delete the media files, open the Orphan Clips folder and the Kept Media subfolder.

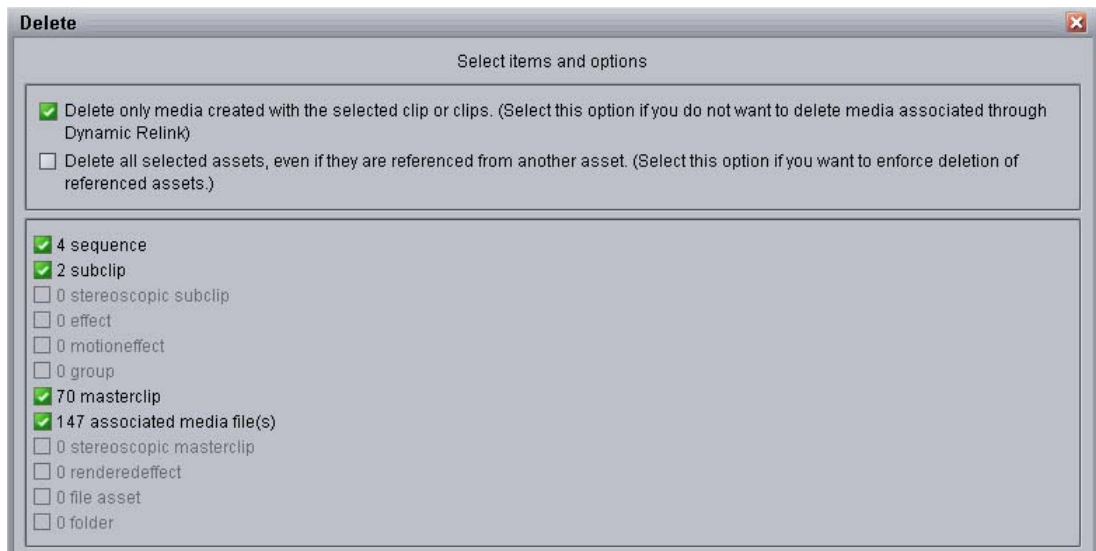


By default, only administrators can view these folders. An administrator can assign read/write/delete permissions for this folder to users or groups using the Managing Database Roles view in the Interplay Administration tool.

The two assets deleted earlier are now in the Kept Media folder (unless the scheduled deletion has not processed them yet). Since the two assets were the last links associated with their media files, they were both moved to the Kept Media folder.



- Select all the clips in the Kept Media folder, right click one of the clips and select Update Status from Media Indexer.
- Select all of the assets in the search results window and delete them. Delete the metadata and the media as shown in the following illustration.



You can use the `delete_kept_media.exe` tool to help clean up the Kept Media folder. For more information, see the *Interplay / Engine and Interplay / Archive Engine Administration Guide*.

Determining How Much Space a Delete Operation Will Free Up

The Delete dialog in Interplay Access provides the best available estimate of the space that will be freed up by a Delete operation. Performing a search in Interplay Access provides a less reliable estimation.

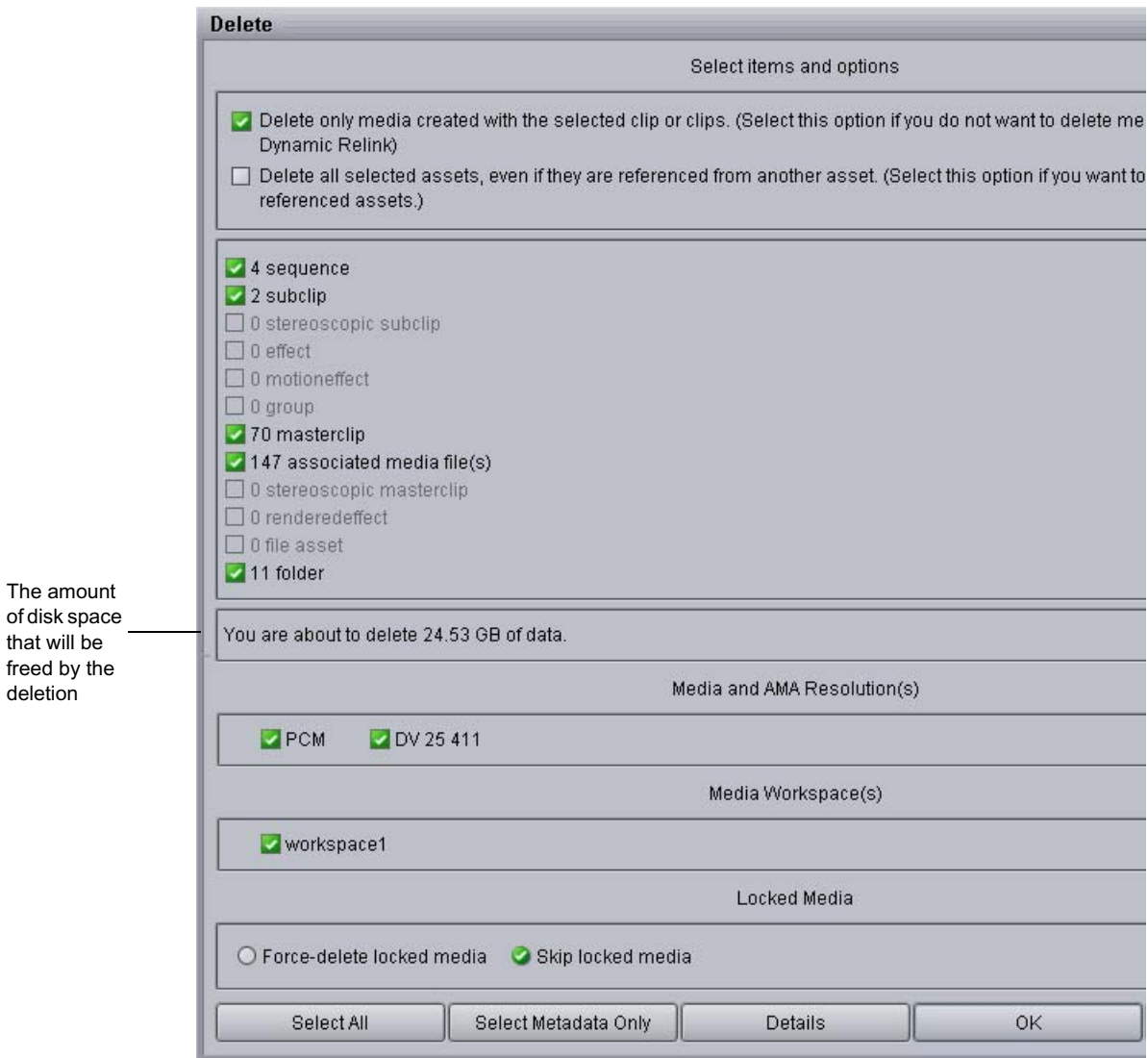
To determine how much space will be freed up by a deletion:

1. Right-click the folder you want to delete and select Delete.

The Delete dialog box opens.

2. Select the assets, files, and resolutions you want to delete.

The value in the center of the dialog box shows the disk space that will be freed up if you perform the deletion.



Note the following:

- The amount of disk space that will be freed up is affected by the option “Delete only media created with the selected clip or clips.”
 - The amount of disk space that will be freed up is not affected by your selections in the Media Resolutions or Media Workspace sections of the dialog box. This is because the Delete dialog provides an estimate for all online media files of the selected assets, not for a specific resolution or workspace.
3. Click OK or Cancel, depending on whether you want to perform the deletion or whether you were just checking on the disk space.

Searching for Media

Interplay Access can search across all projects and bins in the Interplay environment. There are two types of searches:

- A simple search allows you to filter media by name, project, bin, time, and media type.
- An extended search allows you to search by more detailed attributes and provides operator based searching using “and” and “or” type searches.

An extended search gives you more flexibility to create a list of assets that match your criteria for deleting. For example:

- An Offline Material search returns anything that has been deleted from ISIS, but not Interplay Production.
- An Unused Clips search returns any master clips from the Incoming Media folder that are not cut into sequences or subclips.
- A DRM Clips search returns all master clips/subclips/sequences that contain restrictions.

For complete information on searching, see “Finding Assets” in the *Interplay / Access User’s Guide*.

4 Refining Your Search in Interplay | Production

Interplay Production includes a powerful, customizable search. If you are working in a facility with a large database, you can speed up the search process by specifying a particular folder as the search root and by using attributes. For example, by using the Time attribute, you can limit your search to clips modified within the last 24 hours, the last 7 days, and other time periods.

You can search the Interplay Production database from the following applications:

- Interplay Assist
- Avid Instinct
- Interplay Window (in Avid editing systems)
- Interplay Access

All applications use the same search engine, which runs on the Interplay Engine server, and all applications share a core set of attributes. Interplay Assist, Instinct, and the Interplay Window share the same user interface (the Media Search tab) and the same set of attributes. Interplay Access includes these core attributes, but also offers additional options for an extended search. All applications search for Avid assets (clips, subclips, and other media), but only Interplay Access searches for file assets (such as Adobe Photoshop files).

The following topics provide information about Interplay Production search in general as well as specific information for Interplay applications:

- [Using the Search Root and Attributes to Speed Up Interplay | Production Search](#)
- [Basic Interplay | Production Search Attributes](#)
- [Searching in Interplay | Assist, Avid Instinct, and the Interplay Window](#)
- [Searching in Interplay | Access](#)
- [Example: Finding Assets By Using a Unique Identifier](#)

Using the Search Root and Attributes to Speed Up Interplay | Production Search

To speed up your search, make your search as specific as possible. The more specific you can make your search, the faster you can get the results you want. Two ways to make your search more specific are:

- Specifying a search root
- Using attributes

Keep in mind that even if you only change the search root, or search by a single attribute, you can dramatically improve the response time.

Specifying a Search Root

By specifying a particular folder as the search root, you are instructing the search engine to limit the set of assets that it searches to that particular folder and any folders included in that folder's subtree. By default, the search root is the entire database. If you know the folder subtree in which the asset you are searching for is stored, you can select that folder to speed your search. For example, in Interplay Access, you can select the Project folder, or a particular project within that folder.

Using Attributes

By specifying one or more attributes, you are also instructing the search engine to limit the set of assets that it searches. Simply searching for text, without restricting the search root, is the most time-consuming search, because the search engine searches the entire database. However, if you specify a particular attribute (such as the Time attribute), the search engine can quickly limit the database to a small subset and search for text within that subset.

For example, if you wanted to find a clip about the Nobel Peace Prize acceptance speech, you could search for the text "Nobel Peace Prize." In this case, the search engine will search through the entire database of clips, which could take twenty-five seconds for a large database. However, if you know that the speech took place within the past week, you can select the Time attribute and then the option "Last week." The search engine creates a subset of clips that were modified in the past seven days, and searches for the text within that set. Setting this attribute can reduce the time for a search from twenty-five seconds to one second.

Basic Interplay | Production Search Attributes

The following table lists the basic Interplay search attributes. These are available in the Media Search tab in Interplay Assist, Avid Instinct, and the Interplay Window. They are available in a slightly different format in the Simple Search tab of Interplay Access.

Attribute	Options	Description
Text	<i>User-supplied text</i>	Allows you to search all text-based attributes: • Name • Tape • VideoID • Comments • Locator and restriction annotations • User-created fields If you specify more than one word in the Text attribute, the search engine searches for an exact match. For example, if you specify "President speech" the search engine looks for assets with that exact phrase. To search for clips that contain "President" and "speech" anywhere in the text, use the Extended Search in Interplay Access.

Attribute	Options	Description
Text Fields	- Any system property or custom metadata containing a text field - Select All - Clear All	Allows you to search in any text field you select; Name is selected by default.
Search In		Allows you to search in any folder managed by the asset manager
Types	Master Clip Subclip Sequence	Allows you to specify the type of media object for your search (you must select at least one type for each search)
Category	<i>Site-specific list of categories</i>	Allows you to specify which category to search. The specific categories listed for this attribute are defined by the system administrator.
Time	Whenever Last 10 minutes Last hour Last 24 hours Last week Last month Last year	Allows you to set a time parameter for your search (time parameters refer to the date the media object was last modified)

Interplay Access includes additional attributes and options that you can use to search for Avid assets or file assets.

Searching in Interplay | Assist, Avid Instinct, and the Interplay Window

Interplay Assist, Avid Instinct, and the Interplay Window share the same set of attributes and the same user interface: the Media Search tab. They also provide a way to define a folder as a search root. This topic describes how to perform a search in any of these applications.



Interplay Assist includes a Find function that you use to search within clips and shotlists for text in locators and annotations. It is different from the Media Search function, which searches the media database (see Interplay Assist Search Options).



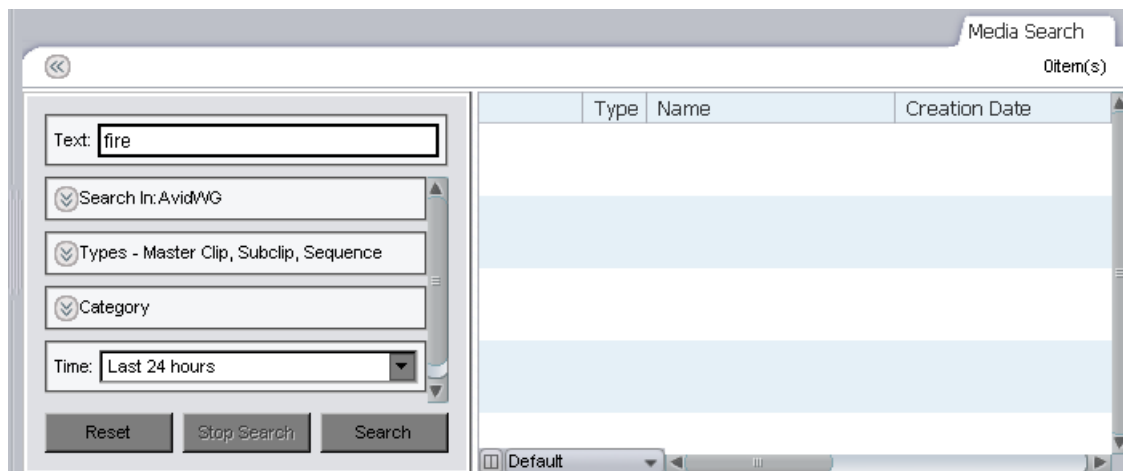
Avid Instinct includes an iNEWS Search function that you use to search for information in the iNEWS database. It is different from the Media Search function that you use to search the media database.

The following example shows how to search for clips that have been logged to reference a fire that took place last night.

To search for clips in Interplay Assist, Instinct, or the Interplay Window:

1. Open a Media Search tab.
 - ▶ In the Interplay Window, a Media Search tab is always displayed.
 - ▶ In Interplay Assist or Instinct, right-click an Interplay folder and select Search.

A Media Search tab opens in the Research panel.



2. In the Text field, type a search term. In this case, type the word “fire.”
You can specify words or characters for your search. Search terms are not case-sensitive and apply to all text attributes of the media objects in your search. If you specify more than one word, the search engine searches for the exact phrase.
3. Click the Text Fields menu, and select the kinds of text fields to search in.
4. (Option) If you know which folder subtree might contain the clips, expand the “Search in” field and navigate to the folder. You can also expand the Types or Category field and select the appropriate option for the search.
5. Click the Time menu, and select a time parameter for your search. In this case you know that the fire took place last night, so you can select “Last 24 hours.”
6. Click Search.
The Interplay search engine limits the database to clips that have been modified (captured or edited) in the last 24 hours, and then searches within that subset for text that contains the word “fire.” Any clips that meet these criteria are displayed in the Research panel.
7. (Option) To start a new search, click Reset and then repeat steps 1 through 5.

For complete information on searching media, see the documentation for your application (Interplay Assist, Instinct, or Avid editing application).

Searching in Interplay | Access

Interplay Access provides more options for refining your search than other Interplay applications. There are two ways to search for assets in Interplay Access:

- Running a text-based search on the database (through the text box on the Interplay Access menu bar)
- Running an attributes-based search on the database or a selected folder subtree (through a Search tab)

Running a text-based search is more convenient, but depending on the size of your database, could take more time than you want. An attributes-based search provides you with more options for refining your search criteria and results in a faster search.

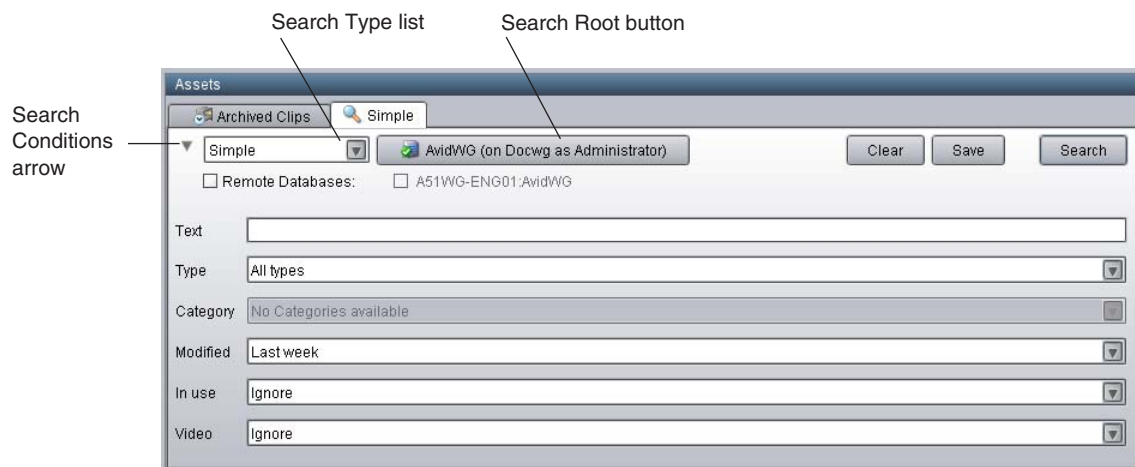
The following examples shows how to search for clips that have been logged to reference a fire that took place last night.

- The first example illustrates how to search for a clip if you know that the word “fire” appears in any text attribute, such as clip name, comment, and frame locators. This example also uses the Modified attribute to limit the search to clips modified in the last 24 hours.
- The second example illustrates how to make the same search more efficient if you know that the word “fire” is used in the clip’s name, because the scope of the search is limited to clip names only. This example also uses the Modified Date attribute to limit the search to clips created after a certain time.

For complete information on searching in Interplay Access, see the *Interplay Access User’s Guide* or the Interplay Access Help.

To search for clips in Interplay Access (Simple search):

1. Do one of the following:
 - ▶ Click the Search tab. Simple search is selected by default.
 - ▶ Select Edit > Search to open a new search tab. You can also press F3 or Ctrl+F (Windows) or Command+F (Macintosh).
 - ▶ Select Simple from the Search Type list.
2. If necessary, click the Search Conditions arrow to show the search criteria.



3. (Option) If you know which folder might contain the clips, click the Search Root button and navigate to the folder.
 The Search root button displays the database or folder you selected to search. Selecting a particular database or folder will make the search faster.
 You can also select a folder to search before you open a Search tab.
4. Type the text for the search. In this case, type the word “fire.”
5. In the Modified field, click the down arrow and select the time period. In this case you know that the fire took place last night, so you can select “Last 24 hours.”
6. (Option) Select a value for the other search attributes by clicking the down arrow and selecting an option.

7. Do one of the following:

- ▶ Press Enter.
- ▶ Click the Search button.

The Search button changes to a Stop button, which you can click if you want to stop the search. The Interplay search engine limits the database to clips that have been modified (captured or edited) in the last 24 hours, and then searches within that subset for text that contains the word “fire.” The results are displayed in the Search Results pane. They are visible until you close Interplay Access or until you perform a new search.

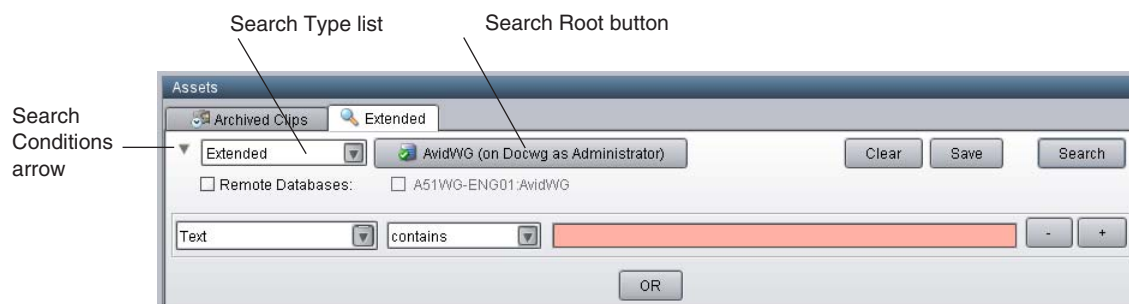
8. (Option) To start a new search, click Clear and repeat steps 1 through 7.

To search for clips in Interplay Access (Extended search):

1. Do one of the following:

- ▶ Select Extended from the Search Type list.
- ▶ Select Edit > Search to open a new Search tab. You can also press F3 or Ctrl+F (Windows) or Command+F (Macintosh). Then select Extended from the Search Type list.

2. If necessary, click the Search Conditions arrow to show the search criteria.



The Extended search uses a set of statements to construct a search. In this example, you can limit the search to the Name field. Even if you are not sure about using other attributes for the search, you can speed your search considerably by limiting the search to just the name of the clip.

3. (Option) If you know which folder might contain the clips, click the Search Root button and navigate to the folder.

The Search root button displays the database or folder you selected to search. Selecting a particular database or folder will make the search faster.

You can also select a folder to search before you open a Search tab.

4. Specify an attribute, in this case, “Name.”

5. Specify an operator, in this case, “contains.”

6. Type the words you want to search for, in this case, “fire.”

7. Add another search statement by clicking the + button and specifying the following:

- Select “Modified Date” as the attribute. The attribute selections include any attribute defined and enabled in the Property Layout view of the Interplay Administrator.
- Select “after” as the operator.
- Specify a date and time. You can edit the default date and time in the field. You can also use the calendar button to select a date.

8. Do one of the following:

- ▶ Press Enter.
- ▶ Click the Search button.

The Search button changes to a Stop button, which you can click if you want to stop the search. The Interplay search engine searches only the names of clips for those that contain the word “fire” that were created after April 22, 2010. The results are displayed in the Search Results pane.

9. (Option) To start a new search, click Clear and repeat steps 2 through 8.

Example: Finding Assets By Using a Unique Identifier

This topic illustrates a workflow in which you assign a unique identifier (numbers, letters, or a combination) to an asset and use that identifier to find and display the asset. For example, you could choose to associate a unique number with each sequence in an Interplay database so that you can quickly find a particular sequence later using that number, even if someone has changed the name of the sequence. This workflow uses a custom property and a custom search for the most efficient and reliable way of finding the asset.

There are alternative methods for associating a unique identifier to the one described in this topic. However, they do not make optimal use of the Interplay search engine. For example, one workflow might be to append an identifier to the end of an asset name so that it can be tracked with the asset. A disadvantage to this approach is that you need to use the full-text search mechanism to find the asset. Another disadvantage is that renaming the asset will remove the identifier unless you add it to the new name.

A more efficient approach is to create a custom property (or *metadata field*) that is used exclusively to hold the identifier. You can then search for the asset by using an exact match rather than a full-text search, and you can rename the asset without needing to add the identifier. You use the Interplay Administrator to create a custom property, and then use Interplay Access to set and search for the identifier.



Adding an identifier to the name of an asset and using full-text search to locate the asset by name causes unnecessary extra load on the Interplay Server, because it searches through much more data than it would if searching on a single custom property. You should always define a custom property if your workflow uses a unique identifier.

The following procedures describe

- How to add and display a custom property
- How to add the identifier
- How to search for the identifier

Prerequisite: A user must have write permission on an asset to add an identifier to an asset.

To add a custom property:

1. In the Site Settings section of the Interplay Administrator window, click the Property Layout icon.
2. In the Property Layout view, select the database for which you want to set properties.
3. Click the Custom Meta Data tab.
4. In the Add Custom Meta Data Field text box, type a name for the custom property. In this case, type the name “Identifier.”

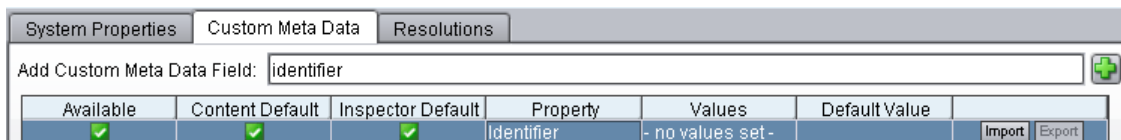


System Properties Custom Meta Data Resolutions

Add Custom Meta Data Field: identifier

5. Click the green plus sign (+).

The custom property appears in the list.



System Properties Custom Meta Data Resolutions

Add Custom Meta Data Field: identifier

Available	Content Default	Inspector Default	Property	Values	Default Value	
✓	✓	✓	Identifier	- no values set -		Import Export

6. Select the Available option, which makes the custom property available for users to add into Interplay Access displays and to use in an Extended search.
7. Select the options to include the property in the Content tab and the Object Inspector’s Properties tab. Users can then add or edit a value for Identifier in Interplay Access.
8. Click Apply to save the added custom property and new layout.

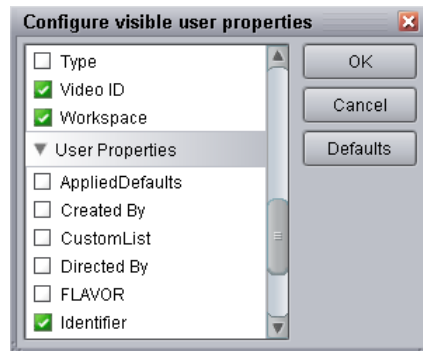
The next time a user logs in to Interplay Access, the Identifier property will be displayed.



Deselect the property in the Available column to turn off custom properties that you do not want displayed in the Interplay Access properties lists. You cannot delete custom properties for this release but you can turn off their display. For more information about working with custom properties, see “Creating Custom Properties” in the Interplay Engine and Interplay Archive Engine Administration Guide.

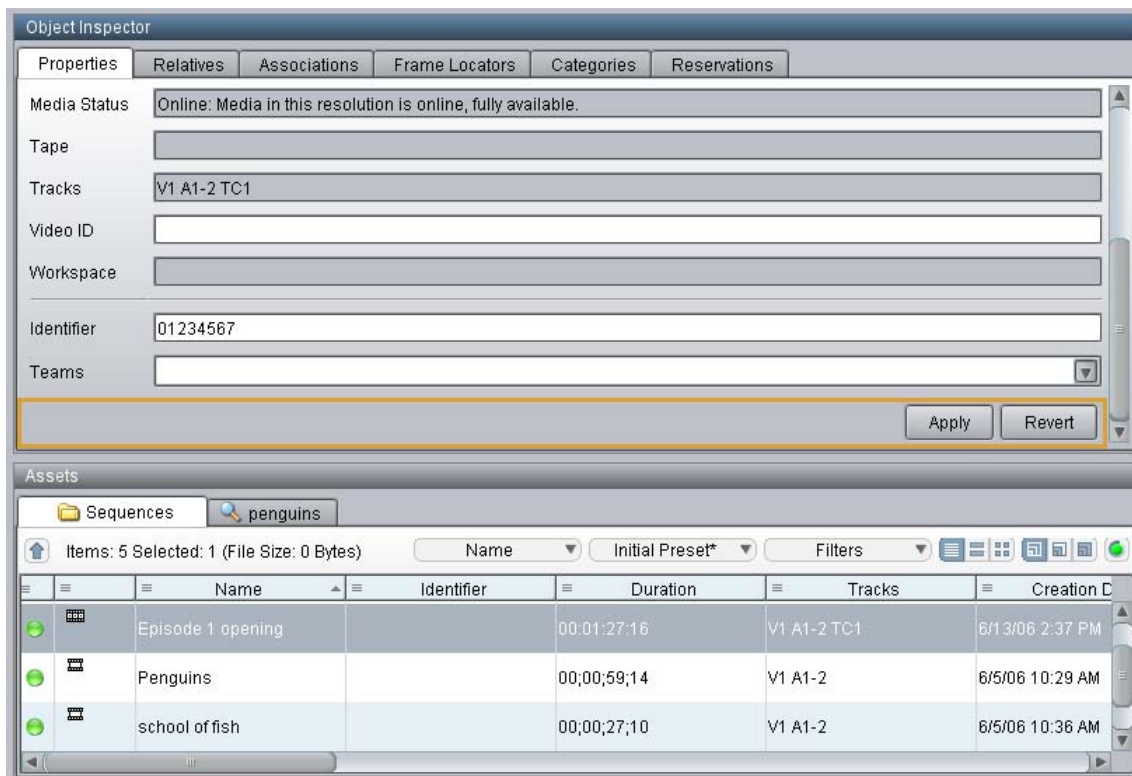
To set an identifier for an asset:

1. In Interplay Access, select the asset for which you want to set the identifier. In this case, select a sequence.
2. The first time you want to set an identifier, do the following:
 - a. Click the Properties tab, scroll down and click the Configure button.
The “Configure visible user properties” dialog box opens.



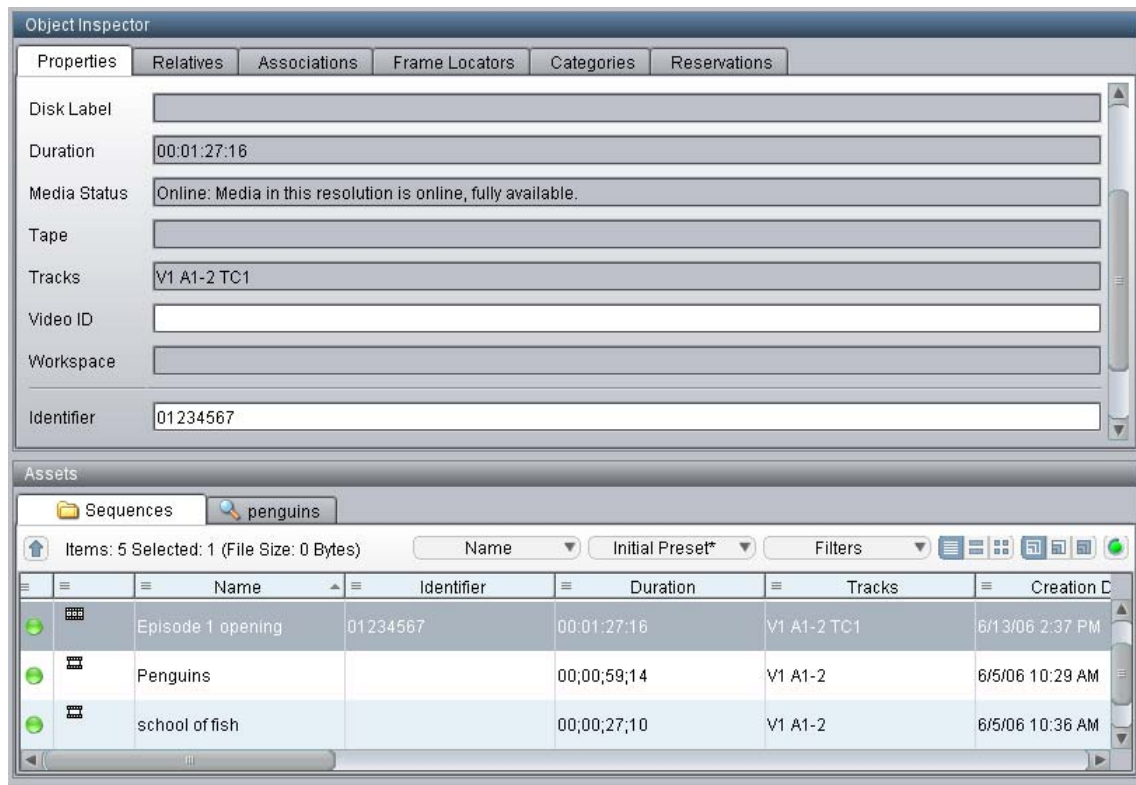
- b. In the User Properties section, select “Identifier” to display the property in the Object Inspector, and click OK.
3. In the Object Inspector, scroll down until you see the new custom property Identifier.
4. Click the text box and type the identifier.

You need write permissions on the asset to add the identifier. An orange box highlights the controls for adding a property value.



5. Click Apply.

The identifier is displayed in the Identifier column in the Contents tab.



To search for an asset using the unique identifier:

1. In Interplay Access, click the Search tab, then select Extended from the Search type list.
2. Click the Attribute Name field, scroll down the list, and select Identifier.



3. In the Operator field, select "is."
4. In the Value field, type the identifier, and click Search.

The asset is displayed in the Search results list.



If you know only part of an identifier, select “contains” from Operator field and type part of the identifier. This is less efficient than an exact match, but is much more efficient than a full-text search on the name.

5 Using AMA Material in Interplay | Production

This chapter contains the following topics:

- [Overview of Working with AMA Material](#)
- [AMA on Interplay | Production Workflow](#)
- [How Interplay | Media Indexer Works with AMA Material](#)
- [Configuring Interplay | Administrator and Interplay | Media Indexer to Manage AMA Media](#)
- [Using Third-Party Storage for AMA Media](#)
- [Viewing AMA Clips in Interplay | Access](#)
- [Deleting AMA Clips Using Interplay | Access](#)
- [Resyncing AMA Media Files](#)

Overview of Working with AMA Material

Interplay v3.0 and later supports checkin and dynamic relink of AMA (Avid Media Access) material. The following are two goals for this feature:

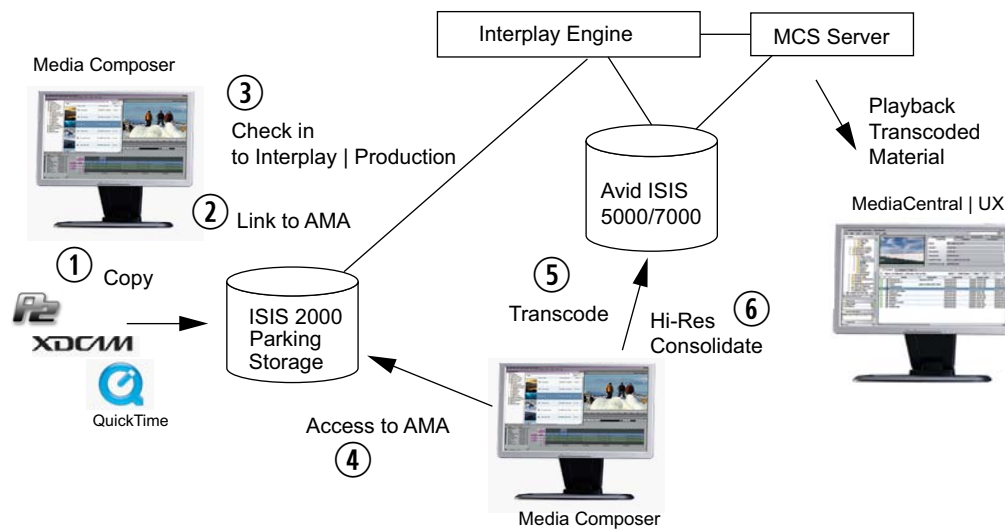
- Improve the Interplay workgroup aspects of the Avid file-based workflow
- Provide better support for workflows that make use of a large ratio of raw footage compared to the size of the final show

For example, some post-production shoot scenarios, such as reality TV, use multiple, heterogeneous cameras, with shoot ratios (unused/used) can that range from 30/1 to 100/1 or more. This means that every hour of programming can involve perhaps 100 hours of raw media to process. Processing includes copying the media from camera storage (Optical disk or SSD) to something more usable by the production system, such as ISIS shared storage.

One way to handle this in the past was to consolidate the recorded AMA media to native MXF files on the ISIS system associated with the Interplay Production environment (such as an ISIS 5000 or ISIS 7000). However, the potentially high shoot ratios mean that a lot of capacity was used to hold media that might be discarded later in the process. To solve this, the AMA workflow introduced in Interplay v3.0 uses a second, low bandwidth ISIS 2000 as a parking storage area for the AMA media. Later, when the material that should be saved is identified, it can be consolidated from the AMA source to the ISIS production system at full resolution.

AMA on Interplay | Production Workflow

The following illustration shows a basic AMA workflow for Interplay Production.



The following sections describes these numbered portions of the AMA workflow for Interplay Production. You can perform the operations individually or you can use the Dynamic Media Folder (DMF) feature in the Avid editing application to automate the process.

See the chapter “Acquisition of File-Based Media (AMA)” in the *Avid Media Composer | Software Editing Guide* or the *Media Composer Help* for information on working with AMA media in an Avid editing application.

Interplay v3.1 and later supports Media | Director, which can automate much of this process. For details, see the Media | Director User’s Guide on the Avid Knowledge Base.

Before You Begin

Before you can perform the third step in the workflow (Check in to Interplay), you must configure the Interplay Administrator and the Media Indexer to recognize the new material. Avid recommends that you dedicate a specific workspace for the .spaaf files. See “[Configuring Interplay | Administrator and Interplay | Media Indexer to Manage AMA Media](#)” on page 87.

1. Copy the AMA Material to Shared Storage

This step involves copying files from the camera media card or disk to shared storage and does not have to be performed on an editing workstation.

In order to be able to perform a dynamic relink to AMA material, the AMA material must reside on shared storage at the time of checkin.

This workflow uses a shared storage system such as an ISIS 2000 as a parking area for the AMA media. This allows you to store all of the original AMA media on shared storage without taking up space on your high bandwidth ISIS 5500 or ISIS 7500. This is particularly useful if you have a high ratio of shooting versus footage used in the final production. In this scenario you would have a multiple ISIS setup; one for the original AMA material and one for your production material.

Interplay v3.1 and later supports third-party storage as parking storage. For details see “[Using Third-Party Storage for AMA Media](#)” on page 93.

Create a folder structure in the parking area that organizes the material by project and use a unique name for each new folder. For example, if you have a series of interviews associated with Project A you could create a folder structure such as Project A/interviews/interview1 and store the media for interview1 in that folder.

Drag and drop all of the material from the media card or disk into the appropriate folder.

2. Create a New Bin and Create a Link to the AMA Media on Shared Storage

Create a new bin on the Avid editing application. From the new bin, use the Link to AMA command in the Avid editing application to link to the AMA material on shared storage. For information on the Link to AMA process, search for “Linking Media with AMA” in the editing guide or Help for your Avid editing system.

After you successfully link to the AMA material, the Avid editing system creates clips in the bin for the media and you can start working with the material in the editor. If you want other editors to be able to access the media, you need to check the AMA clips into Interplay Production.

3. Check the AMA Clips into Interplay Production

You can use the following methods to check the AMA clips into Interplay Production:

- Use menu commands or automatic checkin
- Drag assets to the Interplay Window
- Check the files in automatically as part of the profile associated with a DMF folder

For information on checking assets into Interplay Production, see “Checking Avid Assets into the Interplay Database.” For information on using DMFs, see “Creating Dynamic Media Folders.” Both topics are in the *Avid Media Composer / Software Editing Guide* and the Media Composer Help.

When you check the AMA clips into Interplay Production, the Interplay client library creates files that the Interplay Media Indexer can index. There are two types of files:

- .spaaf (sparse AAF) files. These files are similar to .mxf files that are used for native Avid media. The main difference is that .mxf files contain media but .spaaf files contain pointers to the media.
- AAF files. These files are similar to .aaf files for native Avid media but refer to AMA media.

For information on the files that are created, see [“How Interplay | Media Indexer Works with AMA Material” on page 85](#).



The SPAAF workspace on ISIS must be mounted and accessible by the editing user on the editing machine. On a Mac OS X system, the workspace must be mounted using the same name as the workspace. The checkin will fail if this condition is not met.

4. Access to AMA

After the Interplay Media Indexer indexes the new files, other Avid editing applications can check out the clips and work with them. Interplay Access can also search for the files and display information about the files in the Object Inspector. See [“Viewing AMA Clips in Interplay | Access” on page 97](#). Note that MediaCentral UX, Interplay Assist, and Avid Instinct cannot play the material until you transcode it to an Avid resolution.

5. Transcode the AMA Material

Depending on the bandwidth requirements of the AMA material, you can improve playback and editing performance on the Avid editing systems if you transcode the AMA material you want to work with to an Avid resolution. Then use Dynamic Relink to work with the transcoded material.

Another benefit of transcoding the AMA material is that MediaCentral UX, Interplay Assist, and Instinct users can now work with the clips.



You cannot create an H.264 proxy from the AMA material using the Transcode feature on an Avid editing application. However, you can use Interplay Transcode to create a proxy.

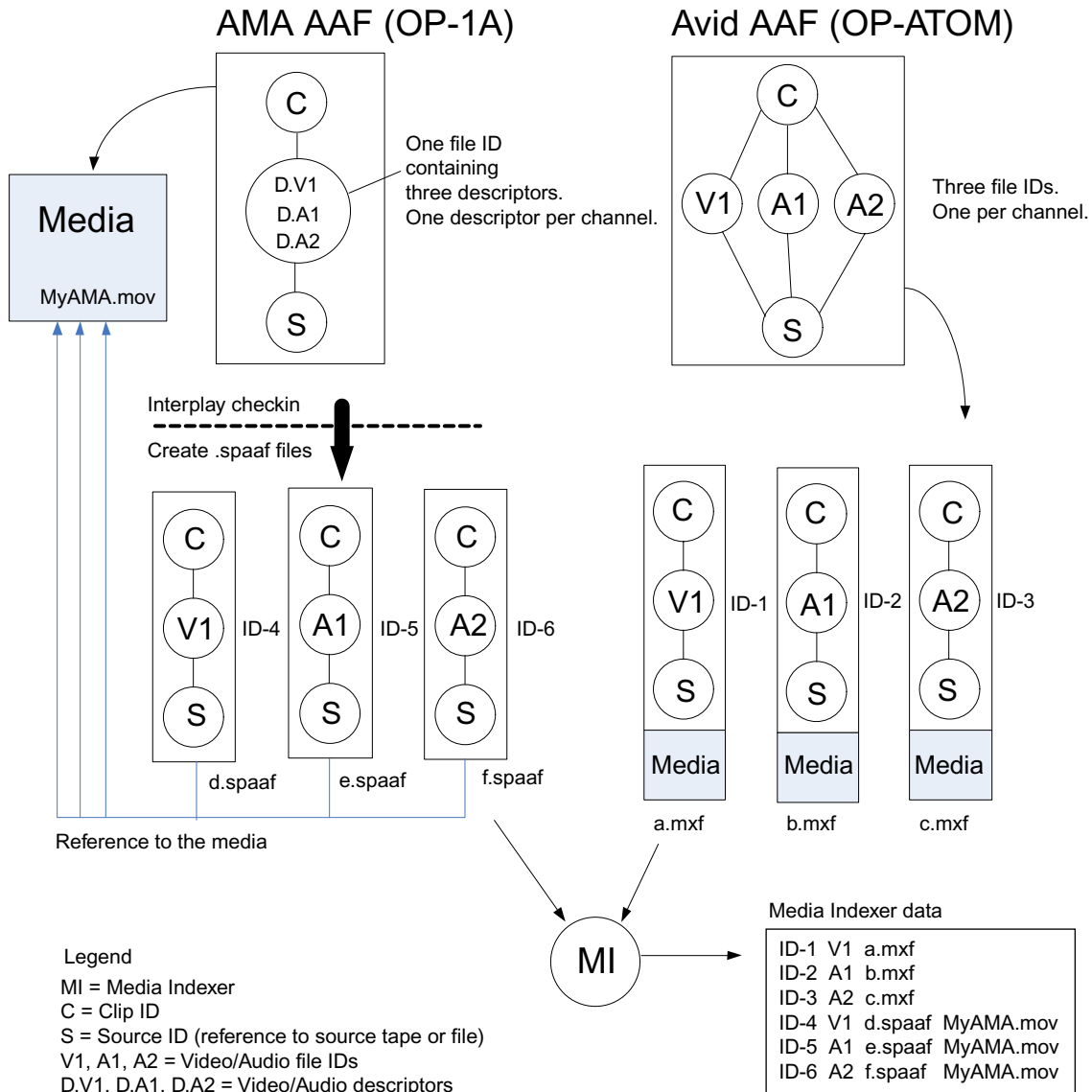
6. High Res Consolidate

If you are working with a transcoded resolution, you can dynamically relink back to the high resolution AMA material when you want to consolidate a high res version of your final sequence to your production ISIS system. This allows you to finish the sequence using the original AMA resolution. In this case only the portions of the AMA material that are used for the final sequence need to be stored on the production ISIS system.

In a broadcast facility, you can transcode this final AMA sequence to the format that you need for a send to playback operation. This allows you to create a final version from the original AMA media. In a post facility, a colorist may use the high resolution AMA material for a final color correction. Then it can be transcoded for the final output.

How Interplay | Media Indexer Works with AMA Material

In releases prior to Interplay v3.0, the Interplay Media Indexer was not able to index AMA material. This limitation is related to the difference between AMA material (OP-1A format) and the OP-ATOM format associated with .mxf material. The following illustration compares OP-1A and OP-ATOM material.



The following sections compare the checkin process for OP-ATOM material with the checkin process for OP-1A material.

Avid AAF (OP-ATOM) Format

The example in the previous illustration uses a clip with one video channel and two audio channels. The top portion of the Avid AAF section is a representation of what you can see in an Avid editing system bin for digitized MXF material. The metadata contains the following:

- A clip ID
- Three file IDs, one for each video and audio track.
- A source ID (for example, a reference to the source file or source tape)

The three file IDs relate directly to the associated .mxf files on shared storage. The .mxf files contain the related media and each contains a File ID that identifies the file as video or audio. In the “Media Indexer data section” of the illustration, the Interplay Media Indexer lists the three files starting with ID-1, ID-2, and ID-3.

AMA AAF (OP-1A) Format

The format for the OP-1A material above the dashed line in the illustration is obviously different. Instead of separate file IDs for video and audio, the information is stored as separate descriptors within one file ID. And the media is store separately without the metadata elements associated with OP-ATOM material.

AMA Checkin to Interplay | Production

The area below the dashed line in the illustration shows the new files created when you check AMA material into Interplay v3.0. The Interplay client library (also known as the JXDK) creates a new file for each channel or track of the AMA material. These new files have a .spaaf extension and are in a format that the Interplay Media Indexer can index. Each file contains a file ID for the associated audio or video channel as well as a pointer to the related portion of the media. The “Media Indexer data” portion of the illustration shows these files as ID-4, ID-5, and ID-6. Each entry in the Interplay Media Indexer database references the .spaaf file and the AMA Media file.

As you check the AMA files into Interplay Production, the system maintains a hierarchy of folders within that folder. You store the AMA media files in a separate location such as an ISIS 2000 parking storage. The Interplay Media Indexer needs to be aware of the location of both sets of files. It does not index the AMA media files but it does index the .spaaf files. This allows you to use dynamic relink to an AMA resolution. See [“AMA on Interplay | Production Workflow” on page 82](#).

For Interplay v3.1, the checkin process copies the OP-ATOM AAF file to the same location that you specified to hold the .spaaf files.

Configuring Interplay | Administrator and Interplay | Media Indexer to Manage AMA Media

The Interplay Administrator and Interplay Media Indexer work together to index and manage AMA media. Starting with Interplay v3.2, the AMA metadata location you specify in the Interplay Administrator is automatically set in the Media Indexer.

See the following topics:

- [“Configuring the AMA Metadata Location” on page 88](#)
- [“Specifying the AMA Media Files Location” on page 89](#)

Special configuration is required for third-party storage. See [“Using Third-Party Storage for AMA Media” on page 93](#).

Configuring the AMA Metadata Location

You need to determine a workspace on ISIS shared storage to hold the .spaaf files and AAF files for AMA media. The following requirements apply to this workspace:

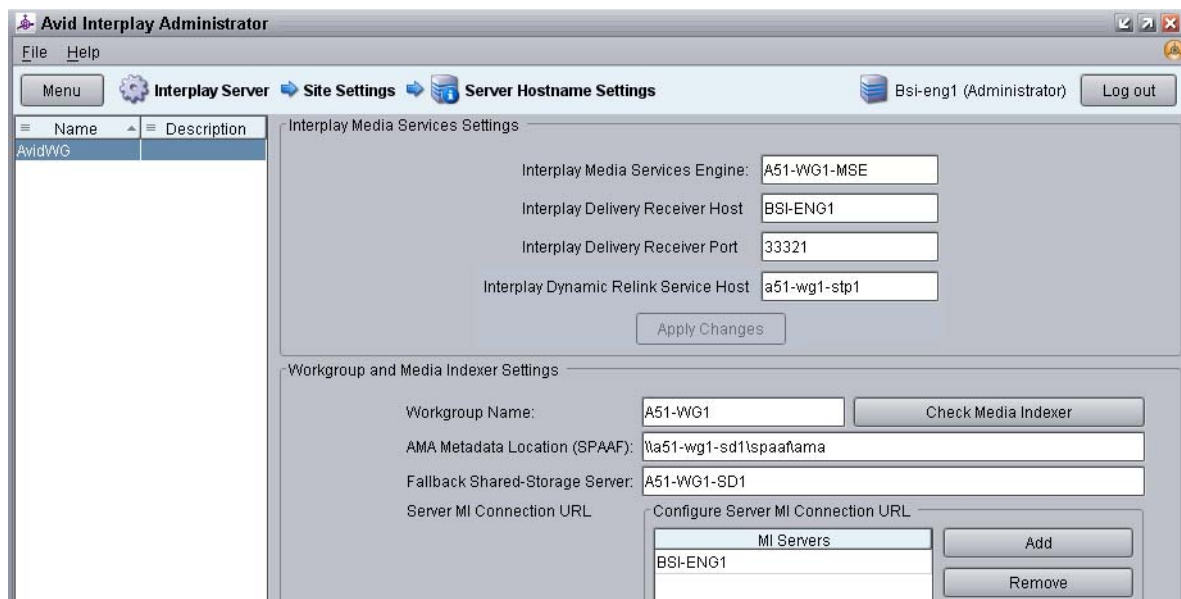
- All of the .spaaf and AAF files will be located under one top-level directory. Avid recommends setting up a specific shared-storage workspace for the spaaf files. Do not add .mxl files to this workspace.
- The workspace containing the .spaaf and AAF files must be accessible to the ISIS client on the editing client. On Macintosh systems, the workspace must be mounted. Checkins will fail if the workspace is not mounted. The workspace does not need to be mounted on Windows systems.
- The editor using Media Composer must have write permission to this workspace.

This workspace is labeled “AMA Metadata Location (SPAAF)” in both the Interplay Administrator and the Media Indexer. Starting with Interplay Administrator v3.2, if you set this location in the Interplay Administrator, it is automatically set in Media Indexer.

To configure the AMA metadata location in the Interplay Administrator:

1. Log in to the Interplay Administrator tool and open the Server Hostname Settings view.

The following illustration shows the Server Hostname Settings view.



2. In the “AMA Metadata Location (SPAAF)” text box, enter the location for the .spaaf and AAF files. Enter the value as a full UNC path (on both Windows and Macintosh systems). In the above example, the name of the ISIS System Director system is a51-wg1-sd1. A specific workspace is set aside for the .spaaf files named spaaf. This location is automatically provided to the Media Indexer and is displayed in the Interplay Media Indexer Web interface (see [“Specifying the AMA Media Files Location” on page 89](#)).
3. Click Apply Changes.

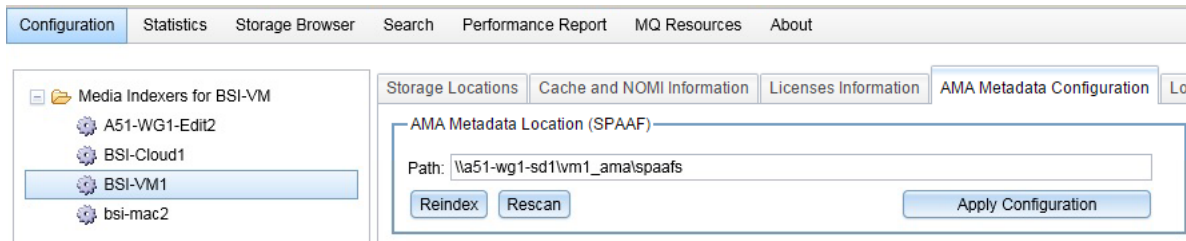
To check the Interplay Media Indexer Web interface for the AMA metadata location:

1. Open the Interplay Media Indexer Web interface by typing the following in a browser window:
`http://<MI system name>:8443/mi/`

where *<MI system name>* can either be the name of the system running the Interplay Media Indexer or its IP address. The string “:8443” identifies the port that the Interplay Media Indexer system uses for web communication.

2. Click the Configuration tab and then click the AMA Metadata Configuration tab.

The path displayed is the top level of the storage location for the .spaaf files and AAF files created when you check an AMA clip into Interplay Production.



The path that is displayed must match the path that you entered in the Interplay Administrator. If it does not match, enter the UNC path as shown in the previous illustration and click Apply Configuration.

Specifying the AMA Media Files Location

You need to specify the storage folder that holds the AMA media files so that Media Indexer can identify the media files. How you specify the folder depends on the type of storage you use:

- For media stored on ISIS shared storage, use the following procedure.
- For media stored on third-party storage, see [“Using Third-Party Storage for AMA Media” on page 93](#).



The Interplay Media Indexer needs to know the location of the AMA media files but it does not index those files. The Interplay Media Indexer indexes the related .spaaf files at the location identified in “Configuring the AMA Metadata Location” on page 88.

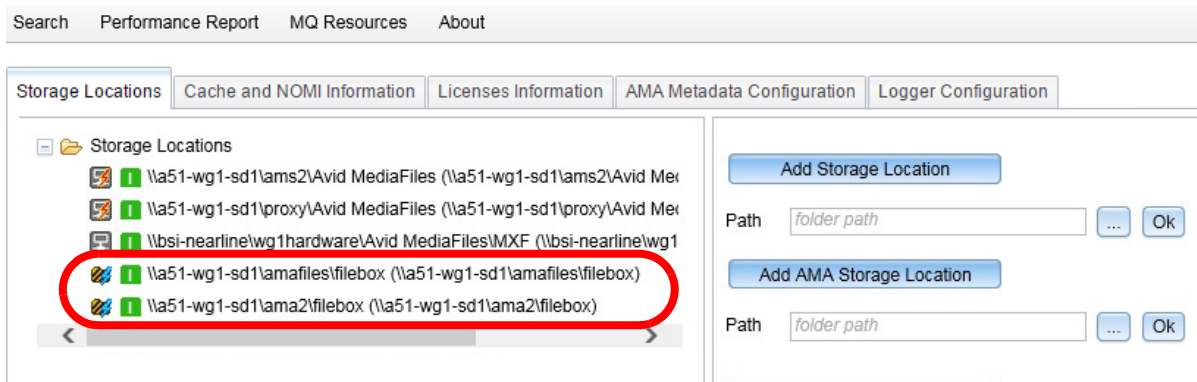
To specify the location of the AMA media files for ISIS shared storage:

1. Open the Interplay Media Indexer Web interface by typing the following in a browser window:
`http://<MI system name>:8443/mi/`

where *<MI system name>* can either be the name of the system running the Interplay Media Indexer or its IP address. The string “:8443” identifies the port that the Interplay Media Indexer system uses for web communication.

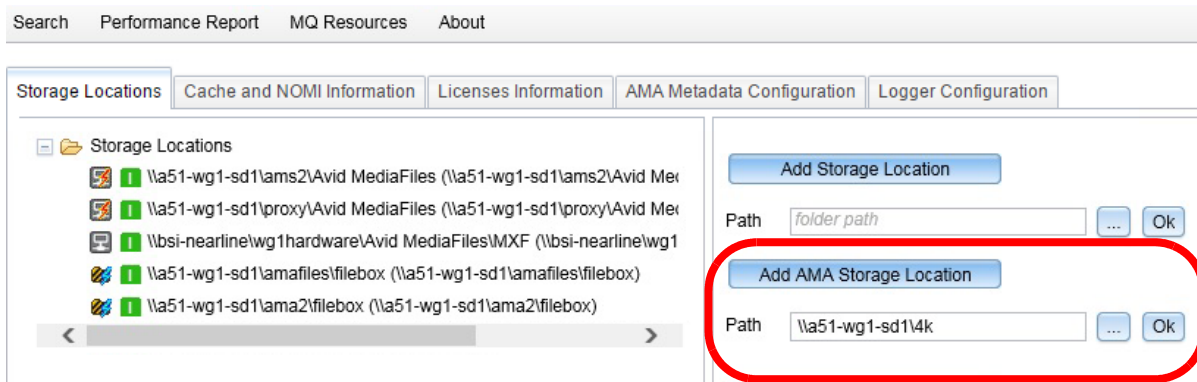
2. Click the Storage Locations tab.

The system displays the locations of .mxf media on shared storage as well as any previously defined AMA storage locations. The following illustration shows three locations for Avid media files and two locations for AMA media files (circled).



3. Click the Add AMA Storage Location button.

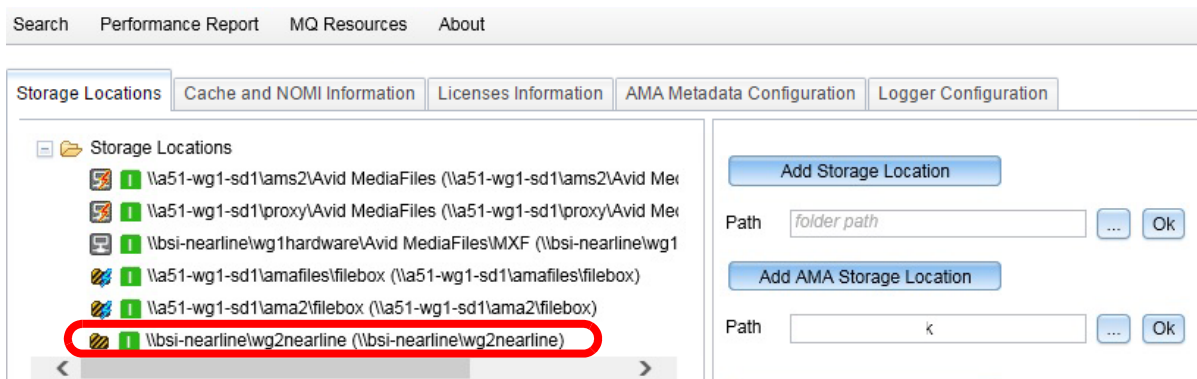
The system displays a path dialog, a browse button, and an OK button.



4. Click the Browse button to navigate to the location of a folder containing AMA media on shared storage. You can have multiple AMA media folders in different locations.

5. Click OK.

The AMA storage location appears in the list of storage locations.



 Do not mix .mxf media files and AMA media files in the same storage location. Also, do not mix .spaf files and AMA media files in the same location.

Viewing Statistics about AMA Media

You can view information about AMA files using the Interplay Media Indexer Statistics page and Storage Browser. Note that this information is not available until you check the AMA media into the Interplay Production database.

To view storage statistics about AMA files:

1. Click the Statistics tab.
2. For comparison, open and (optionally) rearrange the following panes:
 - Storage Statistics
 - AMA Storage Statistics
 - Storage Workspaces
 - AMA Storage Workspaces

Interplay Media Indexer displays separate panes for .mxf files and AMA media files.

The image displays four separate windows from the Interplay Media Indexer interface, each showing different types of statistics. Each window has a title bar, a list of metrics, and a 'Refresh' button at the bottom.

Indexing Workers Thread Pool Statistics

Storage Statistics	
Workspaces:	15
Folders:	374
Files:	163234
Quarantined Files:	110
Unique Media Files (File Mob IDs):	162570
Duplicates:	664
Channel Ids:	67478
Quality Descriptors:	1183
Channel Id Interests:	2581

Ama Storage Workspaces

Workspaces:	1
Workspace:	\\a51-wg2-sd1\ama\filebox
Folders:	124
Files:	1540
Quarantined Files:	0
Quarantined folder size:	0.0 bytes
Total size:	1.043 GB
Unc Path:	\\a51-wg2-sd1\ama\filebox
Runtime Id:	1
Flags:	1

Ama Storage Statistics

Workspaces:	1
Folders:	125
Files:	1540
Quarantined Files:	0
Unique Media Files (File Mob IDs):	1550
Duplicates:	-10
Channel Ids:	7068
Quality Descriptors:	121
Channel Id Interests:	2581

AMA Metadata (SPAAF) Workspace

Workspaces:	1
Workspace:	\\a51-wg2-sd1\ama\spaafs
Folders:	1538
Files:	7149
Removed SplitOff Files:	9
Removed SplitOff folder size:	9.0 bytes
Quarantined Files (excluding removed splitOff files):	362
Quarantined folder size (excluding removed splitOff folder):	56.634 MB
Total size:	1.043 GB
Unc Path:	\\a51-wg2-sd1\ama\spaafs
Runtime Id:	0
Flags:	1

Note that the Total size entries provided in the AMA Storage Workspaces window relates to the total size of the .spaaf files associated with the AMA media files.

To obtain the total file count for the Interplay Media Indexer:

- Add the Files value from the Storage Statistics section (top window in the example above) and add the number of .spaaf files on the ISIS file system. Since the .spaaf files are all in one workspace, you can use Windows OS tools to determine the number of .spaaf files that the Interplay Media Indexer is monitoring.

Note that the Files count listed in the AMA Storage Statistics window shows the number of AMA media files. It does not show the number of .spaaf files.



A separate .spaaf file is created for every channel (track) in an AMA media file. This means that if you check an AMA file with one video track and four audio tracks into Interplay Production, you add five files to the Interplay Media Indexer file count.

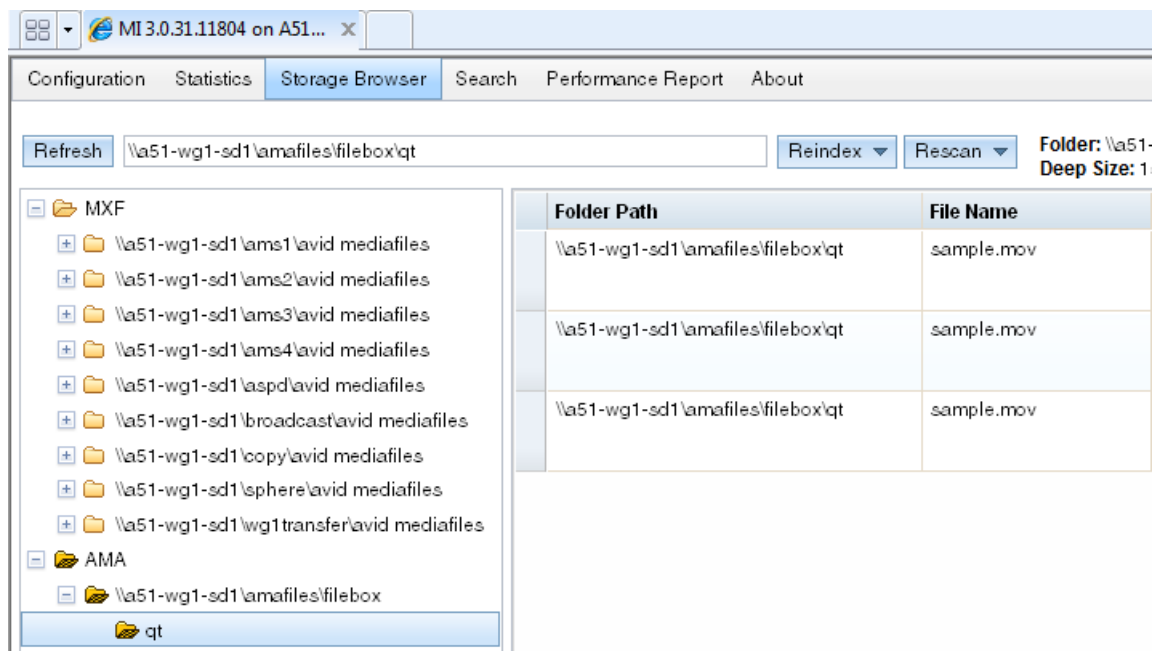
Using the Storage Browser

Starting at Interplay v3.0, the Storage Browser window contains separate sections for MXF media and AMA media.

To view information about AMA media in the Storage Browser:

1. Click the Storage Browser tab.

The Storage Browser window opens to show separate folders for MXF and AMA media.



You can use the storage browser to see how the AMA media files are organized. In this example, the AMA files were copied to a subfolder on a workspace named amafiles.

To view the location of the associated .spaaf files, scroll over to the right in the window.

Using Third-Party Storage for AMA Media

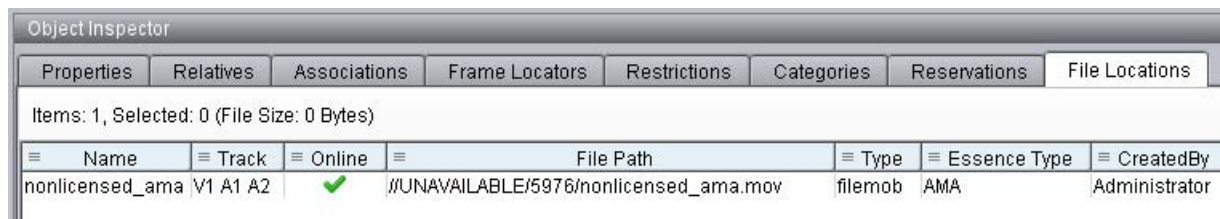
Interplay v3.1 and later supports third-party (3rd party) storage for “parking storage” of AMA material. Parking storage is often used to store AMA sources or other material where real-time playback is not required. For example, parking storage is used as part of a Media | Director workflow where real-time playback is not required and/or where high-density storage offers a better value for large files that may need to be stored for an extended period.

A license is required to use third-party storage. A procedure in this section provides more details.

Note that in Interplay Access the network locator (file path) for an asset that is stored on unlicensed third-party storage is shown by a path that uses the following format:

//UNAVAILABLE/nnnn/filename

where *nnnn* is a variable four-digit number. The following illustration shows an example of how this path is displayed in the Object Inspector.



Object Inspector

Properties | Relatives | Associations | Frame Locators | Restrictions | Categories | Reservations | File Locations

Items: 1, Selected: 0 (File Size: 0 Bytes)

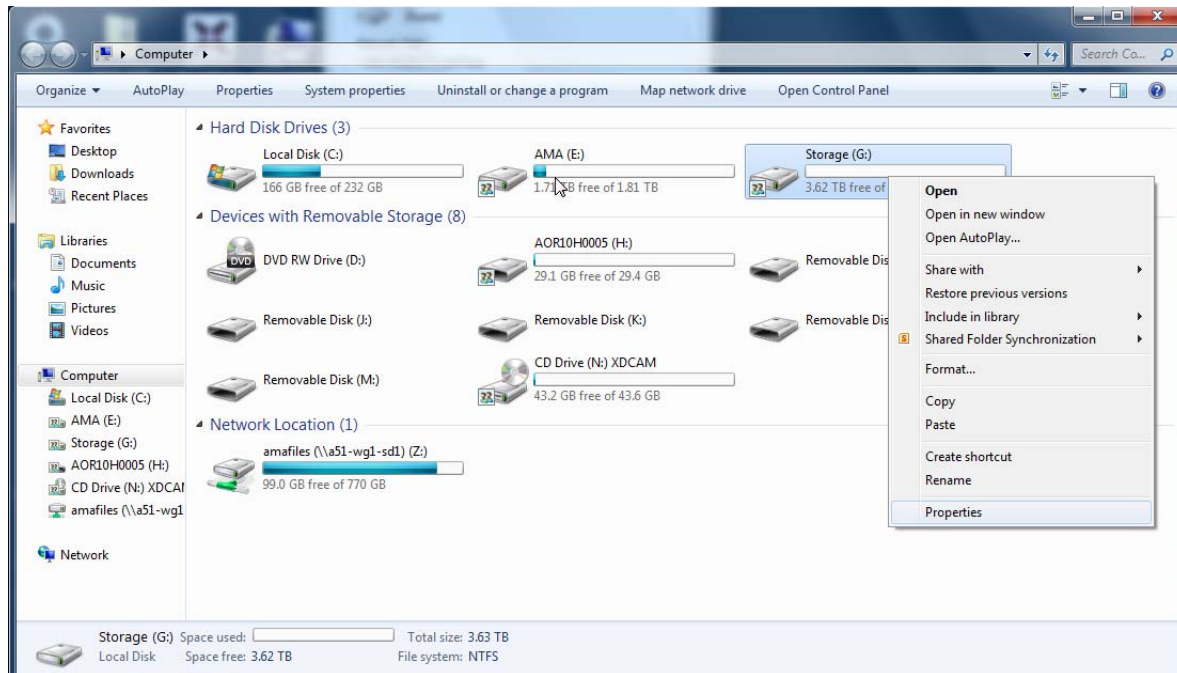
Name	Track	Online	File Path	Type	Essence Type	CreatedBy
nonlicensed_ama	V1 A1 A2	✓	//UNAVAILABLE/5976/nonlicensed_ama.mov	filemob	AMA	Administrator

A full path is shown if the system is properly licensed.

The third-party storage must be defined as a network share that is publicly accessible. It must appear on the network as a UNC path. How a third-party storage and its UNC share path are created on a network depends on the storage product and technology. The following procedure describes how to configure a Windows storage device as a Windows share.

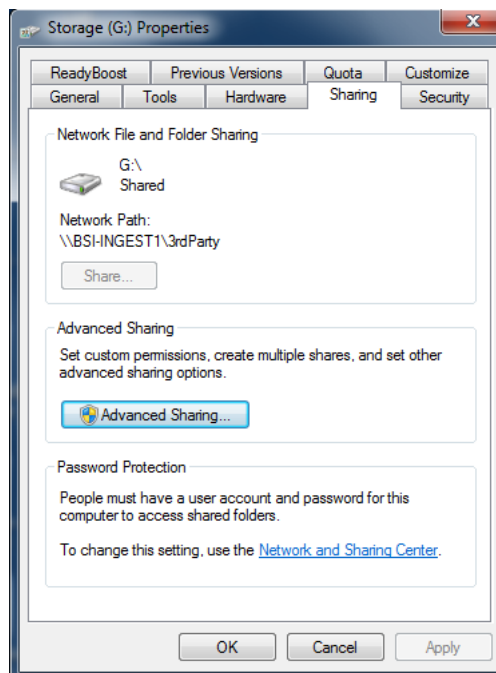
To define a third-party storage share as a Windows share:

1. On the Windows desktop, right-click the storage location you want to use as third-party storage and select Properties.

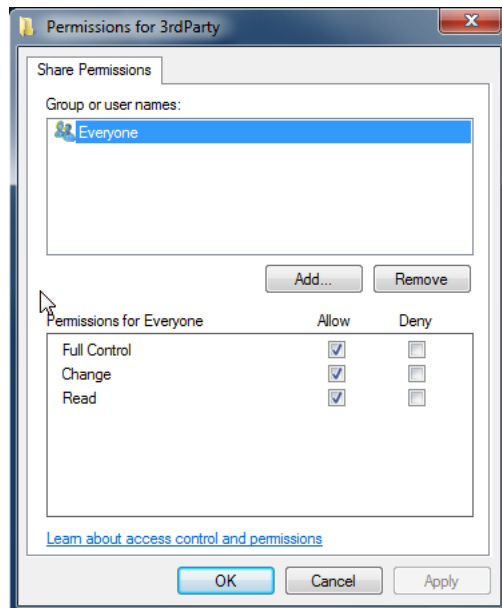


The Storage Properties dialog opens.

2. Select the Sharing tab.



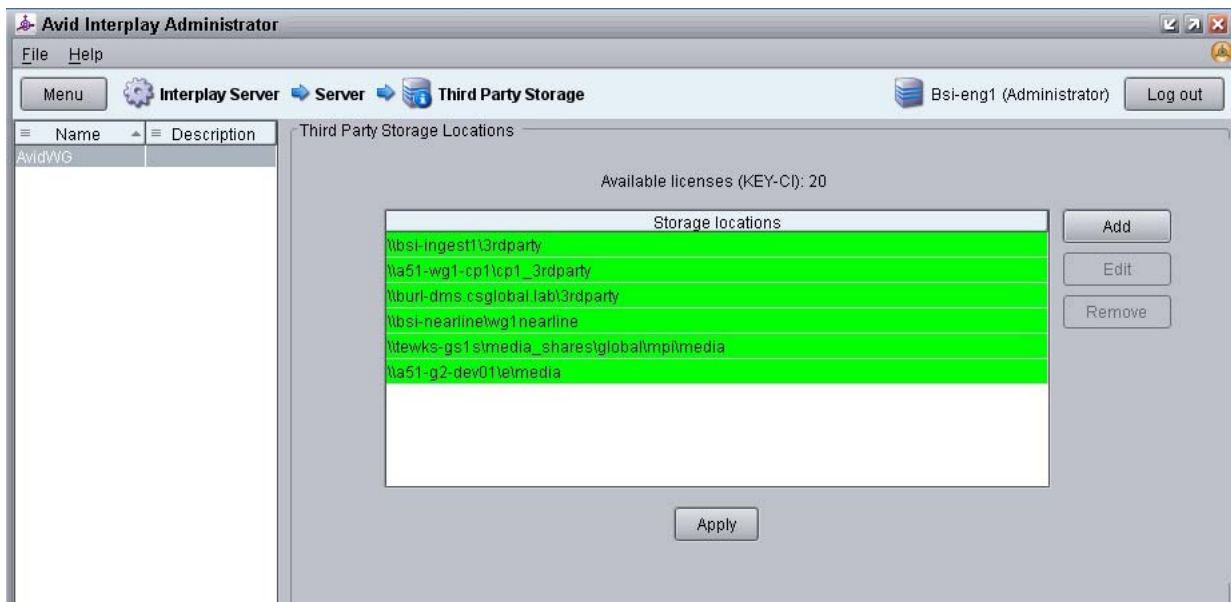
3. Click Advanced Sharing. The Share Permissions dialog opens.



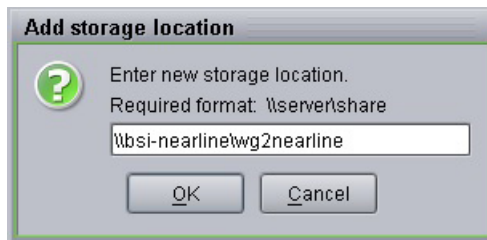
4. Give Everyone full access to the share and click Apply and OK.
The share is ready to be added to the Interplay environment.

To configure the third-party storage share in the Interplay Administrator:

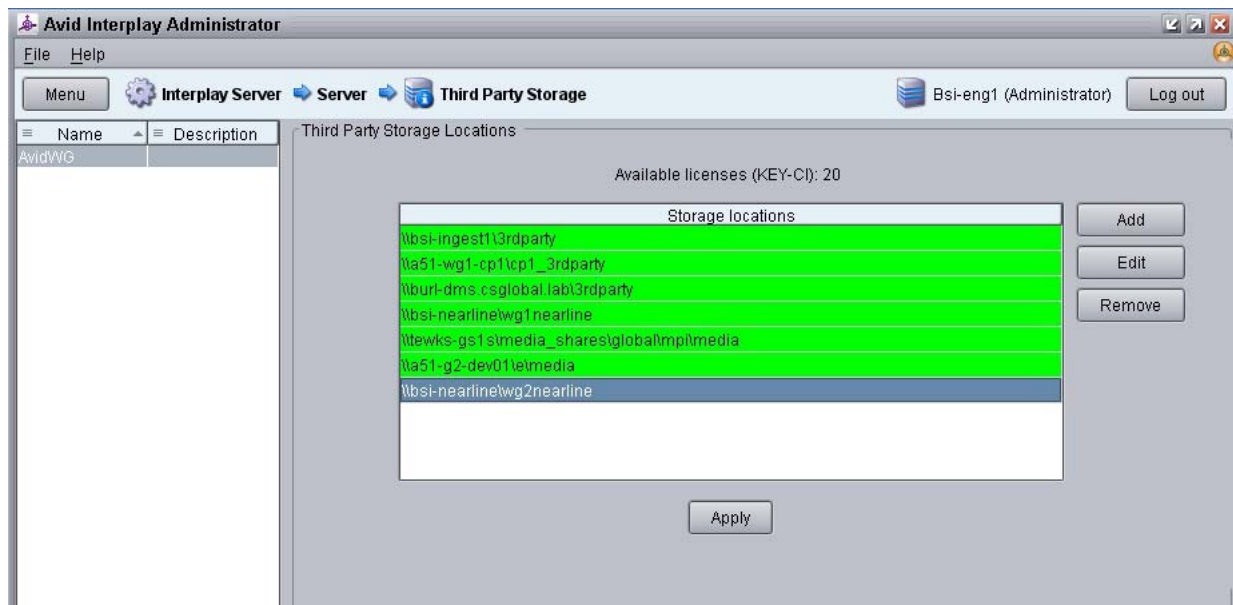
1. Purchase one or more Key-CI licenses from your Avid representative and add the licenses to your Interplay system using the Licenses view in the Interplay Administration tool. You need a license for each third-party location that you configure. See the *Interplay System Administration Guide* for details on obtaining and installing a license.
2. Make sure the network share for the third-party license is properly configured. The previous procedure describes how to create a Windows share.
3. Open the Third Party Storage view in the Server section of the Interplay Administrator tool.



- Click Add, type the network share as a UNC path, and click OK.



The storage location is added to the list.



- Click Apply.

The Interplay Administrator sends both the licenses and the storage information to the Media Indexer. The following procedure describes how to check the storage location.

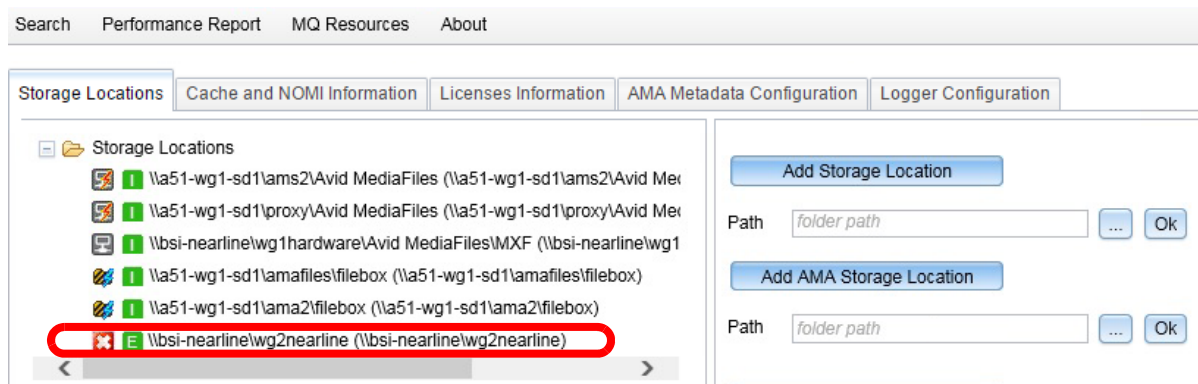
To check the third-party storage share in the Media Indexer:

- Open the Media Indexer web interface in a browser as follows:

`http://<MI system name>:8443/mi/`

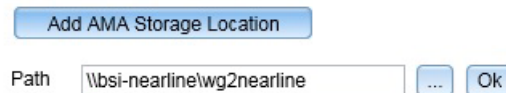
where <MI system name> can either be the name of the system running the Media Indexer or its IP address. The string “:8443” identifies the port that the Media Indexer system uses for web communication.

- Click the Configuration button at the top of the window and click Storage Locations.



The third-party storage location should be listed. If it is not, add it manually, as described in the following step.

3. Click Add AMA Storage Location and add the network share as a UNC path. Then click OK.



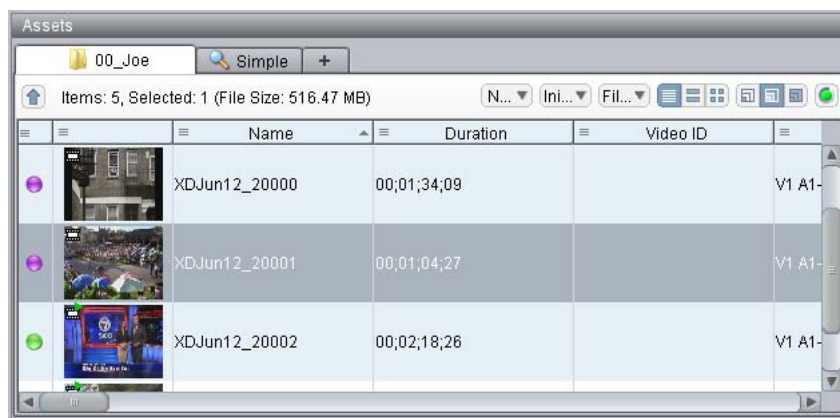
The third-party storage location is added to the list of storage locations.

Note that storage locations with a third-party license associated with them are designated with a green E (stands for explicit license) icon. Other storage locations have an I designation (implicit license).

You can also check information in the Licenses Information tab.

Viewing AMA Clips in Interplay | Access

After you check an AMA clip into Interplay Production you can search for or navigate to the clip in Interplay Access. The following illustration shows an AMA clip in the Interplay Access Object Inspector. Assets that only have AMA media are shown with a purple circle.



The following information is specific to using Interplay Access with AMA material.

- The AMA Media Status column in the Assets view allows you to determine whether the object you are looking at contains AMA material. The column displays one of the values described in the following table.

Value	Dot color	Description
has AMA media linked	Green	The clip is linked to both AMA media and native Avid media
has only AMA media linked	Purple	The clip is linked to AMA media, but is not linked to native Avid media.
no AMA media linked	Green	The clip is not linked to AMA media.

The following illustration shows an example of each value.

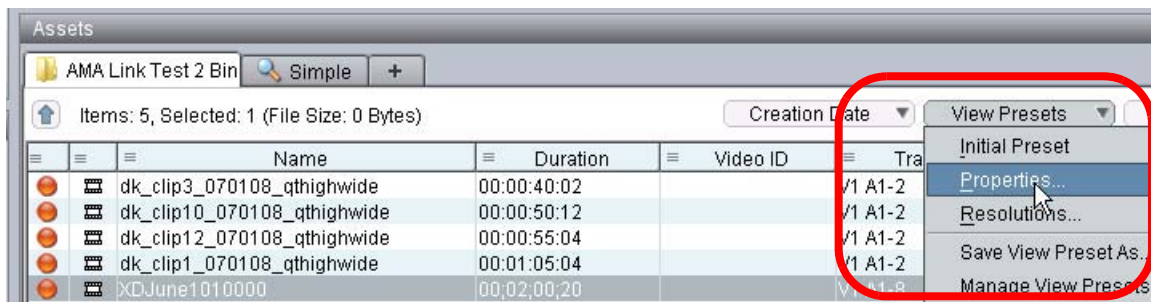
	Name	AMA Media Status
	0017L3	has only AMA media linked
	aa_masterclip_20	no AMA media linked
	aa_masterclip_8	has AMA media linked

- There are several new search criteria for an Interplay Access extended search, as described in the following table.

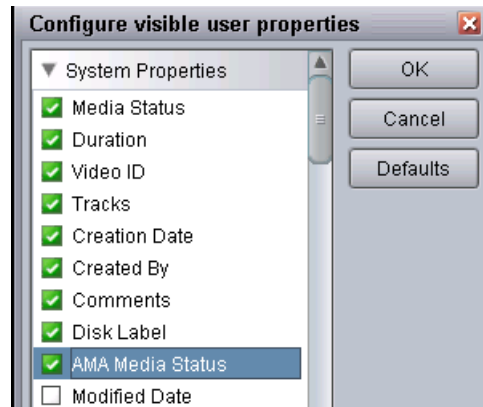
Criterion	Description
has AMA media linked	Search for clips linked to AMA media. These clips can include only AMA media or both AMA media and native Avid media.
has native media linked	Search for clips linked to native Avid media or both native Avid media and AMA media.
has only AMA media linked	Search for clips that are linked to AMA media but not to native Avid media.
has native and AMA media linked	Search for clips that are linked to both AMA media and native Avid media.
does not have AMA media linked	Search for clips that are not linked to AMA media.

To turn on the display of the AMA Media Status column and search fields in Interplay Access:

1. Open the Interplay Administrator tool.
2. Open the Property Layout view in the Site Settings area.
3. Select the check box in all three columns for AMA Media Status and click Apply.
4. In Interplay Access, select Properties from the View Presets menu in the Assets view.



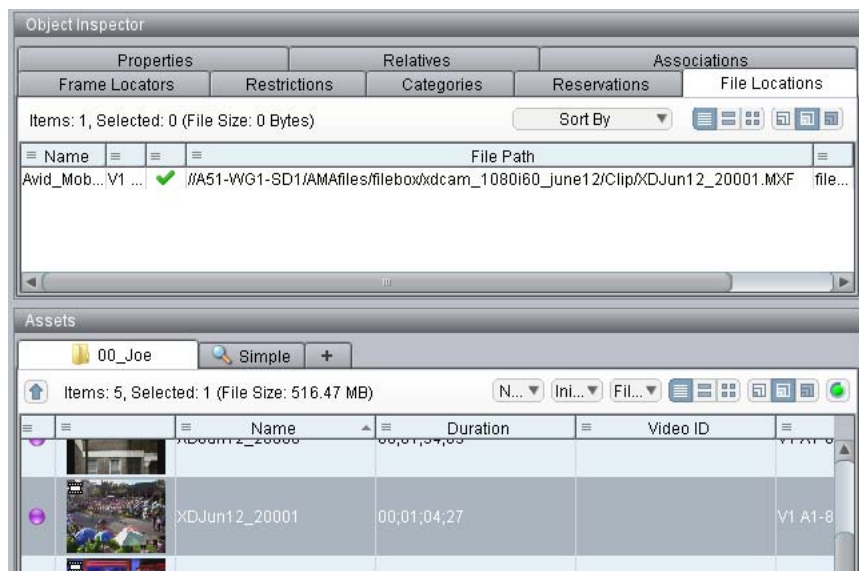
5. Select AMA Media Status and click Apply.



The AMA Media Status column appears in the Assets view and now you can perform an extended search for AMA Media Status values.

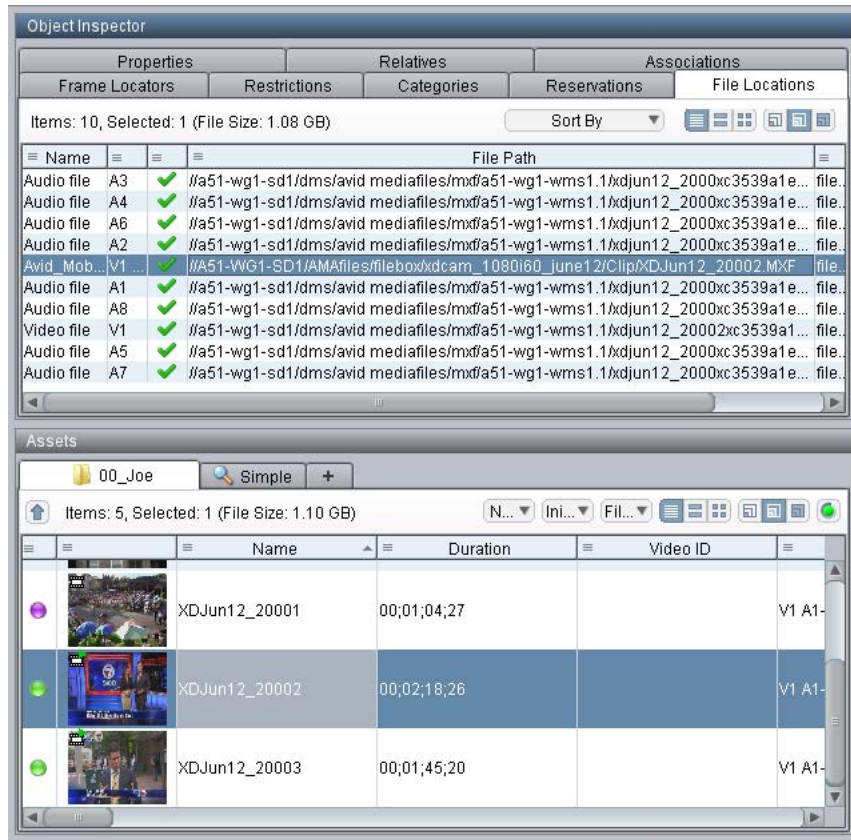
AMA Source File Locations in Access

The Interplay Object Inspector shows the location of the associated AMA source file in the File Path column of the File Locations tab. The following illustration shows a clip that only has AMA media associated with it.



The extension of the AMA source file depends on the AMA plugin used to create the file. For example, QuickTime files have a .mov extension. Note that when you delete the AMA clip in Access you delete only the reference to this source file.

The following illustration shows the file locations for a clip that has both AMA media and proxy media. The AMA source file is highlighted. Note that the AMA media is stored in a separate workspace (named AMAfiles in this example). Following this best practice helps you to organize your AMA material.



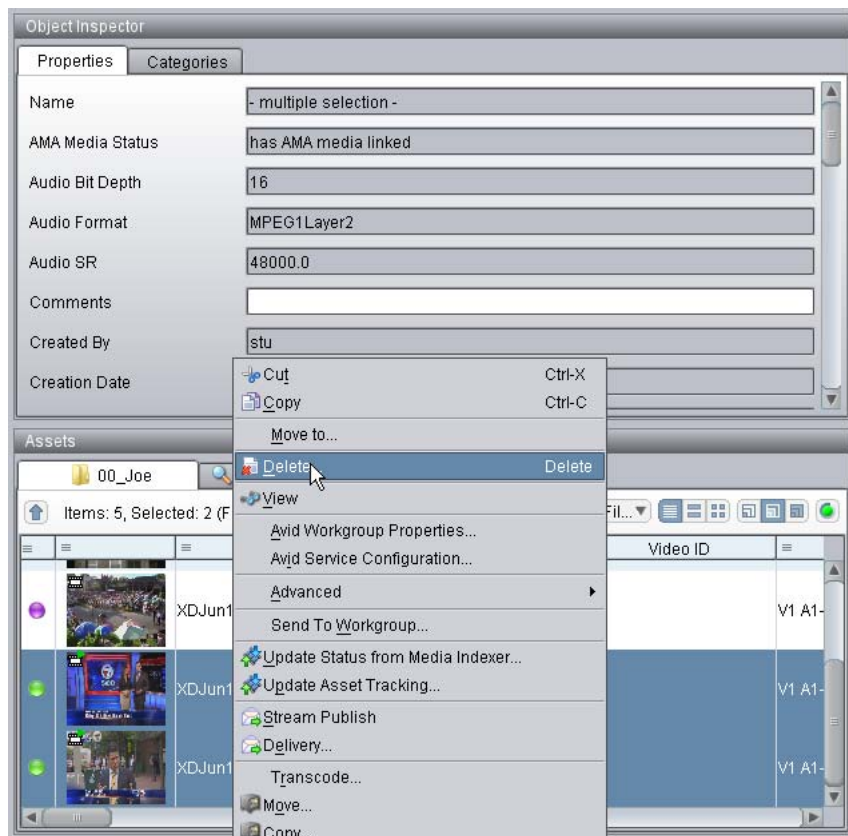
Beginning with Interplay v3.1, there is an additional column in the File Locations tab named Essence Type, which shows if the media file is AMA or not.

Name	Type	Essence Type	CreatedBy	Format	Size
1-win8.1\xdjune27_100xc651cc66ee.17.mxf	filemob	Native	steves	PCM	3,41 MB
1-win8.1\xdjune27_100xc651cc66ee.18.mxf	filemob	Native	steves	PCM	3,41 MB
1-win8.1\xdjune27_100xc651cc66ee.19.mxf	filemob	Native	steves	PCM	3,41 MB
1-win8.1\xdjune27_100xc651cc66ee.20.mxf	filemob	Native	steves	PCM	3,41 MB
1-win8.1\xdjune27_100xc651cc66ee.21.mxf	filemob	Native	steves	PCM	3,41 MB
1-win8.1\xdjune27_100xc651cc66ee.22.mxf	filemob	Native	steves	PCM	3,41 MB
1-win8.1\xdjune27_100xc651cc66ee.23.mxf	filemob	Native	steves	PCM	3,41 MB
1-win8.1\xdjune27_100xc651cc66ee.24.mxf	filemob	Native	steves	PCM	3,41 MB
1-win8.1\xdjune27_10003.nxc651cc66ee.mxf	filemob	Native	steves	XDCAM-HD 50mbps 1080i 60	121,62 MB
XDJune27_10003.MXF	filemob	AMA	mikeb	XDCAM-HD 50mbps 1080i 60	0 Bytes

Deleting AMA Clips Using Interplay | Access

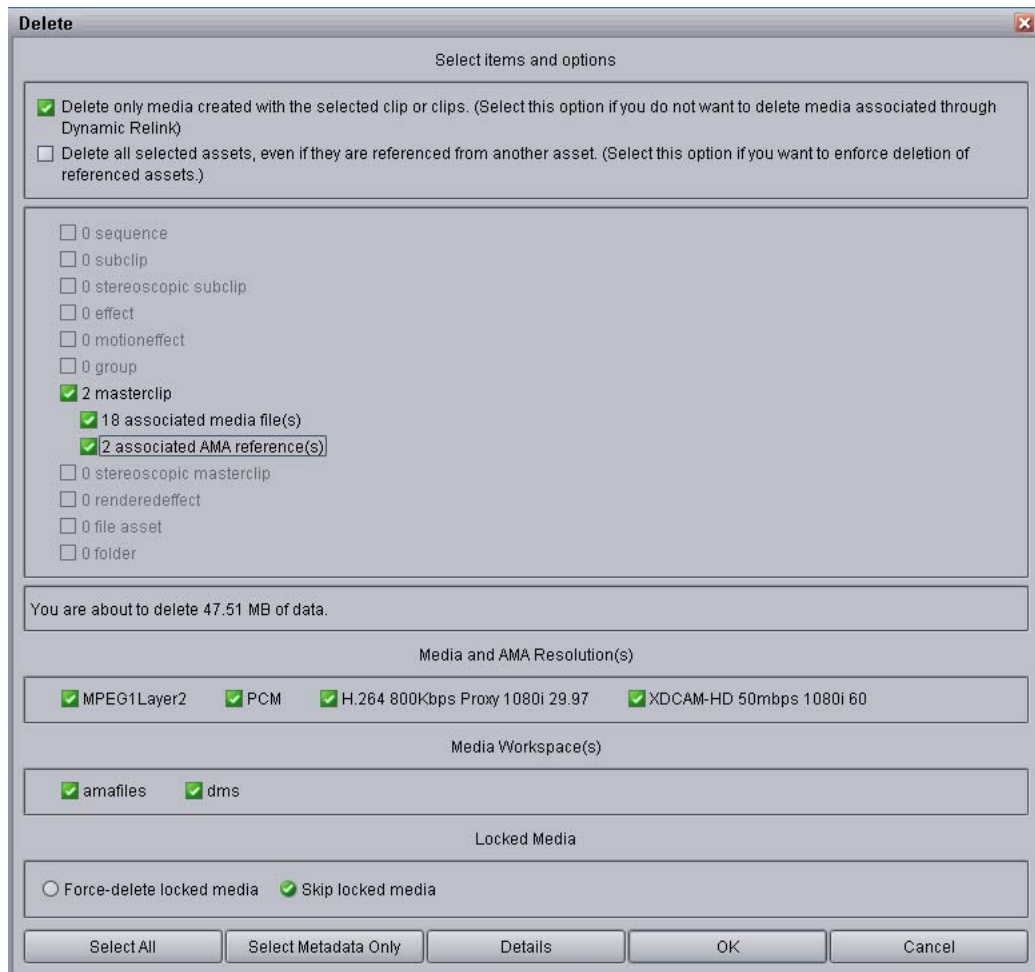
When you delete an AMA clip from the Interplay database, you delete the reference to the AMA source file. This reference includes a number of .spaaf files, one for each track in the AMA source file, and since v3.1 an AMA AAF file. For additional information on .spaaf files, see [“How Interplay | Media Indexer Works with AMA Material”](#) on page 85.

The following illustration shows two clips selected for deletion. Both clips have green dots meaning they are online. In this case the clips have associated AMA and native proxy media.



The following illustration shows the Delete dialog box for this example. It shows that three types of objects will be deleted:

- Two master clips (database objects)
- 18 media files (native proxy media, not native AMA media)
- Two associated AMA references (.spaaf files)



When you delete an AMA clip from the Interplay database the system does not delete the AMA source files. It deletes only the database object and the associated .spaaf files.

Resyncing AMA Media Files

The Resync process, which you initiate from Interplay Access, synchronizes metadata contained in Media Indexer with metadata contained in the Interplay Production database.

For Interplay Production v3.2, the Resync process has been extended to include resync of AMA (Avid Media Access) media files. Resync works with Media Indexer and checks if the AMA media files indexed in the MI (through their associated .spaaf files) are managed in the Interplay database.



Media Indexer v3.2 is required when using Resync from Interplay Access v3.2. This requirement applies to resyncing both AMA media files and Avid media files.

You should run Resync for AMA media in the following cases:

- After restoring the database from backup. (This is also a case in which you should run Resync for Avid media.)
- When upgrading from Interplay v3.0.x to Interplay v3.2. In this case, Resync adds file locations to AMA assets created in v3.0.x.



Starting with Interplay Production v3.1, a copy of the AAF file created by the AMA linking process is copied to the same location as the spaaf files.

Less frequent reasons to run Resync for AMA media are:

- If the complete asset or AMA reference was deleted from the database, but for some reason the .spaaf and AAF files could not be deleted (for example, because of a network issue).
- If the .spaaf and AAF files were manually copied to or deleted from the AMA metadata location.

The Resync process for AMA media works as follows:

- If .spaaf files and an AAF file exist for AMA media, but a corresponding asset does not exist in the database, Resync creates a new asset. This asset is checked into the Resync catalog and can be moved to another folder.
- If only .spaaf files exist for AMA media (without an AAF file), but there is an asset in the database (as is the case with assets checked in prior to Interplay Production v3.1), the media file locations are added to the asset. The location of the media files is shown in the File Locations tab.
- If only .spaaf files exist for AMA media (without an AAF file), and there is no asset in the Interplay database, the .spaaf files are selected for deletion and the user is asked to confirm the deletion.

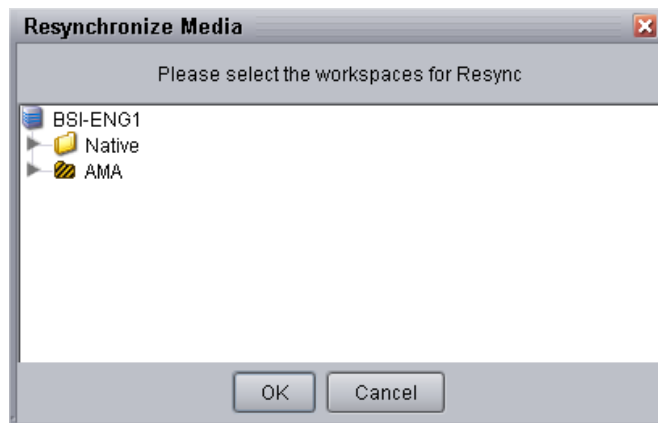


If no .spaaf files exist, Media Indexer does not index the AMA media, and the Resync process has no effect on the Interplay Production database.

To resync the Interplay database with Avid shared-storage workspaces that contain AMA media:

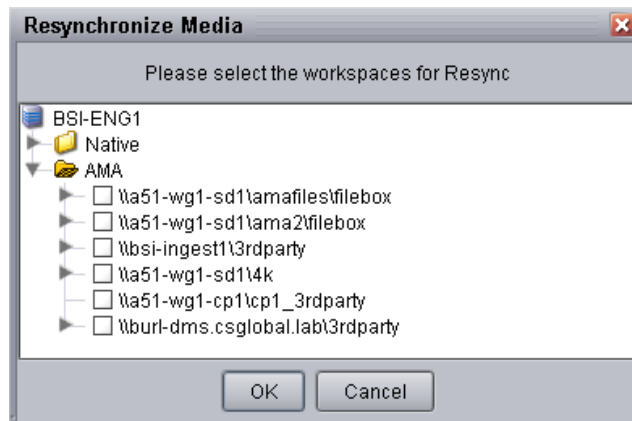
1. Log in to Interplay Access with an Administrator account.
2. In the Asset tree, right-click the database and select Resync.

The Resynchronize Media dialog box opens.

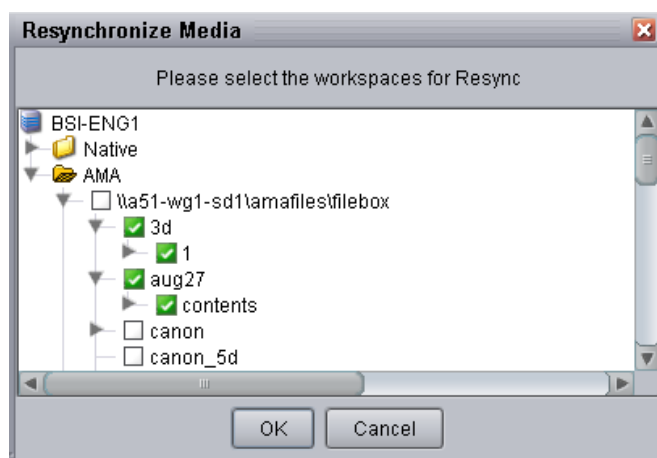


- Click the triangle next to the AMA folder.

All workspaces indexed by the Media Indexer are displayed.

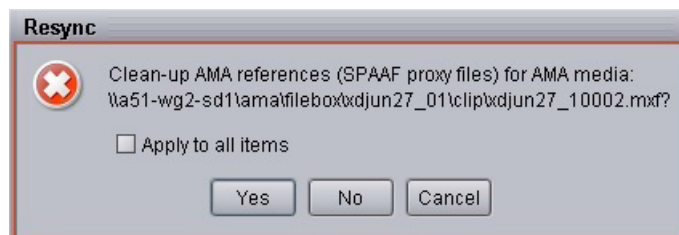


- Select the workspace or workspaces with which you want to resync. Click the triangle next to the workspace name to view folders in the workspace. You can select one or more individual folders or sub-folders.



- Click OK.

A dialog box displays the progress. If Resync discovers .spaaf files to be deleted for an AMA media file, a message is displayed.



Click Yes to delete .spaaf files for this AMA media file, click “No” to preserve them, or click “Cancel” to end the Resync process.

Select “Apply to all items” if you want to apply the action to any additional AMA media files, then click Yes or No. If you do not select this option, a dialog box is displayed for each set of .spaaf files.



The Resync process does not delete media.

6. When the Resync process is complete, click OK.

Any assets created during Resync are checked in and added to the Resync catalog.

6 Long GOP End-to-End Workflows

Interplay supports end-to-end Long GOP workflows for several of the Sony® and Panasonic formats that are used at many broadcast sites. This section provides an overview of Long GOP and describes the supported workflows. For a list of supported formats (also known as operating points), see the following:

- [“XDCAM HD and XDCAM EX Operating Points Supported in Interplay | Production” on page 189](#)
- [“AVC-LongG Formats Supported in Interplay | Production” on page 193](#)
- [“XAVC Long GOP Formats Supported in Interplay | Production” on page 195](#)

This chapter includes the following topics:

- [Long GOP Overview](#)
- [Long GOP Media Wrappers](#)
- [Long GOP Workflows](#)

Long GOP Overview

MPEG (Motion Pictures Expert Group) is an ISO/IEC standards group that has defined a number of standards for digital compression of audio and video signals. MPEG-2 and AVC are some of the standards defined by the group. MPEG-2 and AVC define video and audio compression and decompression technologies (codecs) that support a wide range of SD and HD video applications.

MPEG-2 and AVC codecs can employ either intra-frame compression (“I-frame only” as with IMX) or inter-frame compression (“Long GOP” as with Sony XDCAM HD, Panasonic AVC Ultra LongG, and Sony XAVC Long GOP). GOP stands for the “Group of Pictures” that are compared during compression and decompression of the images.



AVC Ultra LongG is referred to as AVC-LongG in this document.

A GOP contains several different types of compressed frames as follows:

- I frames, which are independently compressed frames.
- P (predictive) frames and B (bidirectional) frames, which depend on the frames around them and require significantly less space to store than I frames.

You cannot decompress an isolated P or B frame because of their dependency on other frames. This leads to some of the limitations associated with Long GOP.

- An IDR (Instantaneous Decoding Refresh) frame marks a point in the picture stream after which no (P or B) picture *following* this frame can have a dependency on any frame *preceding* this frame.

When starting from an IDR location in the stream, decoders will immediately start turning out valid decoded frames. For all other cases the decoders will require a “warm-up” phase, during which they will return non-valid decoded frames due to missing prior reference images. The length of the warm-up phase limits the achievable random-access rate during editing (scrubbing).

Interplay v3.6 and later supports end-to-end workflows for the following Long GOP formats:

- MPEG-2 based formats such as Sony XDCAM HD
- H.264-based formats such as Panasonic AVC-LongG.

Interplay v3.7 and later supports end-to-end workflows for the following Long GOP format:

- H.264-based Sony XAVC Long GOP

For an overview of MPEG-2 and other compression standards, see Symes, Peter. *Digital Video Compression*. McGraw-Hill, 2003.

Long GOP Benefits and Limitations

Benefits

The primary benefits of Long GOP native workflows in a broadcast environment are in the areas of acquisition and distribution efficiency. Lower data rates for a given image quality mean faster network transfer times and lower storage (bandwidth and capacity) requirements.

Limitations

Long GOP native workflows have the following limitations:

- **Processing Times:** Processing Long GOP files takes more overall computational power than I-frame-only files, making most aspects of the editing workflow slower. For example, often during the creation of a program for air, new frames must be created (for example, visual effects), requiring decompression and re-compression of the GOPs. It is often impossible to avoid the time-consuming decompressing and re-compressing of the Long GOP data stream. Scrubbing performance can also be affected because the frame that you want to display may have to be generated from a number of surrounding frames. Note that this limitation is even more prominent in H.264-based formats than in MPEG-2.
- **Image Degradation:** In many cases, Long GOP images might degrade in visual quality very quickly as the stream is decompressed and then re-compressed. Many operations, such as visual effects rendering, can dramatically affect image quality. Rendering the same media several times can increase the degradation. Even cuts-only editing of Long GOP content can affect image quality around the edit points. For this reason, many content creators prefer to use I-frame-only formats during program creation and use Long GOP formats only for acquisition and distribution.

For example, if you perform a large amount of compositing or rendering, you should use DNxHR resolutions for rendering. This creates I-frame only media and minimizes degradation of the rendered media. In this way you will benefit from the storage and bandwidth savings for your captured Long GOP material and only use the less bandwidth-efficient resolutions for rendered media.

Long GOP Media Wrappers

A Media Wrapper format describes how media and metadata coexist in a single file. The following media wrappers are associated with Long GOP material:

- **Avid OP-ATOM:** This is the Avid editing application native media wrapper for Long GOP material. This format stores each audio and video tracks as a separate file.
- **MXF OP-1A:** XDCAM HD devices store media in an MXF OP-1A wrapper. OP-1A files include multiple tracks of audio and video sources that are interleaved into a single file. This is referred to as “Single Item, Single Package” schema. XDCAM HD material is sometimes referred to as disc-based to differentiate it from XDCAM EX (card-based) material.
- **MXF OP-1B:** AVC-LongG files created by Panasonic cameras are typically in an OP-1B wrapper. OP-1B material can contain multiple file packages. For example, the video and audio can be in different file packages in the same file rather than all interleaved in one package. This is referred to as a “Single Item, Ganged Packages” schema. The OP-1B wrapper is defined in SMPTE ST 391:2004.
- **MP4:** XDCAM EX devices store media in MP4 format. This is basically MPEG-2 material with an MP4 wrapper. This format includes multiple tracks of audio and video sources that are interleaved into a single file.

The MP4 format also supports AVC-LongG and XAVC Long GOP. These formats include multiple tracks of audio and video sources in a single file.

You can copy files from the Sony or Panasonic media, link them through AMA, and then consolidate to Avid OP-ATOM. You can also use a variety of methods to convert the files to Avid OP-ATOM during the import or capture process. For details on working with these formats, see [“Long GOP Workflows” on page 108](#).

For a list of supported formats (also known as operating points), see [“AVC-LongG Formats Supported in Interplay | Production” on page 193](#).

Long GOP Workflows

This section describes the basics of an end-to-end Long GOP workflow and lists the Avid applications that are used in the process.

Supported Software Releases

The following table matches Long GOP formats with related minimum software and hardware releases.

Long GOP Format	Media Composer	AirSpeed Hardware	AirSpeed Software	Interplay Production	Media Director
Sony XDCAM HD and XDCAM EX	v7.x and v8.x	AirSpeed MultiStream, AirSpeed 5000, AirSpeed 5500	v2.x	v2.7.5	v1.0
Panasonic AVC-LongG	v8.6	AirSpeed 5500	v2.10	v3.6	v1.5
Sony XAVC Long GOP	v8.7	AirSpeed 5500	v2.11.1	v3.7	v1.6

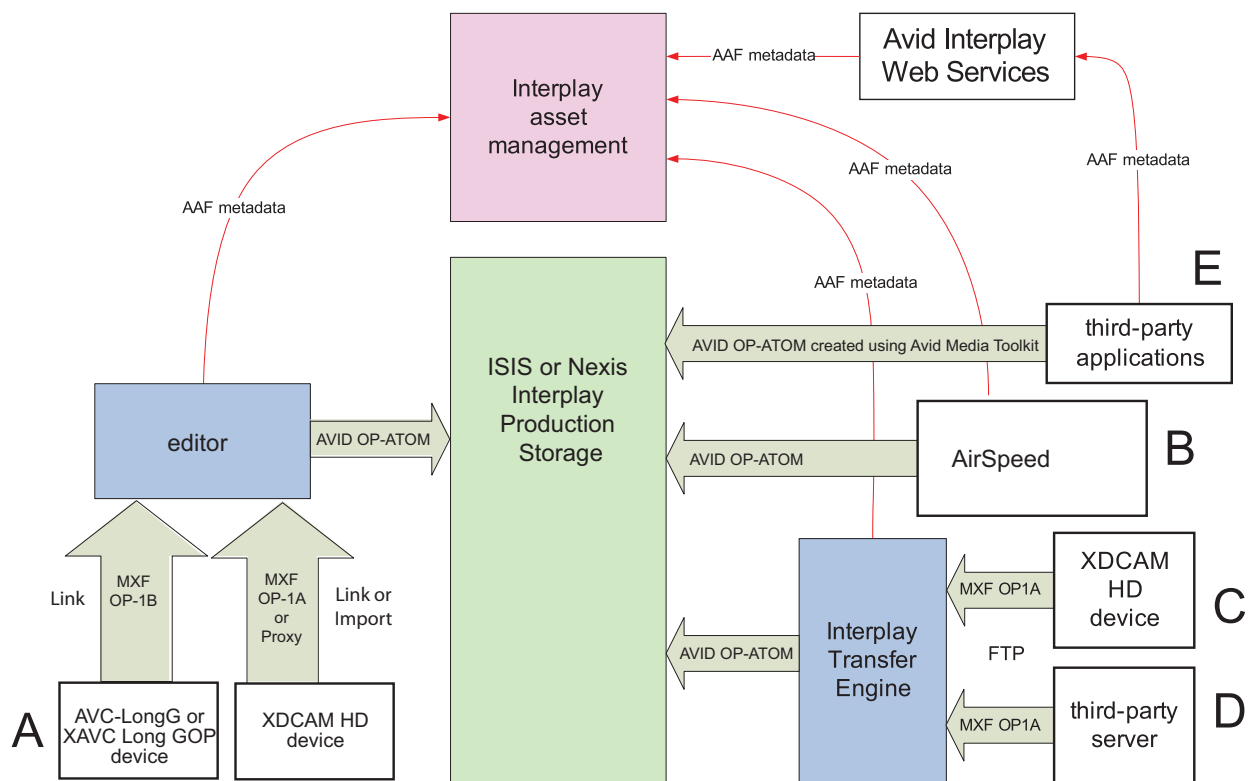
For more information, including supported operating points, see the following:

- “XDCAM HD and XDCAM EX Operating Points Supported in Interplay | Production” on page 189
- “AVC-LongG Formats Supported in Interplay | Production” on page 193
- “XAVC Long GOP Formats Supported in Interplay | Production” on page 195

Capture Workflows

The capture workflow moves the Long GOP material into the collaborative production environment, including shared and local storage and Interplay Production asset management.

The following illustration shows five Long GOP ingest paths. See also “Media | Director Ingest Workflow” on page 112.



The ingest paths in the illustration are marked with the letters A through E. The following sections describe the ingest paths.



Ingest methods C and D are not supported for AVC-LongG or XAVC Long GOP material.

Path A. Editor Import from an XDCAM HD, XDCAM EX, AVC-LongG, or XDCAM Long GOP Device

This path uses the following connections:

- XDCAM HD (disc based) Firewire or USB
- XDCAM EX, AVC-LongG, or XAVC Long GOP (card based) USB only

The following sections describe the Link and Import workflows available directly from the editor. You can also use the Media Director for importing media natively or as part of a Link workflow. See [“Media | Director Ingest Workflow” on page 112](#).

Link Workflow

The Link workflow is also known as the as AMA Link Workflow. AMA stands for Avid Media Access. You can use the Link workflow for all supported Long GOP material.

This method allows you to work directly with the native files on the disc or card. The files open in a bin on your Avid editing application. You can create sequences with the media or select the clips you want to capture. When you are ready to bring the files into the Interplay environment, select the sequence or clips and use the Consolidate command to create new OP-ATOM media files. The Consolidate command copies the new files to shared storage and performs a check-in to Interplay Production.

You can also copy the native files from the disc or card directly to a folder on shared storage and work with them there before you edit the files and perform the consolidation. This provides better editing performance than working with the material on the card or disc. It also frees up the editor from the process of copying the XDCAM, AVC-LongG, or XAVC Long GOP data from the cards or discs. For example, a cameraman can copy the contents of several cards to shared storage and then the editor can edit and consolidate the material at a later time. In this workflow, copy the files to a workspace or folder that is not monitored by the Interplay Media Indexer that monitors shared storage. See the *Interplay Best Practices Guide* for information on using Media Indexer with AMA material.



Support for non-ISIS storage for the native files is a licensable option.

Import Workflow

You can import XDCAM HD and XDCAM EX files directly into the editor using the “Import and Export Workflow” method described in the Avid editing application Help.



You cannot use the Import command in the editor to capture AVC-LongG or XAVC Long GOP material. Use the Link or Media Director workflows.

- For XDCAM HD (disc-based) files, import the MXF OP-1A files into the Avid editing application. During the import process the editing application converts the files to OP-ATOM.
- For XDCAM EX (card-based) files, use a Sony application named ClipBrowser to convert the MP4 files to OP-1A. Then import the OP-1A files into the Avid editing application or drag the clip to a bin. During the import process the editing application converts the files to OP-ATOM.
- You can also use the ClipBrowser application to convert XDCAM EX files directly to OP-ATOM files and AAF metadata files. In this case, copy the OP-ATOM files to the Avid MediaFiles folder on a shared storage workspace and drag the AAF file to a bin.

The Import operation writes the files to shared storage and performs a check-in to Interplay via AAF metadata. For additional information, see “File Based Media” in the Help for your Avid editing application.

For some Import workflows, you might want to import the proxy media first so you can start editing. Once you create your sequence, you can use the batch import function to import only those portions of the high-resolution clips needed for your sequence.

Note that the Import workflow only allows you to import one or more files by selecting them from the editor. The Link and Media Director workflows allow you to select an entire volume at one time depending on the AMA plugin associated with the material.

Path B. Ingest Through AirSpeed

The AirSpeed application performs the following operations:

- Encodes the incoming video to local files
- Copies the material into OP-ATOM files on shared storage
- Performs a check-in to Interplay Production through AAF metadata

AirSpeed also supports the following Interplay Production features:

- Dual ingest
- Frame Chase Editing

For a list of supported Long GOP formats per AirSpeed device, see [“Supported Software Releases” on page 108](#). For additional information, see the Installation Guide for your AirSpeed device.

Paths C. and D. Interplay Transfer Ingest from XDCAM Device or Third-party Servers that Support FTP



The Interplay Transfer ingest method is not supported for AVC-LongG or XAVC Long GOP material.

The Interplay Transfer Engine performs the following operations:

- Pulls the media from the device using the Generic FTP plugin and splits it into separate OP-ATOM media files
- Writes the files to shared storage
- Performs a check-in to Interplay Production through AAF metadata

Paths C and D have the following limitations:

- In the case of a third-party server, the server must support MXF OP-1A at the required rates and formats.
- XDCAM EX devices do not have FTP capability, therefore EX is not supported through this path

For additional information, see the *Interplay Transfer Setup and User's Guide*.

Path E. Third-party Applications that Support the Interplay Web Services API

This path applies to third party applications that take advantage of the Interplay Web Services application programmer interface (API). The third party application performs the following tasks:

- Create separate OP-ATOM media files and AAF metadata files.



Avid strongly recommends that the third party application use the Avid Media Toolkit (AMT) to create the correct Avid OP-ATOM media files.

- Write the media files to shared storage
- Use the Interplay Web Services to perform a check-in to Interplay Production through AAF metadata

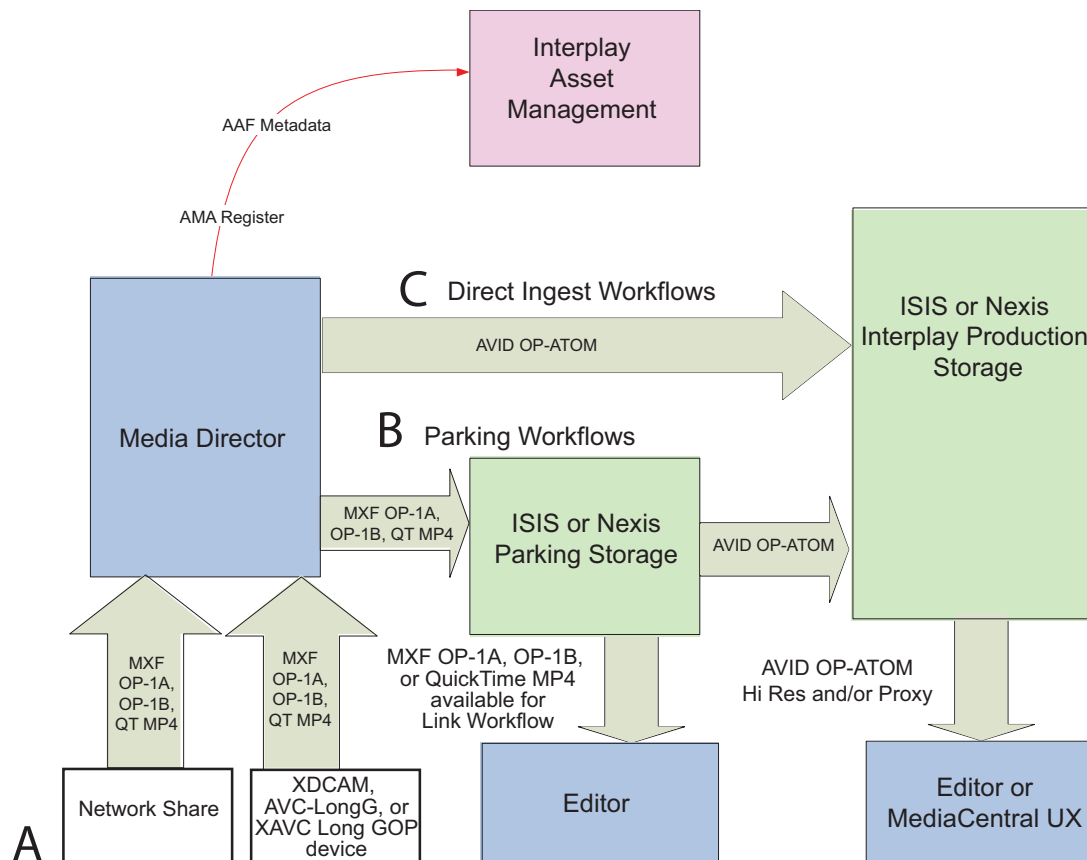
For additional information, see the documentation for your third party application.

Media | Director Ingest Workflow

Media Director is an optional service for Interplay Production that allows workgroups to capture field footage shot in various formats quickly by processing it against user-definable profiles that orchestrate copy, naming and transcode processes via a powerful rules system.

Media Director provides a central server that can process ingest jobs. Individual jobs can be initiated from client machines with LAN access to the server. This frees up valuable editing systems and allows staff to use older client machines to initiate ingest jobs. Operators can prioritize jobs by assigning priorities to a job. This section provides an overview of the Media Director workflow. For additional information, see the *Media Director User Guide*.

The following illustration shows the Media Director ingest paths.



The following sections describe the Media Director ingest paths. For additional information, see the *Media Director User Guide*.

A. Starting a Media Director Workflow

Bring material into the system as follows:

1. Do one of the following:
 - Select one or more sources on a network share or field disk drive.
 - Insert camera-formatted media such as backup disk with camera media, a P2 or Sony card, or XDCAM disk into a reader on the Media Director client.
2. Drag the volume onto the Media Director client and select a predefined Profile.



Support for non-ISIS parking storage is a licensable option.

B. Parking Workflows

Media Director provides the following Parking workflows:

- Park and Register
- Park, Register, and Transcode

For parking workflows, Media Director copies the media from the card or disk to a parking storage system such as an ISIS 2000 or other parking storage. This parking storage provides a single location where you can always find a copy of your original media. This allows Link workflows (also known as AMA Link workflows) to function reliably no matter what happens to the original carrier.

1. Media Director uses an AMA register process to create an AMA-linked asset in the Interplay database. These assets point to the media on the parking storage.

At this point the media is available for a limited number of users that can work directly with high res AMA material. For example, you can now edit the material on a Media Composer system.

2. Next (if the Park, Register and Transcode workflow is used), the Media Director server uses its built in proxy generation to create a proxy. You can also instruct Media Director to consolidate the Hi Res media to OP-ATOM and store it on a particular shared storage workspace.

Once the proxy is available, editors and other Interplay clients can relink to the proxy. This makes the material available to a much larger group of users such as MediaCentral | UX, Interplay Assist, and Media Composer | Cloud.

C. Direct Ingest Workflows

Media Director provides two Direct Ingest workflows:

- Direct Ingest
- Direct Ingest from Cache

The Direct Ingest workflows directly rewrap volumes from the source carrier to Production Storage (ISIS) without copying the AMA clips first. The Transcode portion of the profile contains a “transparent pass-through” option. This option treats native codecs as follows:

- If the native codecs are supported in Interplay Production, they are preserved.
- Some AMA plugins provide only uncompressed frames for media like GoPro and some QuickTime files. As uncompressed frames cannot be consolidated, Media Director automatically transcode these formats to a user configurable format (XDCAM HD 50 by default).



Media Director provides additional workflows such as “Hi-Res Conform from Parked Volume” and “Ingest from OP-1A.” See the Media Director User’s Guide for additional information.

Editing Workflow

Avid editing applications allow full media creation and playback support for all of the specified formats, rates, and modes. This includes the following support:

- Video mixdown
- Consolidation
- Render
- Transcode
- Graphic and Title creation

For information on editing with Long GOP material, see “File Based Media” in the Help for your Avid editing application.

If you use a lot of compositing or rendering in your workflow, consider rendering to one of the DNxHR formats. For more information, see [“Long GOP Benefits and Limitations” on page 107](#).

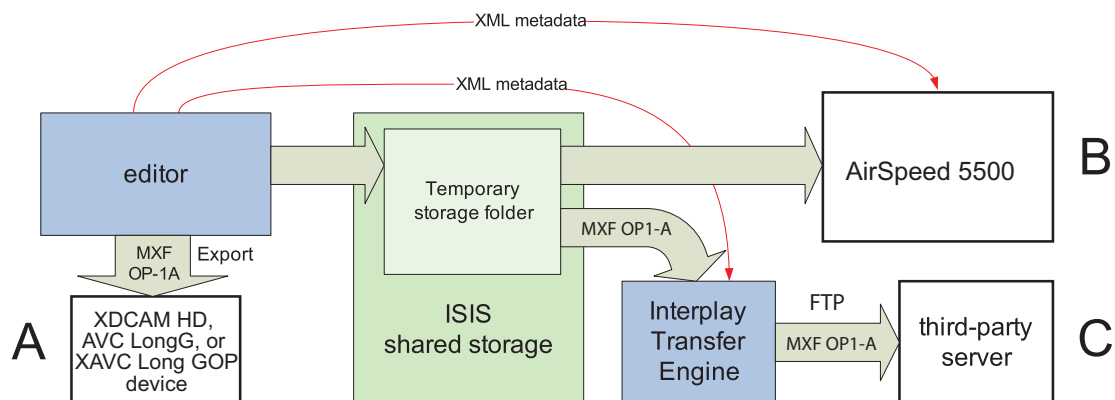
Output Workflow

The output workflow involves the following:

- Export the finished product to either an XDCAM, AVC-LongG, or XAVC Long GOP device
- Send the finished product to a broadcast recording device or to a playback server.

In these workflows, the editing application creates a single media file or volume that is a flattened representation of the edited sequence. This requires the editing application to perform all necessary Long GOP editing, rendering, and multiplexing of streams. The playback device is handed a single integrated stream in the desired format.

The following illustration shows three Long GOP output paths. Also see [“STP Encode Workflow: Background Processing and Transferring of Long GOP Media” on page 116](#).



The FTP transfer to the third party device does not use Generic FTP.

The output devices in the illustration are marked with the letters A through C. The following sections describe the output paths:

Path A. Editor Export to XDCAM HD, AVC-LongG, or XAVC Long GOP Device

The XDCAM HD, AVC-LongG, or XDAVC Long GOP device mounts on the system using Sony or Panasonic drivers, and appears to the Avid editing application as a standard volume. The editing application writes the following directly to the device:

- For Sony XDCAM, the editor exports a single MXF OP-1A file
- For AVC-LongG and XAVC Long GOP, the editor exports a volume



This release does not support export to XDCAM EX devices.



You cannot export Long GOP media as QuickTime Same as Source or QuickTime Reference. If you want to use Long GOP material in an application such as Adobe After Effects, first transcode the media to a DNxHD format.

Paths B. and C. Playout Through AirSpeed or Third-party Server

The following steps describe the workflow for Paths B and C:

- The editor selects a sequence in a bin and chooses the Send To Playback command.
- The editing application performs the following operations:
 - If necessary, perform an audio mixdown
 - If the sequence contains any mixed resolutions, perform a transcode operation.
 - Generate a temporary MXF OP-1A clip and write it to a Temp folder on shared storage. The application automatically selects a Temp folder on the top level of the workspace with the most free space.
 - Send an XML file describing the clip to an Interplay Transfer Engine or AirSpeed. The XML file contains the shared storage location and the target location.
- When using an AirSpeed, the AirSpeed transfers the MXF-OP1A clip file to the target location. The Airspeed deletes the temporary clip from the shared storage system after the transfer is complete.
- The Interplay Transfer Engine or AirSpeed performs the following operations:
 - Initiates a transfer of the temporary MXF OP-1A clip file to the target location.



This step does not use Generic FTP. This Interplay Transfer step is simply a file move operation. The frames are not published to the DHM interface.

- Delete the temporary clip from the shared storage system after the transfer is complete.

The editing application writes the temporary file to a Temp folder at the top level of the workspace with the most free storage space. Since the Interplay Administrator or Media Manager typically configures Interplay Media Indexers to monitor only the Avid MediaFiles folder on a workspace, the Interplay Media Indexer will not be aware of the files in the Temp folder.

Once the Transfer Engine or AirSpeed has finished the transfer, it deletes the clip from the Temp folder on shared storage. In an AirSpeed Studio configuration, several servers might need to transfer the same clip. In this context, the deletion of clips from the temporary folder is coordinated between the servers in the configuration.

If the export operation fails, the applications delete the file from the Temp folder as follows:

- If the operation fails during the export from the Avid editing application, the editing application deletes the file from the Temp folder.
- If the operation fails during the FTP transfer, the Interplay Transfer or AirSpeed application deletes the file from the Temp folder and displays the status in the Interplay Transfer Status window.

For additional information, see the *Interplay Transfer Setup and User's Guide*.

Performance Considerations

When the editing application performs the Send to Playback operation, it performs a mixdown and transcode if necessary. The mixdown and transcode are typically performed in real time or less depending on the complexity of the sequence. See the *Interplay Production ReadMe* for details on any limitations associated with the release.

STP Encode Workflow: Background Processing and Transferring of Long GOP Media

This section describes the portion of the Long GOP workflow that can be handled by the Interplay STP Encode service instead of the Avid editing application or MediaCentral UX.

The STP Encode service provides background processing that off loads the time-consuming processes involved in Long GOP OP1a exporting and transferring from an Avid editing application or MediaCentral UX. Before you can use the STP Encode service, the workgroup needs an FTP Server profile and the STP Encode provider must be registered with the Interplay Production Services Engine.



If an STP Encode service is not available, a send to playback request is handled in the foreground by the Avid editing application.

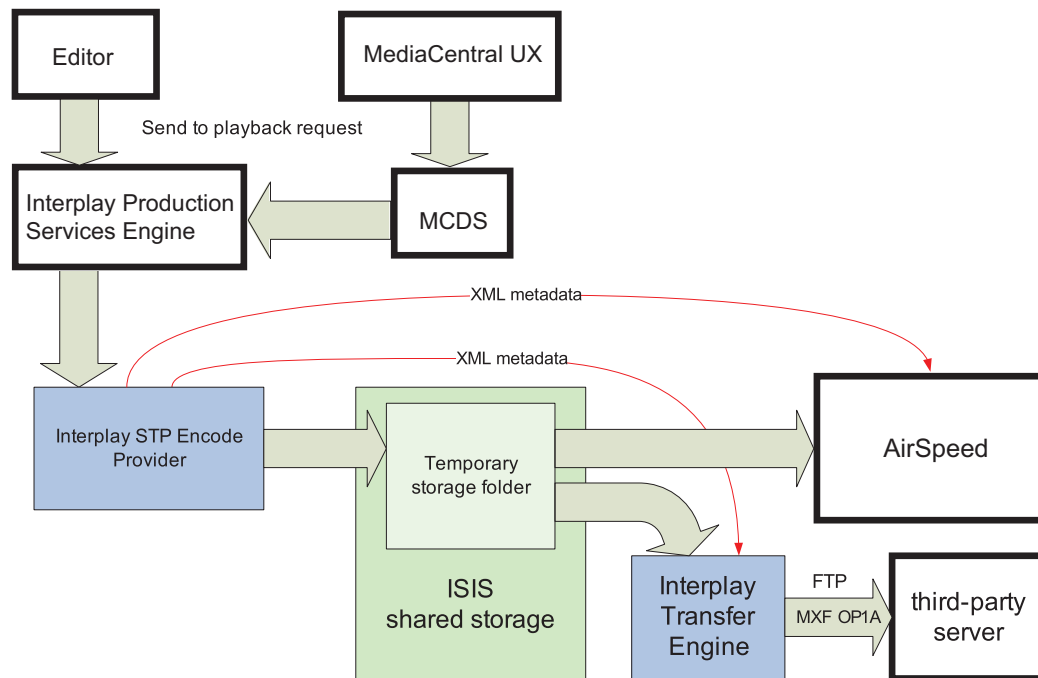
Best practices for when to use the STP Encode service:

- Use foreground send to playback processing (no STP Encode service)—when sending short or urgent sequences to a playback device
- Use STP Encode service for background processing—when sending long or non-urgent sequences to a playback device



Performing the Long GOP processing in the background might not improve on the time it takes to perform the operation in the foreground. However, by using the background process, you can continue to work with the Avid editing application for other tasks.

The following illustration shows the workflow for playout through AirSpeed or a supported third-party server.



The following steps describe the workflow for playout through AirSpeed or a supported third-party server:

- **Editor Send to Playback:** In the case of an Avid editing application, the editor selects a sequence in a bin and chooses the Send To Playback command.
The editing application performs the following operations:
 - If necessary, performs an audio mixdown.
 - If the sequence contains any mixed resolutions, performs a transcode operation.
 - Submits an STP Encode service job for the sequence composition to the Production Services Engine.
 - All audio effects are pre-rendered by the Avid editing application before the sequence is exported and transferred using the STP Encode service. The Interplay STP Encode service can render real-time video effects. However, you have to pre-render any static video effects before exporting the sequence as an STP Encode job.
- **MediaCentral Send to Playback:** In the case of MediaCentral UX, the MediaCentral UX user selects a sequence and chooses the Send to Playback command.
 - The MediaCentral UX application passes the sequence to MCDS (MediaCentral Distribution Service).
 - MCDS determines whether an audio mixdown is required and whether the video must be transcoded.
 - If either a mixdown or transcode is required, MCDS submits the job to the Production Services Engine. The Production Services Engine uses Interplay Transcode to perform the mixdown or transcode.
 - The Production Services Engine then submits the job to STP Encode.

- The Interplay STP Encode provider performs the following operations:
 - For output to Interplay Transfer, STP Encode generates a temporary MXF OP1A clip. For output to AirSpeed, STP Encode uses the clip generated by the Media Composer. STP Encode writes the clip to a Temp folder on shared storage. The application automatically selects a Temp folder on the top level of the workspace with the most free space.
 - Sends an XML file describing the clip to an Interplay Transfer Engine or AirSpeed. The XML file contains the shared storage location and the target location.
- The Interplay Transfer Engine or AirSpeed performs the following operations:
 - Initiates a transfer of the clip to the target location.
 - Deletes the temporary clip from the shared storage system after the transfer is complete.

The STP Encode provider writes the temporary file to a Temp folder at the top level of the workspace with the most free storage space. Since the Interplay Administrator typically configures Interplay Media Indexers to monitor only the Avid MediaFiles folder on a workspace, the Interplay Media Indexer is not aware of the files in the Temp folder.



In general, you should make sure that your Interplay Media Indexers do not monitor entire workspaces. They should point to the specific Avid MediaFiles folders that you want to monitor.

After the Transfer Engine or AirSpeed has finished the transfer, it deletes the clip from the Temp folder on shared storage. In an AirSpeed Studio configuration, several servers might need to transfer the same clip. In this case, the deletion of clips from the temporary folder is coordinated between the servers in the configuration.

If the export operation fails, the applications delete the file from the Temp folder as follows:

- If the operation fails during the export from the STP Encode provider, the STP Encode provider deletes the file from the Temp folder.
- If the operation fails during the transfer, the Interplay Transfer or AirSpeed application deletes the file from the Temp folder and displays the status in the Interplay Transfer Status window.

For additional information on STP Encode, see the *Interplay Production Services Setup and User's Guide*. For additional information, see the *Interplay Transfer Setup and User's Guide*.

7 720p MultiRez Workflow

This chapter describes a 720p MultiRez workflow. It is similar to other MultiRez workflows except that it always requires a transcode operation to create the low-res proxy.

This document covers the following topics:

- [Overview of the 720p MultiRez Workflow](#)
- [Editor MultiRez Workflow for 720p](#)
- [720p MultiRez Workflow Components](#)
- [Setting Up the Production Services Profiles for a 720p MultiRez Workflow](#)

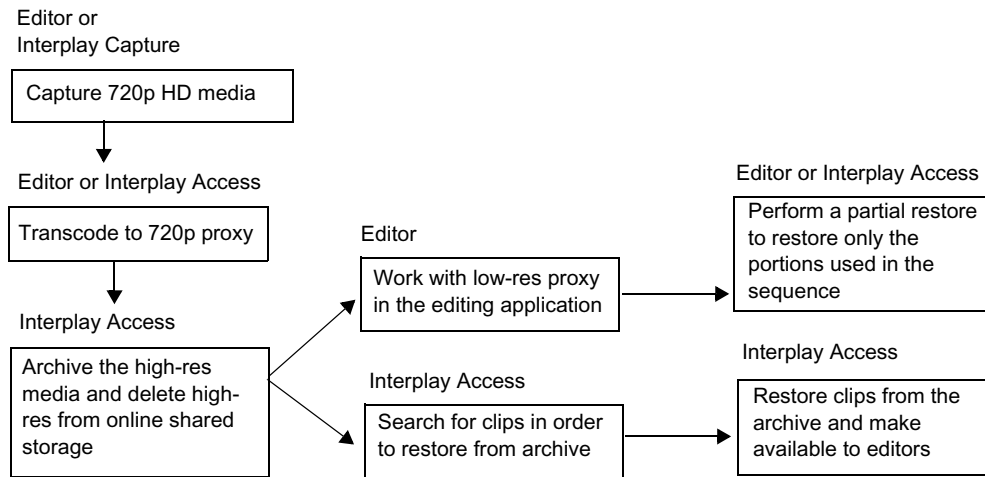
Overview of the 720p MultiRez Workflow

Interplay v2.0 supports low-res proxy media for 720p high-res media. The supported resolutions are as follow. AVCI stands for AVC intra-frame.

Format	Resolution	Avid Editor Interface Name
NTSC	720p/59.94 (H.264)	AVCIBP-BLL3.0 60
PAL	720p/50 (H.264)	AVCIBP-BLL3.0 50

In a MultiRez workflow a media manager or Interplay administrator archives most of the high-res material to a robotic tape library and maintains a low-res version online. For example, a sports station may capture high res media for entire sporting events with a duration of three or four hours each. After archiving, the media manager deletes the online high-res material and maintains the low-res material online. The low-res material is stored on line partly to allow editors to work with it and partly as a reference for the material in the archive.

The following illustration shows the main stages of the workflow.



The above illustration shows two methods for restoring the high-res material. For example, if the media manager gets a request for the entire game played on 6/12/09, he or she can use Interplay Access to search for it and restore the entire clip from the archive. An editor can also search for the clip in the online database and begin working directly with the low-res version. After the editor finishes the sequence, he can perform a partial restore and restore only the portions of the game that are in the sequence, rather than restoring the entire sporting event from archive.

Editor MultiRez Workflow for 720p

The editor can begin working with the media as soon as it is transcoded to the low-res proxy. In fact, the archiving and deleting of the high-res material can go on in the background while the editor is working. This section describes the editor's portion of the workflow. The details of transcoding, archiving, and restoring are covered later in the chapter. See [“Working with a MultiRez Clip in the Editor and Performing a Partial Restore” on page 120](#).

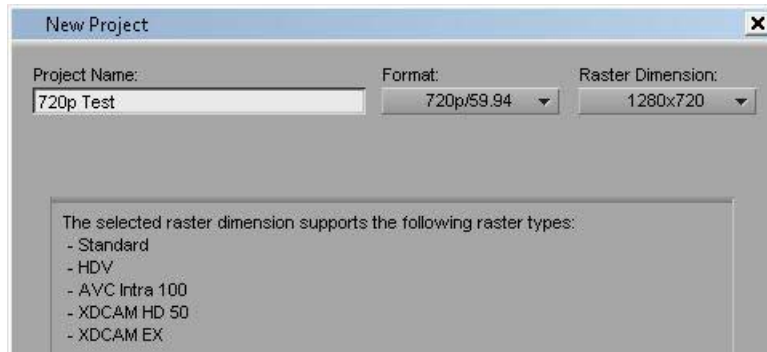
Working with a MultiRez Clip in the Editor and Performing a Partial Restore

This section describes the following topics:

- Opening a 720p project and checking out the clip
- Setting Dynamic Relink to work with the low-res proxy
- Performing a partial restore to restore the portions of the clip used in the sequence.
- Using Dynamic Relink to start working with the high-res material.

To set up a 720p project:

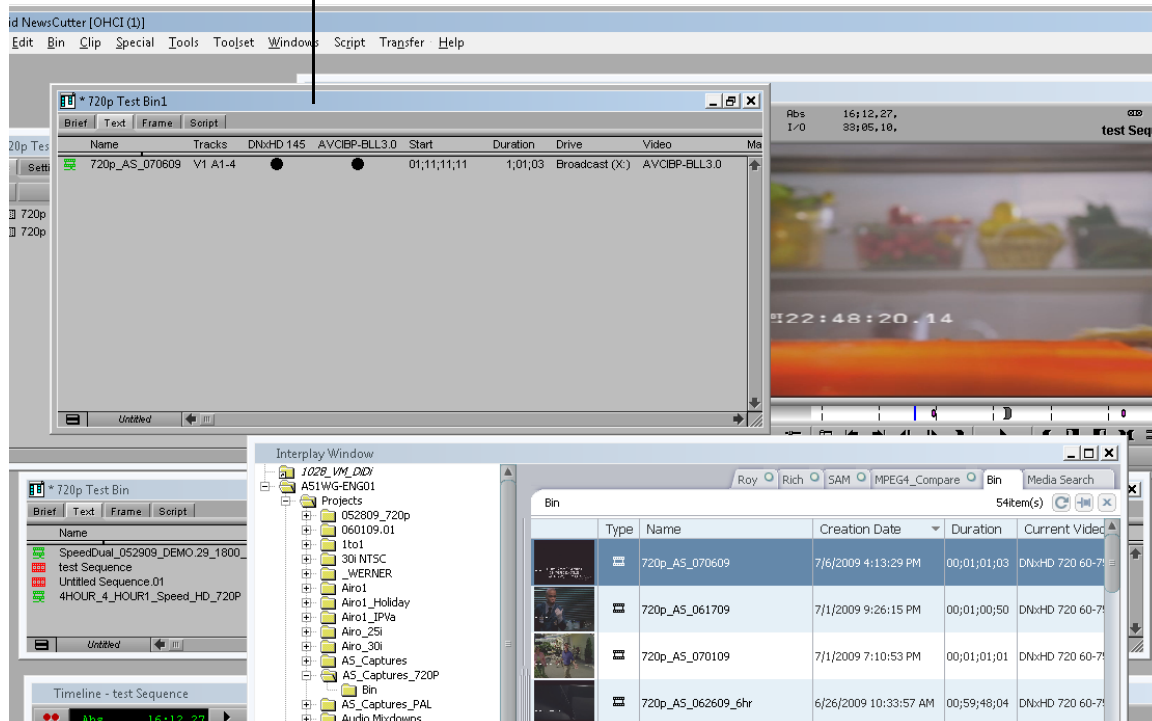
1. When you create the project, make sure it supports the resolution you want to use. For example, the following illustration shows the Project selection dialog box used in this example.

**To check out the clip and set Dynamic Relink:**

1. Select Tools > Interplay Window.
2. Locate the clip in the Interplay Window and drag it to the bin.

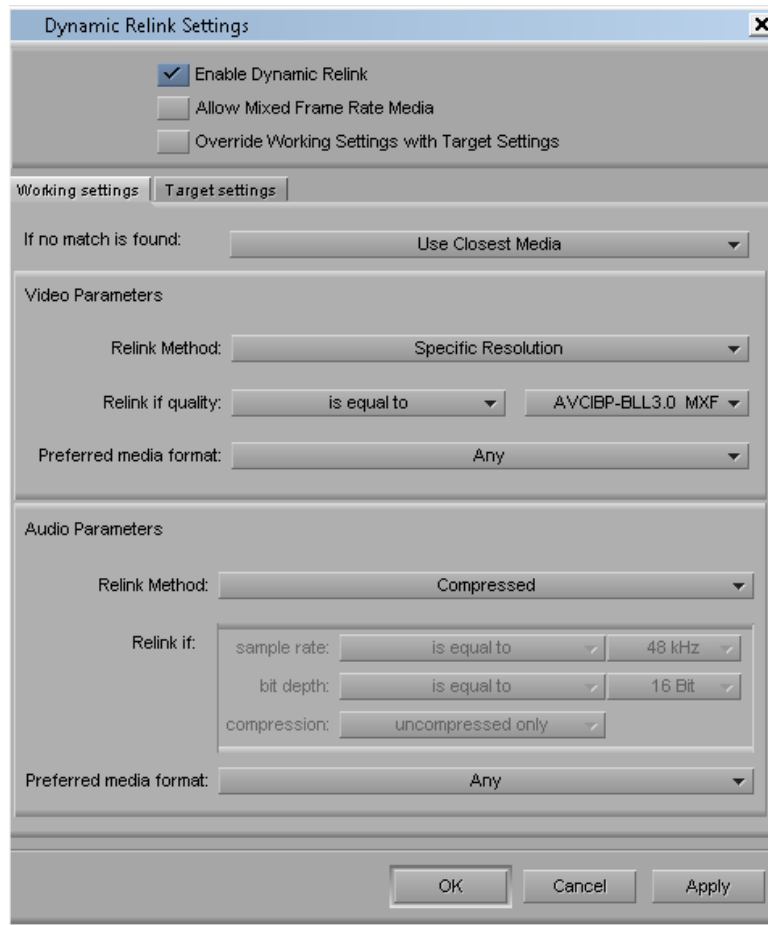
The following illustration shows the clip in the Interplay window and the bin. Notice that the clip has been transcoded and both the high-res and low-res are online.

high-res and low-res media is online



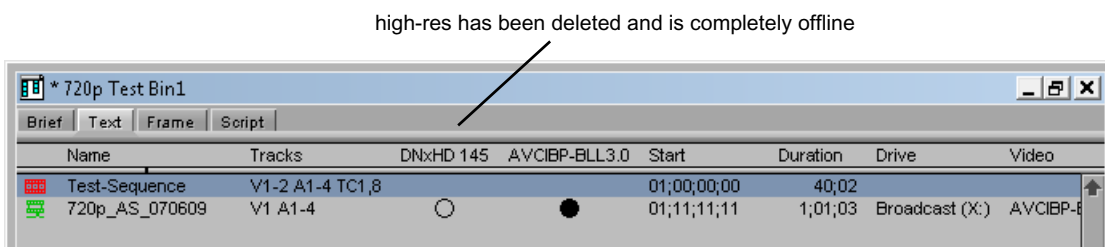
3. Open the Dynamic Relink dialog box and set the working resolution to the low-res proxy.

The following illustration shows the Dynamic Relink values used in this example to link to the low-res media.



4. Edit the low-res media into a sequence.

The following illustration shows the bin after the sequence is created. Notice that at this point the high-res material has been deleted and is showing completely offline.



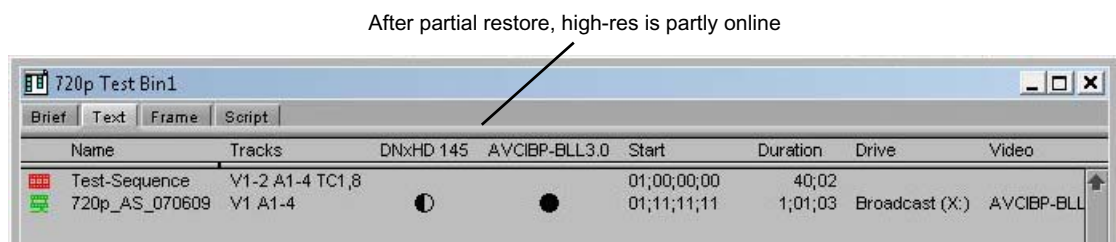
5. When you are finished editing the sequence, right-click the sequence in the bin and select Restore. Select a profile that is set up to perform partial restore. The following illustration shows the Restore profile used in this example.

If the profile that you choose is set up to perform partial restores, the application restores only the portion of the clip that is in the sequence.

Parameters for 720p_DnX145_Partial

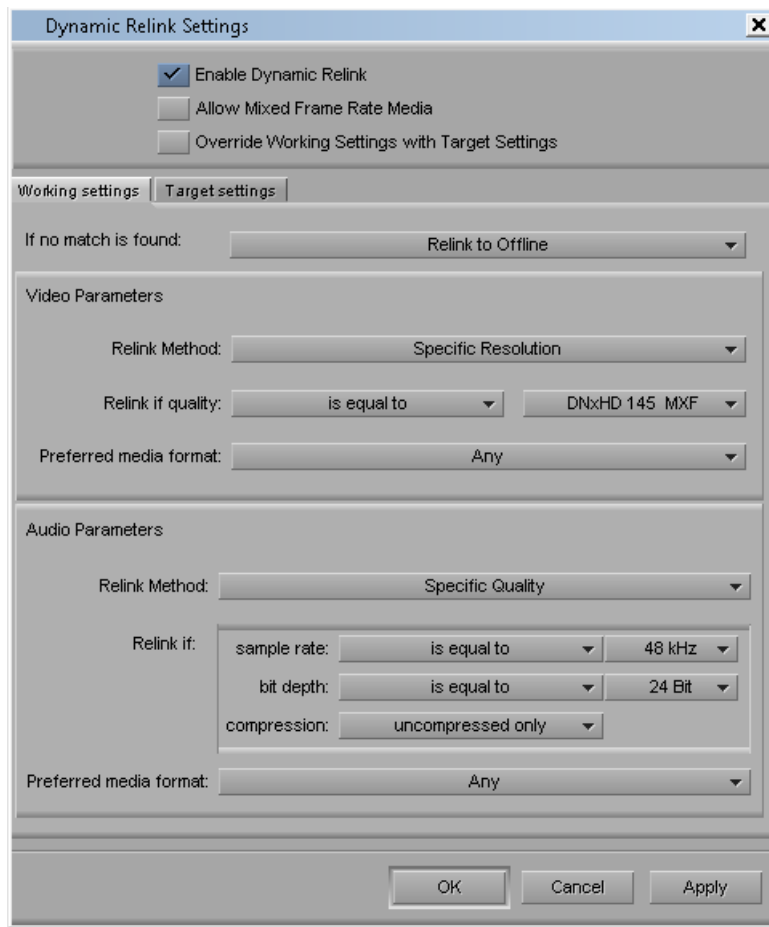
Destination_Path	AvidWG/Catalogs/HighestPartial
Destination_Server	a51wg-eng01
Destination_Workspace	Transfer
Partial	☒
Priority	25
TargetVideoQuality	DNxHD 720 60-75-145

The following illustration shows the bin after the partial restore operation is completed. The high-res media is now partly online.



- Open the Dynamic Relink dialog box and set the working resolution to the high-res version.

The following illustration shows the Dynamic Relink values used to link to the high-res video and audio in this example. This is linking to a specific resolution. You could also link to the highest available resolution.



7. Now you can work with the restored high-res material.

720p MultiRez Workflow Components

The following topics describes the different aspects of the 720p proxy workflow:

- [Capturing High Res 720p Material](#)
- [Transcoding to 720p Proxy](#)
- [Archiving the High Res 720p Material](#)
- [Deleting the High Res Media](#)
- [Searching for and Restoring High Res Master Clips](#)
- [Performing a Partial Restore](#)

Capturing High Res 720p Material

You can capture 720p high-res material directly from the editing application or using Interplay Capture. This example assumes that you are capturing DNx145 720p/59.94 media. For additional information, see the following:

- For information on capturing from within the Avid editing application, see the editing application Help or the Basics Guide for your editing application.
- For information on using Interplay Capture, see the *Interplay / Capture User's Guide*.

Transcoding to 720p Proxy

Transcoding is the only way to obtain 720p proxy material. You have the following choices for transcoding:

- Right-click the clip within the Avid editing application, select Avid Interplay Production Services > Interplay Transcode Service and choose a profile.



The editor Consolidate/Transcode operation does not create the 720p proxy for this release.

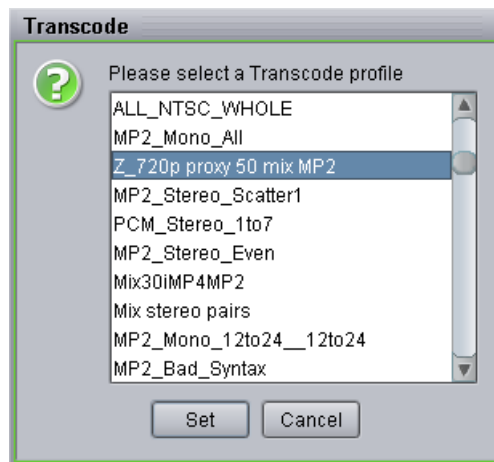
- Right-click the clip within Interplay Access and select Transcode.
- Set up an Auto Transcode folder in Interplay Access and drag and drop clips to the folder.

For additional information, see the Interplay Production Help or the *Interplay Production Services Setup and User's Guide*.

To transcode a clip from within Interplay Access:

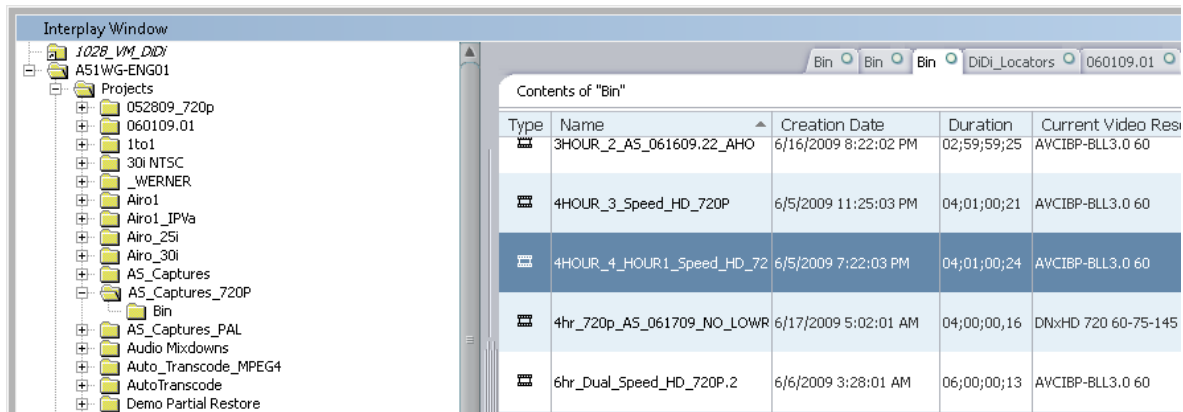
1. Right-click the clip in Interplay Access and select Transcode.

The Transcode profile dialog box opens.



2. Select the profile and click Set.

The Transcode service transcodes the clip and stores the media according to the instructions in the profile. The new media is associated with the original clip metadata. The following illustration shows the low-res proxy in the Interplay Window. You can see that the current resolution for the clip is AVCIBP-BLL3.0.60. This is the Avid editing application representation for 720p/59.94 proxy media.



To begin working with the media, drag it to an Avid editing application bin and modify your Dynamic Relink settings to use the low-res proxy as your working resolution. See [“Editor MultiRez Workflow for 720p” on page 120](#).

Archiving the High Res 720p Material

You can use Interplay Access or the Avid editing application to archive the high-res material: Both techniques use a Production Services profile as described in [“Archive Profile” on page 136](#).

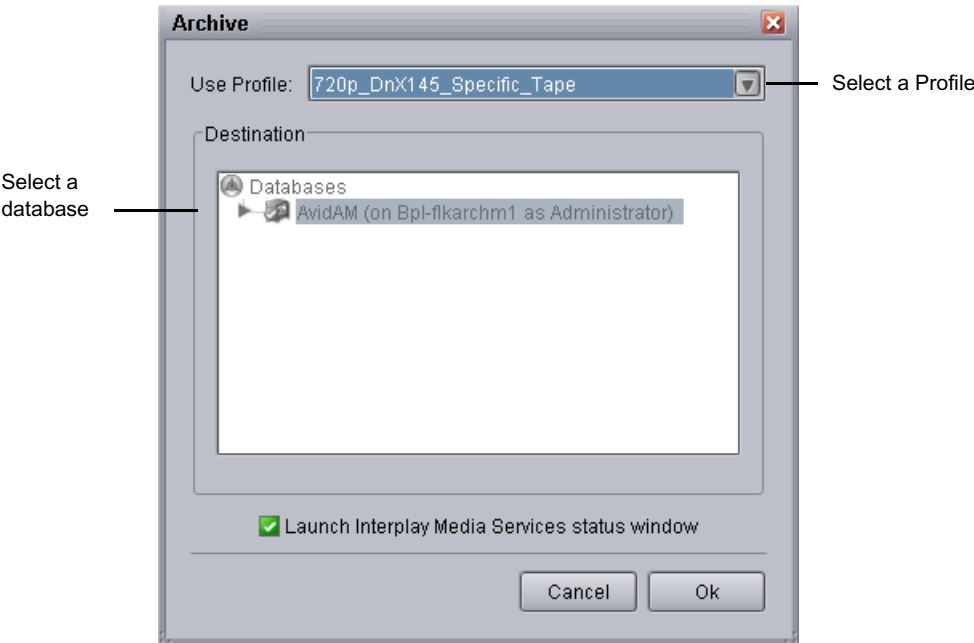


You can also set up an Auto Archive folder and use Interplay Access to drag and drop files to archive into the folder. For additional information, see the Interplay Production Services Setup and User's Guide.

To archive the high-res material:

1. Start Interplay Access.
2. Select the asset that you want to archive from the online database.
3. Select Tools > Archive or right-click the asset and select Archive.

The Archive dialog box opens.



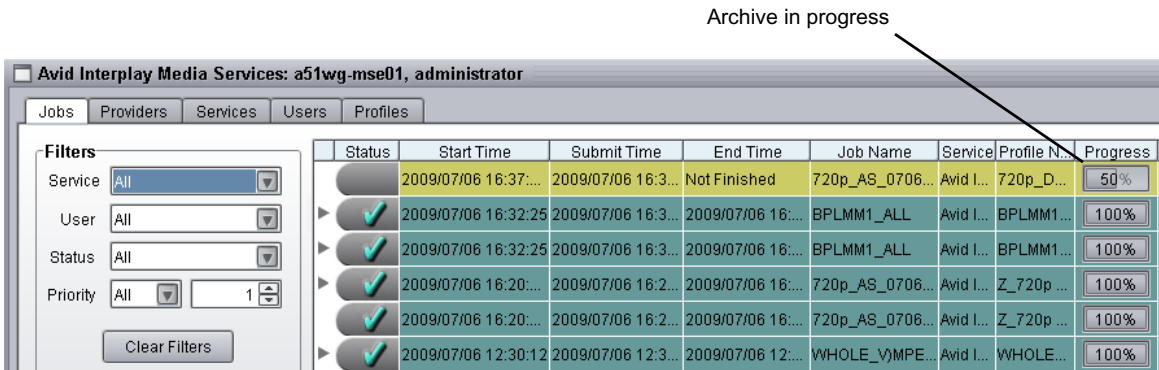
4. Select a profile from the Use Profile menu. See [“Archive Profile” on page 136](#).



Avid recommends that you use profiles for archiving, because profiles define many key settings such as target folders and archive partitions. For example, if you do not use a profile with an SGL archive implementation, the system uses the default SGL archive volume name. If this volume is not defined on the SGL system, the archive operation will fail.

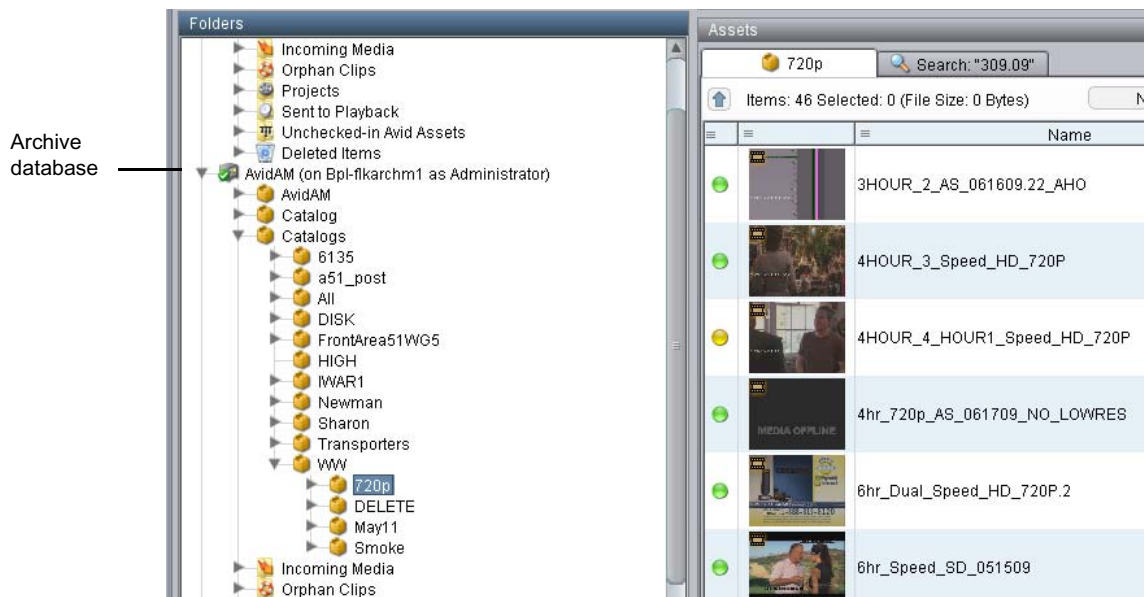
5. (Option) Select “Launch Interplay Production Services Status window.”
6. Click OK.

The system starts the Archive operation. The following illustration shows the Production Services and Transfer Status tool with the archive operation in progress.



7. (Option) For details about the job, select the job and click the Details button.:
8. (Option) Navigate to the destination folder in the archive database and verify that your file has been archived.

The following illustration shows the clip in the Archive database.



Deleting the High Res Media

After the high-res material is archived, you can delete the online version. Use the Delete dialog box in Interplay Access to select Avid assets for deletion. You have the option of deleting the assets only, the media only, or both assets and media. Delete only the high-res media associated with the master clips; do not delete the master clips or the low-res proxy. For additional information on deleting assets and media, see the *Interplay Access User's Guide* and the *Interplay Production Best Practices Guide*.

To delete the high-res media:

1. In the Interplay Access Content tab, select the assets you want to delete.

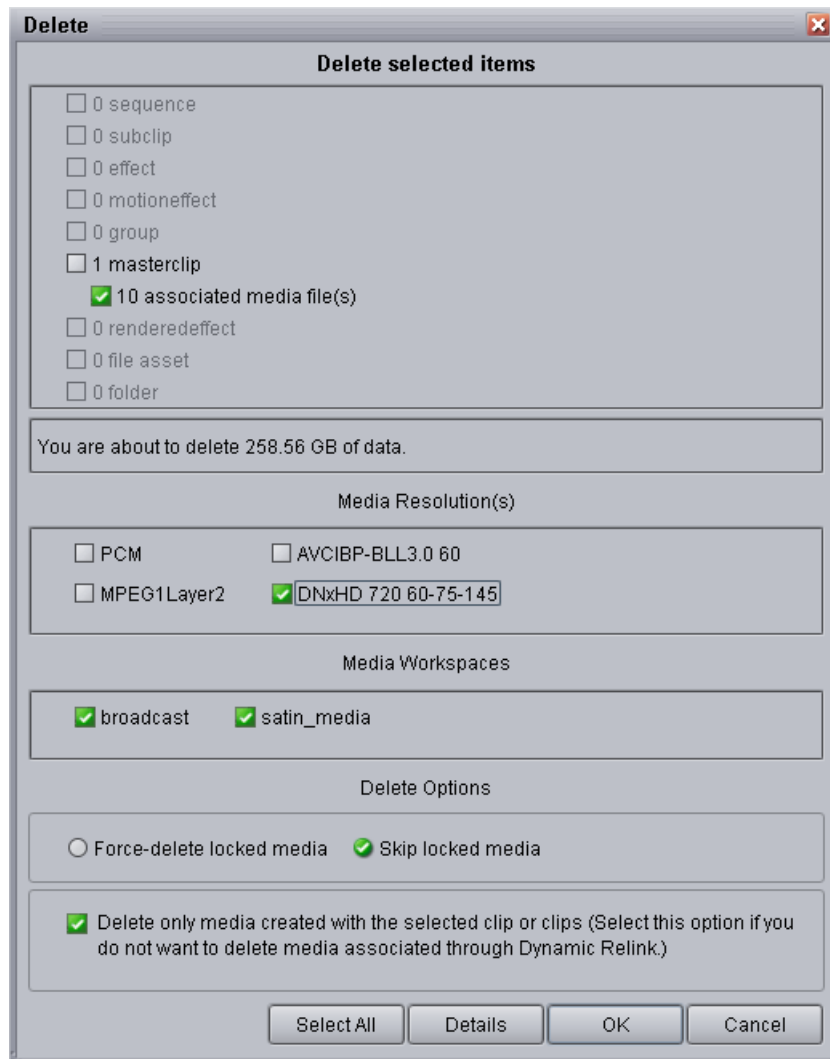
You can select multiple folders or clips for deletion. You can also delete folders by selecting them in the tree structure in the Folders pane.

2. Do one of the following:

- ▶ Right-click and select Delete.
- ▶ Press the Delete key.

The Delete dialog box opens, displaying information about the items you selected. It also displays the amount of disk space that would be freed if you delete the media files. No items are selected for deletion until you select them in the Delete dialog box.

The following illustration shows the results of selecting a master clip for deletion. The master clip is associated with both high-res and low-res media files but you are only deleting the high res material. Leave the master clip and low res material online.



3. In the “Delete selected items” area, deselect the master clip as shown in the above illustration. Keep the associated media files box checked.
4. Specify the media that you want to delete.
 - a. Select the resolutions you want to delete.
 The Media Resolutions section of the dialog box lists all video resolutions and audio formats for the clips you selected, including media associated through the Dynamic Relink feature. All resolutions are selected by default. Deselect the resolutions you want to keep. In this example, deselect the low-res media and keep the high-res DNxHD media selected for deletion.
 - b. Select which workspaces you want to delete media from.
 All workspaces that contain the selected media are listed and are selected by default.
 - c. Select whether to force the deletion of media files locked by an editor or to skip deletion of locked files.
 These options are available only if the user has administrative rights or if they are enabled by an administrator. “Skip locked media” is the default.

- d. Select “Delete only media created with the selected clip or clips.” This ensures that you only delete the media that was originally created with the clip.



Beginning with Interplay v2.0, the “Deselect only media...” option is selected by default.

5. (Option) Click Details to determine if any clips are reserved.

The Clips Reserved dialog box opens. Any clips with reservations are listed, along with information about the clips. If you have the proper privileges, you can override the reservations. Otherwise, click Cancel.

6. When you are ready to delete the files, click OK in the Delete dialog box.

The Delete in Progress dialog box displays the status of the deletion.

7. When the delete operation is completed, click Done.

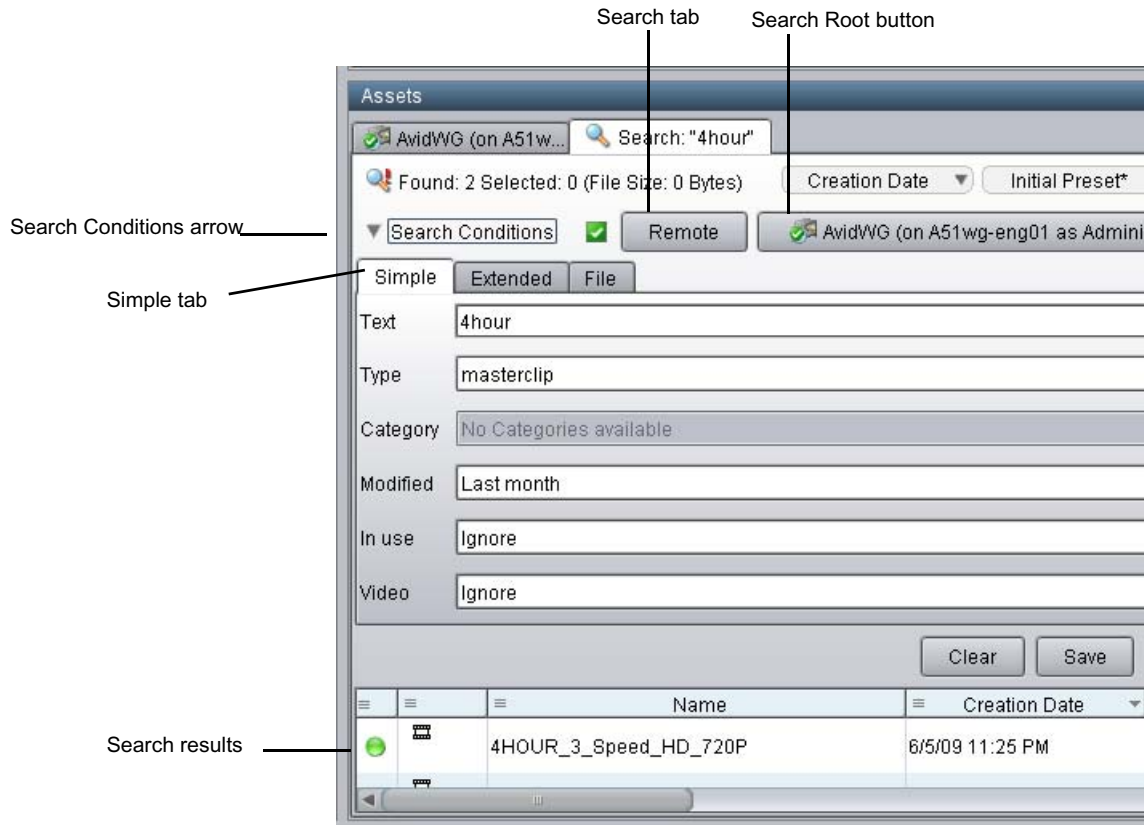
Searching for and Restoring High Res Master Clips

You can select a sequence or master clip for restoring from the archive. This example assumes the following:

- high-res media is in the archive and the corresponding low-res proxy media is online.
- You can search the online database for clips meeting specific criteria, view the low-res proxy, and then decide which clips to restore them from the archive.
- You can also search the archive database for clips to restore.

To search the online database for clips:

1. In the Assets view, click the Search tab.
2. Click the Simple tab. If necessary, click the Search Conditions arrow to show the search criteria.



3. Select a search root folder by doing one of the following:

- ▶ Use the Search Root button to select the archive database.
- ▶ Click a folder in the Tree view and drag it onto the Search Root button.

The Search root button displays the database or folder you selected to search.

4. Type the text or select the attributes and values you want to use for the search. In this example, the desired clip name contains the text “4hour”, it is a master clip, and it was created within the last month.



If you do not select any criteria, the search returns all the records in the database.

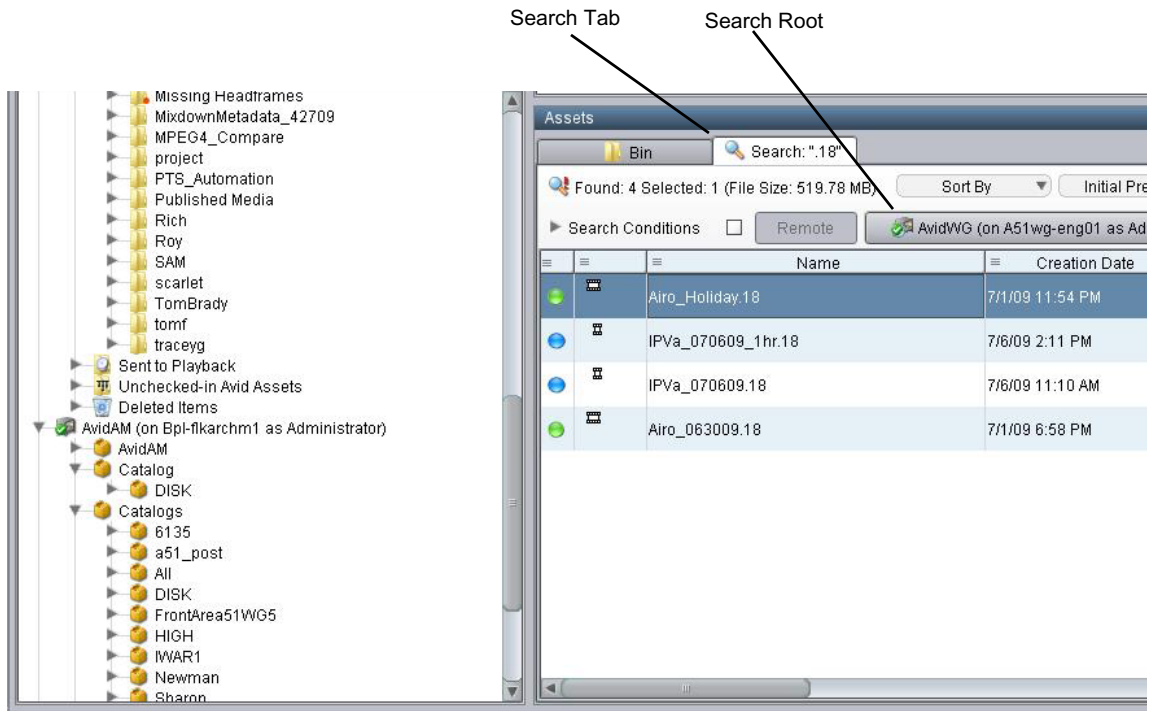
5. Click the Search button.

The Search button changes to a Stop button, which you can click if you want to stop the search. The results are displayed in the Search Results pane. They are visible until you close Interplay Access or until you perform a new search.

For additional information on searching and search criteria, see the *Interplay Access Users Guide*.

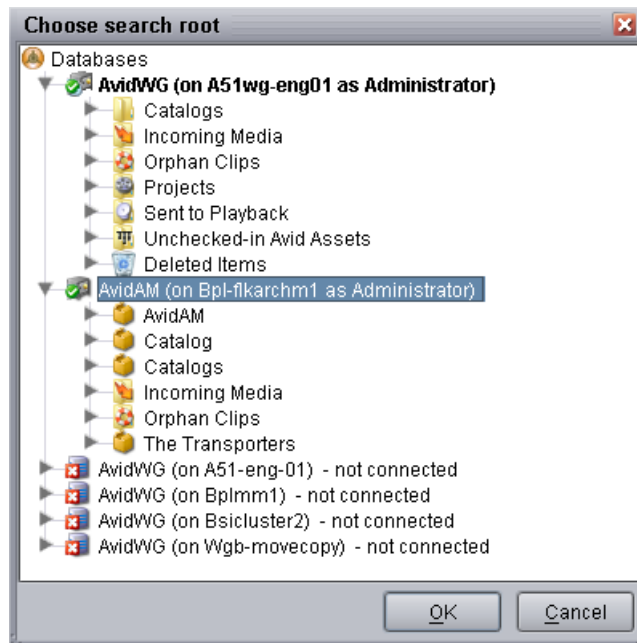
To search for a clip in the archive database:

1. Connect to the archive database.
2. Click the Search tab in the Interplay Access window.



3. Click the Search Root button

The Choose Search Root dialog box opens.



4. Select the Archive database (AvidAM) and click OK.

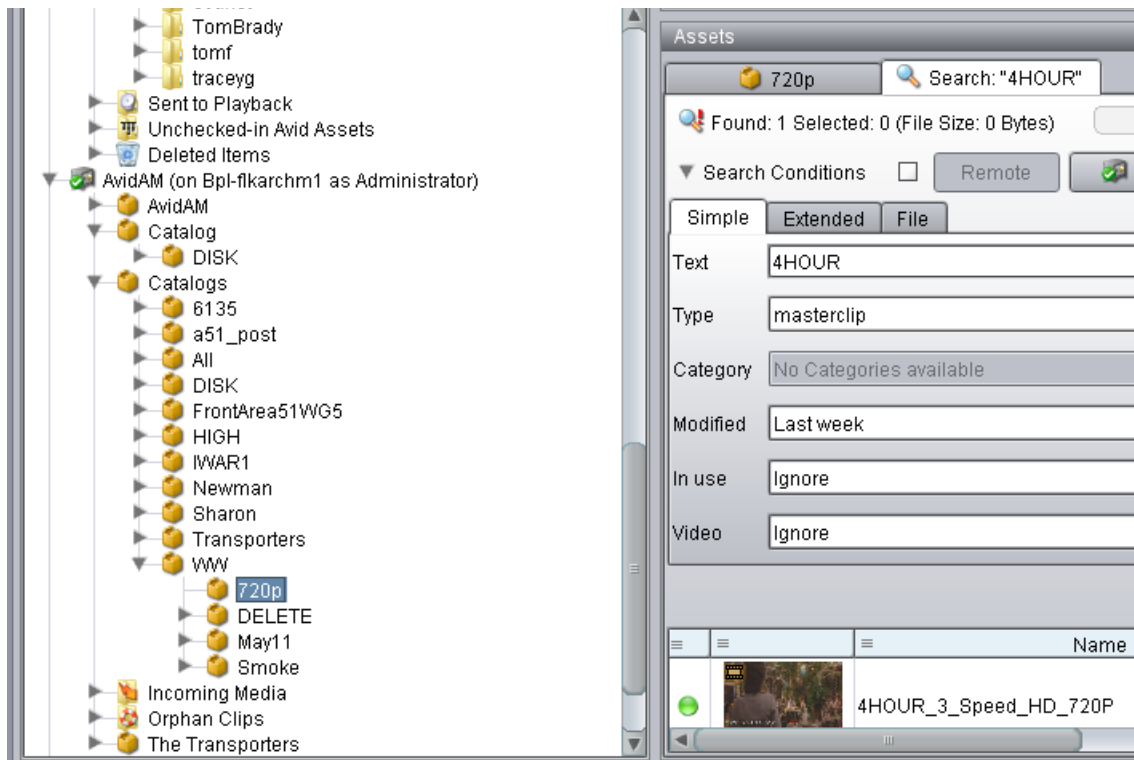


If necessary, expand the bottom of the window to display the Search Results area.

5. Type the name (or a portion of the name) of the clip you want to search for and select the appropriate search criteria.

6. Click the Search button.

The system displays the search results. In this example, one master clip in the archive database has the text “4HOUR” in its name.

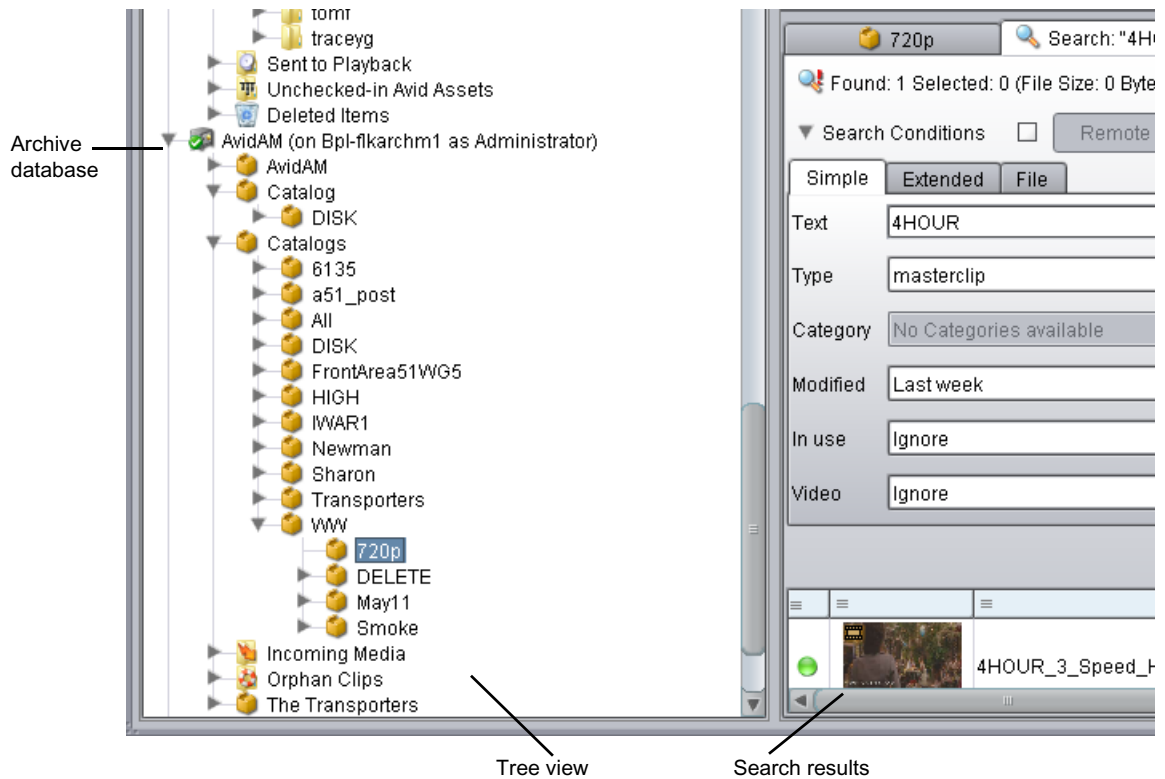


Restoring a Full Clip from the Archive

This example uses the results of a search to select a clip to restore. You can also use the Tree view to locate the clip you want to restore.

To restore a clip located in the Search Results area of Interplay Access:

1. Make sure the Production Services Engine application and Interplay Archive service are connected. See “Verifying a Service Provider Is Connected” in the Interplay Production Help or the *Interplay Production Services Setup and User’s Guide*.
2. Right-click the clip in the Search Results area and choose Restore.



The Restore dialog box opens.

3. Select a profile from the Use Profile menu. For information about profiles, see [“Creating Profiles” on page 135](#).
4. Click OK.

The system performs the restore operation. You can use the Production Services Status window to view the progress and results of the restore operation.

5. (Option) Navigate to the destination folder in the Interplay Access window and verify that the clip was restored.
6. Make sure Enable Dynamic Relink is selected to view the restored media.

Performing a Partial Restore

For a description of how to perform a partial restore, see [“Working with a MultiRez Clip in the Editor and Performing a Partial Restore” on page 120](#).

Setting Up the Production Services Profiles for a 720p MultiRez Workflow

This workflow requires a number of Production Services profiles for the transcode, archive, and restore operations. This section describes the profiles. For details on creating a profile see the Interplay Production Help or the *Interplay Production Services Setup and User Guide*.

Creating Profiles

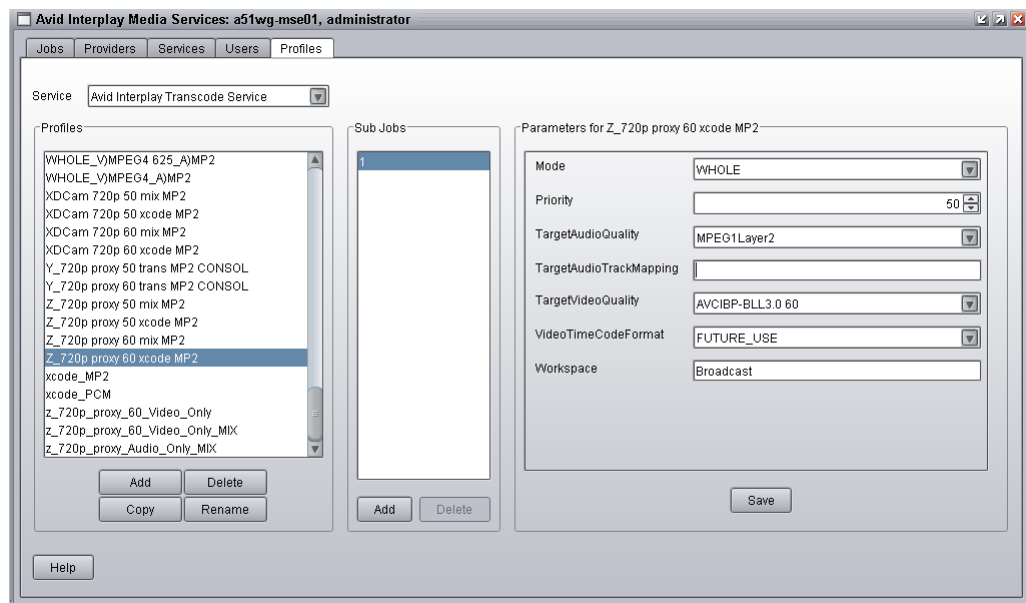
This example uses the following profiles:

- Transcode high-res DNxHD 720p material to 720p proxy
- Archive the high-res material
- Restore high-res clips from the archive
- Perform a partial restore of clips or subclips that are part of a sequence.

To use the Production Services application to create a profile:

1. In Interplay Access, select View > Production Services Status.
2. Click the Profiles tab and select the Provider from the Service menu in the top left-hand window.

The following illustration shows the Interplay Transcode Service provider selected.



3. Click Add and type in a name for the new profile.
4. Add the appropriate values and click Save.

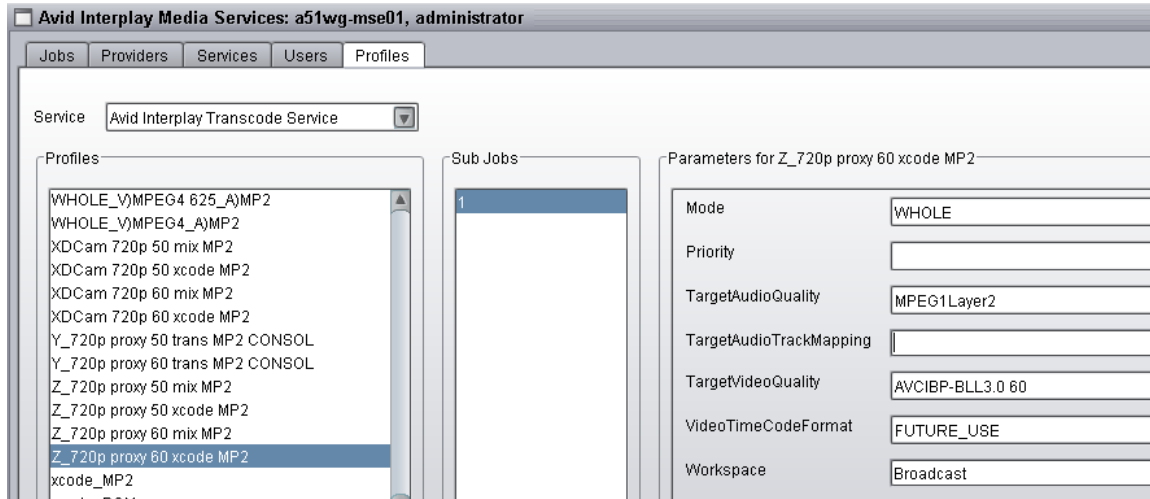
For more information on creating profiles, see the Interplay Production Help or the *Interplay Production Services User and Setup Guide*. For examples of the profiles used in this workflow, see [“Interplay | Production Service Profile Examples” on page 135](#).

Interplay | Production Service Profile Examples

This section shows examples of the profiles used in this workflow. For information on parameter values, see the Interplay Production Help or the *Interplay Production Services User and Setup Guide*.

Transcode Profile

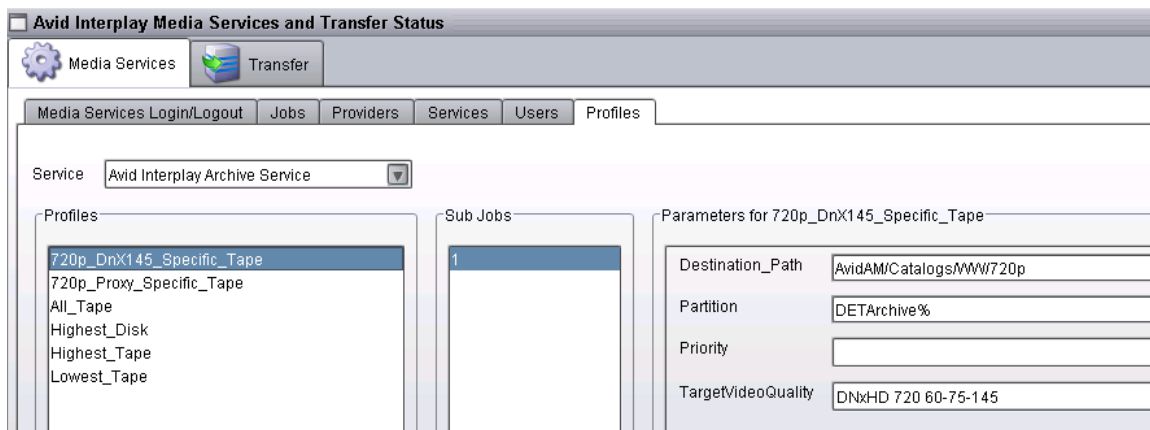
The following illustration shows a typical 720p Transcode profile. This profile creates low-res proxy video and compressed audio.



Parameter	Value
Mode	Whole
Priority	50
TargetAudioQuality	MPEG1LAYER2
TargetAudioTrackMapping	blank
TargetVideoQuality	AVCIBP-BLL3.0 60
VideoTimeCodeFormat	FUTURE_USE
Workspace	Broadcast

Archive Profile

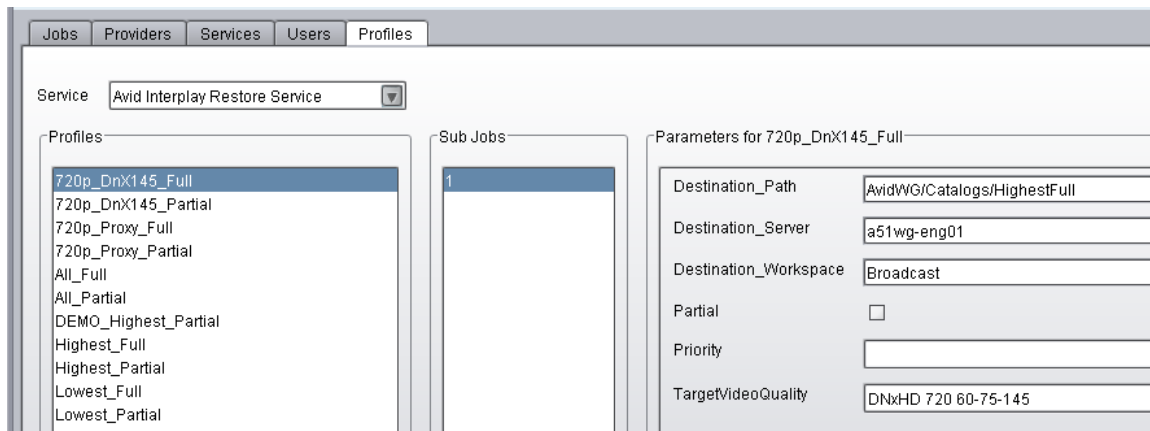
The following illustration shows an archive profile.



Parameter	Value
Destination_Path	AvidAM/Catalogs/WW/720p
Partition	DETArchive%
Priority	50
TargetVideoQuality	DNxHD 720 60-75-145

Full Restore Profile

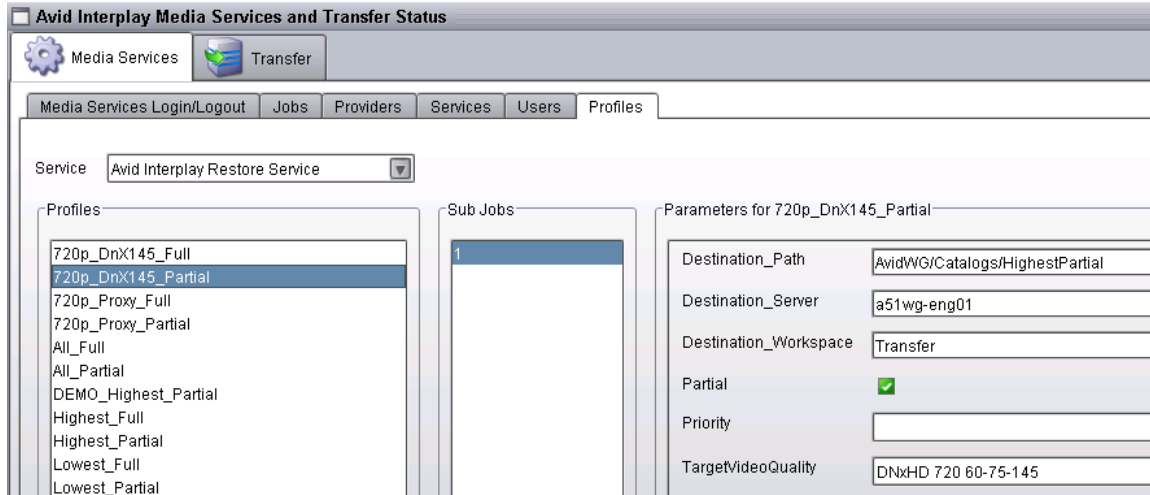
The following illustration shows a profile used for restoring full clips from the archive. Note that Partial is not checked.



Parameter	Value
Destination_Path	AvidWG/Catalogs/HighestFull
Destination_Server	a51wg-eng01
Destination_Workspace	Transfer
Partial	Not checked
Priority	50
TargetVideoQuality	DNxHD 720 60-75-145

Partial Restore Profile

The following illustration shows a profile used to perform a partial restore of 720p high-res material. This profile only restores a specific resolution: DNxHD 720 60-75-145. You could also select “Highest” and restore the highest available resolution.



Parameter	Value
Destination_Path	AvidWG/Catalogs/HighestPartial
Destination_Server	a51wg-eng01
Destination_Workspace	Transfer
Partial	Checked
Priority	50
TargetVideoQuality	DNxHD 720 60-75-145

8 Determining Interplay Database Scalability

This document describes how to use the size of the Interplay Engine database and the amount of available RAM to make decisions about how much larger your database can grow while maintaining your current workflow and hardware. This document contains the following topics:

- [Definition of Terms](#)
- [Total Used Database Pages and Estimated Database Page Limit](#)
- [Comparing Database Page Count and Cache Size](#)
- [Tips for Improving Database Performance](#)

Definition of Terms

Database Page

A *database page* is a fixed size, contiguous block of data that contains persisted database content. The metric of *total used database pages* provides a consistent way to determine whether you are operating within the scalability limits of your system. For additional information, see [“Total Used Database Pages and Estimated Database Page Limit” on page 140](#).

Maximum Cache Size

The *maximum cache size* refers to the maximum number of database pages the Interplay Engine can keep cached in RAM. This value is dependent on the amount of RAM available on the Interplay Engine system or by manual configuration (in the case of the Interplay Bundle).

If the database size exceeds the maximum cache size, the Interplay Engine may need to swap out pages whenever it needs to load new database pages into RAM. The probability of swapping pages increases as you continue to exceed the maximum cache size. Slow downs can occur when you perform large searches, deletions, or add a large number of new assets to the database.

Database Object

A *database object* contains the metadata that defines an asset in the Interplay Production database. One database object can reference many other objects. For example, a sequence is represented as a database object and each sequence references other database objects such as master clips and audio tracks.

The number of database objects that an Interplay environment can support is highly dependent on the workflow at a particular site. The following areas can be affected by the workflow:

- *Highly linked objects*: For example, if your workflow involves adding one short introductory clip to the beginning of every sequence, that clip might be associated with hundreds of database objects. See [“Tips for Improving Database Performance” on page 144](#).
- *Dependency complexity*: For example, if you have a complex sequence with many rendered effects, master clips, subclips, and locators, the database object representing the sequence can be complex. Different versions of the sequence and multiple resolutions associated with master clips also adds complexity.
- *Object size*: Depends on the type of object and the number of (custom) properties associated with the object. For example, if you store a large amount of text within locators this can contribute to the size of the object.

For additional information, see [“Establish a Stable Ratio Between Database Objects and Database Pages” on page 145](#).

Total Used Database Pages and Estimated Database Page Limit

The metric of *total used database pages* provides a consistent way to determine whether you are operating within the scalability limits of your system. A *database page* is a fixed size, contiguous block of data that contains persisted database content. To maintain an acceptable level of performance, the total number of used database pages should not exceed the recommended database page limit for your system. Recommended values are listed in the [“Interplay | Engine Database Page Limits” on page 141](#).

You can view the total number of database pages in use in the Database Information view of the Interplay Administration tool. See [“Comparing Database Page Count and Cache Size” on page 142](#). The same information is available (for all releases) in the log file named NXNServer_<date>.log in the following directory:

C:\Program Files\Avid\Interplay Engine\Logs\Machines\<machine name>

In the NXN log file, search for the text string “AvidWG pages total” for Interplay Engines and search for “AvidAM pages total” for Archive Engines.

The *estimated database page limit* is the maximum amount of pages the Interplay Engine can keep cached in RAM (maximum cache size) plus an estimated permissible percentage of overusage.

This “overusage” is possible because all databases contain data that is never or very seldom accessed and because a low frequency of swapping cached pages does not have a perceptible performance impact. The permissible overusage value is typically 20% of the size of the cache. The values for different hardware configurations are estimated by extracting cache usage data from customer databases and by performing internal testing with databases that are slightly larger than recommended.



The permissible overusage depends on the customer's workflows and might exceed or be lower than the estimated 20%. Therefore it is important that you closely monitor the performance of the Interplay Engine once the total used database pages exceed the Maximum Cache size.

If the total used pages exceeds the recommended limit for your system, performance may degrade and slow downs may occur. In this case, do one of the following:

- Delete assets in the Interplay Production database to bring the total number of pages down below the recommended limit for your system.
- Increase the amount of RAM or move to a more powerful system as shown in the table under [“Interplay | Engine Database Page Limits” on page 141](#).

Interplay | Engine Database Page Limits

This section lists the maximum cache size and the estimated database page limits supported by an Interplay Engine running on supported servers for Interplay v3.x or higher. For additional information, see [“Total Used Database Pages and Estimated Database Page Limit” on page 140](#).

System	RAM	Operating System	Maximum Cache Size	Estimated Database Page Limit
Standalone Interplay Engine and Interplay Engine Cluster				
AS3000	12 GB	W2K8 R2 Standard W2K3 R2 Enterprise	1.8 million	2.2 million
	36 GB ^a (standalone)	W2K8 R2 Standard	6.3 million	7.8 million
	36 GB (cluster)	W2K8 R2 Enterprise	7.3 million	9.1 million
	48 GB (cluster)	W2K8 R2 Enterprise	10.4 million	12.4 million
Dell R630 or HP DL360	32 GB	W2K12 R2 Standard	6.3 million	7.8 million
	48 GB	W2K12 R2 Standard	10.4 million	12.4 million
	64 GB	W2K12 R2 Standard	14.6 million	16.6 million

- a. Windows Server 2008 Standard limitation: 36GB installed and 32GB usable by the Windows Server 2008 OS. Windows Server 2008 Enterprise can use more than 32GB.

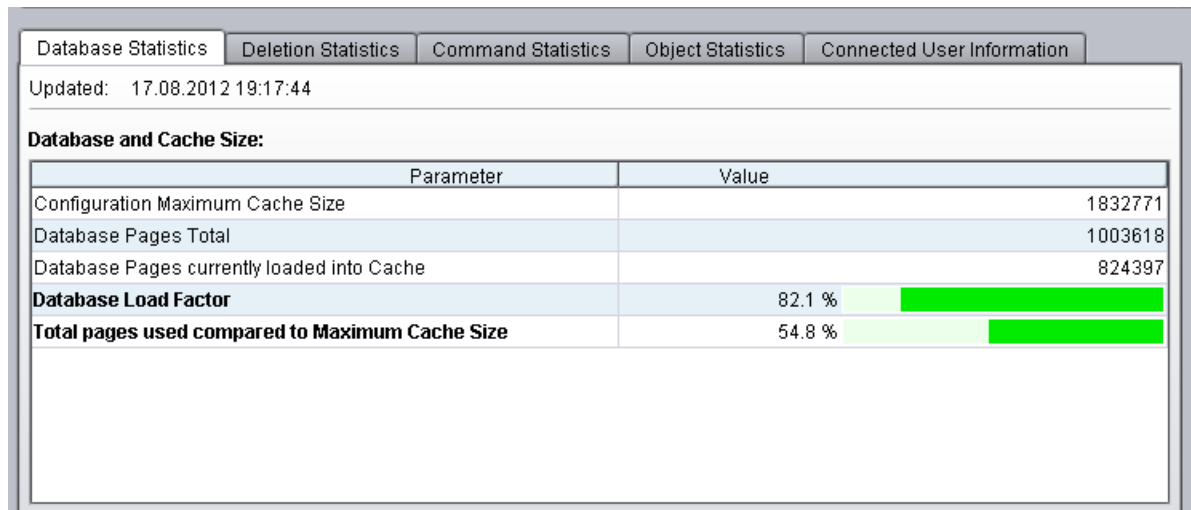
The following table also shows values for supported Interplay Bundle systems. An Interplay Bundle consists of an Interplay Engine, Interplay Media Indexer, and Lookup Service running on the same system. Note that maximum cache size does not apply to Bundle configurations. Maximum cache size is based on the total RAM available on the system but the Interplay Engine must share the RAM with other applications on a Bundle system. Also note that 12 GB RAM is not supported for Interplay Bundle v3.6 and later because MI v3.6 requires a minimum of 32 GB RAM.

System	RAM	Operating System	Estimated Database Page Limit
Interplay Bundle Configurations			
AS3000	12 GB Interplay v3.5 and earlier	W2K8 R2 Standard	1.7 million
	36 GB ^a Interplay v3.6.x and later	W2K8 R2 Standard	2.5 million
Dell R630 or HP DL360	32 GB ^b	W2K12 R2 Standard	2.5 million

- a. Windows Server 2008 Standard limitation: 36GB installed and 32GB usable by the Windows Server 2008 OS. Windows Server 2008 Enterprise can use more than 32GB.
- b. You can increase the database page limit by increasing the amount of RAM.

Comparing Database Page Count and Cache Size

The following illustration shows an example of the Database Information view for Interplay v2.6.



The Database and Cache Size parameters are described as follows:

- **Configuration Maximum Cache Size:** The number of database pages that can be loaded into RAM for this system. This value corresponds directly to the “Maximum Cache Size” column in the tables listed in [“Interplay | Engine Database Page Limits” on page 141](#). In this example, the RAM cache can hold 1.8 million pages. This corresponds to an SR2500 or AS3000 with 12 GB of RAM.
- **Database Pages Total:** The number of pages in this Interplay database. This is the current size of the database.

- Database Pages currently loaded into Cache: The number of database pages currently loaded into RAM. In this example, the database contains 1,003,618 pages and 824,397 of those pages are loaded into RAM.
- The fourth entry in the display differs depending on whether you are using Interplay v2.6 or v2.7 and higher:

- Interplay v2.7: Database Load Factor. Shows the percentage of the database that is currently loaded into RAM. On a completely loaded database the ratio would be 100%, but since some pages of a database are never loaded, it will always stay below 100%. If the operations/workflows executed against this database do not require the whole database to be loaded into cache, this ratio will stay farther below 100%.

This number can be used to judge the "warmness" of a database (whether the database is fully operational). For example, the value may steadily increase after starting up the Interplay Engine after a maintenance window.

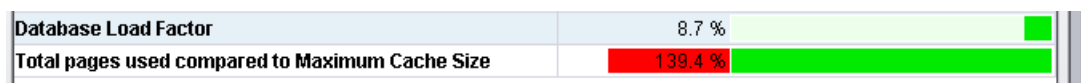
Note that if the "Total pages used compared to Maximum Cache Size" of the database exceeds 100%, then the load factor can never reach 100%. The bigger the database grows over the maximum cache size, the less database pages can be loaded into the cache.

- Interplay v2.6: Cache Utilization. Shows a graphical representation of how much RAM you are currently using. As the number of cached pages gets very close to the maximum cache size, performance degrades during times when large database activities occur. The slow downs occur because the Interplay Engine must swap pages in and out of RAM in order to make room for the new information

This value cannot be larger than 100% because the amount of RAM is fixed. Note that when you get close to 100% the Interplay Engine automatically starts releasing pages to allow for new requests. If you reach 100%, the Interplay Engine runs out of memory and you will see a corresponding error in the NXNserver.log file.

- Total pages used compared to Maximum Cache Size: Shows a graphical representation of how close the current database size (total pages used) is to the maximum cache size. This graphs shows whether the entire database can fit into RAM.

This value can be larger than 100% because it is possible for your database to be larger than the number of pages that can be loaded into RAM. If that occurs, the excess portion of the graph turns red as shown in the following illustration.



The value can be up to 120% without causing performance problems. Recall that the recommended database page limit is the maximum cache size plus 20%. See [“Interplay | Engine Database Page Limits” on page 141](#). If the total number of database pages exceeds 120%, performance slowdown can be dramatic (exponential). In this case you should either delete assets to reduce the size of your database or increase the cache size by adding RAM or moving to a more powerful system.

As mentioned earlier, the permissible overusage depends on the customer's workflows and might exceed or be lower than the estimated 20%. Therefore it is important that you closely monitor the performance of the Interplay Engine once the total used database pages exceed the Maximum Cache size.

Database Activity Information

The following illustration shows the Database Activity parameters at the bottom of the Database Statistics tab.

Database Activity:			
Period	Pages Created	Pages Written	Pages Read
Since last Engine start	0	86	53443
In the last 180 second interval	0	0	3064
End of last 180 second interval: 30.05.2012 16:45:53			

The page activity counters are accumulated for two different intervals:

- The time interval since the last time the Interplay Engine was started
- The three minute interval that ended at the time displayed at the bottom of the screen. The Interplay Engine collects the information every three minutes and displays the results at the bottom of the screen.

The Database Activity parameters are described as follows:

- Pages created: Gives an indication of how much new information has been added to the database.
- Pages Written: An indication of how much has changed in the database.
- Pages Read: An indication of the requests that have been made of the database. This shows how many pages have been read from disk. After pages are loaded into RAM, this number usually decreases.

At the bottom of the screen the system shows the date and time of the last data collection. Time is measured by the system time of the Interplay Engine.

Tips for Improving Database Performance

The following list provides some tips for improving performance:

- Determine the ratio between the number of objects in your Interplay database and the number of total pages used in the database to understand how the number of objects stored in the database relate to the size of your database. See [“Establish a Stable Ratio Between Database Objects and Database Pages” on page 145](#).
- Reduce the number of highly linked objects.

For example, a broadcast site may have one high level Projects folder that contains an Intro_clip that is used by all groups at the site. You could consolidate that clip to create several copies and then move the copies into separate folders, one for each group. For example, you could create a Promo_Intro_clip and a News_Intro_Clip, and put those clips in folders that would be accessed by the different groups. Delete the old Intro_clip asset and have the groups start using the new versions. This can dramatically reduce the number of highly linked objects.

- Develop an efficient deletion workflow and only perform large deletions using a custom deletion schedule. For additional information, see the *Interplay Production Best Practices Guide*.

- Use search criteria to avoid searching the entire database.

If you only use the simple “Contains” text search option, the Interplay Engine searches the entire database for the asset. To improve search performance, limit the search to a particular folder or use an attribute such as a particular time frame to narrow down the search. Note that for simple text searches the search time increases linearly as your database grows. For example, if the database is twice as large, it can take twice as much time to perform the same simple text search. For additional information, see [“Refining Your Search in Interplay | Production” on page 71](#).

- Keep the number of files per folder on ISIS under 5000. As the number of files in a folder grows, it takes the Interplay Media Indexer a longer time to traverse the folder. Traversing a folder with more than 5000 files can result in a performance degradation. Note that this does not affect the Interplay database performance directly but it does affect the performance of the system as a whole.

Apply this same limitation to the number of assets in a folder in the Interplay database.

Establish a Stable Ratio Between Database Objects and Database Pages

The number of database objects that an Interplay environment can support is dependent on the workflow at a particular site. For additional information, see [“Database Object” on page 139](#).

For a given workflow, the ratio of database objects per total number of database pages remains fairly constant. Stable ratios are often in the range of 0.4 to 1.5 but this can vary depending on your workflow. Calculate the ratio when you feel you have an established workflow. Keep track of the following values:

- Total number of objects in the database
- Total number of database pages in use
- Ratio of objects per total number of pages (*total objects / total database pages*)

If you do this periodically while you maintain a constant workflow, you will start to see a pattern. Once you determine a stable ratio for your current workflow you can start to make predictions about how much you can increase the database size while maintaining the same workflow.

Calculating the Ratio Between Pages and Objects

The method for determining the ratio differs depending on your Interplay Production version. The following table shows where to locate the data for each version.

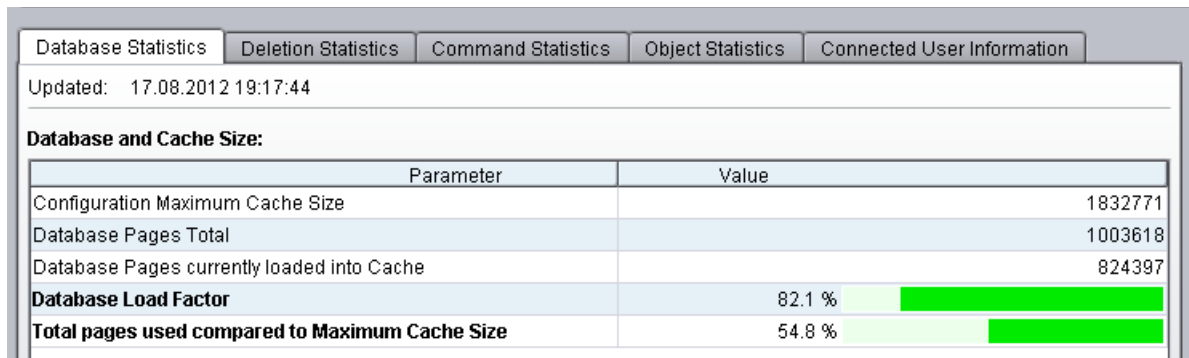
Interplay Production version	Total Objects	Total Database Pages	Ratio
v2.7 and higher	Objects Statistics tab	Database Statistics tab	Automatically calculated and displayed on the Objects Statistics tab
v2.6	Objects Statistics tab	Database Statistics tab	calculate: total objects/total database pages

Interplay Production version	Total Objects	Total Database Pages	Ratio
v2.5 and earlier	Objects Statistics tab	NXNServer_<date>.log file See “Total Used Database Pages and Estimated Database Page Limit” on page 140	calculate: total objects/total database pages

The following section presents an example of how to use the ratio.

Example: Using the Ratio to Plan for Database Growth

The following illustration shows an example database.



The following statistics apply to this database:

- Maximum cache size: For this hardware configuration, the maximum number of database pages that can fit into RAM is 1.8 million. For additional information, see [“Interplay | Engine Database Page Limits”](#) on page 141.
- Database Pages: The total number of database pages is currently 1,003,618.
- The number of database pages currently loaded into cache (RAM) is 824,397.
- The fourth entry in the list differs depending on whether you are using Interplay v2.6 or v2.7 and higher:
 - Interplay v2.7: Database Load Factor. Shows the percentage of the database that is currently loaded into RAM.
 - Interplay v2.6: Cache Utilization. Shows how much of the cache is currently being used.
- Total pages used compared to Maximum Cache Size: If the complete database (1,003,618 pages) was loaded into cache, it would consume 54.8% of the cache.

According to the final list item, the database can grow by almost 45% and still be within the cache limits. Calculate the number of pages that can be added as follows:

$$\text{maximum cache size} - \text{total database pages} = 1,832,771 - 1,003,618 = 829,153$$

This means that you can add 829,153 database pages and the database should still load into RAM.

Growing the database by a specific number of pages is a relatively abstract concept. The following sections describe how to correlate this information to the more concrete terms of database objects such as master clips and sequences.



You can add an additional 20% of pages to the database once the number of pages reaches the maximum cache size. This is the “overusage” value described in “Total Used Database Pages and Estimated Database Page Limit” on page 140. The extra 20% is not included in these calculations and you can consider it as an additional buffer.

Calculating the Objects per Page Ratio

Interplay v2.7 automatically calculates and displays the objects per database page ratio on the Objects Statistics tab. For earlier releases, you can calculate the ratio of database objects to database pages. For information on where to obtain the data, see “Calculating the Ratio Between Pages and Objects” on page 145.

The objects per page ratio can help determine how much you can grow your database while maintaining the same workflow. When your workflow is stable, the ratio tends to be constant over time. If you change your workflow then you need to calculate a new ratio. The objects per page ratio is calculated as follows:

total number of database objects / total number of database pages

The following illustration shows the Object Statistics tab for this example. The total number of database objects is 2,925,082.

Database Statistics		Deletion Statistics	Command Statistics	Object Statistics	Connected User Information
Total number of database objects:		2925082			
Database objects per database page:		2.91			
Database Object Type		#			
masterclip		276279			
sequence		11392			
subclip		7681			
group		2235			
motioneffect		1264			
effect		7527			
renderedeffect		487907			
filemob		1156037			
avid asset (in browse tree)		870323			
folder		8487			
user		304			
other objects		95646			

The objects per page ratio in this example is $2,925,082 / 1003618 = 2.91$.

The ratio is 1:2.91. This means that there are approximately 2.91 database objects per database page.

In this example, the database can grow by 45% or roughly 829K database pages. Using a ratio of 1:2.91, that means you can add 2.4 million objects and the database should still be able to load into RAM. ($829K \times 2.91 = 2,412,390$)

Using the Ratio with Masterclip Objects

This example takes the calculation one step farther to predict how many additional masterclips can be added to the Interplay database and still keep the database within the cache limits. First calculate the percentage of database objects that includes masterclips:

$$\text{Masterclip count} / \text{total database objects} = 276,279 / 2,925,082 = 0.9 = 9\%$$

In this example, masterclips make up approximately 10% of the total number of database objects.

Since the database can grow by 2.4 million objects, this means that you can add approximately 240K masterclips (10% of 2.4 million).

In summary, you can increase the size of the database by adding 240K masterclips and the database should still be able to load into RAM. If you change the workflow you will need to calculate a new ratio and recalculate the number of masterclips you can add.

H.264 End-to-End MultiRez Workflow

The following topics describe stages in a MultiRez workflow that uses H.264 low-res proxy media for editing and high-res media for output, both ingested through an AirSpeed Multi Stream server. They also describe a workflow that uses Interplay Transcode to create the low-res proxy media.

- [H.264 Resolutions Supported by Interplay | Production](#)
- [Installation and Configuration for an H.264 Workflow](#)
- [Acquiring Media for an H.264 Workflow](#)
- [Editing with H.264 and High-Res](#)
- [Archiving, Restoring, and Deleting H.264 and High-Res Media](#)

There are specific settings for ingest of H.264 proxy media through an AirSpeed Multi Stream server. Otherwise, this dual-ingest workflow is similar to a dual-ingest workflow with two devices and MPEG-4 proxy media.

H.264 Resolutions Supported by Interplay | Production

H.264 is a low-res proxy format intended for low bit-rate applications. The proxy is a Baseline profile, Level 3.0 H.264 bit stream, which limits the raster size and bit rate of the compressed data. Interplay Production supports both 800 Kbps and 2.0 Mbps versions of H.264 media.

H.264 800 Kbps Proxy Resolutions

Interplay version 2.3 and later supports an end-to-end workflow for H.264 800 Kbps proxy resolutions. You can ingest H.264 800 Kbps material (through AirSpeed Multi Stream only), work with it in an Avid editing application, check it in and out of the Interplay database, and process it through Interplay Production Services and Interplay Transfer. You can use Interplay Transcode to transcode from another resolution to H.264 800 Kbps.



Avid editing applications cannot capture H.264 proxy resolutions. They can be created only through dual ingest from AirSpeed Multi Stream or Interplay Capture, or by transcoding or mixing down a clip or sequence to one of the Avid H.264 proxy resolutions.

Note the following limitation:

- Partial delivery and partial restore are not supported for H.264 proxy resolutions.

The following table lists the supported H.264 800 Kbps resolutions.

Editor Project Type	Edit Rate (frames/sec)	Resolution Name in Interplay Access and Interplay Transcode	KBit/Sec	Aspect Ratio	Proxy Raster Size
25i PAL	25	H.264 800Kbps Proxy 625	800 Kbps	4:3	352x288
30i NTSC	29.97	H.264 800Kbps Proxy 525	800 Kbps	4:3	352x240
720p 23.976	23.976	H.264 800Kbps Proxy 720p 23.976	800 Kbps	16:9	320x180
720p 50	50	H.264 800Kbps Proxy 720p 50	800 Kbps	16:9	320x180
720p 59.94	59.94	H.264 800Kbps Proxy 720p 59.94	800 Kbps	16:9	320x180
1080i 50	25	H.264 800Kbps Proxy 1080i 25	800 Kbps	16:9	480x270
1080i 59.94	29.97	H.264 800Kbps Proxy 1080i 29.97	800 Kbps	16:9	480x270
1080p 23.976	23.976	H.264 800 Kbps Proxy 1080p 23.976	800 Kbps	16:9	480x270
1080p 24	24	H.264 800Kbps Proxy 1080p 24	800 Kbps	16:9	480x270
1080p 25	25	H.264 800Kbps Proxy 1080p 25	800 Kbps	16:9	480x270
1080p 29.97	29.97	H.264 800Kbps Proxy 1080p 29.97	800 Kbps	16:9	480x270

Support includes eight channels of compressed MPEG1 Layer 2 audio media (96k per channel).



See the Interplay Production ReadMe for limitations on 24p proxy resolutions.



If you perform a partial restore operation on a sequence that includes H.264 media, H.264 media is fully restored. Other resolutions are partially restored.

H.264 2.0 Mbps Proxy Resolutions

Interplay version 2.7 and later supports workflows for H.264 2.0 Mbps (HD) and 1500 Kbps (SD) proxy resolutions. These resolutions are created by a Media Composer Cloud editing system or by an Interplay Transcode provider. You can work with H.264 2.0 Mbps material in an Avid editing application, check it in and out of the Interplay database, and process it through Interplay Production Services and Interplay Transfer.

Note the following limitations:

- Partial archive and partial restore are not supported for H.264 proxy resolutions.

The following table lists the supported H.264 2.0 Mbps resolutions.

Editor Project Type	Edit Rate (frames/sec)	Resolution Name in Interplay Access and Interplay Transcode	Bit/Sec	Aspect Ratio	Proxy Raster Size
25i PAL	25	H.264 1500Kbps Proxy 625	1500 Kbps	4:3	352x288
30i NTSC	29.97	H.264 1500Kbps Proxy 525	1500 Kbps	4:3	352x240
720p 50	50	H.264 2.0Mbps Proxy 720p 50	2.0 Mbps	16:9	320x180
720p 59.94	59.94	H.264 2.0Mbps Proxy 720p 59.94	2.0 Mbps	16:9	320x180
1080i 50	25	H.264 2.0Mbps Proxy 1080i 25	2.0 Mbps	16:9	480x270
1080i 59.94	29.97	H.264 2.0Mbps Proxy 1080i 29.97	2.0 Mbps	16:9	480x270
1080p 50	50	H.264 2.0Mbps Proxy 720p 50	2.0 Mbps	16:9	320x180
1080p 59.94	59.94	H.264 2.0Mbps Proxy 720p 59.94	2.0 Mbps	16:9	320x180

Support includes eight channels of compressed MPEG1 Layer 2 audio media (96k per channel).



If a clip has both 800 Kbps and 2Mbps proxies, Interplay Access plays the 800 Kbps version.

Installation and Configuration for an H.264 Workflow

The following topics describe installation and configuration that is specific to an H.264 workflow:

- [H.264 Workflow Overview](#)
- [H.264 Card for AirSpeed Multi Stream](#)
- [Folders and Shared-Storage Workspaces for an H.264 Workflow](#)
- [Interplay | Production Services Profiles for an H.264 Workflow](#)
- [Configuring Interplay | Production Clients for an H.264 MultiRez Workflow](#)

H.264 Workflow Overview

Interplay version 2.3 and later supports an end-to-end workflow that includes H.264 resolutions. This topic describes two workflows:

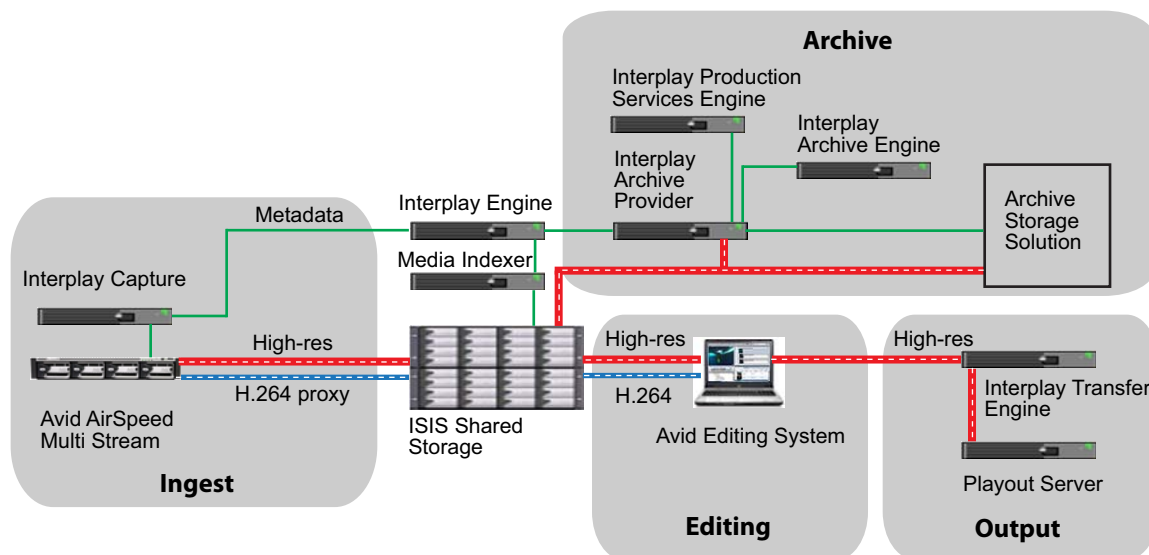
- AirSpeed MultiStream server to ingest both high-res and H.264 proxy media (H.264 800 Kbps only)
- Interplay Transcode to create H.264 media (either H.264 800 Kbps or H.264 2.0 Mbps)



This topic describes two possible workflows. For help in defining a workflow for your facility, consult an Avid workflow specialist.

Dual-Ingest Using AirSpeed MultiStream

The following illustration shows the main stages of a workflow that uses an AirSpeed Multi Stream system for ingest of both high-res media and H.264 800 Kbps proxy media.



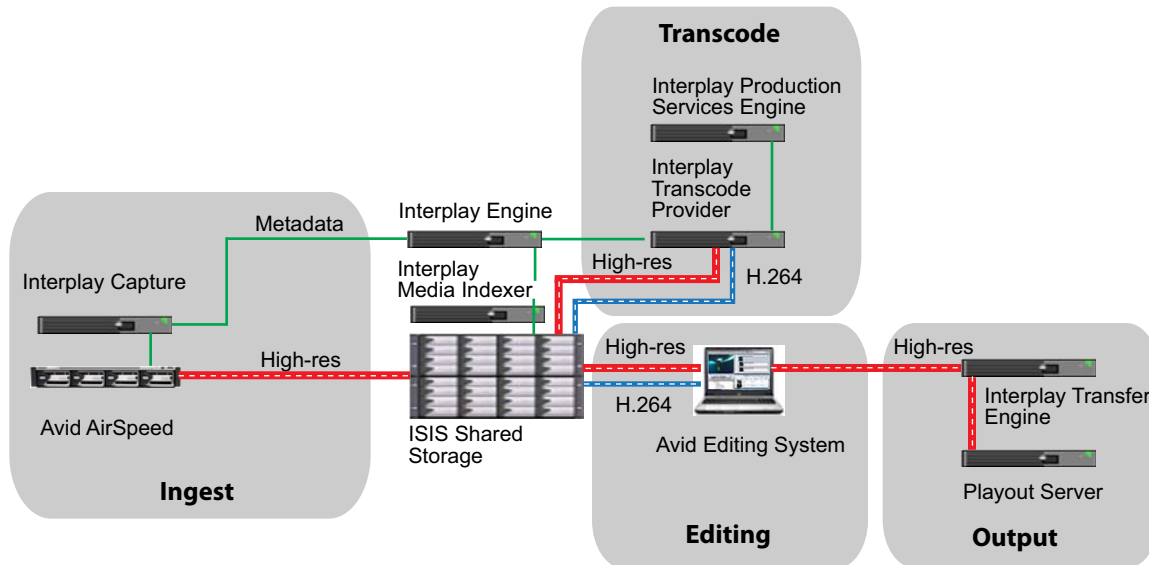
1. **Ingest:** Interplay Capture manages dual ingest from the AirSpeed Multi Stream (high-res media and H.264 proxy media) and checks the metadata into the Interplay database on the Interplay Engine.
2. **Editing:** A user on an Avid editing system edits with H.264 proxy media (working resolution).
3. **Output:** When the sequence is complete, the user dynamically relinks to the high-res media (target resolution) and sends the sequence to playback.

For information about input and output workflows for XDCAM HD media, see [“Long GOP End-to-End Workflows”](#) on page 106.

4. **Archive:** From an Avid editing system, a user consolidates the sequence and uses the Interplay Archive Provider to send metadata to the Interplay Archive Engine and send media to a third-party storage solution. A user can also use the Interplay Transcode provider (from an Avid editing system or from Interplay Access) to create a mixdown and then archive the metadata and media.

Transcoding Through Interplay Transcode

You can also use the Interplay Transcode service to create H.264 proxy media. The following illustration shows the main stages of a multi-resolution workflow that uses Interplay Transcode to create H.264 proxy media from high-res media ingested through an AirSpeed server.



1. **Ingest:** Interplay Capture manages ingest of high-res media from the AirSpeed server and checks the metadata into the Interplay database on the Interplay Engine. (You can also transcode media captured through an Avid editing system.)
2. **Transcode:** From Interplay Access or an Avid editing system, a user uses Interplay Transcode to create H.264 proxy media. Interplay Transcode stores media on the Avid shared-storage system and associate the high-res clips with the H.264 proxy media in the Interplay database.
3. **Editing:** A user on an Avid editing system edits with H.264 proxy media (working resolution).
4. **Output:** When the sequence is complete, the user dynamically relinks to the high-res media (target resolution) and sends the sequence to playback.

For information about input and output workflows for XDCAM HD media, see [“Long GOP End-to-End Workflows” on page 106](#).

5. **Archive:** From an Avid editing system, a user consolidates the sequence and uses the Interplay Archive Provider to send metadata to the Interplay Archive Engine and send media to a third-party storage solution. (See [“Dual-Ingest Using AirSpeed MultiStream” on page 151](#) for an illustration.) A user can also use the Interplay Transcode provider (from an Avid editing system or from Interplay Access) to create a mixdown and then archive the metadata and media.

Dynamic Relink and MPEG1 Layer 2 Audio

MPEG1 Layer 2 audio is a compressed audio format that is used as proxy media in a multi-resolution environment. MPEG1 Layer 2 uses a sample rate of 48 kHz with 16-bit depth. Interplay Production supports eight channels of MPEG1 Layer 2 audio at 96k per channel.

To explicitly link to proxy audio, select “Compressed” as the audio relink method in the Dynamic Relink settings. If you select “Compressed,” the sample rate and bit depth are inactive. This setting is valid only for Working settings. Compressed audio is not supported in Target settings, to avoid sending compressed audio to a playback device.

H.264 Card for AirSpeed Multi Stream

An AirSpeed Multi Stream system uses an encoder card to ingest H.264 800 Kbps media. This card is shipped with new systems or can be installed in the field. For information about how to install the H.264 card, see the *Avid AirSpeed Multi Stream H.264 Option Kit Installation Guide*.

Folders and Shared-Storage Workspaces for an H.264 Workflow

As part of the configuration for acquiring media through Air Speed Multi Stream and an ingest automation server — for example, Interplay Capture — you need to specify the folders in which you want to store the clip metadata, the shared-storage workspace in which you want to store the media, and other information. The following topics describe the necessary settings:

- [Configuring AirSpeed Multi Stream for an H.264 800Kbps Workflow](#)
- [Specifying the Folder Location in Interplay Capture for H.264 Multi-Res Clips](#)

Configuring AirSpeed Multi Stream for an H.264 800Kbps Workflow

To configure AirSpeed Multi Stream to ingest H.264 800 Kbps clips and media, you need to specify settings on four tabs in the Avid Service Configuration application:

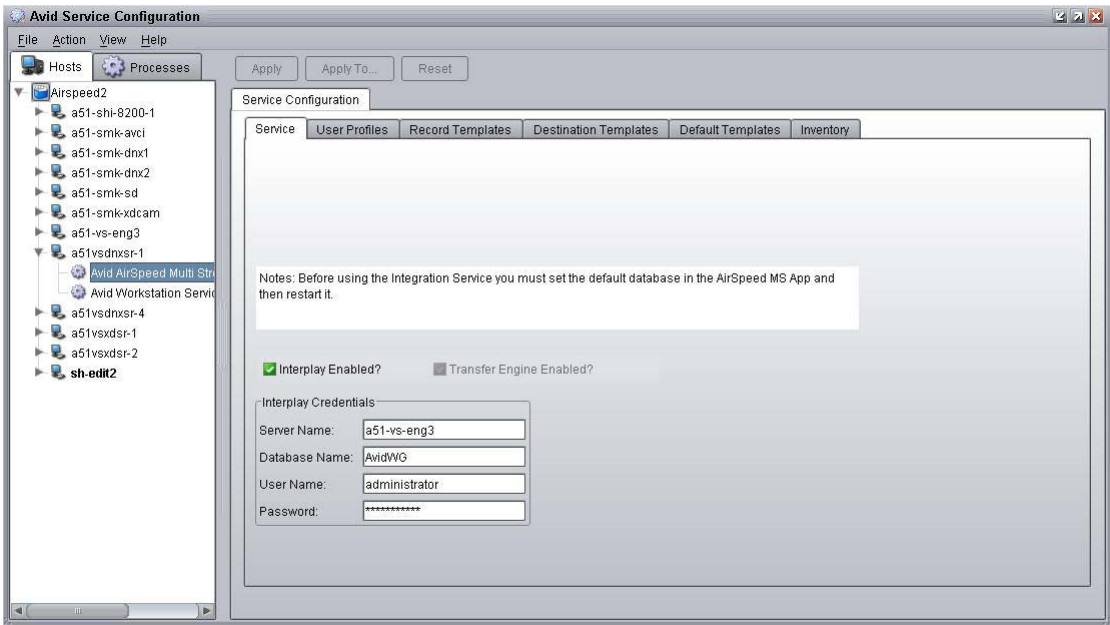
- Record Templates
- Destination Templates
- Default Templates
- Inventory

You can open the Avid Service Configuration on any workstation in the workgroup that includes your AirSpeed Multi Stream server, if the Avid Service Framework is installed on the workstation.

To open the Avid Service Configuration dialog box:

1. Select Start > Programs > Avid > Avid Framework > Avid Service Configuration.
2. Click your AirSpeed Multi Stream server name and select AirSpeed Multi Stream Integration Service.

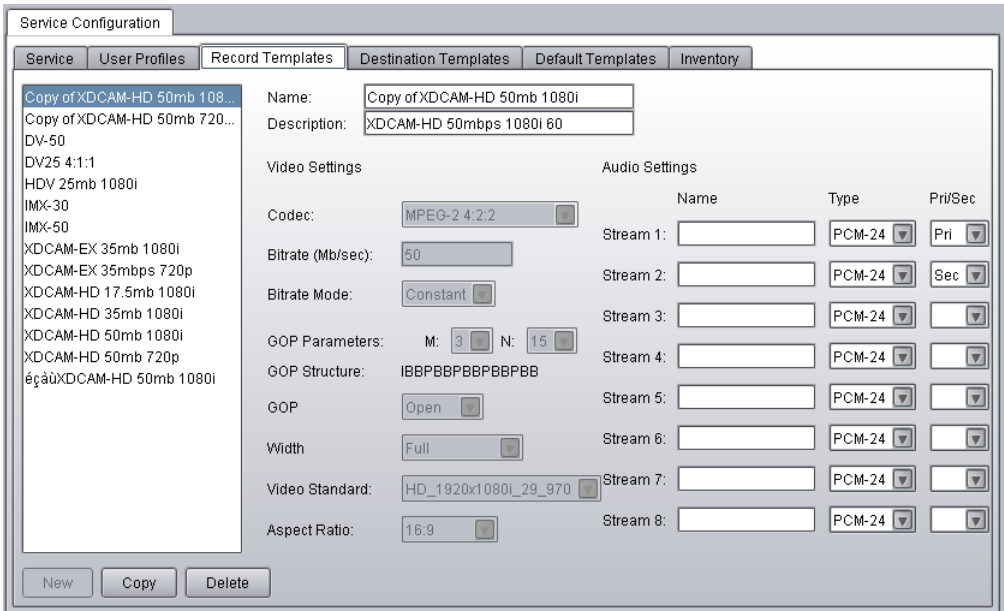
The Avid Service Configuration dialog box opens.



 For complete information on configuring AirSpeed Multi Stream in a workgroup environment, see the Avid AirSpeed Multi Stream Installation and User's Guide.

For an H.264 workflow, use the settings described in the following sections.

Record Templates Tab



Use the Record Templates tab to specify the audio settings for each template that you use to record a high-res resolution and H.264 proxy resolution. In the Type column, select an audio format for each high-resolution stream you are recording. In the Pri/Sec (Primary/Secondary) column, select one stream as Pri to record H.264 proxy audio. You must have at least one stream selected as Pri. You can select a second stream as Sec (secondary).

Destination Templates Tab

Service Configuration

Service User Profiles Record Templates **Destination Templates** Default Templates Inventory

Airo-1_New
Airo-1_XDCam
Airo2_DNx
Airo3_AVCI

Name: Airo-1_New
Description:
Clip ID: Airo-1
Clip Name: Airo-1_H264_1080i.\$n
Tape Name: Tape
Interplay Folder Path: 1080i50 Project
Proxy Interplay Folder Path:
Workspace: WA51-WG1-SD1VAiro1
Proxy Workspace: WA51-WG1-SD1VPV
Categories:

New Copy Delete

Put '\$n' in the 'Clip Name' field to generate a unique name in Interplay, for example, "HD Clip.\$n".

Use the Destination Templates tab to specify the folder in which AirSpeed Multi Stream should store ingested clips and the workspace in which it should store the ingested media.

- **Interplay Folder Path:** Type the name of a subfolder in the Interplay database Projects folder in which to store the ingested clips. This is optional and is not necessary if you manage ingest through Interplay Capture (Interplay Capture overwrites this setting).

If you specify a folder in the Interplay Folder Path field, the clip that is created in that folder is associated with both the low-res and high-res media and can be used for dynamic relink. In most cases, do not specify a folder in the Proxy Interplay Folder Path field.

- **Workspace:** Specify the workspace or workspaces in which AirSpeed Multi Stream should store the ingested media. You must specify a workspace in the Workspace field. You have the option of specifying a Proxy Workspace if you want to store the high-res media and the H.264 proxy media in different Avid ISIS workspaces.

Default Templates Tab

Service Configuration

Service User Profiles Record Templates Destination Templates Default Templates Inventory

	Record	Proxy	Destination
Channel 1:	XDCAM-HD 50mb 1080i	H.264 Proxy 1080i 29.97 800	Airo-1_New
Channel 2:	(none)	(none)	(none)
Channel 5:	(none)	(none)	(none)
Channel 6:	(none)	(none)	(none)
NTSC:	(none)	H.264 Proxy 525	
720p 59.94:	(none)	H.264 Proxy 720p 59.94 800	
1080i 29.97:	(none)	H.264 Proxy 1080i 29.97 800	
XDCAM send-to-playback format based on:	XDCAM-HD 50mb 1080i		
Ingest transfer workspace/project based on:			
Master clip location (Standalone):			

Note: If you change the 'HD send-to-playback format', or the 'Ingest transfer workspace', or the contents of the templates, you MUST restart the 'Interplay Transfer Engine'.

Use the Default Templates tab to select the Record template, proxy resolution, and Destination template for each channel you are going to record. In the Record column, select the Record template for the high-res resolution. In the Proxy column, select an H.264 proxy resolution. (The correct corresponding proxy is displayed in the list automatically.) In the Destination column, select the Destination template for the clips and media.

Inventory Tab

Service Configuration

Service User Profiles Record Templates Destination Templates Default Templates Inventory

☒ Allow Play While Transfer?

☒ Allow Overwrite on Send to Playback?

☒ Delete XDCAM Temporary Files after Transfer?

☒ Transfer Ancillary Streams to Interplay?

☒ Transfer Compressed Audio Streams to Interplay?

☐ Generate local MPEG-1 proxy after playback transfer completes?

☐ Delete after ingest transfer completes?

☐ Begin delete

When used space reaches: 75 %

Until used space reaches: 70 %

☐ Delete all inventory daily at (24-hour HH:MM) : 00:00

☐ Restart transfer engine daily at: 00:00

Use the Inventory tab to specify if you want to capture the audio for the H.264 proxy video. Select the option “Transfer Compressed Audio Streams to Interplay?”



The setting “Generate local MPEG-1 proxy after playback transfer completes?” applies only to local playback on the AirSpeed Multi Stream. It is not related to H.264 proxy media.

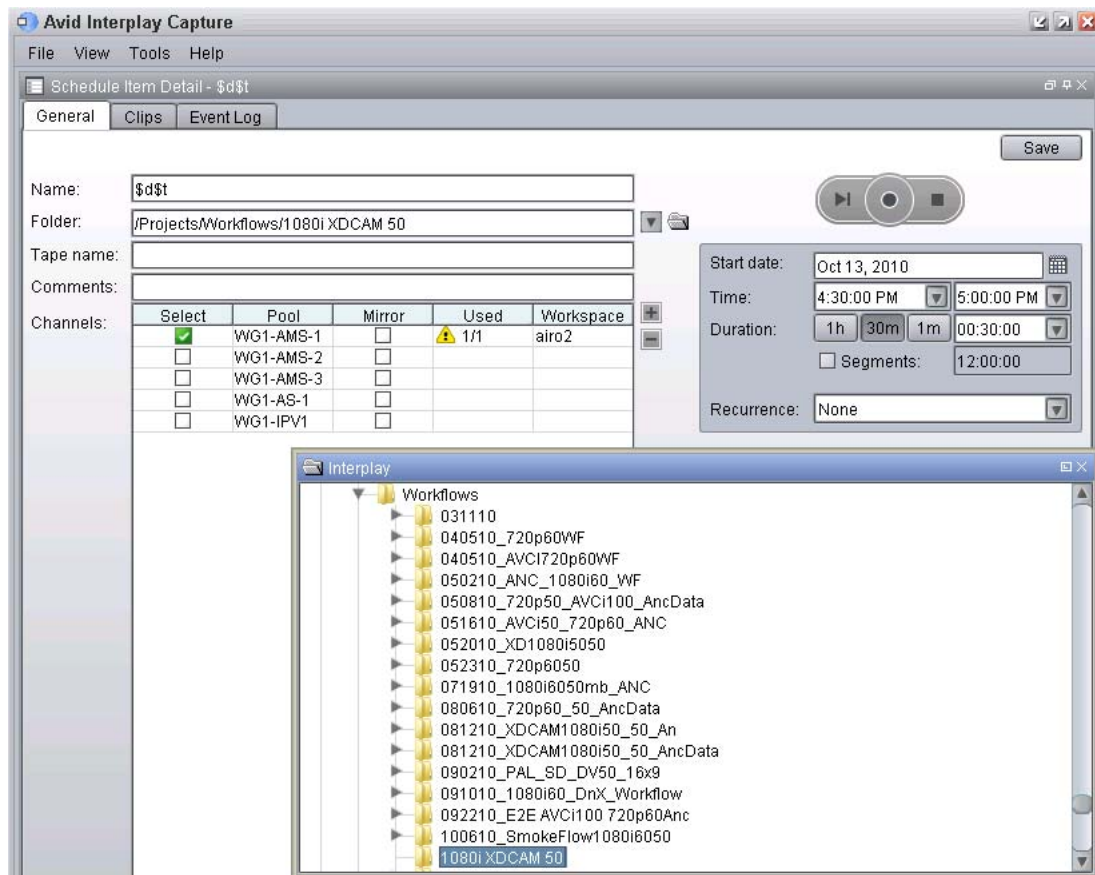
Specifying the Folder Location in Interplay Capture for H.264 Multi-Res Clips

If you are using Interplay Capture, you set the folder for storing ingested clips in the General tab of the Schedule Item Detail panel. Select a folder location from the Interplay Production directory tree. You can access a list of recently used folders by clicking the down-arrow icon next to the Folder field.



For complete information on configuring Interplay Capture, see the *Interplay Capture User’s Guide*.

The following figure shows Interplay Capture set to store clips in the Interplay folder /Projects/Workflows/1080iXDCAM50.



After a capture is completed, the editor should work with the clip that is stored in the Projects folder. See “[Acquiring Media for an H.264 Workflow](#)” on page 162.

Interplay | Production Services Profiles for an H.264 Workflow

Depending on your workflow, you may need to create profiles for the following Interplay Production Services. You set these profiles in the Profiles tab of the Interplay Production Services and Transfer Status tool. For information about supported H.264 resolutions, see [“H.264 Resolutions Supported by Interplay | Production” on page 149](#). For more information about creating profiles, see the *Interplay Production Services Setup and User’s Guide*.

- **Interplay Transcode Service:** If you are transcoding high-res media to H.264 media, select your desired H.264 resolution from the TargetVideoQuality list. You can select any mode (WHOLE, CONSOLIDATE, MIXDOWN, DUALMIXDOWN, CROSSRATE).

Parameters for A_H264_525

Mode	WHOLE
Priority	50
TargetAudioQuality	MPEG1Layer2
TargetAudioQualityDual	None
TargetAudioTrackMapping	
TargetProjectType	
TargetVideoQuality	H.264 800Kbps Proxy 525
TargetVideoQualityDual	None
VideoTimeCodeFormat	FUTURE_USE
Workspace	broadcast
WorkspaceDual	

- **Interplay Archive Service:** If you plan to archive H.264 media, select your desired resolution from the TargetVideoQuality list. For H.264 media, you can select All, Lowest, or a specific resolution.

Parameters for 30i_Proxy_Specific_DET

Destination_Path: AvidAM/Catalogs/DET

Partition: a51-wg6-ww

Priority: 25

TargetVideoQuality: H.264 800Kbps Proxy 525

Save

- Interplay Restore Service. If you plan to restore H.264 media from archive, select your desired resolution from the TargetVideoQuality list. For H.264 media, you can select All, Lowest, or a specific resolution.



If you use a profile in which Partial is selected, H.264 media is fully restored.

Parameters for DET_H264

Destination_Path: AvidWG/Catalogs/DET/Partial_AVC3.0

Destination_Server: a51-wg6-eng1

Destination_Workspace: Broadcast

Partial: ☐

Priority: 25

TargetVideoQuality: H.264 800Kbps Proxy 525

Save

Configuring Interplay | Production Clients for an H.264 MultiRez Workflow

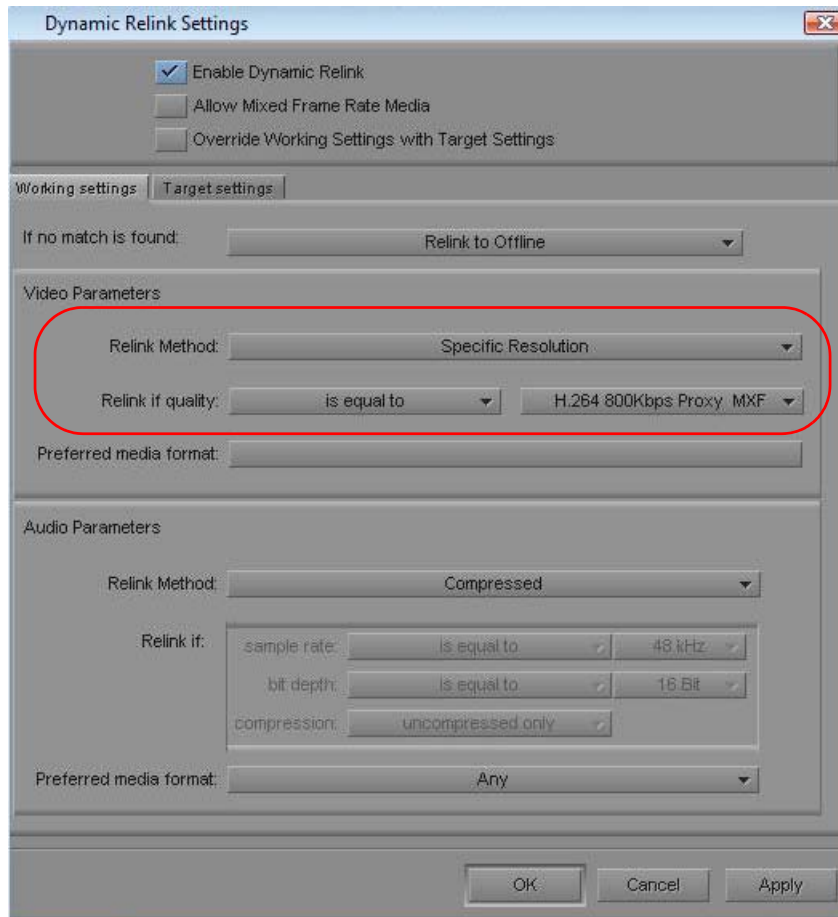
To prepare for editing with H.264 proxy media, specify your desired working and target settings in the Dynamic Relink Settings dialog box in your Avid editing application. The following are typical settings for a multirez workflow with H.264 proxy media:

- Working Settings: Relink to the specific resolution H.264 800Kbps
- Target Settings: Relink to the specific resolution XDCAM HD 50 Mbit

Other settings are possible, depending on your specific workflow.

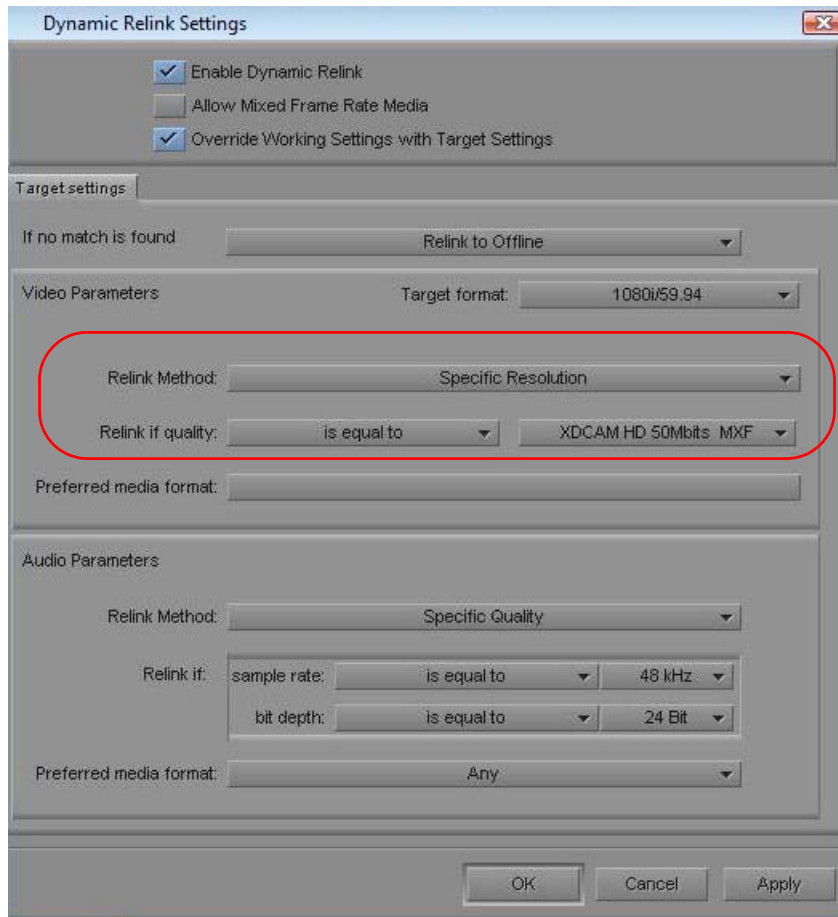
The following illustration shows the Working Settings tab for relinking to an H.264 resolution.

Working Settings: H.264



The following illustration shows the Target Settings tab for relinking to an XDCAM HD resolution.

Target Settings: 1080i/59.94 XDCAM HD 50



For more information on MultiRez, see “Using MultiRez and Dynamic Relink” in the Interplay Production Help or the Help for your Avid editing application.

Acquiring Media for an H.264 Workflow

The following topics describe two methods for acquiring H.264 media:

- [Dual-Ingest for High-Res Media and H.264 Proxy Media](#)
- [Transcoding to Create H.264 Media](#)

Dual-Ingest for High-Res Media and H.264 Proxy Media

An AirSpeed Multi Stream server with an H.264 encoder card can simultaneously ingest both high-res media and low-res H.264 800 Kbps media. You can use Interplay Capture to manage the ingesting process. For specific H.264 settings, see “[Folders and Shared-Storage Workspaces for an H.264 Workflow](#)” on page 154.

Interplay Capture works with AirSpeed MultiStream to create a clip in a Projects folder that you specify. This clip is associated with both the high-res media and the low-res media, and you should use this clip for editing.

However, Interplay Capture creates additional clips and links, as described in the following section.

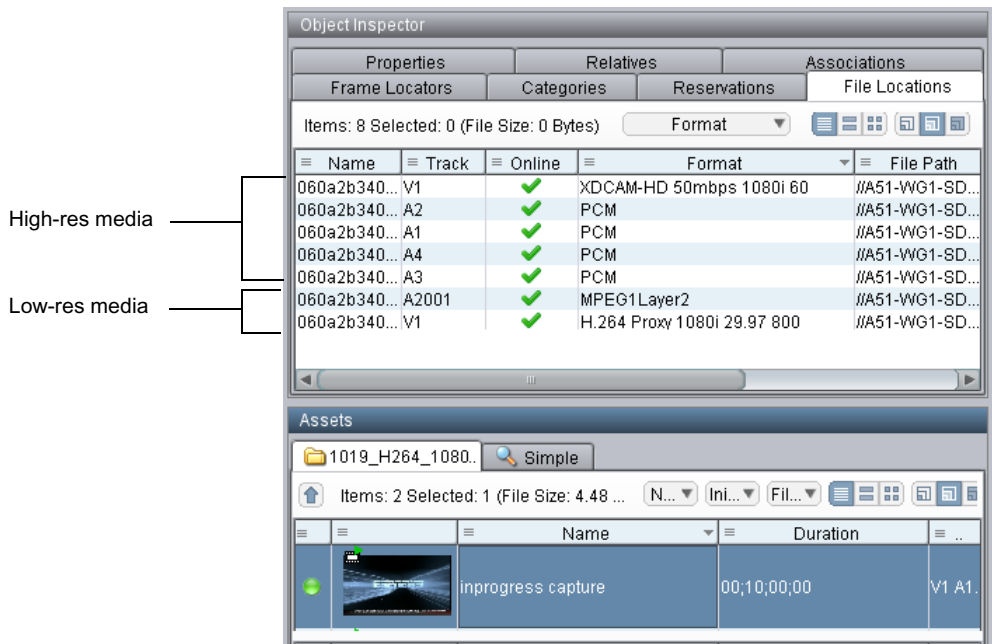
Interplay Capture and AirSpeed MultiStream

Interplay Capture and AirSpeed MultiStream create three different but related clips in the following locations:

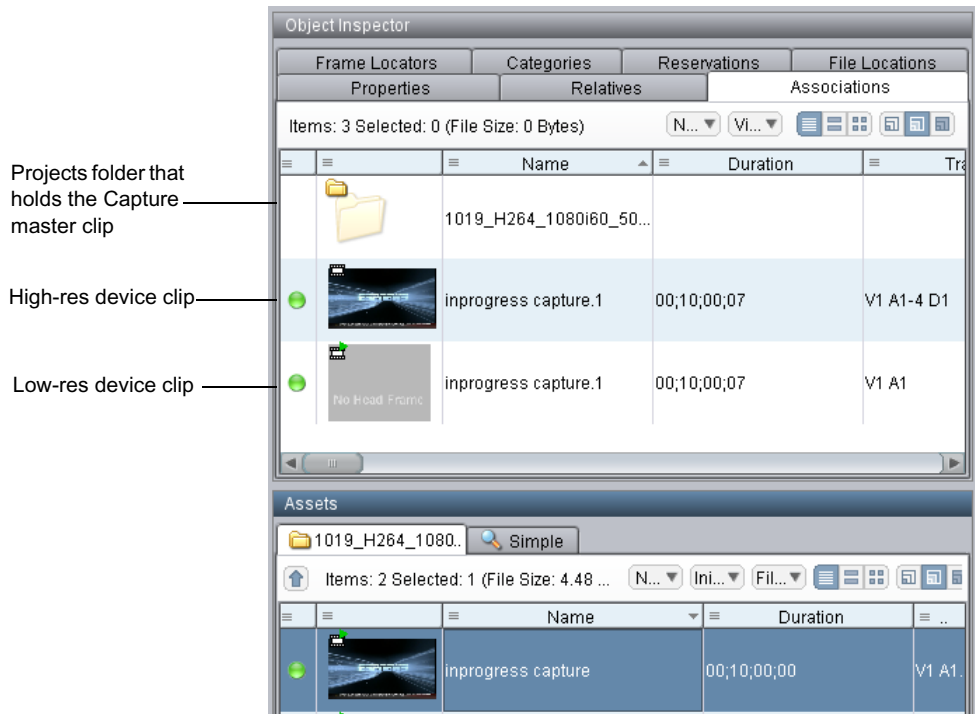
- Capture master clip: This clip is checked into the folder Projects*folder_name*, where *folder_name* refers to the path you set in Interplay Capture (in the General tab of the Schedule Item Detail panel). This folder contains a clip that is associated with both high-res and low-res media. You should use this clip for editing.
- Device clips: When working with an AirSpeed Multistream server, Capture creates two additional clips: one associated with high-res media and one associated with low-res media. These clips are checked into the following folders:
 - Incoming Media\workspace_name, where workspace_name refers to the shared-storage workspace you set in the AirSpeed Multi Stream Destination template. You must specify a workspace in this template to perform an ingest operation, and this workspace name is used to create this folder.
 - Catalogs\server_name, where server_name is the network name of the AirSpeed Multi Stream server. No setting is required.
 - Projects\name, where name is the folder name you set in the General tab of the Capture Settings plugin in the Interplay Administrator. Make sure to distinguish this folder from the folder you set for the Capture master clip.

Avid recommends that these three folders be hidden from all users except administrators, because the multi-res associations are managed only on the Capture master clip. For more information, see *Interplay Capture Best Practices*.

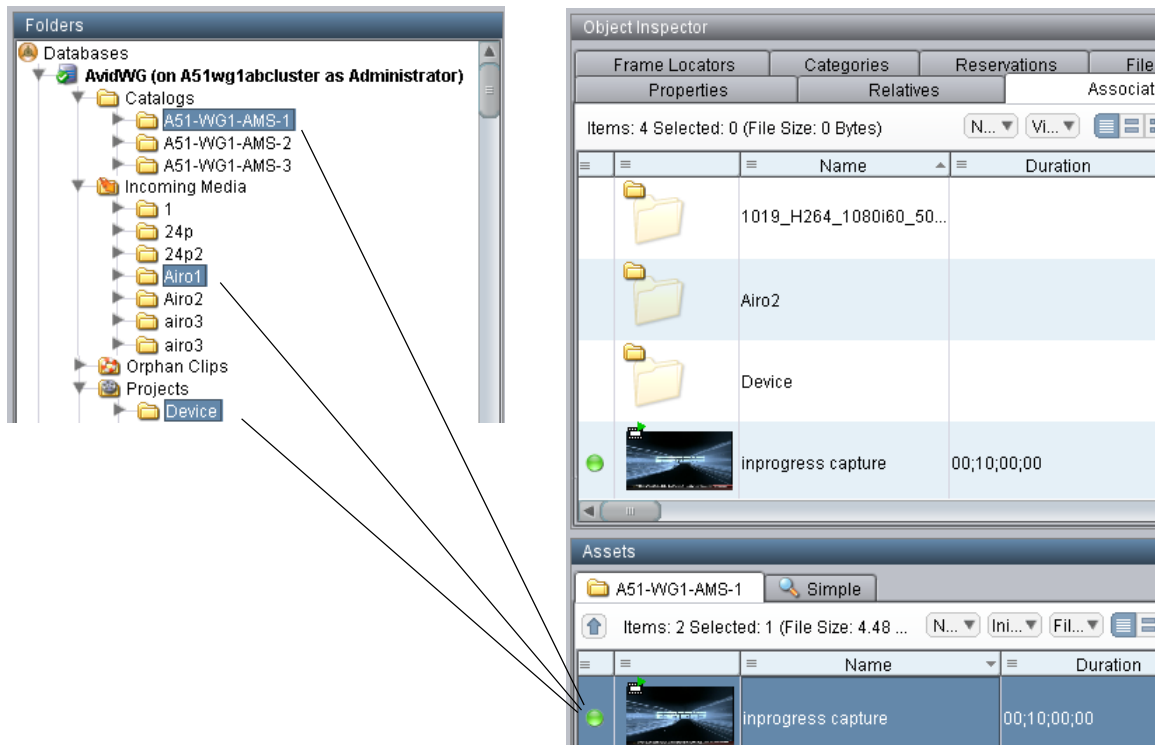
The following illustration shows the high-res media and the low-res media associated with the Interplay Capture master clip, as displayed in Interplay Access.



The following illustration shows the associations for the Interplay Capture master clip. Because it is a multi-res clip, it is associated with both the low-res and the high-res device clips. Interplay Capture appends a .1 to the low-res and the high-res device clips.



The following illustration shows the associations for the high-res device clip: three folders and the Interplay Capture master clip. Both high-res and low-res device clips are located in these three folders.



Transcoding to Create H.264 Media

If you do not use a dual-ingest configuration, you can use the Interplay Transcode service to create H.264 media for proxy editing. Transcoding to create H.264 media requires a Transcode profile (see [“Interplay | Production Services Profiles for an H.264 Workflow” on page 159](#) and [“H.264 Resolutions Supported by Interplay | Production” on page 149](#)). You can use the Auto-Transcode service to set up folders for automatic transcoding to H.264.

For information about using the Interplay Transcode and the Interplay Auto-Transcode services, see the *Interplay Production Services Setup and User’s Guide* or the Interplay Production Help.

Editing with H.264 and High-Res

After the H.264 media is acquired, you can use the MultiRez and Dynamic Relink features to edit with H.264 low-res proxy media and output high-res media.

When editing with H.264 media in an Avid editing system, for the best performance during playback, right-click the Video Quality menu button below the Timeline and select Best Performance (yellow/yellow).



Video Quality Menu Button

For more information, see the documentation for your Avid editing system.

Archiving, Restoring, and Deleting H.264 and High-Res Media

Procedures for archiving, restoring, and deleting H.264 and high-res media are similar to procedures you use for other multirez workflows. However, If you perform a partial restore operation on a sequence that includes H.264 media, H.264 media is fully restored. Other resolutions are partially restored.

Two possible workflows are:

- Archiving high-res media after it is ingested:
 - Use the Interplay Archive service to archive the high-res media to archive storage, while keeping the low-res media online on shared storage. Alternatively, you can archive the low-res media also.
 - Delete the high-res online media.
 - Create a sequence using low-res H.264 media.
 - Use the Interplay Production Restore service to perform a partial restore to bring the necessary portions of the high-res media online.
 - Use Dynamic Relink to create the final output.
- Archiving the final sequence and its media.
 - Create a sequence using H.264 media.
 - Use Dynamic Relink to create the final output.
 - Consolidate the sequence to create only the media used in the sequence.
 - Use Interplay Archive to archive the sequence and the consolidated media.

For information about delete workflows, see [“Developing an Interplay Delete Strategy” on page 38](#).

10 HD 23.976p and HD 24p Support

The following topics describe Interplay Production support for HD 23.976p and HD 24p formats and their associated proxy resolutions:

- [Interplay Production Support for Native HD 23.976p and HD 24p Formats](#)
- [Interplay Production Support for HD 23.976p and HD 24p Proxy Resolutions](#)

Interplay Production Support for Native HD 23.976p and HD 24p Formats

Interplay version 2.3 and later supports management of assets acquired in the native frame rate for the following HD formats:

Format	Type of Acquisition (in Avid editing systems)
1080p/23.976	Camera, deck, file-based clips through AMA, import
1080p/24	Camera, deck, file-based clips through AMA, import
720p/23.976	File-based clips through AMA, import

In an Avid editing project, Interplay Production support requires the following:

- The media must be acquired directly in one of the supported formats, such as through a camera that is recording HD 1080p/23.976 without pulldown. In this case, HD 1080p/23.976 is referred to as the “native frame rate.”
- The native frame rate of the media must match the project frame rate.

This is especially important for dynamic relink. For example, if you are working in a 1080p/23.976 project, your media must be captured directly as 1080p/23.976. You can then use dynamic relink between DNxHD 1080 36 and 1:1 MXF or with any other resolutions that are supported for the 1080p/23.976 format. See the table below for the supported resolutions for each project type.



Interplay Production does not support dynamic relink between sources that use different edit rates. For example, you cannot dynamically relink a 1080p/23.976 master clip to 720p/23.976 media. Interplay Production also does not support dynamic relink between NTSC sources from which pulldown has been removed (23.976p or 24p projects) and their HD 23.976p and 24p equivalents.



You cannot use Dynamic Relink with clips that are captured in a 720p 23.976 project from a deck set to 720p 59.94 or 720p 60. This configuration creates media from which pulldown has been removed, which is not supported for Dynamic Relink in Interplay Production.

Interplay Production support for native HD 23.976p and 24p formats includes the following:

- Checkin to and checkout from the Interplay database
- Dynamic relink (only as described above)

- Logging clips in Interplay Assist
- Archive and restore through Production Services
- Transcode through Interplay Production Services (with the exceptions noted in the table below)
- Transfer to another workgroup (Send to Workgroup)
- Output directly to a 24p or 23.976p device (camera or file-based media player). Other file-based output, such as Send to Playback, requires modifying the sequence in the Avid editing application to the desired format. For example, in a 1080p/23.976p project, modify a 1080p/23.976p sequence to 1080i/59.94, then open it in a 1080i/59.94 project.
- Output to tape can be supported through Avid hardware, through the tape device, or by transcoding the sequence in the Avid editing application.

The following table lists supported resolutions:

Project Type and Raster Size	Resolution	Notes
1080p/23.976 1920x1080	1:1 MXF	Not a supported target for Interplay Transcode
	1:1p 10b MXF	Not a supported target for Interplay Transcode
	AVC-Intra 50 1080p 23.976	
	AVC-Intra 100 1080p 23.976	
	DNxHD 1080 36	
	DNxHD 1080 115-120-145	
	DNxHD 1080 175-185-220	
	DNxHD 1080 175X-185X-220X	
	DNxHD 444 1080 350X-365X-440X	RGB color space
	DNxHD 1080p 100	
	XDCAM-EX 35mbps 1080p 23.976	
	XDCAM-HD 50mbps 1080p 23.976	
	Apple ProRes Proxy 1080p 23.976	
	Apple ProRes LT 1080p 23.976	
	Apple ProRes 1080p 23.976	
	Apple ProRes HQ 1080p 23.976	
	Apple ProRes 4444 1080p 23.976	
	JPEG 2000 1080p 23.976	
1080p/24 1920x1080	1:1 MXF	Not a supported target for Interplay Transcode
	1:1p 10b MXF	Not a supported target for Interplay Transcode
	DNxHD 1080 36	

Project Type and Raster Size	Resolution	Notes
	DNxHD 1080 115-120-145	
	DNxHD 1080 175-185-220	
	DNxHD 1080 175X-185X-220X	
	DNxHD 444 1080 350X-365X-440X	RGB color space
	DNxHD 1080p 100	
	Apple ProRes Proxy 1080p 24	
	Apple ProRes LT 1080p 24	
	Apple ProRes 1080p 24	
	Apple ProRes HQ 1080p 24	
	Apple ProRes 4444 1080p 24	
	JPEG 2000 1080p 24	
720p/23.976 1280x720	1:1 MXF	Not a supported target for Interplay Transcode
	1:1p 10b MXF	Not a supported target for Interplay Transcode
	AVC-Intra 50 720p 23.976	
	AVC-Intra 100 720p 29.97	
	DNxHD 720 60-75-145	
	DNxHD 720 90-110-220	
	DNxHD 720 90X-110X-220X	
	DNxHD 720p 100	
	XDCAM-EX 35mbps 720p 23.976	
	DVCPRO HD	
	Apple ProRes Proxy 720p 23.976	
	Apple ProRes LT 720p 23.976	
	Apple ProRes 720p 23.976	
	Apple ProRes HQ 720p 23.976	
	JPEG 2000 720p 23.976	

Support for HD RGB Media

Interplay Production supports two HD RGB resolutions for native HD 23.976p and HD 24p:

- 1:1 10-bit HD RGB. This resolution is supported for check in, check out, and deletion. It is not supported for Production Services operations.
- DNxHD 444 RGB. This resolution is supported for all Interplay Production operations.

Film Option When Creating a New Project

If you want to use Dynamic Relink in any of the following project types:

- 1080p 23.976
- 1080p 24
- 720p 23.976

do not select the Film option in the Create New Project dialog box. Selecting the Film option disables Dynamic Relink.

Interplay Production Support for HD 23.976p and HD 24p Proxy Resolutions

Interplay version 2.5 and later includes support for HD 23.976p and HD 24p low-res proxy media. You can work with this proxy media in an Avid editing application, check it in and out of the Interplay database, and process it through Interplay Production Services and Interplay Transfer.



This proxy media is supported only in workgroups that use Interplay Engine v2.5 or later.

The following table describes the HD formats supported by Interplay Production and the corresponding proxy formats, as they are labeled in the Production Services Profiles tab in the Production Services and Transfer Status tool. Use the Profiles tab to create a Transcode profile that specifies the required proxy format.

HD Format	Type of Acquisition (in Avid editing systems)	Proxy Format
1080p/23.976	Camera, deck, file-based clips through AMA, import	H.264 800Kbps Proxy 1080p 23.976
1080p/24	Camera, deck, file-based clips through AMA, import	H.264 800Kbps Proxy 1080p 24
720p/23.976	File-based clips through AMA, import	H.264 800Kbps Proxy 720p 23.976



Avid editing applications cannot create HD 23.976p or HD 24p H.264 proxy media. This media can be created only by using the Interplay Production Services to transcode or mix down a clip or sequence to one of the Avid H.264 proxy resolutions.

Because these proxy resolutions use the H.264 format, you can play clips and sequences through Interplay Access over a WAN.

In an Avid editing system, you can use the low-res proxy media to edit a sequence, and then use dynamic relink to link to the high-res media for conforming and finishing.

For important limitations, see the latest version of the *Interplay Production ReadMe*, available on the Customer Support Knowledge Base at www.avid.com/readme.

11 Apple ProRes Resolutions Supported in Interplay | Production

Interplay v2.6 supports workflows for clips and sequences that use Apple ProRes resolutions, with some limitations. You can work with Apple ProRes clips in an Avid editing application, check them in and out of the Interplay database, process them through Interplay Production Services, and use Interplay Transfer for workgroup-to-workgroup transfers.

Note the following limitations:

- Apple ProRes 4444 is not supported for 720p projects.
- SD progressive resolutions are not supported in Interplay Production.
- Partial archive and partial restore are not supported for Apple ProRes resolutions.
- Sequences composed of Apple ProRes clips are not supported for Send to Playback.

The following table lists the supported resolutions. These resolutions are supported as sources and targets for Interplay Transcode.

Project	Edit Rate	Resolution Name in Interplay Access and Interplay Transcode	Mbit/sec	Raster Size
720p	23.976	Apple ProRes Proxy 720p	18	1280 x 720
		Apple ProRes LT 720p	41	1280 x 720
		Apple ProRes 720p	59	1280 x 720
		Apple ProRes HQ 720p	88	1280 x 720
720p	50	Apple ProRes Proxy 720p	38	1280 x 720
		Apple ProRes LT 720p	84	1280 x 720
		Apple ProRes 720p	122	1280 x 720
		Apple ProRes HQ 720p	184	1280 x 720
720p	59.94	Apple ProRes Proxy 720p	45	1280 x 720
		Apple ProRes LT 720p	101	1280 x 720
		Apple ProRes 720p	147	1280 x 720
		Apple ProRes HQ 720p	220	1280 x 720
1080p	23.976	Apple ProRes Proxy 1080p	36	1920 x 1080

Project	Edit Rate	Resolution Name in Interplay Access and Interplay Transcode	Mbit/sec	Raster Size
1080p	24	Apple ProRes LT 1080p	82	1920 x 1080
		Apple ProRes 1080p	117	1920 x 1080
		Apple ProRes HQ 1080p	176	1920 x 1080
		Apple ProRes 4444 1080p	264	1920 x 1080
		Apple ProRes Proxy 1080p	36	1920 x 1080
		Apple ProRes LT 1080p	82	1920 x 1080
		Apple ProRes 1080p	117	1920 x 1080
		Apple ProRes HQ 1080p	176	1920 x 1080
		Apple ProRes 4444 1080p	264	1920 x 1080
1080i	50	Apple ProRes Proxy 1080i	38	1920 x 1080
		Apple ProRes LT 1080i	85	1920 x 1080
		Apple ProRes 1080i	122	1920 x 1080
		Apple ProRes HQ 1080i	184	1920 x 1080
		Apple ProRes 4444 1080i	275	1920 x 1080
1080i	59.94	Apple ProRes Proxy 1080i	45	1920 x 1080
		Apple ProRes LT 1080i	102	1920 x 1080
		Apple ProRes 1080i	147	1920 x 1080
		Apple ProRes HQ 1080i	220	1920 x 1080
		Apple ProRes 4444 1080i	330	1920 x 1080
25i PAL	25	Apple ProRes Proxy 25i	12	720 x 576
		Apple ProRes LT 25i	28	720 x 576
		Apple ProRes 25i	41	720 x 576
		Apple ProRes HQ 25i	61	720 x 576
		Apple ProRes 4444 25i	92	720 x 576

Project	Edit Rate	Resolution Name in Interplay Access and Interplay Transcode	Mbit/sec	Raster Size
30i NTSC	29.97	Apple ProRes Proxy 30i	12	720 x 486
		Apple ProRes LT 30i	29	720 x 486
		Apple ProRes 30i	42	720 x 486
		Apple ProRes HQ 30i	63	720 x 486
		Apple ProRes 4444 30i	94?	720 x 486

12 JPEG 2000 Resolutions Supported in Interplay | Production

Interplay v2.6 supports workflows for clips and sequences that use JPEG 2000 (J2K) resolutions, with some limitations. You can work with JPEG 2000 material in an Avid editing application, check it in and out of the Interplay Production database, and process it using Interplay Production Services and use Interplay Transfer for workgroup-to-workgroup transfers.

JPEG 2000 resolutions are supported in Media Composer v6.1 and later.

Note the following limitation:

- Sequences composed of JPEG 2000 clips are not currently supported for Send to Playback.

The following table lists the supported resolutions. These resolutions are supported as sources and targets for Interplay Transcode.

Project	Edit Rate	Resolution Name in Interplay Access and Interplay Transcode	Average Video Bit Rate (Mbit/Sec)	Maximum Video Bit Rate (Mbit/Sec)	Raster Size
30i	29.97	J2K NTSCi	50 Mbit	120 Mbit	720x486
25i	25	J2K PALi	50 Mbit	120 Mbit	720x592
720p	23.976	J2K 720p	60 Mbit	150 Mbit	1280x720
	50	J2K 720p	125 Mbit	250 Mbit	1280x720
	59.94	J2K 720p	150 Mbit	250 Mbit	1280x720
1080i	50	J2K1080i	125 Mbit	250 Mbit	1920x1080
	59.94	J2K1080i	150 Mbit	250 Mbit	1920x1080
1080p	23.976	J2K1080p	120 Mbit	250 Mbit	1920x1080
	24	J2K1080p	120 Mbit	250 Mbit	1920x1080

13 Support for Stereoscopic 3D Workflows

The following topics describe support for stereoscopic 3D workflows in Interplay Production:

- [Overview of Stereoscopic Support](#)
- [Stereoscopic Media and Stereoscopic Clips](#)
- [Stereoscopic 3D Support in Interplay | Access](#)
- [Stereoscopic 3D Support in Interplay | Transcode](#)
- [Stereoscopic 3D Support in the Interplay Window, Interplay | Assist, and Avid Instinct](#)
- [Properties for Stereoscopic Clips](#)

Overview of Stereoscopic Support

Interplay v2.6 and later supports workflows for clips and sequences that use stereoscopic 3D formats, with some limitations. Stereoscopic metadata is preserved when you check stereoscopic objects in and out of the Interplay database, process them through Interplay Production Services, and use Interplay Transfer for workgroup-to-workgroup transfers. Interplay Transcode supports transcoding of stereoscopic media from one format to another.

Note the following limitations:

- Stereoscopic clips are supported in Transcode WHOLE and CONSOLIDATE mode but are not supported in DUALMIXDOWN or CROSSRATE mode. MIXDOWN mode is supported with limitations. See [“Stereoscopic 3D Support in Interplay | Transcode” on page 180](#).
- You cannot transcode Stereoscopic clipsto H.263 or H.264 target resolutions.
- Sequences composed of stereoscopic clips are not supported for Send to Playback.
- Stereoscopic clips are not supported for partial delivery or partial restore.
- Stereoscopic clips created from a simultaneous left/right capture do not load in an editing system if dynamic relink is enabled. An error message is displayed. As a workaround, disable dynamic relink. Note: Clips created manually or from QuickTime movies load and play correctly.

The following project types and resolutions are supported:

- 1080p 23.976
 - DNxHD 36, 115, 175, 175x, 444
 - AVC-Intra 50 and 100
 - 8-bit HD Uncompressed (1:1 MXF in Avid editing systems)
- 1080i 60
 - DNxHD 145, 220, 220x
 - AVC-Intra 50 and 100

- 8-bit HD Uncompressed (1:1 MXF in Avid editing systems)
- 1080i 50
 - DNxHD 120, 185, 185x
 - AVC-Intra 50 and 100
 - 8-bit HD Uncompressed (1:1 MXF in Avid editing systems)

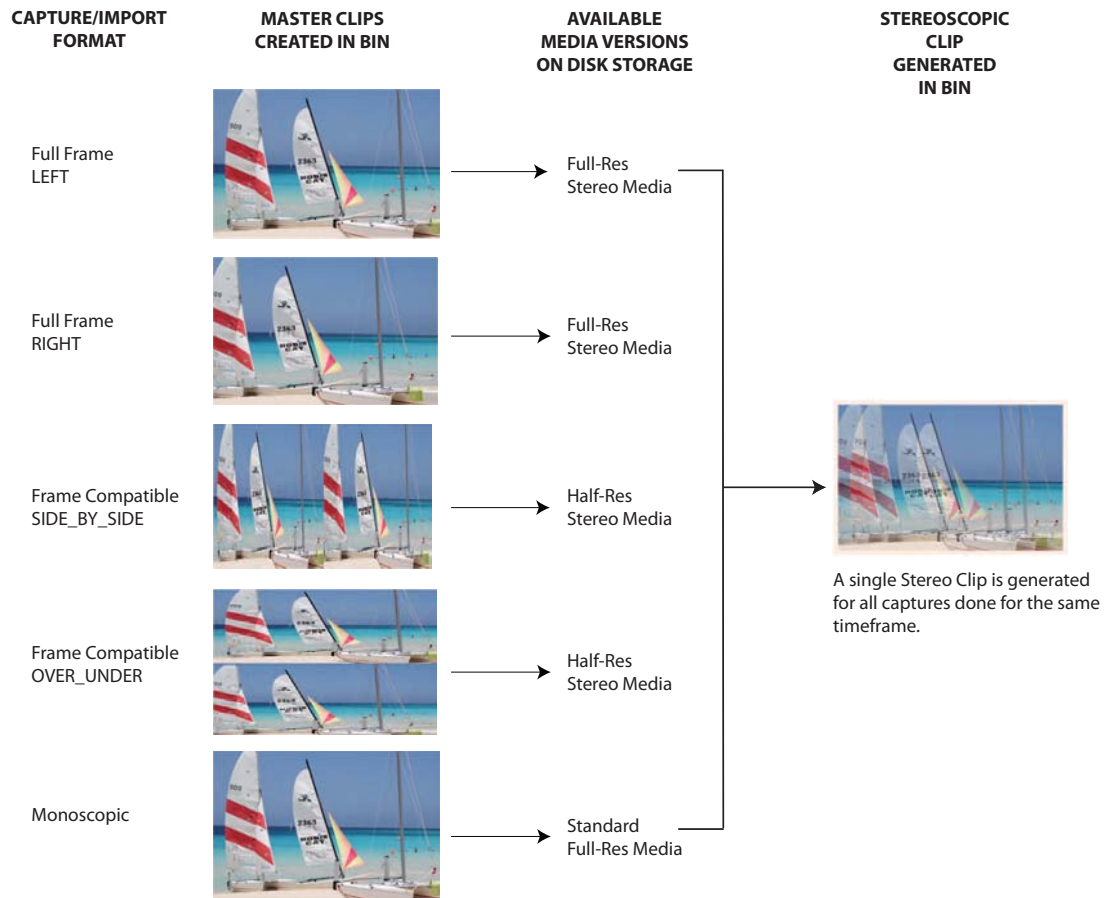
Stereoscopic Media and Stereoscopic Clips

Stereoscopic 3D editing requires two synchronized images, representing the left and right human eyes. When these left and right images are captured into an Avid editing application, they can be combined to produce a special type of clip called a *stereoscopic clip*, which allows you to more easily edit stereoscopic material.

In an Avid editing application, stereoscopic media can be edited in one of two formats: *full frame* or *frame compatible*.

- Full frame describes a format that represents each eye of the stereoscopic pair at full resolution. During the capture process, one master clip is created for the left eye and one master clip is created for the right eye. A stereoscopic clip is created to reference these master clips.
- Frame compatible describes a format where the images from both the eyes are squeezed into a single video image, either side by side or over/under. During the capture process, only one master clip is created. A stereoscopic clip is created to reference this master clip.

The following illustration shows the four different types of 3D master clips. It also shows how all 3D master clips and a monoscopic clip can be associated with the same stereoscopic clip.



The following illustration shows a stereoscopic clip and two full-frame master clips in a Media Composer bin. The stereoscopic clip is represented by a unique icon. The illustration also shows some of the bin columns that are specifically used for stereoscopic objects.

* S3D_Full_1080i50a_02						
* S3D_Full_1080i50a_02			✕			
	Name	S3D Contributors	S3D Leading Eye	S3D Channel	S3D Group Name	S3D Clip Name
☐	Full_2_1080i50avci50	Left, Right, Audio	Right		Full_2_1080i50avci50	
☐	Full_Right_2_1080i50avci50			Right Eye	Full_2_1080i50avci50	Full_2_1080i50avci50
☐	Full_Left_2_1080i50avci50			Left Eye	Full_2_1080i50avci50	Full_2_1080i50avci50



For more information about working with stereoscopic clips on an Avid editing system, see the Avid Stereoscopic 3D Editing Workflow Guide, which is available on the Customer Support Knowledge Base.

Stereoscopic 3D Support in Interplay | Access

Interplay Access v2.6 and later includes the following features that support workflows for stereoscopic media and clips:

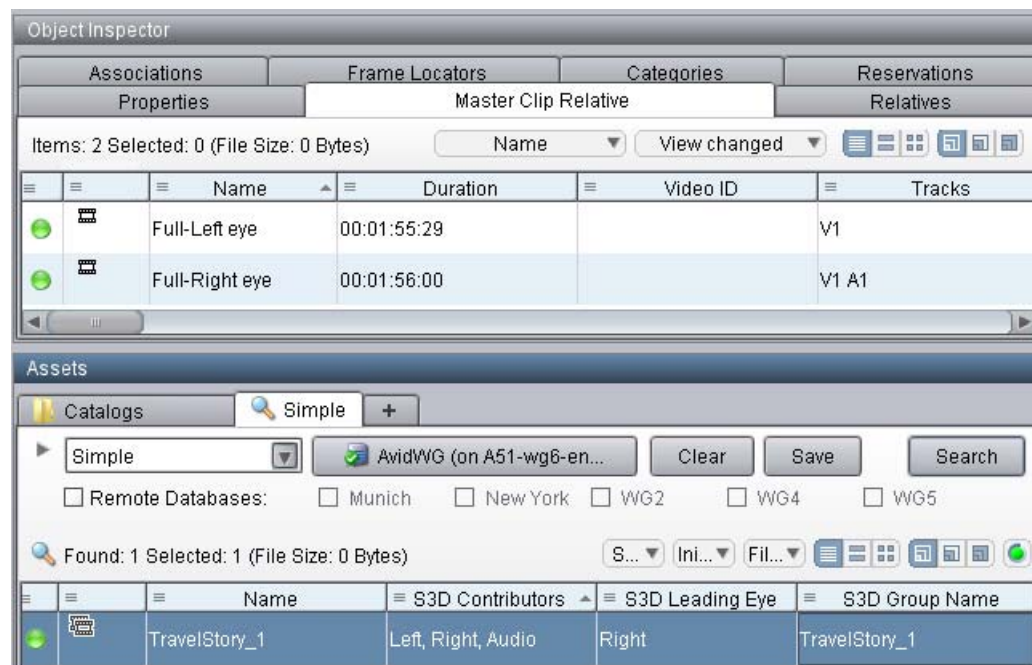
- Stereoscopic icons
- Stereoscopic properties
- Stereoscopic search criteria

Stereoscopic Icons and Properties

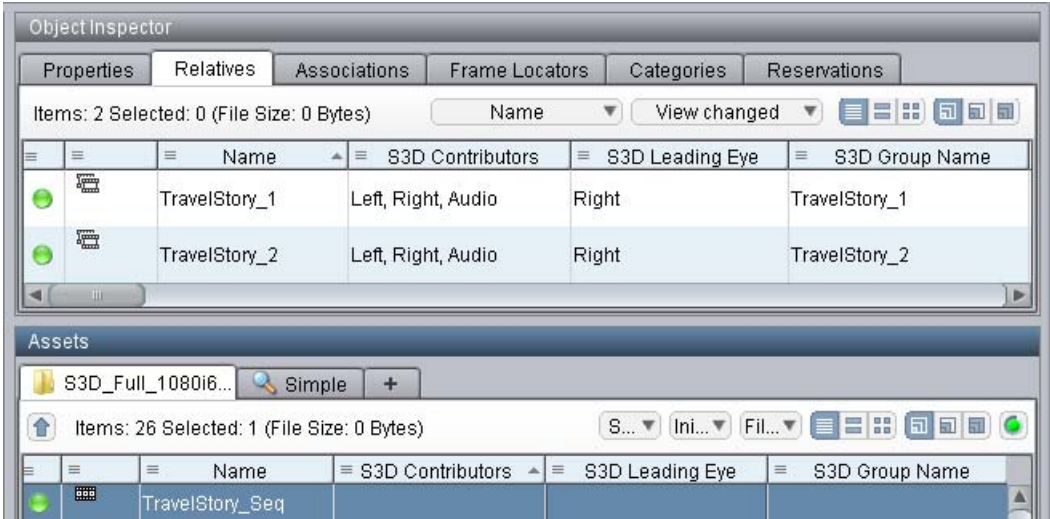
Interplay Access uses the same icons for stereoscopic clips as Avid editing systems, as described in the following table.

Icon	Description
	Stereoscopic clip
	Stereoscopic subclip

The following illustration shows a stereoscopic clip in the Assets window and the related master clips (full left and full right) in the Master Clip Relative tab of the Object Inspector. The Assets window shows some of the stereoscopic property columns.



The following illustration shows a sequence in the Assets window and its referenced stereoscopic clip in the Object Inspector.



For a list of all stereoscopic properties available for display, see [“Properties for Stereoscopic Clips” on page 181](#).

Stereoscopic Search Criteria

Interplay Access Search now includes the following criteria. To display the extended criteria, you need to enable them for display in the Property Layout view in the Interplay Administrator.

Type of Search	Criteria
Simple (Type field)	All stereoscopic clips
	Stereoscopic master clips
	Stereoscopic subclips
Extended	S3D Alignment
	S3D Channel
	S3D Clip Name
	S3D Contributors
	S3D Eye Ordering
	S3D Group Name
	S3D Inversion
	S3D InversionR
	S3D Leading Eye

Stereoscopic 3D Support in Interplay | Transcode

Interplay Transcode v2.6 includes the following stereoscopic 3D support:

- Transcoding of stereoscopic clips, stereoscopic subclips, and sequences containing stereoscopic clips and subclips, in WHOLE and CONSOLIDATE modes.
- Mixdown of sequences containing only stereoscopic clips, stereoscopic subclips, or both. Metadata properties (column data) and DRM (restrictions) are preserved.

You can use MIXDOWN mode to change the stereoscopic project type of the sequence when creating a mixed-down clip. Options for creating a MIXDOWN profile now include an option labeled TargetStereoScopicProjectType.

The following table lists the stereoscopic project types you can select and the resulting mixed-down clips.

Stereoscopic Project Type	Master Clips Created	Stereoscopic Clip Created?
Monoscopic	One master clip	No
Leading Eye	One master clip for the Leading Eye footage only	No
Left eye only	One master clip for the Left Eye footage only	No
Right Eye only	One master clip for the Right Eye footage only	No
Side by side	One master clip with Left and Right images side by side	Yes
Over / Under	One master clip with the Left image above the Right image	Yes
Full	One master clip for the Left eye, one master clip for the Right eye	Yes

Note the following limitations:

- DUALMIXDOWN mode is not supported for sequences containing stereoscopic clips, stereoscopic subclips, or both.
- CROSSRATE mode is not supported for stereoscopic clips or subclips.
- Frame locators (markers) are not preserved during mixdowns.
- H.263 and H.264 resolutions are not supported as target resolutions for transcoding.
- Source resolutions and target resolutions for mixdowns are limited to the following:
 - 8-bit HD Uncompressed (1:1 MXF in Avid editing systems)
 - DNxHD (all resolutions, including DNxHD 1080p 444)
 - AVC-Intra 50 and AVC-Intra100

See [“Overview of Stereoscopic Support” on page 175](#) for specific project types and resolutions.



For information about working with stereoscopic project types on an Avid editing system, see the Avid Stereoscopic 3D Editing Workflow Guide, which is available on the Customer Support Knowledge Base.



For information about creating a Transcode profile, see the Interplay Production Services Setup and Users Guide or the Interplay Production Help.

Stereoscopic 3D Support in the Interplay Window, Interplay | Assist, and Avid Instinct

Stereoscopic 3D clips are supported in the Interplay Window, Interplay Assist, and Avid Instinct in the same way as other master clips are supported. Playback of full S3D clips in Interplay Assist and Instinct uses the leading eye only, as set in the Avid editing application.

Properties for Stereoscopic Clips

The following properties are available for display in Interplay Access. To select properties for display, open the Property Layout view in the Interplay Administrator.

Bin column	Description
S3D Alignment	Displays whether or not the media has already been aligned by a third-party application. Presently, this is just an indicator for the editor. There are no software functions associated with this field. • None (Default) • Full: Indicates that media has been pre-aligned by an external application.
S3D Channel	Displays which channel this master clip will deliver into the stereoscopic clip. (Left eye, right eye and monoscopic are full frame channels, while Side by Side and Over/Under are frame compatible channels.) Left Eye: Indicates that the contents of this clip are to be used as the Left eye. Right Eye: Indicates that the contents of this clip are to be used as the Right eye. Over/Under: Indicates that this is frame compatible media in an over/under format. Side by side: Indicates that this is frame compatible media in a side by side format. Monoscopic: Indicates that this is a standard 2D clip (not stereoscopic).
S3D Clip name	Displays the name that was used to name the stereoscopic clip when it is created. The name only needs to be set for one of the contributing master clips. If no name is entered for any of the contributing clips, then the S3D Group name will be used instead.
S3D Contributors (master clip only)	Lists the S3D Channels (master clips) that feed into the stereoscopic clip.

Bin column	Description
S3D Eye Ordering	<i>Only available if the media is in a frame compatible format with the S3D Channel set to either Over/Under or Side by Side.</i> Specifies which half of the image the eye is displayed. Left-Right: - For Side by Side format, the Left eye is on the left, the Right eye is on the right. - For Over/Under format, the Left eye is in the top half of the image, Right eye is in the bottom. Right-Left: - For Side by Side format, the Right eye is on the left, Left eye is on the right. - For Over/Under format, the Right eye is in the top half of the image, Left eye is in the bottom.
S3D Group name	A common name used to identify all master clips originating from the same stereoscopic footage. The group name is used by the application to create a single stereoscopic clip that references these master clips. Therefore, this name must be exactly the same for all clips that need to be grouped into a stereoscopic clip. When material is captured from tape, this field is automatically populated for all the left and right eye master clips. When material is imported/linked from file, this data may not be present. If so, you can select the clips in the bin and request for auto-population of the group name. If an S3D Clip name is not specified, then the group name is used to name the stereoscopic clip when it is created.
S3D Inversion	Displays if and how the contents of this clip should be inverted before being used. This is useful when beam-splitter rigs are used during filming, and a mirror is used to create the image for one of the eyes. In this case, one of the eyes will be inverted so it needs to be corrected None: Leaves the image as is. Horizontal: Inverts the image horizontally. Vertical: Inverts the image vertically. Horizontal/Vertical: Inverts the image horizontally and vertically.
S3D InversionR	<i>Only available if the media is in a frame compatible format with the S3D Channel set to either Over/Under or Side by Side.</i> Displays how the part of the image that provides the right eye should be inverted before being used. None: Leaves the image as is. Horizontal: Inverts the right eye image horizontally. Vertical: Inverts the right eye image vertically. Horizontal/Vertical: Inverts the right eye image horizontally and vertically.

Bin column	Description
S3D Leading Eye (master clip only)	Displays which eye image should be used when editing with the leading eye. This is useful in scenarios where a beam-splitter rig was used during filming. In such a case, the leading eye is typically the non-reflected image. Left: (Default) Indicates that the left eye image is the leading eye. Right: Indicates that the right eye image is the leading eye.

14 DNxHD Resolutions Supported in Interplay | Production

Interplay v3.2 and later supports workflows for clips and sequences that use many of the DNxHD resolutions. Supported workflows include the following:

- Capture media on an AirSpeed system or an Avid editing application
- Work with media in an Avid editing application
- Check assets in to and out from the Interplay database
- Process assets using Interplay Production Services (including partial restore and partial delivery)
- Transfer assets from workgroup-to-workgroup with Interplay Transfer
- Send a sequence composed of DNxHD clips to a playback device (STP). Progressive resolutions are not supported.

The following tables list the supported resolutions..

720p Projects

Project	Edit Rate (frames/ sec)	Resolution Name in Avid Editing Application	Resolution Name in Interplay Access and Interplay Transcode	Approx. Video Bit Rate (Mbit/Sec)	Raster Size
720p/23.976	23.976	DNxHD 40	DNxHD 720p 100	40 Mbit	960x720
		DNxHD 90x	DNxHD 720 90X-110X-220X 1 Field	88 Mbit	1280x720
		DNxHD 90	DNxHD 720 90-110-220 1 Field	88 Mbit	
		DNxHD 60	DNxHD 720 60-75-145 1 Field	60 Mbit	
720p/50	50	DNxHD 85	DNxHD 720p 100	85 Mbit	960x720
		DNxHD 185x	DNxHD 720 90X-110X-220X 1 Field	184 Mbit	1280x720
		DNxHD 185	DNxHD 720 90-110-220 1 Field	184 Mbit	
		DNxHD 120	DNxHD 720 60-75-145 1 Field	121 Mbit	
720p/59.94	59.94	DNxHD 100	DNxHD 720p 100	100Mbit	960x720
		DNxHD 220x	DNxHD 720 90X-110X-220X 1 Field	220 Mbit	1280x720
		DNxHD 220	DNxHD 720 90-110-220 1 Field	220 Mbit	
		DNxHD 145	DNxHD 720 60-75-145 1 Field	145 Mbit	

1080p Projects

Project	Edit Rate (frames/ sec)	Resolution Name in Avid Editing Application	Resolution Name in Interplay Access and Interplay Transcode	Approx. Video Bit Rate (Mbit/Sec)	Raster Size
1080p/23.97	23.976	DNxHD 80	DNxHD 1080p 100	80 Mbit	1440x1080
		DNxHD 444 350x	DNxHD 1080p 444	352 Mbit	1920x1080
		DNxHD 175x	DNxHD 1080 175X-185X-220X 1 Field	176 Mbit	
		DNxHD 175	DNxHD 1080 175-185-220 1 Field	176 Mbit	
		DNxHD 115	DNxHD 1080 115-120-145 1 Field	116 Mbit	
		DNxHD 36	DNxHD 1080 36	36 Mbit	
1080p/24	24	DNxHD 80	DNxHD 1080p 100	80 Mbit	1440x1080
		DNxHD 444 350x	DNxHD 1080p 444	352 Mbit	1920x1080
		DNxHD 175x	DNxHD 1080 175X-185X-220X 1 Field	176 Mbit	
		DNxHD 175	DNxHD 1080 175-185-220 1 Field	176 Mbit	
		DNxHD 115	DNxHD 1080 115-120-145 1 Field	116 Mbit	
		DNxHD 36	DNxHD 1080 36	36 Mbit	
1080p/25	25	DNxHD 444 365x	DNxHD 1080p 444	367 Mbit	1920x1080
		DNxHD 185x	DNxHD 1080 175X-185X-220X 1 Field	184 Mbit	
		DNxHD 185	DNxHD 1080 175-185-220 1 Field	184 Mbit	
		DNxHD 120	DNxHD 1080 115-120-145 1 Field	121Mbit	
		DNxHD 36	DNxHD 1080 36	38 Mbit	
1080p/29.97	29.97	DNxHD 444 440x	DNxHD 1080p 444	440 Mbit	1920x1080
		DNxHD 220x	DNxHD 1080 175X-185X-220X 1 Field	220 Mbit	
		DNxHD 220	DNxHD 1080 175-185-220 1 Field	220 Mbit	
		DNxHD 145	DNxHD 1080 115-120-145 1 Field	145 Mbit	
		DNxHD 45	DNxHD 1080 36	45 Mbit	
1080p/50	50	DNxHD 444 735x	DNxHD 1080p 444	734 Mbit	1920x1080
		DNxHD 365x	DNxHD 1080 175X-185X-220X 1 Field	367 Mbit	
		DNxHD 365	DNxHD 1080 175-185-220 1 Field	367 Mbit	
		DNxHD 240	DNxHD 1080 115-120-145 1 Field	242 Mbit	
		DNxHD 75	DNxHD 1080 36	75 Mbit	
1080p/59.94	59.94	DNxHD 444 880x	DNxHD 1080p 444	880 Mbit	1920x1080
		DNxHD 440x	DNxHD 1080 175X-185X-220X 1 Field	440 Mbit	
		DNxHD 440	DNxHD 1080 175-185-220 1 Field	440Mbit	
		DNxHD 290	DNxHD 1080 115-120-145 1 Field	291Mbit	
		DNxHD 90	DNxHD 1080 36	90 Mbit	

1080i Projects

Project	Edit Rate (frames/ sec)	Resolution Name in Avid Editing Application	Resolution Name in Interplay Access and Interplay Transcode	Approx. Video Bit Rate (Mbit/Sec)	Raster Size
1080i/50	25	DNxHD 85	DNxHD 1080i 100	85 Mbit	1440x1080
		DNxHD 185x	DNxHD 1080 175X-185X-220X	184 Mbit	1920x1080
		DNxHD 185	DNxHD 1080 175-185-220	184 Mbit	
		DNxHD 120	DNxHD 1080 115-120-145	121 Mbit	
1080i/59.94	29.97	DNxHD 100	DNxHD 1080i 100	100 Mbit	1440x1080
		DNxHD 220x	DNxHD 1080 175X-185X-220X	220 Mbit	1920x1080
		DNxHD 220	DNxHD 1080 175-185-220	220 Mbit	
		DNxHD 145	DNxHD 1080 115-120-145	145 Mbit	

15 DNxHR Resolutions Supported in Interplay | Production

Interplay v3.3 and later supports workflows for clips and sequences that use the following DNxHR resolutions for UHDTV1 (Ultra HD) formats:

- DNxHR LB (low bandwidth)
- DNxHR SQ (standard quality)
- DNxHR HQ (high quality)
- DNxHR HQX (high quality 12-bit)
- DNxHR 444 (cinema quality 12-bit)

Supported workflows include the following:

- Capture media on an Avid editing application
- Work with media in an Avid editing application
- Check assets in to and out from the Interplay database
- Process assets using Interplay Production Services (including partial restore and partial delivery)
- Transfer assets from workgroup-to-workgroup with Interplay Transfer
- Send a sequence composed of DNxHR clips to a playback device (STP).

Note the following are unsupported:

- DNxHR LB 1/16 and 1/4
- DNxHR media with alpha channels
- DNxHR in Interplay Assist (Assist v3.3 has a clear error message; earlier versions do not)

The following table lists the supported resolutions.

Label in Interplay Access v3.8 and later	Label in Production Services	Video Bit Rate (Mbit/Sec)	Frame Rate	Raster
DNxHR LB UHD1 23.98	DNxHR LB	143	23.976	3840 x 2160
DNxHR SQ UHD1 23.98	DNxHR SQ	462		
DNxHR HQ UHD1 23.98	DNxHR HQ	699		
DNxHR HQX UHD1 23.98	DNxHR HQX	699		
DNxHR 444 UHD1 23.98	DNxHR 444	1398		
DNxHR LB UHD1 24	DNxHR LB	144	24	3840 x 2160
DNxHR SQ UHD1 24	DNxHR SQ	462		
DNxHR HQ UHD1 24	DNxHR HQ	699		
DNxHR HQX UHD1 24	DNxHR HQX	699		
DNxHR 444 UHD1 24	DNxHR 444	1399		

Label in Interplay Access v3.8 and later	Label in Production Services	Video Bit Rate (Mbit/Sec)	Frame Rate	Raster
DNxHR LB UHD1 25	DNxHR LB	150	25	3840 x 2160
DNxHR SQ UHD1 25	DNxHR SQ	481		
DNxHR HQ UHD1 25	DNxHR HQ	729		
DNxHR HQX UHD1 25	DNxHR HQX	729		
DNxHR 444 UHD1 25	DNxHR 444	1457		
DNxHR LB UHD1 29.97	DNxHR LB	179	29.97	3840 x 2160
DNxHR SQ UHD1 29.97	DNxHR SQ	577		
DNxHR HQ UHD1 29.97	DNxHR HQ	873		
DNxHR HQX UHD1 29.97	DNxHR HQX	873		
DNxHR 444 UHD1 29.97	DNxHR 444	1747		
DNxHR LB UHD1 50	DNxHR LB	299	50	3840 x 2160
DNxHR SQ UHD1 50	DNxHR SQ	963		
DNxHR HQ UHD1 50	DNxHR HQ	1457		
DNxHR HQX UHD1 50	DNxHR HQX	1457		
DNxHR 444 UHD1 50	DNxHR 444	2914		
DNxHR LB UHD1 59.94	DNxHR LB	359	59.94	3840 x 2160
DNxHR SQ UHD1 59.94	DNxHR SQ	1154		
DNxHR HQ UHD1 59.94	DNxHR HQ	1747		
DNxHR HQX UHD1 59.94	DNxHR HQX	1747		
DNxHR 444 UHD1 59.94	DNxHR 444	3494		

16 XDCAM HD and XDCAM EX Operating Points Supported in Interplay | Production

“Long GOP End-to-End Workflows” on page 106 describes the end-to-end workflow for Sony XDCAM HD and Sony XDCAM EX Long GOP material. This section lists the operating points that are supported for this Interplay release.

The supported MPEG-2 operating points are based on the Sony XDCAM and XDCAM EX systems. For the end-to-end workflow, Interplay supports 14 operating points that are the basic rates for broadcast applications. Any additional XDCAM or XDCAM EX resolutions are supported in a standalone editor configuration only.



Interplay supports all 14 operating points. However, not all of the operating points have been qualified on video servers. See “Operating Points Qualified for Omneon Servers” on page 191 and “Operating Points Qualified for AirSpeed Multi Stream” on page 191.

The following table shows all of the operating points supported by Interplay. Audio is as specified in XDCAM HD/XDCAM EX: PCM uncompressed audio sampled at 48 kHz. Bit widths can be either 16 bit and 24 bit. The audio is encoded as monaural tracks (not stereo pairs).

H	V	Interlaced or Progressive	Frame Rate	MPEG Profile and Level	Bit Rate	Sub-sample	Sony Product
1920	1080	Interlaced	59.94	422P@HL	50	CBR 4.2.2	XDCAM HD Phase III
1440	1080	Interlaced	59.94	MP@HL	35	VBR 4.2.0	XDCAM HD
1920	1080	Interlaced	59.94	MP@HL	35	VBR 4.2.0	XDCAM EX
1440	1080	Interlaced	59.94	MP@H-14	25	CBR 4.2.0	XDCAM HD & EX (HDV)
1440	1080	Interlaced	59.94	MP@HL	18	VBR 4.2.0	XDCAM HD
1920	1080	Interlaced	50	422P@HL	50	CBR 4.2.2	XDCAM HD Phase III
1440	1080	Interlaced	50	MP@HL	35	VBR 4.2.0	XDCAM HD
1920	1080	Interlaced	50	MP@HL	35	VBR 4.2.0	XDCAM EX
1440	1080	Interlaced	50	MP@H-14	25	CBR 4.2.0	XDCAM HD & EX (HDV)
1440	1080	Interlaced	50	MP@HL	18	VBR 4.2.0	XDCAM HD
1280	720	progressive	59.94	422P@HL	50	CBR 4.2.2	XDCAM HD Phase III
1280	720	progressive	59.94	MP@HL	35	VBR 4.2.0	XDCAM EX
1280	720	progressive	50	422P@HL	50	CBR 4.2.2	XDCAM HD Phase III

H	V	Interlaced or Progressive	Frame Rate	MPEG Profile and Level	Bit Rate	Sub- sample	Sony Product
1280	720	progressive	50	MP@HL	35	VBR 4.2.0	XDCAM EX

The following table lists how these operating points are displayed in different applications:

Video Format	Frame Rate	Bit Rate	Avid Editing Application Name ^a	AirSpeed Multi Stream Record Template Name (Service Framework) ^b	Interplay Production Services Profile Name
1080i	59.94	50	XDCAM HD 50Mbits MXF	XDCAM-HD 50mb 1080i	XDCAM-HD 50mbps 1080i 60
1080i	59.94	35	XDCAM HD 35Mbits MXF	XDCAM-HD 35mb 1080i	XDCAM-HD 35 60i
1080i	59.94	35	XDCAM EX 35Mbits MXF	XDCAM-EX 35mb 1080i	XDCAM-EX 35mbps 1080i 60
1080i	59.94	25	HDV 1080i MXF	HDV 25mb 1080i	HDV 1080i 60
1080i	59.94	18	XDCAM HD 17.5 Mbits MXF	XDCAM-HD 17.5mb 1080i	XDCAM-HD 17.5 60i
1080i	50	50	XDCAM HD 50Mbits MXF	XDCAM-HD 50mb 1080i	XDCAM-HD 50mbps 1080i 50
1080i	50	35	XDCAM HD 35Mbits MXF	XDCAM-HD 35mb 1080i	XDCAM-HD 35 50i
1080i	50	35	XDCAM EX 35Mbits MXF	XDCAM-EX 35mb 1080i	XDCAM-EX 35mbps 1080i 50
1080i	50	25	HDV 1080i MXF	HDV 25mb 1080i	HDV 1080i 50
1080i	50	18	XDCAM HD 17.5Mbits MXF	XDCAM-HD 17.5mb 1080i	XDCAM-HD 17.5 50i
720p	59.94	50	XDCAM HD 50Mbits MXF	XDCAM-HD 50mb 720p	XDCAM-HD 50mbps 720p 60
720p	59.94	35	XDCAM EX 35Mbits MXF	XDCAM-EX 35mb 720p	XDCAM-EX 35mbps 720p 50
720p	50	50	XDCAM HD 50Mbits MXF	XDCAM-HD 50mb 720p	XDCAM-HD 50mbps 720p 50
720p	50	35	XDCAM EX 35Mbits MXF	XDCAM-EX 35mb 720p	XDCAM-HD 35mbps 720p 50

a. For Avid editing applications, the video format and rate are determined by the project settings.

b. For AirSpeed Multi Stream record templates, the frame rate is determined by the channel configuration.

Operating Points Qualified for Omneon Servers

The following table lists the operating points qualified for Omneon servers in an Interplay environment.

H	V	Interlaced or Progressive	Frame Rate	MPEG Profile and Level	Bit Rate	Sub- sample	Sony Product
1920	1080	Interlaced	59.94	422P@HL	50	CBR 4.2.2	XDCAM HD Phase III
1920	1080	Interlaced	50	422P@HL	50	CBR 4.2.2	XDCAM HD Phase III
1280	720	progressive	59.94	422P@HL	50	CBR 4.2.2	XDCAM HD Phase III
1280	720	progressive	50	422P@HL	50	CBR 4.2.2	XDCAM HD Phase III

Operating Points Qualified for AirSpeed Multi Stream

The following table lists the operating points qualified for AirSpeed Multi Stream in an Interplay environment.

H	V	Interlaced or Progressive	Frame Rate	MPEG Profile and Level	Bit Rate	Sub- sample	Sony Product
1920	1080	Interlaced	59.94	422P@HL	50	CBR 4.2.2	XDCAM HD Phase III
1440	1080	Interlaced	59.94	MP@HL	35	VBR 4.2.0	XDCAM HD
1920	1080	Interlaced	59.94	MP@HL	35	VBR 4.2.0	XDCAM EX
1440	1080	Interlaced	59.94	MP@H-14	25	CBR 4.2.0	XDCAM HD & EX (HDV)
1440	1080	Interlaced	59.94	MP@HL	18	VBR 4.2.0	XDCAM HD
1920	1080	Interlaced	50	422P@HL	50	CBR 4.2.2	XDCAM HD Phase III
1440	1080	Interlaced	50	MP@HL	35	VBR 4.2.0	XDCAM HD
1920	1080	Interlaced	50	MP@HL	35	VBR 4.2.0	XDCAM EX
1440	1080	Interlaced	50	MP@H-14	25	CBR 4.2.0	XDCAM HD & EX (HDV)
1440	1080	Interlaced	50	MP@HL	18	VBR 4.2.0	XDCAM HD
1280	720	progressive	59.94	422P@HL	50	CBR 4.2.2	XDCAM HD Phase III
1280	720	progressive	50	422P@HL	50	CBR 4.2.2	XDCAM HD Phase III

17 XAVC-Intra HD Resolutions Supported in Interplay | Production

Interplay v3.2 and later supports workflows for clips and sequences that use XAVC-Intra HD resolutions. Supported workflows include the following:

- Capture media on an AirSpeed system or an Avid editing application
- Work with media in an Avid editing application
- Check assets in to and out from the Interplay database
- Process assets using Interplay Production Services (including partial restore and partial delivery)
- Transfer assets from workgroup-to-workgroup with Interplay Transfer
- Send a sequence composed of XAVC-Intra HD clips to a playback device (STP). Support is limited to:
 - XAVC Intra 100 1080i 50
 - XAVC Intra 100 1080i 60

The following table lists the supported resolutions.

Project	Edit Rate	Resolution Name in Interplay Access	Video Bit Rate (Mbit/Sec)	Raster Size
1080i	50	XAVC Intra 50 1080i 50	50 Mbit	1440x1080
		XAVC Intra 100 1080i 50	100 Mbit	1920x1080
	59.94	XAVC Intra 50 1080i 60	50 Mbit	1440x1080
		XAVC Intra 100 1080i 60	100 Mbit	1920x1080
1080p	23.976	XAVC Intra 50 1080p 23.976	50 Mbit	1440x1080
		XAVC Intra 100 1080p 23.976	100 Mbit	1920x1080

18 AVC-LongG Formats Supported in Interplay | Production

Interplay Production v3.6 and later supports workflows that use the Panasonic AVC-LongG format. Supported workflows include the following:

- Import media through Media Composer v8.6 or later
- Capture media through AirSpeed 5500 v2.10 or later
- Work with media in these Avid applications:
 - Media Composer v8.6 or later
 - MediaCentral UX v2.7 or later
 - Interplay Assist v3.6 or later
- Check assets in to and out from the Interplay Production database
- Process assets using Interplay Production Services

Interplay Transcode can use AVC-LongG resolutions as both sources and targets.

Partial Restore and Partial Delivery are not supported.

- Transfer assets from workgroup-to-workgroup with Interplay Transfer
- Send a sequence composed of AVC-LongG to a playback device (STP)

For more information, see [“Long GOP End-to-End Workflows” on page 106](#). For possible limitations, see the *Interplay | Production ReadMe*.

The Avid names for AVC-LongG formats are composed of the following elements:

Family Name	Bit Rate	Raster	Frame Rate
AVC Long GOP	12	1080i	29.97

The following table lists the currently supported operating points.

Name in Interplay Production	Bit Rate	Raster	Frame Rate	Chroma Sampling	Bit Depth	Panasonic Name
AVC Long GOP 12 1080i 29.97	12	1920x1080i	29.97*	4.2.0	8	AVC-LongG 12
AVC Long GOP 12 1080i 25	12	1920x1080i	25.00**	4.2.0	8	AVC-LongG 12
AVC Long GOP 12 720p 59.94	12	1280x720p	59.94	4.2.0	8	AVC-LongG 12
AVC Long GOP 12 720p 50	12	1280x720p	50.00	4.2.0	8	AVC-LongG 12
AVC Long GOP 25 1080i 29.97	25	1920x1080i	29.97*	4.2.2	10	AVC-LongG 25
AVC Long GOP 25 1080i 25	25	1920x1080i	25.00**	4.2.2	10	AVC-LongG 25

Name in Interplay Production	Bit Rate	Raster	Frame Rate	Chroma Sampling	Bit Depth	Panasonic Name
AVC Long GOP 25 720p 59.94	25	1280x720p	59.94	4.2.2	10	AVC-LongG 25
AVC Long GOP 25 720p 50	25	1280x720p	50.00	4.2.2	10	AVC-LongG 25
AVC Long GOP 50 1080i 29.97	50	1920x1080i	29.97*	4.2.2	10	AVC-LongG 50
AVC Long GOP 50 1080i 25	50	1920x1080i	25.00**	4.2.2	10	AVC-LongG 50
AVC Long GOP 50 720p 59.94	50	1280x720p	59.94	4.2.2	10	AVC-LongG 50
AVC Long GOP 50 720p 50	50	1280x720p	50.00	4.2.2	10	AVC-LongG 50

* Sample rate of 59.94 fields per second.

** Sample rate of 50 fields per second.

19 XAVC Long GOP Formats Supported in Interplay | Production

Interplay Production v3.8 and later supports workflows that use the Sony XAVC Long GOP format. Supported workflows include the following:

- Link or import media through Media Composer v8.7 or later
 - Ingest media through AirSpeed 5500, v2.11.1 or later
 - Ingest media through Media Director v1.6 or later (direct or AMA)
 - Ingest media through Interplay Transfer v3.7 Generic FTP or later.
 - Store media on NEXIS shared storage 7.0 or later or ISIS 4.7.11 or later
 - Work with media in these Avid applications:
 - Media Composer v8.7 or later
 - MediaCentral UX v2.9 or later
 - Interplay Assist v3.7 or later
 - Check assets in to and out from the Interplay Production database
 - Process assets using Interplay Production Services
- Interplay Transcode can use XAVC Long GOP resolutions as both sources and targets. Partial Restore and Partial Delivery are not supported.
- Transfer assets from workgroup-to-workgroup with Interplay Transfer
 - Send a sequence composed of XAVC Long GOP media to a playback device (STP)
 - Export clips and sequences using Media Director v1.6 or later

Supported Operating Points

The following table lists the currently supported operating points. Note that target resolutions for Interplay Transcode specify Sony XAVC.

Label in Interplay Production (except for Interplay Transcode)	Label for Target Video Quality in Interplay Transcode	Raster	Frame Rate	Chroma Sampling	Bit Depth
AVC Long GOP 25 1080i 29.97	XAVC Long GOP 25 1080i 29.97	1920x1080i	29.97*	4.2.2	10
AVC Long GOP 35 1080i 29.97	XAVC Long GOP 35 1080i 29.97				
AVC Long GOP 50 1080i 29.97	XAVC Long GOP 50 1080i 29.97				
AVC Long GOP 25 1080i 25	XAVC Long GOP 25 1080i 25	1920x1080i	25.00**	4.2.2	10
AVC Long GOP 35 1080i 25	XAVC Long GOP 35 1080i 25				
AVC Long GOP 50 1080i 25	XAVC Long GOP 50 1080i 25				
AVC Long GOP 50 720p 59.94	XAVC Long GOP 50 720p 59.94	1280x720p	59.94	4.2.2	10
AVC Long GOP 50 720p 50	XAVC Long GOP 50 720p 50	1280x720p	50.00	4.2.2	10

* Sample rate of 59.94 fields per second.

** Sample rate of 50 fields per second.

Changed Display Labels in Interplay Production

Starting with Interplay Production v3.8, the display labels used for Sony XAVC Long GOP resolutions include the bit rate. This labeling matches the format used for Panasonic AVC-LongG resolutions. Labels used for XAVC in v3.7 did not include the bit rate. The following table lists the changed labels.

Vendor-Specific Operating Point	V3.7 Label	V3.8 Label
Sony XAVC Long GOP 25 1080i 25	AVC Long GOP 1080i 25	AVC Long GOP 25 1080i 25*
Panasonic AVC Long GOP 25 1080i 25	AVC Long GOP 25 1080i 25	
Sony XAVC Long GOP 25 1080i 29.97	AVC Long GOP 1080i 29.97	AVC Long GOP 25 1080i 29.97*
Panasonic AVC Long GOP 25 1080i 29.97	AVC Long GOP 25 1080i 29.97	
Sony XAVC Long GOP 35 1080i 25	AVC Long GOP 1080i 25	AVC Long GOP 35 1080i 25
Sony XAVC Long GOP 35 1080i 29.97	AVC Long GOP 1080i 29.97	AVC Long GOP 35 1080i 29.97
Sony XAVC Long GOP 50 1080i 25	AVC Long GOP 1080i 25	AVC Long GOP 50 1080i 25*
Panasonic AVC Long GOP 50 1080i 25	AVC Long GOP 50 1080i 25	
Sony XAVC Long GOP 50 1080i 29.97	AVC Long GOP 1080i 29.97	AVC Long GOP 50 1080i 29.97*
Panasonic AVC Long GOP 50 1080i 29.97	AVC Long GOP 50 1080i 29.97	
Sony XAVC Long GOP 50 720p 50	AVC Long GOP 720p 50	AVC Long GOP 50 720p 50*
Panasonic AVC Long GOP 50 720p 50	AVC Long GOP 50 720p 50	
Sony XAVC Long GOP 50 720p 59.94	AVC Long GOP 720p 59.94	AVC Long GOP 50 720p 59.94*
Panasonic AVC Long GOP 50 720p 59.94	AVC Long GOP 50 720p 59.94	

* For v3.8, these labels apply to both Sony and Panasonic media. Associating both Sony and Panasonic media with the same master clip is not supported.

Avid highly recommends that facilities that use Sony XAVC Long GOP media upgrade the entire workgroup to Interplay Production v3.8. If some components remain at v3.7 and others at v3.8 (a “mixed environment”) the same asset could be displayed as either a v3.7 label or a v3.8 label, depending on the particular component or workflow.

For example, a discrepancy between an Interplay Engine label and a Production Services profile label might cause a job to fail. In Interplay Access, running “Update Status from MI” changes v3.7 labels to v3.8 labels for online media files.



Interplay Assist supports XAVC Long GOP resolutions with Interplay v3.7 components only.



For more information about Panasonic AVC-LongG, see “AVC-LongG Formats Supported in Interplay / Production” on page 193.



For general Long GOP information, see “Long GOP End-to-End Workflows” on page 106.

20 Using Frame Chase Editing

This chapter describes how to use Frame Chase™ editing. Frame Chase editing (sometimes called “edit while capture”) lets an Avid editing application that is part of an Interplay workgroup view and edit clips while they are being captured. Frame Chase editing is supported only when media is captured from a supported external device through Interplay Transfer.

- [Frame Chase Editing and Frame Chase Capture](#)
- [Understanding Frame Chase Editing](#)
- [Workflow for Frame Chase Editing](#)
- [Limitations When Working With In-Progress Clips](#)

Frame Chase Editing and Frame Chase Capture

Frame Chase editing as described in this chapter always involves Interplay Transfer. You can also use the Capture tool on an Avid editing application to capture to shared storage. This technique is referred to as Frame Chase Capture. Frame Chase Capture also creates in-progress clips that are available for viewing and editing as soon as the initial check-in to Interplay Production takes place. For more information, search for “Frame Chase Capture” in your Avid editing application Help.

Understanding Frame Chase Editing

Frame Chase editing is a time-saving feature that lets an editor view and edit media while it is being captured from a line feed or ingest device. After the capture starts, the editor can use the captured media to create a sequence, view the media as it is captured, and complete a sequence even before the capture has finished.

Clips that are in the process of being captured are referred to as in-progress clips. In-progress clips are indicated by the following icons:



In-progress master clips



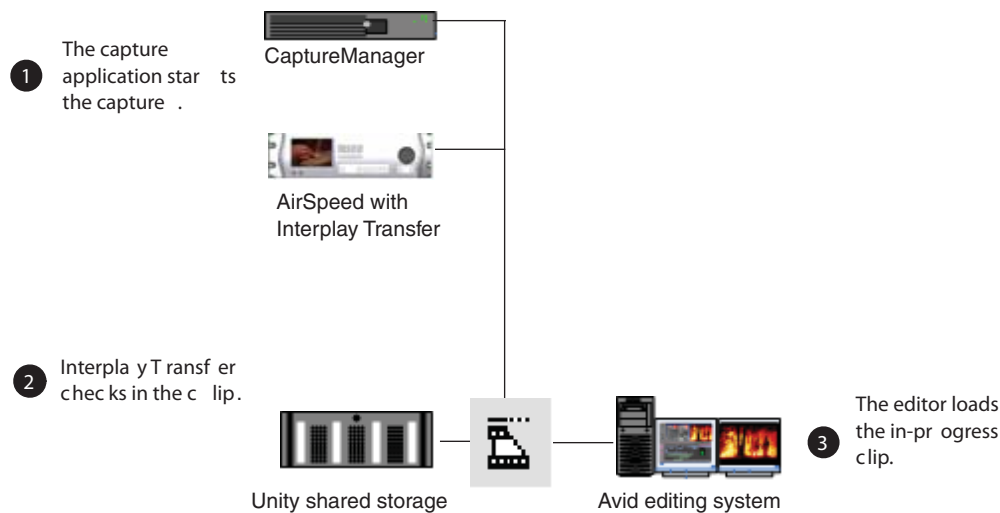
In-progress audio clips

When the capture is complete, the icon is replaced by the standard master clip or audio clip icon. To view the updated icon in the Interplay Window, press F5. To view the updated icon in a bin, select Bin > Update from Interplay.

Frame Chase editing is available only in an Interplay Production workgroup that includes Interplay Transfer and a supported external device. Media must be in the MXF format. The process consists of three basic parts:

- A capture is started on a line feed or ingest device (for example, an Avid AirSpeed video server). This capture is controlled by an application such as Interplay Capture. Depending on the device, you can also “hot record” a clip by using controls on the device itself.
- After the capture begins, the Interplay Transfer software on the AirSpeed checks the in-progress clip into the Interplay database. Interplay Transfer also manages the transfer of media to Avid shared storage.
- The editor or journalist loads the in-progress clip. You can work with in-progress clips on an Avid editing system, Avid Instinct, or Interplay Assist. On an Avid editing system, you can work directly from the Interplay window, or you can check out a clip to a bin.

The following diagram illustrates these basic parts of the process.



The length of an in-progress clip is based on the expected duration of the capture. For on-the-fly or open-ended captures, the length of the clip is based on the maximum capture time set by the capture application. The Frame Chase process creates one media file for each track of the captured media. As a capture continues, the size of the associated media file increases. After the capture is complete, the clip is updated to reflect the actual duration of the capture and the in-progress icon is replaced by the standard icon.



The default duration on an AirSpeed server is one hour (01:00:00:00). When the capture is complete, the duration is updated. If a clip goes beyond one hour, the default duration is two hours (02:00:00:00).

Frame Chase editing is supported in a dual-ingest configuration, in which both a low-res and a high-res version of a clip is created. When an editor loads an in-process clip, the version that is displayed is determined by one of the following:

- If dynamic relink is enabled, the dynamic relink settings determine which version is displayed.
- If dynamic relink is not enabled, the latest version checked in to the Interplay database is displayed.

For more information about dual-ingest configurations, see the *Interplay Capture Installation and Configuration Guide*. For more information about dynamic relink, see “Using MultiRez and Dynamic Relink” in the Help for your Avid editing system..

Workflow for Frame Chase Editing

The following workflow is one example of how to use Frame Chase editing. This workflow includes the following components:

- An Avid editing system that is an Interplay Production client. This workflow uses Avid Media Composer. It also uses the Interplay Window to access the Interplay Production database.
- A supported line-feed or ingest device. This workflow uses an Avid AirSpeed server.
- A capture application. This workflow uses Interplay Capture software, which schedules the ingest on the AirSpeed server.
- Interplay Transfer software. In this workflow, the AirSpeed server contains its own version of Interplay Transfer.

You can adapt this workflow to suit the requirements of your working environment.

In this example workflow, an editor uses Media Composer to create a news story from a timed feed. Interplay Capture has been set up to work with an AirSpeed server to automatically create clips of a set duration from the feed, as shown in the following illustration.

Scheduled Record - A

Content Information

Source: A [Configure...]

Source Route: <none>

Folder: FEED [...]

Name: Disaster

Tape Name: Sat 2

Video ID:

Notes:

Comments: Not for 6PM show

Time Period

☒ Start Time: 03:20:00:00 ☐ End Time: 03:20:00:00

☒ Duration: 01:00:00:00

Expiration

☒ Delete after: 7 days

Visibility/Ownership

Owner: admin

☐ Only Visible to Owner

Recurrence

☒ One-Time Feed

Occuring on: Wednesday, December 28, 2006

☐ Recurring Feed [Change...]

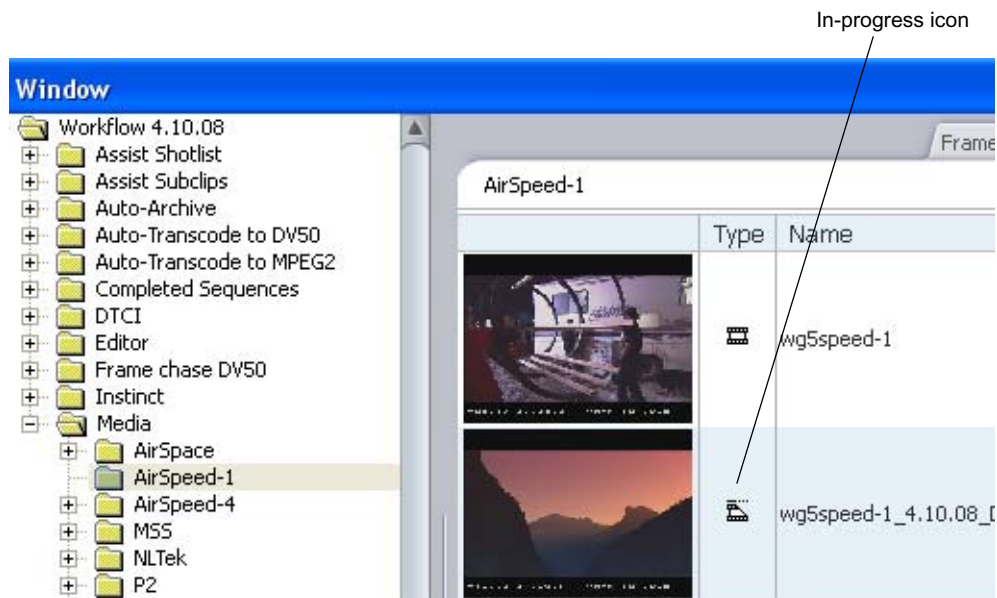
Options

☐ Prevent Auto Archive

OK Cancel

To use Frame Chase editing:

1. Make sure that the ingest device is properly connected and set up for recording and transfer.
2. On the Avid editing system, make sure an Avid shared storage workspace is mounted.
3. Open Media Composer, open a project, and open the Interplay Window.
4. In the Interplay Window, navigate to the folder that holds the clip that you want to use for editing. If you know some information about the clip, you can use the Search feature.
A clip that is being captured displays an in-progress icon.

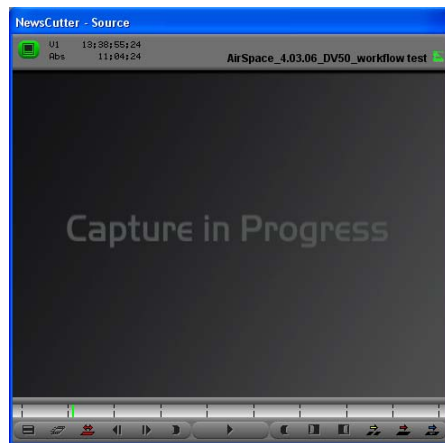


5. Load the clip into the Source monitor and play the clip.

The clip plays in the Source monitor at the same time the media is being captured.



Portions of a clip that are not currently available display a “Capture in Progress” slide until the sections are captured.



In Avid Instinct or Interplay Assist, press Alt+F5 to update the media displayed in the monitor.

6. Create a sequence in a bin on Media Composer.
7. Use standard editing techniques to build a sequence from the in-progress clip.
8. When the clip is completely captured, select Bin > Update from Interplay.

This command ensures that the latest version of the clip is used in your sequence. For more information, see “Updating Remote Assets in a Bin” in the Help for your Avid editing system.

9. Save the sequence and prepare it for playback.

Limitations When Working With In-Progress Clips

During the capture process the final length of the clip is unknown until the capture is complete and the clip information is updated. Because of this, there are some limitations when you work with in-progress clips.

The following editing functions are limited to working with the available captured media of an in-progress clip:

- **Editing in-progress clips into a sequence** — You can edit the captured portions of in-progress clips into a sequence.
- **Subclips** — When creating a subclip from an in-progress clip you must place the IN and OUT points where media has already been captured and is available.
- **Trimming** — The right-side trim limit of an in-progress clip is determined by the currently available media. When you trim an in-progress clip to the end of available media the trim functions as if it had reached the end of the clip. As more media is captured the trim limit increases. Performing the trim at a later time might allow more trimming as more media might become available.

The following editing functions are not available when working with in-progress clips:

- **Clip duplication** — Duplication of an in-progress clip is not allowed because the duplicated clip loses its relationship with the original in-progress master clip.
- **Group clips** — You cannot use an in-progress clip in a group clip, because the length of an in-progress clip is unknown until the capture is complete. Group clips are based on the length of the contributing clips. This limitation applies to all of the related grouping operations, such as AutoSync, Group Clips, and MultiGroup.
- **Consolidate, Transcode, and Audio Conversion** — You cannot use consolidate, transcode, or audio conversion with an in-progress clip. All the media must be available and the length of the clip known before you can perform these functions. However, you can use consolidate and transcode on subclips and sequences that reference or contain in-progress clips. In these cases, any right-side handles are restricted by the known media duration of the in-progress clip at the time of the operation.

For example, if you set 2 second handles but there is only 1 second of media available for a handle at the time the consolidate or transcode operation is executed, then the right-side handle is restricted to 1 second.

- **Decompose** — The restrictions for decompose are the same as consolidate and transcode. The right-side handles cannot exceed the known media duration of the clip.
- **Non-Check-in export** — You cannot export an in-progress clip or a sequence that contains in-progress clips because the final clip length might be different than the clip length at the time of the export.
- **Variable-bit-rate media** — The only variable-bit-rate (variable-frame-size) media supported for Frame Chase editing is low-res long-GOP MPEG-2 media captured by the Interplay Production low-res encoder. Only media captured by supported ingest devices is supported for Frame Chase editing.
- **Using the Capture tool**—Frame Chase editing is not available on an Avid editing system that is using the Capture tool to capture media.

21 Additional Interplay | Production Topics

This chapter covers the following topics:

- [Changing Source Information for a Clip in Interplay | Production](#)
- [Ingest and Playout of Ancillary Data](#)
- [Dynamic Relink: Working with a Single Resolution](#)
- [Disconnecting an Avid Editing Application from the Interplay Environment](#)

Changing Source Information for a Clip in Interplay | Production

There are two types of source information that usually should not be changed for a master clip in an Interplay environment:

- Source tape name
- Timecode format (changing between drop frame and non-drop frame)

The tape name and timecode format are part of the source information for a clip and changing them may cause the clip to become offline whenever Dynamic Relink is activated. Once this happens, the relation of the clip to its media can be permanently lost.

If it is necessary to change the tape name or timecode format for a master clip in an Interplay environment, use the procedure described in this section.

You only need to perform this procedure if both of the following statements apply:

- If your site uses Dynamic Relink. If your site does not use Dynamic Relink there is no need to perform the procedure.
- If you are associating a master clip with a different tape name or timecode format. If you are simply changing the name of an existing tape you don't have to perform the procedure.

To change the tape name or timecode format for a clip in an Interplay environment:

1. Disable Dynamic Relink. See the Avid editing application Help or the Interplay Production Help for information on Dynamic Relink.
2. Right-click the master clip in the bin and select Modify > Set Source.
3. To change the tape name, do the following:
 - a. Select Set Source from the Modify dialog box popup menu. The Select Tape dialog box opens.
 - b. Change the source tape name. Use an existing tape name or create a new one.

4. To change the timecode format to drop or non-drop frame, do the following:
 - a. Select "Set Timecode Drop/Non-drop" from the Modify dialog box popup menu.
 - b. Select the format and click OK.



The media will remain online at this point because Dynamic Relink is turned off.

5. Consolidate the clip to generate new media that will be associated with the newly selected tape name. Use the following options:
 - Uncheck "Skip media files already on the target drive"
 - (Option) check "Delete original media files when done"
 - Use the default settings for all other conversion parameters

The system generates new media files that will reference the correct tape name or timecode format. See the Avid editing application Help for additional information on the Consolidate command.
6. Re-enable Dynamic Relink.
7. Double-click the clip in the bin to trigger the relink. The media will be online and linked correctly to the adjusted tape name.

Ingest and Playout of Ancillary Data

Interplay v2.2.1 and later provides limited support for ancillary data. (Closed Captioning conforming to SMPTE 436-M is a common form of ancillary data.) The following limitations apply:

- Ancillary data is only supported with HD material
- Interplay Production supports one track of ancillary data
- VBI-based ancillary data is not supported
- For successful playback, sequences must be made up of clips captured by the same vendor's hardware. See "Playback Limitations" below.

The following table lists editor and AirSpeed Multi Stream ingest and playout capabilities for HD resolutions that support ancillary data in an Interplay environment.

Resolution with Ancillary Data	Editor Capture^a	AirSpeed Multi Stream Ingest/STP	Supported with Ancillary Data for Interplay v2.2.1^b
XDCAM HD	Yes	Yes	Yes
AVC-Intra	Yes	Yes	Yes
DNxHD	Yes	Yes	Yes

- a. Captured with DX hardware and a device that supports ancillary data. For details, see "Ancillary Data and Avid Editing Functions" in the Avid editing application Help.
- b. Support includes check in and check out to Interplay database, Interplay Media Indexer support, and send to playback.

Playback Limitations

Since there are variations in how ancillary data is encoded and decoded with different vendor's hardware, composition sequences created for a Send To Playback destination must use ancillary data captured or ingested using the SAME vendor's hardware and tools. Sequences with mixed-vendor ancillary data are not supported since mixed-vendor ancillary data is not guaranteed to decode and play correctly.

Support for AFD Ancillary Data

Active Format Description (AFD) is a standard set of codes through which the preferred display format for media can be embedded in video signals. Media Composer v6.1 and later allows the creation of AFD values for sequences in an ancillary data track (D1).



For more information, see the Media Composer documentation.

Interplay v2.7 and later supports workflows for sequences that include modified AFD values. AFD values are preserved when you check sequences in and out of the Interplay database, process them through Interplay Production Services, and use Interplay Transfer for workgroup-to-workgroup transfers and send-to-playback operations. AFD values are displayed in the metadata property labeled AFD, which can be displayed in an Avid editing application bin and in Interplay Access.

* 1080i59.94_1 Bin1		
	Name	AFD
☐	1080i60_J2K_to_XDCAM50Mb	
☐	dnx145_33minSEQ1	16x9 0000
☐	DNX_10min_copy	
☐	xd1080i6050_30sec13SEQ1	16x9 0000
☐	xd1080i6050_30sec12SEQ1	16x9 0100

Found: 12 (Indexed), Selected: 0 (File Size: 0 Bytes)		
	Name	AFD
☒	AFD_CC_FillerxdcamSEQ...	16x9 0100
☒	AFD_CC_FillerxdcamSEQ...	16x9 0100
☒	afdNONEto0100.transfer	16x9 0100
☒	afdNONEto0100.transfer	16x9 0100
☒	xd1080i6050_30sec12SE...	16x9 0100

Working with AFD Values on Avid Editing Systems

When you add an AFD value to a Media Composer sequence, you add the value to the entire sequence. If there is not already a D1 track in the sequence Timeline, Media Composer creates one.

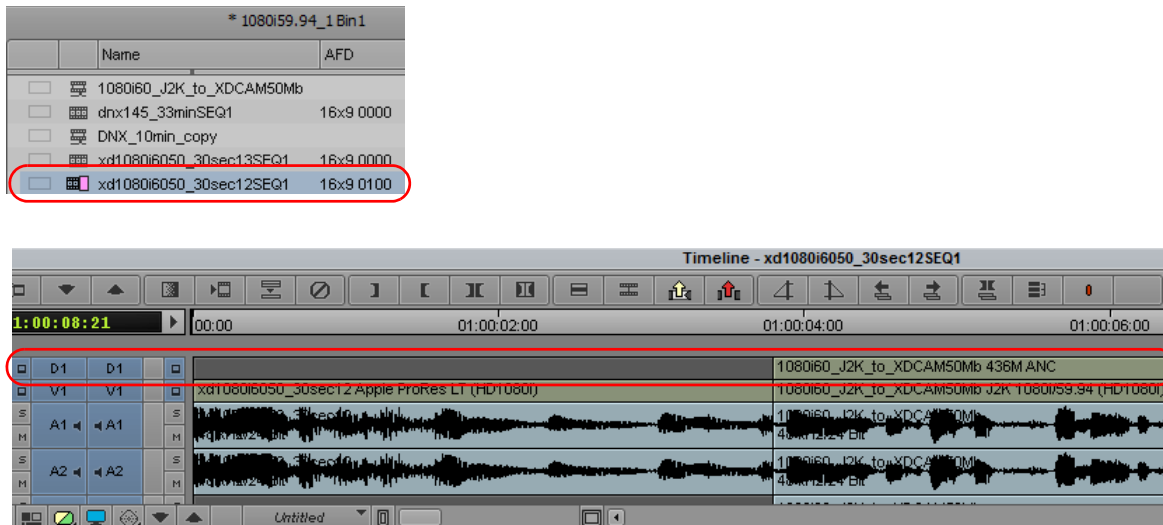
The D1 track must be present and monitored during playback to display the AFD value when the sequence is played back through an ancillary data decoder. Deleting the D1 track prevents display of the AFD value, but it does not remove the assigned AFD value from the sequence.

When you add an AFD value to a sequence, the Timeline display depends on whether there is already a D1 track in the Timeline.

Sequences that do not have a D1 track: When an AFD value is assigned to a sequence without a D1 track, an empty D1 track is created in the sequence to store the AFD value. The D1 track in the Timeline does not display essence for the AFD value.

Sequences that already have AFD values in a D1 track: When you assign an AFD value to a sequence whose master clips already have AFD values in a D1 track, the sequence in the Timeline does not display essence for the new AFD value. However, during playback, the new AFD value supersedes the AFD values for the master clips.

The following illustration shows a Timeline in which an AFD value has been applied to the sequence. The D1 track for the first master clip is empty: it does not include essence. The D1 track for the second master clip includes essence with an AFD value. This value is superseded by the AFD value applied to the sequence.



AFD Values and Send to Playback

The modified AFD value is preserved through a send-to-playback operation, resulting in an asset on the destination playback server that holds the AFD value of the sequence in a D1 track with data essence. If the user on an Avid editing system assigns an AFD value, then deletes a sequence's D1 track before sending it to the playback server, the AFD value will be written on a newly created D1 data track.

Dynamic Relink: Working with a Single Resolution

The term MultiRez refers to having a single master clip associated with multiple resolutions. You use Dynamic Relink to allow the system to automatically switch between the different resolutions depending on whether you are working with your working resolution or target (output) resolution. If you are not using MultiRez in your Interplay environment, you can either specify a single resolution for the Avid editing application or turn the feature off altogether. For Interplay Assist and Avid Instinct you can turn the feature off in the Interplay Administration tool.

Restricting Clips to a Single Resolution

This section describes how to set values in the Dynamic Relink dialog box if you know that you will only be working with one resolution.

For Sites that Use Only One Resolution

If your site never creates multiple resolutions of the same master clip, you can disable Dynamic Relink in the editing applications.

To turn off Dynamic Relink:

1. In the Avid editing application, open the Project window and click Settings.
2. Open the Dynamic Relink dialog box and deselect the Enable Dynamic Relink checkbox.

For Sites that Use Multiple Resolutions of the Same Master Clip

If your site uses multiple resolutions of the same master clip but you always want to use the same resolution in your Avid editing application, you should enable Dynamic Relink and use the “Override Working and Target Setting” option.

To use just one resolution in an Avid editing application:

1. Open the Dynamic Relink dialog box and click “Override Working Settings with Target Settings”. This makes the working and target resolutions the same.
2. Set the target resolution to the resolution you want to use.

The problem with disabling Dynamic Relink in this case is that if there are different resolutions of the same clip used at your site, you always get the “currently checked in resolution” when you check a clip out of the Interplay Window. For example, if another user had transcoded your clip to a different resolution and checked it in, you would get that resolution the next time you checked it out. When you use the Override option you are assured that you will be working with the resolution that you want.

Turning Off Dynamic Relink for Interplay | Assist and Avid Instinct

Interplay Assist supports the ability to disable dynamic relink. Dynamic relink allows you to link your working media to a target resolution for send-to-playback operations. If your workflow doesn’t require dynamic relinking, you can disable this feature.

You set dynamic relink options in the Interplay Administrator in the Editor Database settings and in the Instinct/Assist User settings. If you enable dynamic relink, you must also change the following relink options from the default value (“not set”) before the functionality is applied to a site or a user.

Option	Setting	Description
Send to playback (Editor Database Settings)	Target resolution, primary	Sets a resolution for output to a playback device. The default is DV 25 411.
	Audio Target Resolution	Sets an audio resolution for output to a playback device. The default is PCM. Also sets the Sample Rate and Bit Depth of the audio for playback.
Video Settings (Instinct/ Assist User Settings)	Working Resolution	Sets the user’s working video resolution.
Audio Settings (Instinct/ Assist User Settings)	Working Resolution	Sets the user’s working audio resolution. If the working resolution is set to PCM, you can select a sample rate and a bit depth for audio clips.
Format	Video Format	(Avid Instinct only) Sets the default video format for your sequences. Select a format or select Any. If you select Any, Avid Instinct allows you to select a video format for your sequence when you create a new story.

Disconnecting an Avid Editing Application from the Interplay Environment

Starting at Interplay v1.1.1, Interplay Production supports a Disconnected Client mode that allows you to work while disconnected from the Interplay Production environment.

There are two basic cases when you might want to voluntarily disconnect an Avid editing application from the Interplay environment:

- If you have a laptop that you want to disconnect from the network and then use it to work with media on your local drives.
- If the Service Framework is not available and you want to continue working on your local drives.

Your system administrator might also perform a forced disconnect on your system in case of software installation or system update.

For details on the capabilities and limitations of disconnected mode, see the Interplay Production Help and the ReadMe for your Avid editing application.

22 Working with Interplay | Media Indexers v3.5 and Later

This chapter covers the following topics:

- [MongoDB Database](#)
- [MI Memory and Disk Space Requirements](#)
- [Determining the Number of Monitored Data Sources](#)
- [Support for AVC-LongG Media](#)
- [Media Indexer Independence from ASF](#)
- [Interplay | Media Indexer Overview](#)
- [Web-Based User Interface](#)
- [Web Interface Components](#)
- [Managing Interplay | Media Indexers in an Interplay Environment](#)
- [Troubleshooting Interplay | Media Indexer](#)
- [Prebuilding a Media Indexer Cache During an Upgrade](#)
- [Indexing Files Starting with a Period](#)

MongoDB Database

MI v3.6 and later uses a MongoDB database as well as its classic, legacy database. The MongoDB is a document-based database that runs in its own process on the same machine as the Media Indexer. This MongoDB is automatically installed with the Media Indexer and runs on each MI server and client.



MI does not use any of the sharding features available with MongoDB.

Starting with Media Indexer v3.6, indexed data is stored in two locations on the MI servers and clients:

- The MongoDB database stores information about all resolutions. This includes information about format independent resolutions such as AVC-LongG.
- The classic MI database stores information about all resolutions except for format independent resolutions such as AVC-LongG. The database is stored in memory and is reflected in the MI cache files.

The reason for running both versions of the database is to allow older Media Composer clients to still connect to MI v3.6 and later. Media Composer v8.6 and later clients use the MongoDB database.

Installing and Maintaining the MongoDB Database

The method for connecting to the MI from the editor has not changed for MI v3.6 and later. You still identify the MI servers in the Interplay Administrator tool. Use the MI web interface to configure the Network of Media Indexers (referred to as a HAG in releases earlier than MI v3.5). See [“Configuring Media Indexer Servers” on page 235](#).

The MongoDB database is automatically installed when you install the Media Indexer application. The following table lists some important points about the MongoDB database.

System	Service	Automatically starts	To determine if service is running	To start or stop service
Windows servers and clients	Avid Interplay Media Indexer MongoDB	Yes when you start the MI	Web UI or check Windows Services	Use Windows Services or use start/stop command (see below)
Mac OS 10 clients	Mongod	Yes when you start the Mac.	Web UI or check “mongod” value in Activity Monitor	Use launchctl command (see below)



Note that the MongoDB service will not exist right after the installation of the MI. The service (for MI MongoDB) will be created (and recreated) upon MI startup.

To start or stop MongoDB service on a Windows system:

1. Start or stop the following service in the Services window:
Avid Interplay Media Indexer MongoDB
2. To start the process, execute the following command in a command window:
`sc start "Avid Interplay Media Indexer MongoDB"`
3. To stop the process, execute the following command in a command window:
`sc stop "Avid Interplay Media Indexer MongoDB"`

To start or stop MongoDB service on a Mac system:

1. Do one of the following:
 - To start the process, execute the following command in a command window:
`sudo launchctl load /Library/LaunchDaemon/com.avid.mediaindexer.mongod.plist`
 - To stop the process, execute the following command in a command window:
`sudo launchctl unload /Library/LaunchDaemon/com.avid.mediaindexer.mongod.plist`
- Note that you will need the correct administrator permission to start or stop the process.

Restarting the Media Indexer Service on a Mac OS X System

This section describes how to restart the Media Indexer service. Note that every time you restart the Avid editing application in Media Composer on a Windows system, the system automatically restarts the Media Indexer service. On a Mac OS X system you must manually restart the service.

To restart the Media Indexer process from a Terminal window:

1. Open a terminal window and navigate to the Media Indexer/bin directory as follows:
`cd /Applications/Avid/Media\ Indexer/bin`

2. Type the following commands in the bin directory:

```
sudo ./mi.sh remove
sudo ./mi.sh install
```

where:

- sudo instructs the system to run the command as “super user” or root.
- the characters ./ instruct the system to “run the following command that resides in the current directory”.



You must have an Administrator login to perform commands that begin with “sudo.”

Log and Data File Locations

The following table compares the location of log and data files for the two database models.

Database	Log and Data Files
MongoDB	Windows Server 2008/20012 and Windows 7, 8, and 10 clients: C:\ProgramData\Avid\AvidMI\mongodb-log C:\ProgramData\Avid\AvidMI\mongodb-data Mac clients: See the mongodb-log and mongodb-data folders under this directory: /Applications/Avid/Media Indexer/state
Classic MI	Windows Server 2008/20012 and Windows 7/8/10 clients: C:\ProgramData\Avid\AvidMI C:\ProgramData\Avid\AvidMI\log Mac client cache files: /Applications/Avid/Media Indexer/state

MI Memory and Disk Space Requirements

Memory Requirements

Starting at Interplay v3.6.x, the RAM requirements for a Media Indexer server have increased to a minimum of 32 GB. The increased memory requirements also affect the number of data sources that an MI can store in its database as shown in the following table.

Hardware Platform	Operating System	RAM	Number of Data Sources
Dell R630 or HP DL360	Windows Server 2012 R2	32 GB	5 million
	Windows Server 2012 R2	48 GB or 64 ^a GB	10 million
AS3000	Windows Server 2008 Standard R2	32 GB (36 installed)	5 million
Interplay Bundle ^b - above platforms	See above	See above	500,000

- a. 64 GB is recommended for best performance.
- b. For Interplay Bundle requirements, see the *Interplay Software Installation and Configuration Guide*.

Disk Space Requirements

A Media Indexer server only requires a C: partition. However, MI servers are often configured with a D: partition for storing installers and backup log files. Because the MongoDB database stores its documents on the C: partition, it's good to allocate as much space as possible for the C: partition.

In addition, the MI can save logs with lots of data, especially when set to finer logging levels for trouble shooting. If you are setting up a new server, allocate at least half the available space for the C: partition.

Determining the Number of Monitored Data Sources

When discussing the number of objects that a Media Indexer can store in its database, the term “data sources” is the most accurate term for these objects. A data source can be defined as follows:

- There is usually one data source per mxf file.
- For multitrack audio, there is one data source per audio track in the audio mxf file
- For AMA media there is one data source per track.

Starting at MI v3.6.x, the number of data sources currently stored in the database is listed in the FiDataSource section of the “MongoDB information” view. This information combines the data source information for both MXF files and AMA files. The view is available on the Statistics tab of the MI web interface. The following illustration shows the Items Count for the FiDataSource Information in the MongoDB information view.

MongoDB information

✓ **MongoDB Information**

Status:	Available
Version:	3.2.4
Address:	muc-mibbeken7:27017
Database Name:	indexer
Collections:	3
Items Count:	13342
Average Item Size(in KB):	8
Data Size (in KB):	119289
Storage Size (in KB):	24404
Indexes:	16
Index Size (in KB):	1664

FiDataSource Information

Collection Name:	fiDataSource
Schema Version:	1.0
Items Count:	11999
Storage Size (in KB):	24208
Indexes:	10
Index Size (in KB):	1420

FolderTree Information

Collection Name:	foldertree
Schema Version:	1.0
MXF Items Count:	1333
AMA Items Count:	3
SPAAF Items count:	5
Storage Size (in KB):	180
Indexes:	4
Index Size (in KB):	212

Refresh Check MongoDB connection

For all MI v3.x versions, you can also obtain a rough estimate by combining the Files count values in the Storage Statistics and AMA Storage Statistics views. This shows a rough estimate of the number of objects stored in the classic Media Indexer database. The following illustration shows the two views. In this case the total is 11991.

Storage Statistics	
Workspaces:	1
Folders:	1199
Files:	11990
Quarantined Files:	20
Unique Media Files (File Mob IDs):	1
Duplicates:	11989
Channel Ids:	1
Quality Descriptors:	69
Channel Id Interests:	0
Refresh	

Ama Storage Statistics	
Workspaces:	1
Folders:	1
Files:	1
Quarantined Files:	0
Unique Media Files (File Mob IDs):	1
Duplicates:	0
Channel Ids:	9
Quality Descriptors:	4
Channel Id Interests:	0
Refresh	

Note that the file storage statistics (MXF plus AMA) views detail the load of the classic in-memory database and the MonogoDB Information view details the load of the MongoDB database. They are separate and there could be situations where one database has a different content or load compared to the other. However, in general they should be similar.



For information on how Media Indexer re-indexes the files it is monitoring, see *“Estimating the Indexing Time”* on page 257 and *“Monitoring the Indexing Process”* on page 258.

Support for AVC-LongG Media

See *“AVC-LongG Formats Supported in Interplay | Production”* on page 193.

Media Indexer Independence from ASF

Starting at Media Indexer (MI) v3.5, Avid Service Framework (ASF) is no longer required for the Media Indexer clients or servers. This change has been occurring over time for MI since the introduction of Interplay v3.x and is described in the following sections.

Discontinued support for the Service Configuration tool for MI

The connection between the local MI on an editor and the MI servers is specified through the Interplay Administrator tool. This method of connection has been available since Interplay v3.x and is not changing. In previous releases the MI servers appeared in the Service Configuration tool even though that method of connection was not recommended. Starting at v3.5, the MI servers are no longer available in the ASF Service Configuration tool.

Since Interplay v3.x, the preferred method to configure the connection between the Media Indexer servers has been through the MI web interface. Starting at MI v3.5, this is the only way to configure the servers.

Version support for Media Indexer v3.5

The following describes the version dependencies for MI v3.5.

- Media Composer v6.x and the matching local MI version 2.7.x do not work with MI v3.5 and later. This is because the local MI v2.7.x will not be able to connect to the MI v3.5 server via ASF. This means that all remote media will be offline in the editor. See the Editor and Interplay compatibility matrix on the Avid Knowledge Base for the supported editor versions.
- The following components are not able to connect to Media Indexer v3.5:
 - MediaCentral earlier than v2.5
 - Interplay Access earlier than v3.1
 - Interplay Assist earlier than v3.0

Note that some of the above configurations have not been officially supported in Interplay but were known to work in some situations. Starting at Interplay v3.5 they will no longer be expected to work.

Concept of ASF Workgroup no longer applies to MI

In previous releases you used the ASF Workgroup Properties tool to make sure the local MI could find the Lookup Server. This is no longer necessary and has not been necessary since Interplay v3.x.

Media Indexer no longer registers with the Avid Lookup Service. The ASF tools do not show Media Indexer 3.5 or newer versions in the service or process tree that is usually found at the left hand side view of those tools.

Note that the Workgroup Properties tool still lists the Media Indexer service that is installed on the same machine. It shows whether the MI is available and whether it is started.

Media Indexer Service Startup

In previous releases the Avid Service Starter Service was required to start the Interplay Media Indexer service.

Now Media Indexer Service on Windows will auto start and no longer requires the Avid Service Starter Service.

Note that the Media Indexer Windows service by default will not be configured to take any special recovery activities in case of service failure. If that is desired, the Media Indexer windows service can be configured manually. For example, you can configure it to just restart on the first and second service failure.

Limitation: Media Indexer Service Restart after Shutdown

You can shut down the Media Indexer using the Web interface. However, a shutdown via the MI Web interface will not automatically restart the service, even though the web interface specifies that a restart should be expected.

As a workaround, start the MI from Microsoft Windows Services. Or change the MI service startup type to Manual if Avid Service Starter Service is installed on the machine.

Network of Media Indexers Replaces High Availability Group

Starting at MI v3.5, the High Availability Group (HAG) is replaced by the Network of Media Indexers (NOMI). The user interface for the NOMI in the MI web interface is similar to the HAG interface. The main difference is that the NOMI does not use ASF.

The NOMI contains the Media Indexer servers in the Network of Media Indexers. It does not contain the local Media Indexers on the editor clients. The editor clients continue to use the Hostname Settings view in the Interplay Administrator tool to create a connection to the Media Indexer servers in the NOMI.

Note that if you want to maintain an MI that monitors the storage locations as an active standby node, you can configure it to monitor the storage locations but do not add it to the NOMI.

You can have multiple NOMIs in one Interplay environment, each with its own unique name. The MIs in a particular NOMI must all connect using the same NOMI name. The MI servers in a NOMI use multicast to communicate with each other as described in the next section.

Multicast Connection for MI Servers

The Media Indexer servers in the Network of Media Indexers (NOMI) communicate via Multicast. This has been the case since Interplay v3.x. It usually does not require any special configuration because the MI servers in a NOMI typically reside in the same network segment (subnet). However if the members of the NOMI connect to separate switches for any reason, additional configuration may be required to propagate multicast packets (at layer 2) between switches. Refer to Appendix B in *Network Requirements for ISIS and Interplay PAM and MAM* on the Avid Knowledge Base.

Note that the local Media Indexers on the editors do not require Multicast communication with the servers. Multicast communication is only required between the Media Indexer servers. The multicast address used is 239.255.2.3 on UDP port 6155.

Upgrading an MI HAG to an MI v3.5 or Later NOMI

If you have an MI HAG configured when you update the Media Indexers to v3.5 or later, the installer will maintain the structure of the HAG and use the name of the HAG for the new Network of Media Indexers. The weight and HAG status is also brought forward for the individual Media Indexer servers. So changing the MI settings is only required if you want to change the NOMI name or change the MI weight in the NOMI. Or if a HAG was not configured before the upgrade.

Note that the update to MI v3.5 or later requires a re-index of all the configured storage locations, so plan your time accordingly. See [“Prebuilding a Media Indexer Cache During an Upgrade” on page 259](#).



If you upgrade an MI to v3.5 or later and then downgrade back to an earlier release (before v3.5), the MI will revert to its original settings (such as the weight, HAG status, HAG Name) in the HAG. This will occur even if you had changed the settings while in the v3.5 or later NOMI.

Interplay | Media Indexer Overview

The Interplay Media Indexer keeps track of media files in storage locations that you identify. Other applications and Interplay components communicate with the Media Indexers servers including Media Composer, MediaCentral UX, Interplay Transcode, Interplay Transfer, and Interplay Access.

In an Interplay Production environment, each Avid editing system has its own local Media Indexer that manages the local storage areas on that system. Each Interplay Production environment also has at least one Interplay Media Indexer that manages media on shared storage. The servers are referred to as the Network of Media Indexers (NOMI). When the Avid editing system needs to access media on shared storage, it communicates with its local Interplay Media Indexer, which then communicates with the Network of Media Indexers.

For information on configuring the Network of Media Indexers and connecting Avid editing clients to the Network of Media Indexers, see [“Configuring Media Indexer Servers” on page 235](#).

Web-Based User Interface

Media Indexer uses a Web-based user interface for configuring and gathering information about Interplay Media Indexers and the storage locations they monitor. You connect to the Web UI through a secure HTTPS port 8443 using the following URL:

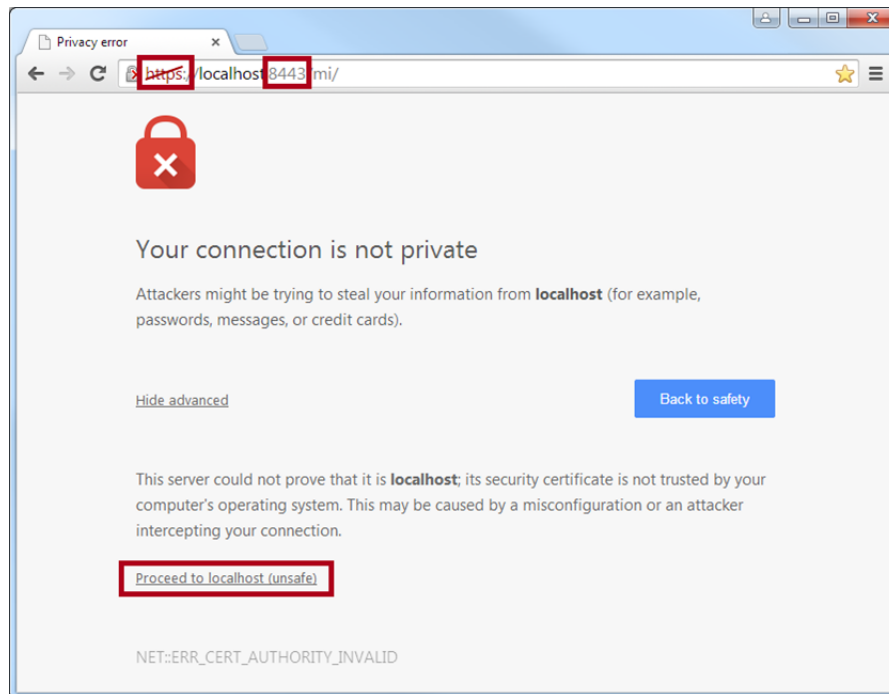
`https://<MI system name>:8443/mi`

Each Interplay Media Indexer hosts its own web-based user interface. This includes MIs on editor systems. The Web-based user interface is only available while the Interplay Media Indexer application is running.

Note the following:

- Earlier MI releases used the following URL `http://<MI system name>:8888/mi`. This URL now redirects to the new URL using HTTPS port 8443.
- Access through the secure port uses the Secure Sockets Layer (SSL), which requires the server to have a digital certificate. Media Indexer uses a self-signed certificate.
- With a self-signed certificate, the user is initially presented with a warning message. The exact message depends on the browser, but in all cases the user will be able to access Media Indexer.

The following illustration shows the warning page in the Chrome Browser when you connect to Media Indexer on the machine that hosts the Web UI. Although the message seems serious, the https connection is secure and there is little security risk for a properly configured Media Indexer system. You can click on “Proceed to localhost (unsafe)” and view the Media Indexer Web UI.



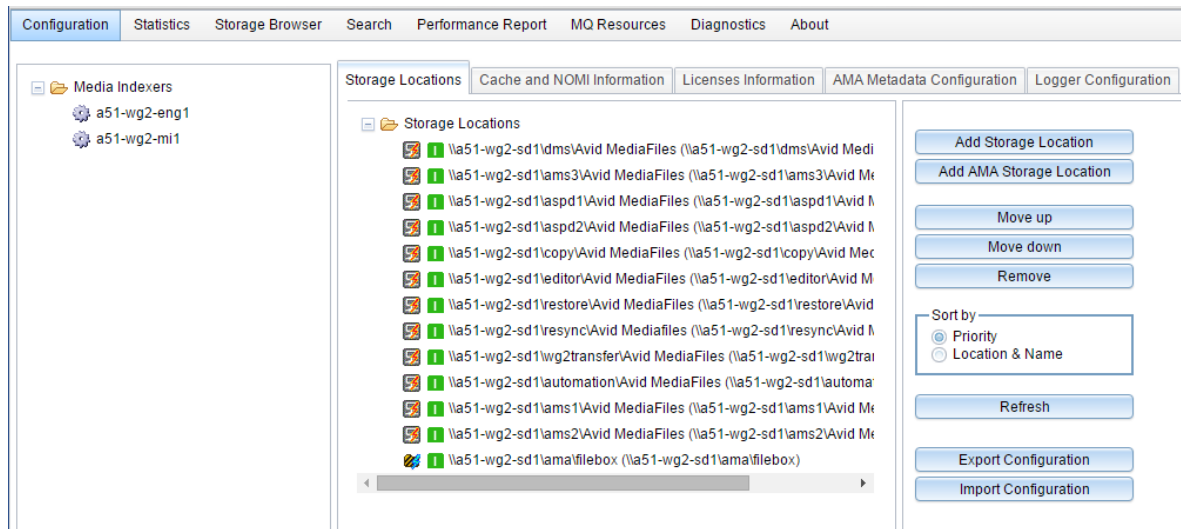
Web Interface Components

The following sections describe the different tabs on the Web Interface.

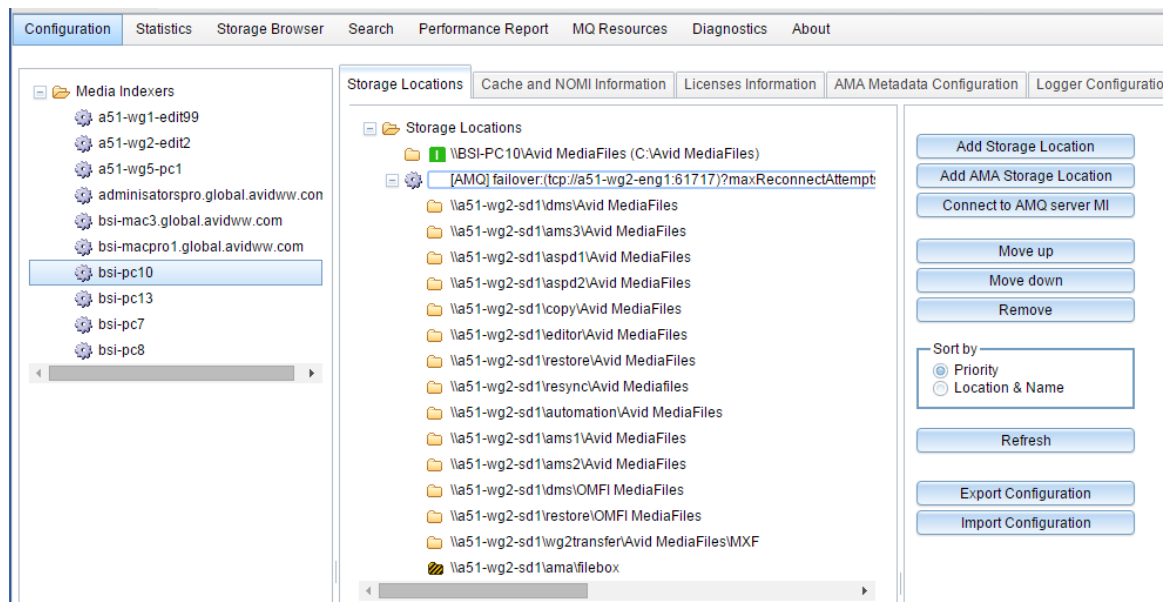
- “Configuration Tab” on page 220
- “Statistics Tab” on page 223
- “Storage Browser Tab” on page 229
- “Search Tab” on page 234
- “Performance Report Tab” on page 234
- “Media Indexer MQ Resource Tab” on page 235

Configuration Tab

Use the Configuration tab to configure a Media Indexer server. The following illustration shows the Configuration window.



In the case of local storage on an editor, the Interplay Media Indexer displays the full UNC path to the storage location as well as the local path to the media files. The following illustration shows the configuration for an editor named bsi-pc10 that has local storage on the C: drive. It is connected to the Network of Media Indexers via the Interplay Media Indexer running on the system named a51-wg2-eng1. The illustration also shows the storage locations configured for the Network of Media Indexers.



For information on configuring the Network of Media Indexers, see [“Configuring Media Indexer Servers”](#) on page 235.

Note that you can add a storage location before it is available online. MI will rescan automatically once the storage location becomes available. MI automatically resets unavailable workspaces each 30 seconds.

To manually rescan or reindex a folder, select the folder in the Storage Browser and select Rescan or Reindex. See [“Reindex and Rescan Buttons” on page 231](#).

Updating the Media Indexer Cache and Configuring the Network of Media Indexers

Use the “Cache and NOMI Information” tab to update the Interplay Media Indexer cache and configure the Network of Media Indexers (NOMI). The “Join to Network of Media Indexers” option is not available on a local Media Indexer.

For information on configuring the Network of Media Indexers, see [“Configuring Media Indexer Servers” on page 235](#).

Storage Location Icons

The icons used for MI storage locations depend on the storage type (ISIS, local, or 3rd party) and whether the storage location is used for AMA or for non-AMA media. ISIS Storage types are further delineated by the ISIS family. For example, ISIS 7500 is defined as Realtime and ISIS 2500 is defined as Nearline.



You can view a tool tip explanation of the icon by hovering the mouse over it.

The following table shows the Media Indexer web interface icons for Interplay v3.1 and later.

MI Web Interface Icon	Description	Comments
	AMA Realtime Storage	This includes AMA media stored on ISIS 7500, 7000, 5500, and 5000.
	AMA Nearline Storage	This includes AMA media stored on ISIS 2500, ISIS 2000, and licensed 3rd party storage.

MI Web Interface Icon	Description	Comments
	MXF Realtime Storage	This includes MXF media stored on ISIS 7500, 7000, 5500, and 5000. These locations are actively indexed by Media Indexer.
	MXF Nearline	This includes MXF material stored on ISIS 2500, and ISIS 2000. These locations are actively indexed by Media Indexer.
	MI Cannot Access this Folder	Media Indexer cannot access the storage location. - For AMA storage locations, this might not be a serious problem because Media Indexer does not index the AMA media directly. It indexes the related .spaaf files. - For MXF storage locations, this indicates a serious problem. The X symbol can also indicate that the folder was mistyped or that the storage location is not available. In this case, Media Indexer might have access to the storage when it is available.

In addition to the above icons, an additional icon identifies the license status of the folder as follows:

- I - Implicit license - the material in this folder does not require a license
- E - Explicit license - the material in this folder requires a license and one has been applied
- U - Unlicensed - the material in this folder requires a license and one has not been applied

The following illustration shows examples of implicitly and explicitly licensed storage locations.



Statistics Tab

The Statistics tab presents information that was previously covered by the ASF Health Monitor. You can leave sections open and move sections around on the window. Your browser retains the arrangement the next time you open the window. This section describes the Statistics views.

Memory Usage Information

This view displays memory usage for different Java memory pools as shown in the following illustration.

Memory usage information	
Code Cache	
Max (in KB):	49152
Used (in KB):	4677
Committed (in KB):	4864
Used memory after last GC collection (in KB)	0
Par Eden Space	
Max (in KB):	545344
Used (in KB):	49814
Committed (in KB):	139776
Used memory after last GC collection (in KB)	0
Par Survivor Space	
Max (in KB):	68096
Used (in KB):	1952
Committed (in KB):	17472
Used memory after last GC collection (in KB)	1952
CMS Old Gen	
Max (in KB):	1366464
Used (in KB):	41688
Committed (in KB):	349568
Used memory after last GC collection (in KB)	33132
CMS Perm Gen	
Max (in KB):	262144
Used (in KB):	48036
Committed (in KB):	74108
Used memory after last GC collection (in KB)	44462
Refresh	

Network of Media Indexers Statistics

Starting at MI v3.5, the Redundancy Statistics view is replaced by the Network of Media Indexers Statistics view as shown in the following illustration.

The screenshot shows the 'Statistics' tab in the web interface. The 'Network of Media Indexers Statistics' section is expanded, displaying the following information:

NOMI Role:	Master
Master in the NOMI:	A51-WG2-ENG1
Weight:	10
Mirroring State:	Not Required
Last Mirroring Time:	None yet
NOMI Size:	1
Last message published:	1/4/16 4:19 PM Eastern Standard Time
Last message received:	None yet
NOMI Members Versions:	3.5.0.27126

Below the table is a 'Refresh' button. To the right of the 'Network of Media Indexers Statistics' section, there is a list of other statistics sections, each with a collapse icon:

- Storage Workspaces
- Ama Storage Workspaces
- AMA Metadata (SPAAF) Workspace
- Local Adapters
- AMA Metadata Location (SPAAF) Adapter
- Amq Server Adapters
- Process Information
- Amq Clients Ids
- Network Information

Other statistics sections visible on the left include JVM Information, Memory usage information, Environment Statistics, Server Statistics, Performance Statistics, Scanning Workers Thread Pool Statistics, Indexing Workers Thread Pool Statistics, Storage Statistics, and Ama Storage Statistics.

This view displays the following information about the currently selected MI:

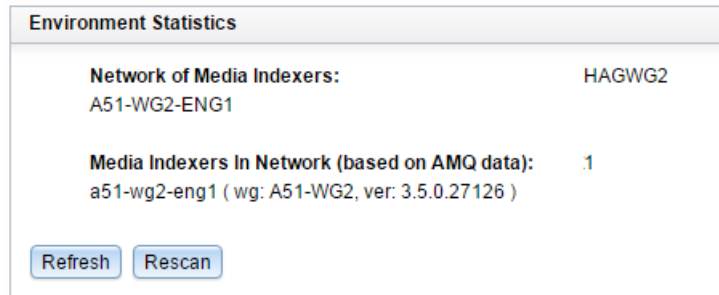
- Identifies the role of the selected MI in the network - master or slave.
- Specifies the machine name of the master of the NOMI.
- The weight of the current node in the network. The node with the highest weight is the master.
- The mirroring state. This example only has one node in the NOMI so the mirroring state is listed as “Not Required” on the master. If there were more than one node in the NOMI, the mirroring state on the slave would be listed as Pending or Done.
- If mirroring is happening, the system shows the timestamps of the last mirroring activity.
- The number of MIs in this NOMI.
- Troubleshooting information regarding the last time this MI published or received a message.
- The versions of the MIs in the NOMI. If all MIs in the NOMI have same version, then only one version is listed. If the versions are different then each version is listed separately.

Environment Statistics

Starting at MI v3.5, the Environment Statistics view contains information about the following:

- The name of the Network of Media Indexers (NOMI) and a list of the MIs in the NOMI.
- Local and server Media Indexers in the network (same subnet) along with the ASF workgroup name and the MI version running on each system.

The following illustrations shows there is one Media Indexer server application in the Network of Media Indexers named HAGWG2. It is running on the server named a51-wg2-eng1. The MI is a member of the ASF workgroup named A51-WG2 and it is running MI version v3.5.0.27126.



Server Statistics

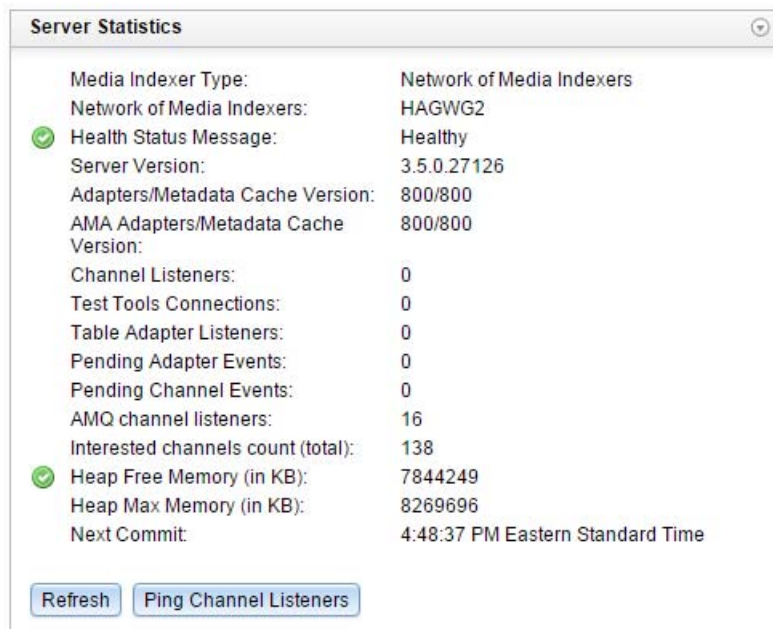
The Server Statistics view contains information about the selected MI.

Possible values for Media Indexer Type:

- Local Media Indexer
- Standby Media Indexer (server MI that is not a member of NOMI)
- Network of Media Indexers (server MI that is a member of NOMI)

Possible values for Network of Media Indexers:

- Not a part of a Network of Media Indexers (for Local and Standby MIs)
- NOMI Name (for MI that is a member of NOMI) In this example, the name of the NOMI is HAGWG2.



Performance Statistics

Starting at MI v3.5, the Table Statistics view is replaced by the Performance Statistics view. The view displays the Average and Maximum time for services requests in milliseconds as shown in the following illustration.

Performance Statistics		
ChannelIdService (in ms)	Avg	Max
generateChannelIdsForSource	3	86
generateChannelIds	3	20
MediaAvailabilityService (in ms)	Avg	Max
getClipsMediaAvailability	2	42
getClipMediaAvailability	2	12
MediaLookupService (in ms)	Avg	Max
findPathByFileMobId	1	80
findMediaByFileMobId	2	10
findMediaBySourceId	0	0
findMediaByChannelId	1	243
findMediaByCriteria	0	0
findChannelById	1	25
findChannelBySourceId	0	0
findChannelByFmobId	1	13
findMediaByQualityId	0	0
ResyncDataProviderService (in ms)	Avg	Max
listRootStorageLocationsByType	0	0
listDirectSubFoldersByPathByType	1	4
listDeepSubFoldersByPathByType	0	0
listMediaByPathPagedByType	48	623
Refresh		



Starting at MI v3.8, the Performance Statistics view contains a more complete list of average and maximum execution times for Media Indexer. Much more complete than the above illustration.

When Java Monitoring is disabled on the Performance Report tab, performance statistics are not available and the system displays the following message:

“Performance statistics are not available. Java monitoring is disabled.”

Scanning Workers and Indexing Workers - File Scanning Features

Media Indexer scans files on a per folder basis. Media Indexer maintains a pool of active workers and the workers select from a queue of folders that must be scanned. When a worker finishes scanning a folder it selects another folder from the queue.

You can use the Media Indexer Web-based user interface to view the progress of the file scanning. This value (Current Queue Length) is displayed in the Statistics window as shown in the following illustration:

Table Statistics	
Scanning Workers Thread Pool Statistics	
Currently Active Worker Threads:	0
Currently Idle Worker Threads:	8
Current Queue Length:	0
Started Worker Threads:	8
✓ Inconsistencies:	0
Accepted Tasks:	1101
Ignored Tasks:	0
Completed Tasks:	1101
Failed Tasks:	0
Average Worker Thread Execution Time:	64
Refresh	
Indexing Workers Thread Pool Statistics	
Currently Active Worker Threads:	0
Currently Idle Worker Threads:	16
Current Queue Length:	0
Started Worker Threads:	23
✓ Inconsistencies:	0
Accepted Tasks:	20732
Ignored Tasks:	7
Completed Tasks:	20635
Failed Tasks:	97
Average Worker Thread Execution Time:	256
Refresh	

There are two Thread Pool Statistics views for the classic database and two for the MongoDB database (designated as FI).

The Scanning Workers views show the activity for scanning folders. The Indexing Workers views show the activity for scanning files.

The Current Queue Length value shows the number of files or folders that must still be scanned.



The display is not dynamic. Click the Refresh button to update the value.

Avid recommends that you keep no more than 5,000 files in an ISIS folder that is being indexed by an Interplay Media Indexer. See [“Improving Interplay | Media Indexer Performance by Limiting the File Count in Media File Folders”](#) on page 254.

Local Adapters

See [“Heartbeat Information”](#) on page 245.

AMQ Server Adapters

AMQ server connection is displayed in the AMQ Server Adapters section

Amq Server Adapters

Adapters:	1
Runtime Id:	2
 Health Status Message:	OK
MXF workspace count	1
AMA workspace count	0
URL:	failover:(tcp://gl-mi-as3000-2:61717)? maxReconnectAttempts=5&reconnectDelay=1000&initialReconnectDelay=1000

Refresh

Network Information

In the Network Information view, a green check mark is displayed when the current and initial IP addresses are the same. A warning triangle is displayed when current and initial IP addresses are different, as in the following illustration.

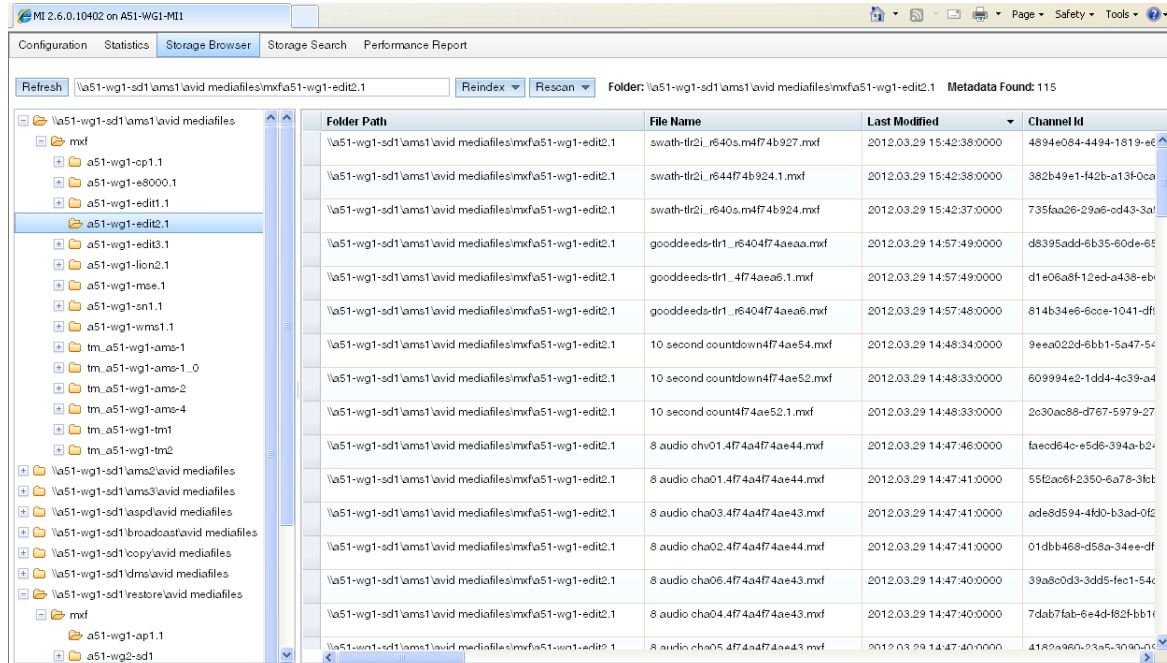
Network Information

	Current IP address:	KBP1-dhp-f54135/192.168.56.1
	Init IP address:	KBP1-dhp-f54135/172.22.206.111
	All IPV4 addresses:	2
		127.0.0.1
		192.168.56.1
	All IPV6 addresses:	2
		0:0:0:0:0:0:1
		fe80:0:0:0:4c86:2c2e:48ee:7342%15

Refresh

Storage Browser Tab

The Storage Browser allows you to browse the storage locations and examine the media files that are being indexed. Click on a folder to view details about the media files. The following illustration shows the contents of a selected folder in the Storage Browser.



The Storage Browser does not update dynamically. Do one of the following to update the display:

- If a storage location is added after you open the page, press the Refresh button to show the new information.
- To see changes within an existing folder, select the folder again.

The Storage Browser displays a variety of information about a media file including the following:

- Full path to the file (Folder Path)
- Channel ID
- FilemobID, original FilemobID, and MasterMobID
- Clip size (in bytes)
- Creation time, start time, duration, and time last modified

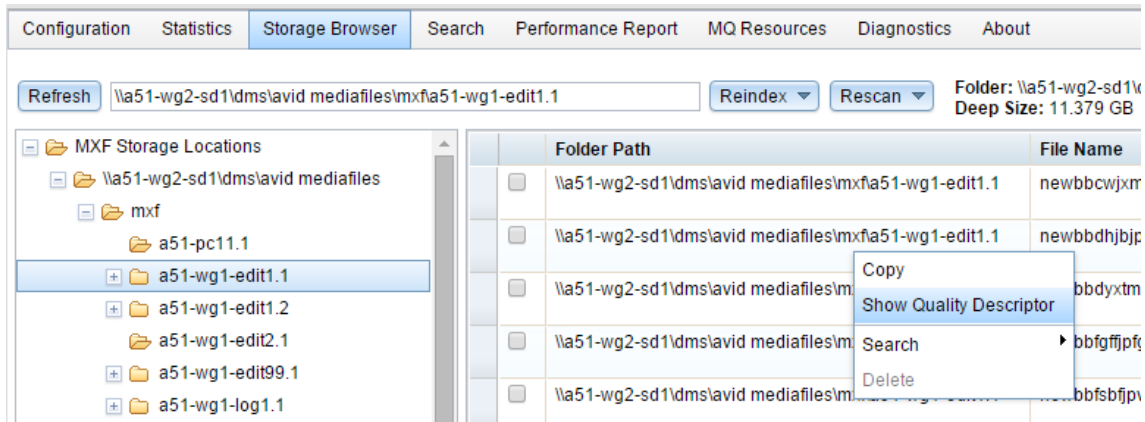
You can right-click an item in the list to view detailed information about the object including codec, quality information, VCID, and timecode information.

Possible uses of the Storage Browser for troubleshooting include the following:

- When investigating dynamic relink and media offline issues, check for problems with the start time, duration, offset, and channel ID.
- When investigating media offline issues in the editor, perform a Reveal File in the editor application to determine the path to the MXF file, then check the Interplay Media Indexer Storage Browser to determine whether the Interplay Media Indexer can locate the file.

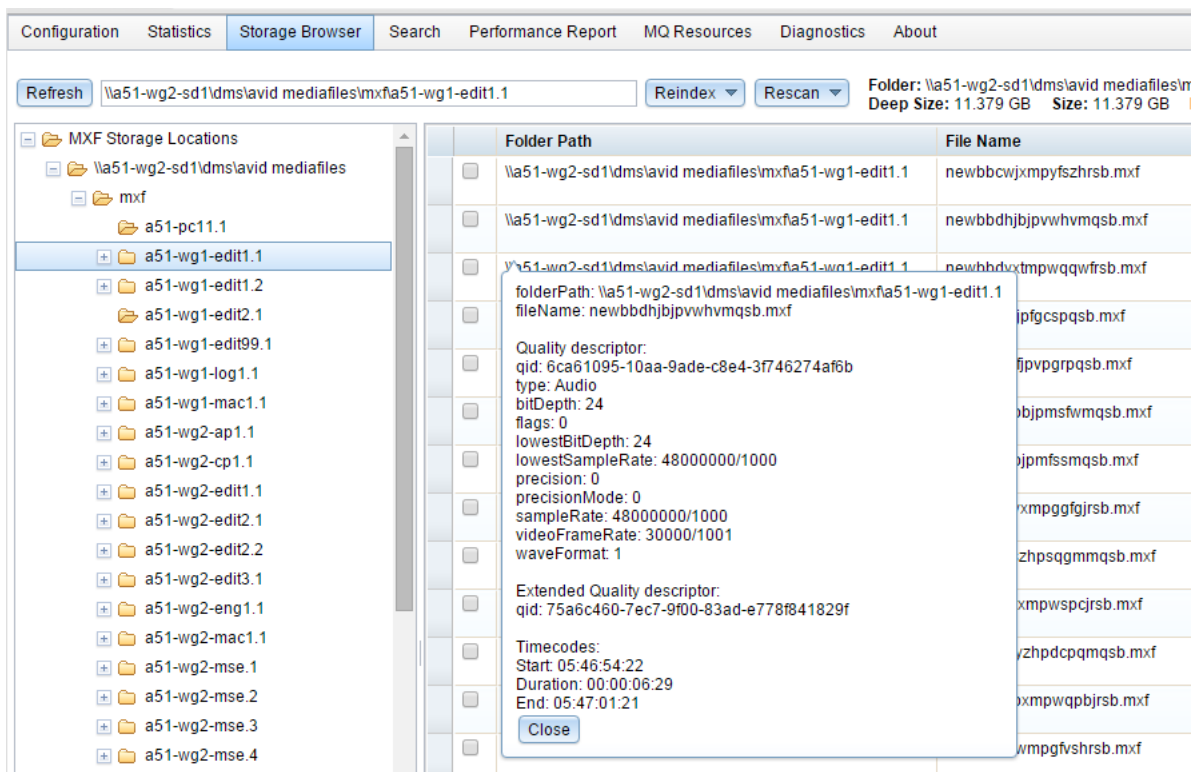
Copy and Search Features

This section describes the context menu options that appear when you right-click an item in the Storage Browser tab.



The following list describes the context menu options:

- Copy allows you to copy the selected entry and for example, paste it into a text file for trouble shooting.
- Show Quality Descriptor displays information about the selected item as shown in the following illustration.

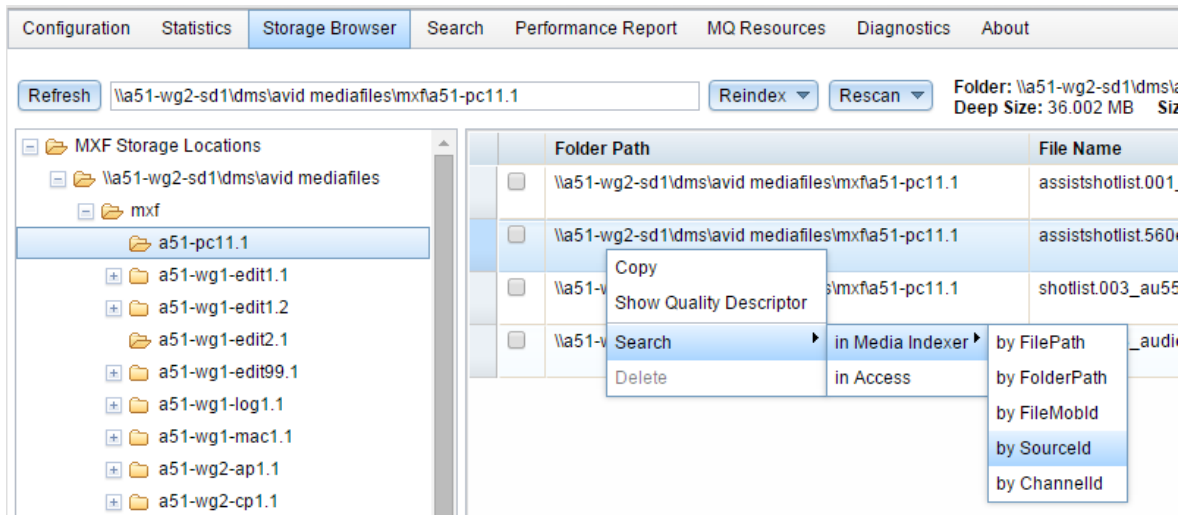


- Search - allows you to automatically search the Media Indexer database for the selected item using several different parameters.
- Delete - allows you to delete spaaf files that belong to certain AMA essences. In most normal workflows spaaf files should be deleted only from Interplay Access.

The following procedure describes how to use the Search feature.

To use the Search context menu item:

1. Right click an item as shown in the following illustration.



The “Search > in Access” item is planned for a future release and does not work for this release.

2. Select the criteria to search. In this example, Sourced ID is selected.

The MI displays a new search tab using the selected parameters and executes the search.

3. Optionally select a item from the search result and perform another search.

This search feature saves time during trouble shooting. In previous releases you had to select an item and then paste the item into a new search tab. Now this step is performed automatically.

This search feature is useful when you want to compare items in Access and MI. It is also useful when you want to search for items that share the similar source ID or the same path, etc.

Reindex and Rescan Buttons

The Reindex and Rescan buttons at the top of the Storage Browser window allow you to perform a reindex or rescan of a particular workspace or folder. You can perform the operation on the top level folder (shallow) or all folders underneath (deep).

The following explains the difference between reindexing and rescanning a folder:

- **ReIndex:** Interplay Media Indexer deletes all indexing information for the files in that folder and then starts indexing the files again. This can create a time period in which no indexing information is available for files in that folder.
- **Rescan:** Interplay Media Indexer reindexes all of the files that have been changed or added since the last time Interplay Media Indexer indexed the folder. It also removes all files that are no longer available on the file system.

A rescan is usually not needed because the Interplay Media Indexer will catch up with changes on the file system by notifications sent from ISIS. However, there are situations, such as network problems, where those notifications might get lost. In that case, a rescan will pick up missed changes.

If a file shows as media offline and it reappears in the web interface after you perform a rescan, it is often a good indication that there may be problems with either your network connection or your connection to ISIS.

See also, [“Changing the Interplay | Media Indexer Full Resync Time”](#) on page 256.

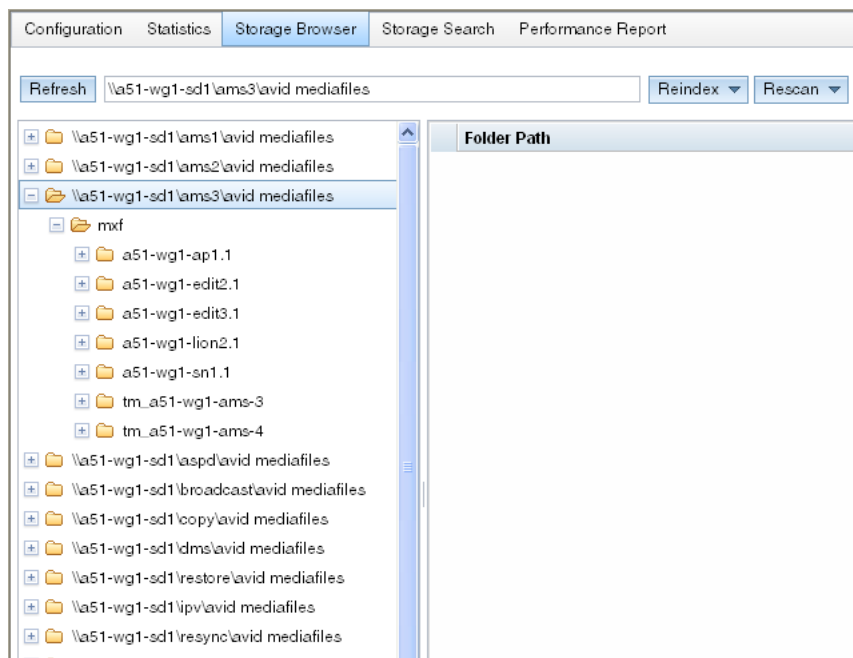


These are CPU intensive operations. Interplay Media Indexer responsiveness will decrease for the duration of the operation. The operation will run in the background (not visible from the web user interface).



For information on how Media Indexer reindexes the files it is monitoring, see [“Estimating the Indexing Time”](#) on page 257 and [“Monitoring the Indexing Process”](#) on page 258.

The following illustration shows an example with multiple workspaces. To rescan the entire workspace named ams3, select the workspace and choose Rescan > Deep. Note that you can also right-click the workspace and select Rescan > Deep.



In the above example, Shallow would have no use since there are no individual files at the top level of the workspace. In the following example, to rescan the folder named a51-wg1-ap1.1, select the folder and choose either Shallow or Deep.

There are two Thread Pool Statistics views for the classic database and two for the MongoDB database (designated as FI).

The Scanning Workers views show the activity for scanning folders. The Indexing Workers views show the activity for scanning files.

The Current Queue Length value shows the number of files or folders that must still be scanned. When the Indexing Workers Queue Length is zero, the MI is done scanning or reindexing.



The display is not dynamic. Click the Refresh button to update the value.

Search Tab

The Storage Search window allows you to search for the following:

- folder path
- file path (supply the full path to a file)
- File mob ID
- Channel ID
- Source ID
- Quality ID (datasource)
- Quality ID (quality descriptor)

The following illustration shows the results of a search for a File mob ID.

Configuration Statistics Storage Browser Search Performance Report MQ Resources Diagnostics About Ci				
060a2b340101010101010f0013-000000-551d813c65c00017-060e2b347f7f-2a80		by file mob id	AMA ☒ MXF ☒ Only licensed ☐	Search Metadata Found: 1
	Folder Path	File Name	Last Modified	Channel Id
☐	\\a51-wg2-sd1\ams3\avid mediafiles\mxf\bsi-pc8.1	bowtie.tif142799698551d813c.mxf	2015.04.02 13:49:31:0000	d141df8c-e76d-4eb5-b9b



You can right-click an item in the results section and create another search. See “Copy and Search Features” on page 230.

Performance Report Tab

The Performance Report tab provides access to Interplay Media Indexer specific performance reports based on JAMon. JAMon stands for Java Application Monitor. It allows developers to monitor applications such as Interplay Media Indexer. This page is intended for Customer Support to help investigate problems occurring on a Interplay Media Indexer system.

JAMon allows you to view data on the web page and to export the data using various formats. For additional information on JAMon, see the following web page:

<http://jamonapi.sourceforge.net/>

Note that the JAMon monitoring is on by default, The performance effect is very low because the application only collects statistical data.

Media Indexer MQ Resource Tab

The MQ Resources tab provides technical details about the embedded messaging infrastructure as shown in the following illustration.

Configuration

Statistics

Storage Browser

Search

Performance Report

MQ Resources

About

Destinations

Refresh

Queues

Queue Name	Producer Count	Consumer Count	Messages Count
queue://queueWithStaleMessages	3	0	3
queue://mq.ChannelIdService.v.3.1.generateChannelIds	0	1	0
queue://mq.ChannelIdService.v.3.1.generateChannelIdsForSource	0	1	0
queue://mq.ChannelInterestRetrieve.v.3.1.getAllClientInterests	0	1	0
queue://mq.ChannelInterestRetrieve.v.3.1.getAllClientInterestsEx	0	1	0
queue://mq.Configuration.v.3.1.change			
queue://mq.Configuration.v.3.1.connect			
queue://mq.Configuration.v.3.1.create			
queue://mq.Configuration.v.3.1.create			
queue://mq.Configuration.v.3.1.disconnect			
queue://mq.Configuration.v.3.1.generate			
queue://mq.Configuration.v.3.1.getConsumer			
queue://mq.Configuration.v.3.1.getMessage			
queue://mq.Configuration.v.3.1.getMessage			
queue://mq.Configuration.v.3.1.isBlocked			

Broker Info

Broker statistics

Broker Version:	5.9.0	Consumer Count	73
VM url:	vm://KBP1-dhp-f54135	Producer Count	3
Job Scheduler Store(%):	0	Total Messages Enqueue:	505
Memory Limit:	524288000	Total Messages Dequeue:	212
Memory Usage(%):	0	Total Messages Count:	3
Store Usage(%):	0	Total Messages Cached:	0
Temp Usage(%):	0		

Topics

Topic Name
topic://ActiveMQ.Advisory.Connection
topic://ActiveMQ.Advisory.NetworkBridge
topic://ActiveMQ.Advisory.Queue
topic://mq.channel.subscriptions
topic://ActiveMQ.Advisory.ConsumerCount
topic://ActiveMQ.Advisory.ConsumerCountSource
topic://ActiveMQ.Advisory.ConsumerCountStats
topic://ActiveMQ.Advisory.ConsumerCountStatsEx
topic://ActiveMQ.Advisory.ConsumerCountStat
topic://ActiveMQ.Advisory.ConsumerCount
topic://ActiveMQ.Advisory.ConsumerCount
topic://ActiveMQ.Advisory.ConsumerCount

queue://queueWithStaleMessages

Refresh

Destination name: queue://queueWithStaleMessages

Memory Limit:	524288000	Enqueue Count:	3
Memory Usage(%):	0	Dequeue Count:	0
Memory Usage:	3096	Expired Count:	0
Consumer Count:	0	Cursor Full:	false
Producer Count:	3	Cursor Size:	0
Producer Flow Control:	true	Cursor Memory Usage:	3096
In Flight Count:	0	Cursor Memory Usage(%):	0

Consumers:

Messages:

- ActiveMQTextMessage (commandId = 5, responseRequired = true, messageId = ID:KBP1-dhp-f54135-64895-1417543191147-0:1:1, originalDestination = null, originalTransactionId = null, producerId = ID:KBP1-dhp-f54135-64895-1417543191147-0:1:1, destination = queue://queueWithStaleMessages, transactionId = null, expiration = 0, timestamp = 1417543191339, arrival = 0, brokerInTime = 1417543191339, brokerOutTime = 0, correlationId = null, replyTo = null, persistent = true, type = null, priority = 4, groupId = null, groupSequence = 0, targetConsumerId = null, compressed = false, userId = null, content = org.apache.activemq.util.ByteSequence@27921471, marshalledProperties = null, dataStructure = null, redeliveryCounter = null, size = 1032, properties = null, readOnlyProperties = false, readOnlyBody = true, droppable = false, jmsXGroupFirstForConsumer = false, text = test)
- ActiveMQTextMessage (commandId = 5, responseRequired = true, messageId = ID:KBP1-dhp-f54135-64895-1417543191147-0:1:1, originalDestination = null, originalTransactionId = null, producerId = ID:KBP1-dhp-f54135-64895-1417543191147-0:1:1, destination = queue://queueWithStaleMessages, transactionId = null, expiration = 0, timestamp = 1417543191339, arrival = 0, brokerInTime = 1417543191339, brokerOutTime = 0, correlationId = null, replyTo = null, persistent = true, type = null, priority = 4, groupId = null, groupSequence = 0, targetConsumerId = null, compressed = false, userId = null, content = org.apache.activemq.util.ByteSequence@27921471, marshalledProperties = null, dataStructure = null, redeliveryCounter = null, size = 1032, properties = null, readOnlyProperties = false, readOnlyBody = true, droppable = false, jmsXGroupFirstForConsumer = false, text = test)

Configuring Media Indexer Servers and Clients

This section describes how to use the Web interface to configure Media Indexer serves and clients.

- [Configuring Media Indexer Servers](#)
- [Identifying the Media Indexer in the Interplay Administrator Tool](#)
- [Configuring the Local Media Indexer on an Avid Editing System](#)

Configuring Media Indexer Servers

To configure the Media Indexer server:

1. Open the web interface on the Media Indexer server using the following URL:

http://<MI system name>:8443/mi/

where *<MI system name>* is the server name or IP address of the system running the Media Indexer. If you are logged onto the MI server, you can use `localhost` for the MI server name.

- Click the Configuration tab and then click the Cache and NOMI Information tab.

The following illustration shows a completed section.

The screenshot shows the 'Configuration' tab selected in the top navigation bar. On the left, under 'Media Indexers', two servers are listed: 'a51-wg2-eng1' and 'a51-wg2-mi1'. The right pane shows the 'Cache and NOMI Information' sub-tab. It contains a checked checkbox for 'Join to Network of Media Indexers (NOMI)'. Below this are four input fields: 'Name' (HAGWG2), 'Weight' (10), 'Save Period (min)' (120), and 'Maximum Files' (10,000). At the bottom of the right pane are four buttons: 'Commit Cache (Changed Only)', 'Commit Cache (Force)', 'Apply Configuration', and 'Refresh'.

- Specify a name for the Network of Media Indexers and click the check box for “Join to Network of Media Indexers (NOMI)”.

Note that in this example the NOMI is named HAGWG2 since it was inherited as part of an upgrade to the MI v3.5 release.

- Enter the weight of this Media Indexer in the NOMI. The Media Indexer with the highest weight is the master of the NOMI.
- Accept the defaults and click Apply Configuration.

To Add Storage Locations to the Media Indexer Server:

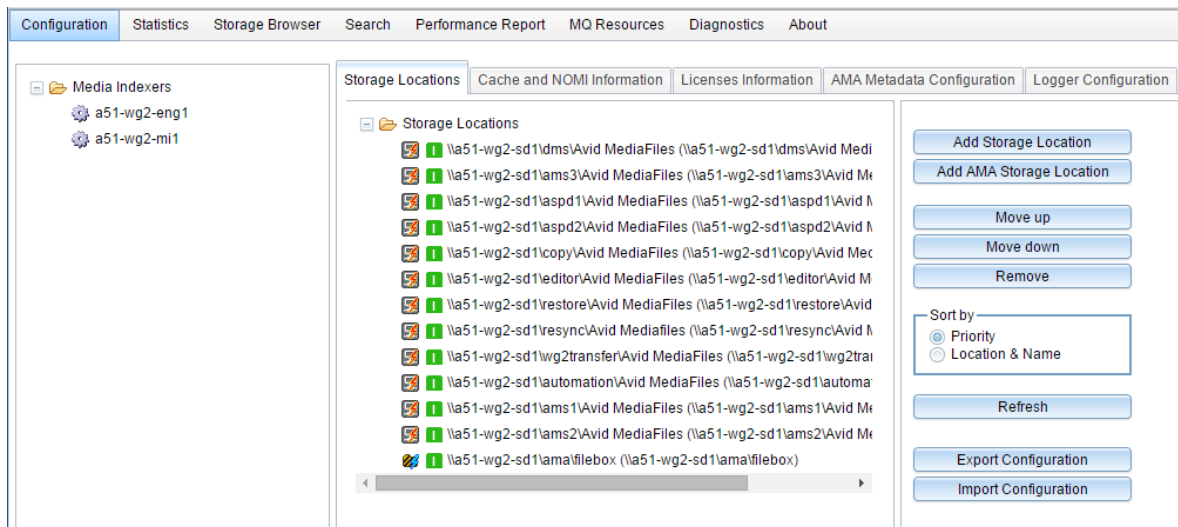
- Click either the Add Storage Location or Add AMA Storage Location button as follows:
 - Use Add Storage Locations for .mxf files on shared storage.
 - Use Add AMA Storage Location for AMA media.



You can only add or modify storage locations on the NOMI leader.

- Click OK to save the configuration.
- Add additional storage locations as necessary. Use UNC path notation for adding locations to index or to describe AMA media locations when pointing to shared storage (ISIS).

The following illustration shows an MI configured with multiple .mxf storage locations.



Adding an Additional Media Indexer to the Network of Media Indexers

You can use the MI web interface to add Media Indexers to the NOMI for redundancy.

To use the web interface to add an additional Media Indexer:

1. Open the web interface for the second MI server.
2. Click the Configuration tab and then click the Cache and NOMI Information tab.
3. Click the “Join to Network of Media Indexers (NOMI)” check box and enter a weight lower than the leader.

If you click the Storage Locations tab you should see the same storages locations as the leader. In this case they will be Read Only.

4. Add the new Media Indexer to the Interplay Administrator as described in [“Identifying the Media Indexer in the Interplay Administrator Tool” on page 237](#).

Identifying the Media Indexer in the Interplay Administrator Tool

The following applications use the Server MI Connection URL setting in the Interplay Administrator tool to locate the Media Indexer servers.

- Avid editing applications
- Production Services Transcode provider
- STP Encode
- Interplay Transfer engine
- Interplay Assist



Starting at Interplay v3.1, Interplay Assist no longer requires the Workgroup Name to connect to the Media Indexer.



Starting in Interplay v3.5, Interplay Access no longer needs the Workgroup Name portion of the Interplay Administration tool to connect to the Media Indexer.

To identify the MI Connection URL in the Interplay Administration tool:

1. Open the Interplay Administrator tool and open the Server Hostname Settings view.
2. In the Workgroup and Media Indexer Settings section, locate the Server MI Connection URL section.
3. Click the Add button to add a Media Indexer server.

The following illustration shows a completed section.

Site Settings **Server Hostname Settings** A51-wg2-eng1 (Administrator) Log

Interplay Production Services Settings

Interplay Production Services Engine: a51-wg2-eng1

Interplay Delivery Receiver Host: a51-wg2-eng1

Interplay Delivery Receiver Port: 33321

Interplay Dynamic Relink Service Host: a51-wg2-stp1

Apply Changes

Workgroup and Media Indexer Settings

Workgroup Name: A51-WG2

AMA Metadata Location (SPAAF): \\a51-wg2-sd1\\AMA\\spaafs

Fallback Shared-Storage Server: a51-wg2-sd1

Server MI Connection URL

Configure Server MI Connection URL

MI Servers
a51-wg2-eng1

failover:(tcp://a51-wg2-eng1:61717)?maxReconnectAttempts=5&rec

Check Media Indexer

Apply Changes

4. Add the name of the MI server and click Apply Changes. Note that this example uses an Interplay Bundle setup where the Interplay Engine, Media Indexer, and Interplay Production Engine are all running on the same system.
5. If you have more than one MI in the Network of Media Indexers, add the other MI servers also. Note that additional Media Indexer servers must join the NOMI in order to connect to the URL.

The system creates a URL for the Media Indexer server and displays the URL below the MI Servers window. The URL contains the MI server name and port 61717. This is the default port used to connect to the Media Indexer server.

Configuring the Local Media Indexer on an Avid Editing System

You do not explicitly add the Network of Media Indexers to the local Media Indexer configuration. If the “Server MI Configuration URL” is correctly set in the Interplay Administration tool, the Avid editing application will automatically add the NOMI connection to the local MI configuration when you log into Interplay.

Do not use the “Connect to AMQ server MI” button on the web interface to make an explicit connection to the Network of Media Indexers. This option is overridden by the value in the Interplay Administrator tool at login time.

To connect an Avid editing application to the Network of Media Indexers:

1. Configure the Administrator tool to recognize the NOMI. See [“Identifying the Media Indexer in the Interplay Administrator Tool” on page 237](#).
2. On your Avid editing system, start the editor and then log into the Interplay database using one of the following methods:
 - Log in using the Project Select dialog that opens when you start the editor.
 - Log in using the Interplay Server and Interplay User dialogs in the Settings window.

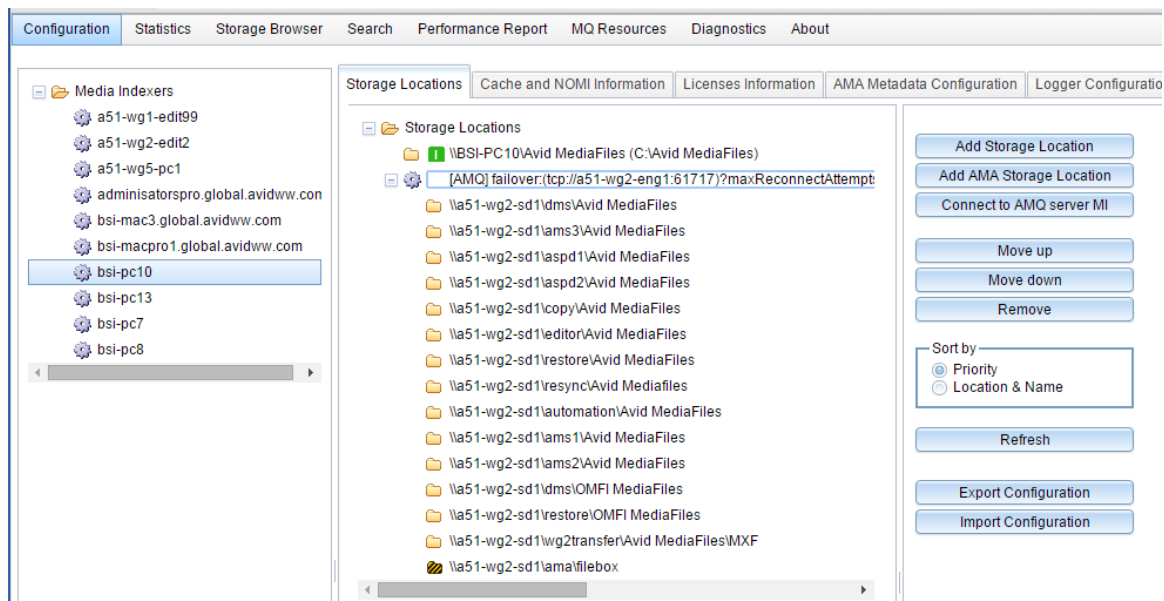
After you log into the Interplay database, your local Media Indexer will automatically connect to the Media Indexer in the Network of Media Indexers



When you start the Avid editing application, there is an initialization period named “Connecting to the Media Indexer.” This process connects to the local Media Indexer.

3. (Option) Open the Media Indexer web interface, select the Avid editing system, and view the connection to the Network of Media Indexers.

The system displays the local storage locations as well as the storage locations for the Media Indexer server as shown in the following illustration. In this example the name of the local editor system is bsi-pc10.



Managing Interplay | Media Indexers in an Interplay Environment

This section covers the following topics:

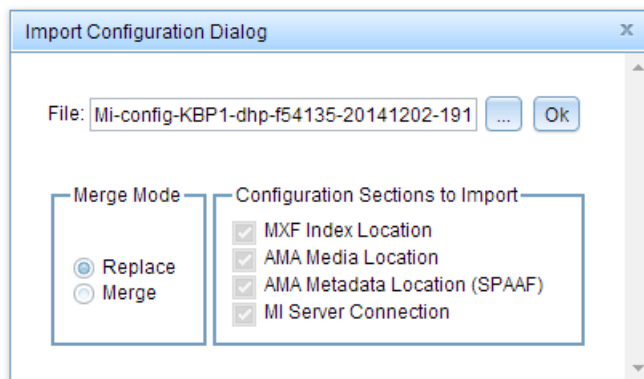
- [Import and Export MI Configuration Files](#)
- [Media Indexer Web UI Security](#)
- [Media Indexer Web UI Shutdown](#)
- [Multiple ISIS Support](#)
- [Managing the Order of Storage Locations](#)
- [Viewing the Size of Indexed Workspaces and Folders](#)

Import and Export MI Configuration Files

You can export and import a Media Indexer configuration, which can save time and effort when setting up another instance of Media Indexer. The following information is included in the exported .cfg file:

- MXF Index Locations
- AMA Media Location
- AMA Metadata (SPAAF) Location
- MI Server Connection

When importing a configuration file, you have options for which configuration information to import and whether to replace or merge the configuration.

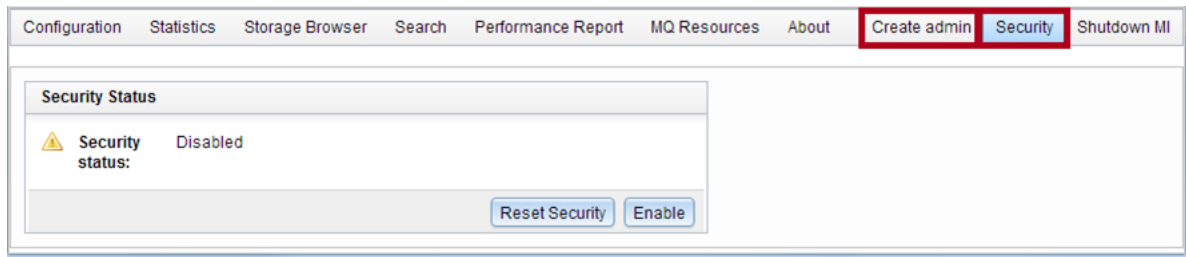


Configuration files created by the MI web interface are incompatible with files previously created by ASF.

Media Indexer Web UI Security

Use the Security page to add security to the MI web interface as follows:

- Click the Enable button to enable security. Security is disabled by default.
- Create an Administrator account for secure access. This is the only account available. The password is stored in userCache.cache. The security status and password can be reset when the Media Indexer Web UI is opened on the machine on which it was installed.
- If Security is enabled, you must log in as administrator is to make changes to the UI.



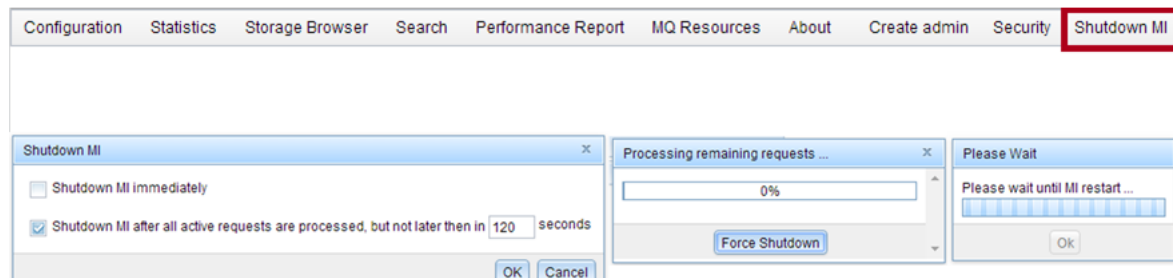
Media Indexer Web UI Shutdown

You can shut down Media Indexer from the Web UI. Shutdown is available only when logged on to the system displaying the Web UI.

There are two options for shutdown:

- Shutdown immediately
- Shutdown after all requests are processed.

After a shutdown, the browser page will be automatically loaded when Media Indexer starts up again.



Multiple ISIS Support

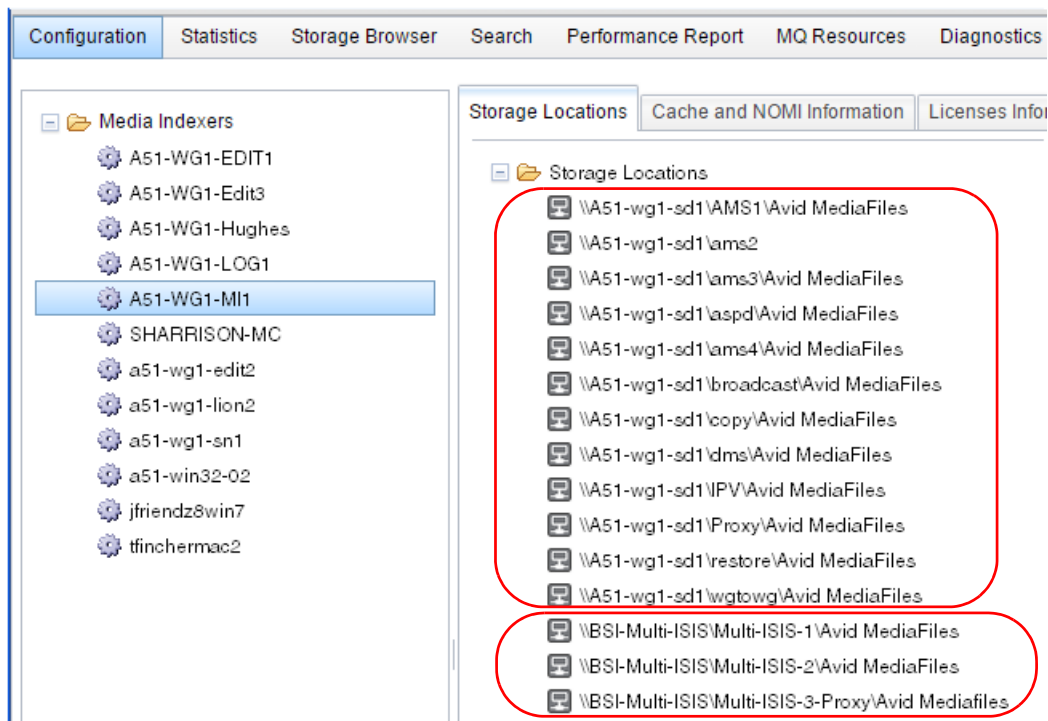
One Interplay Media Indexer can index two or more ISIS systems in the same Interplay workgroup. The following limitations apply:

- Each Interplay Media Indexer in the Network of Media Indexers must have access to all of the available ISIS workspaces. For information on setting up and configuring ISIS systems, see the Setup Guide for your ISIS product.

- Configure the ISIS client on the Interplay Media Indexer to recognize all available ISIS systems. Use the Remote Host Settings button to connect to additional ISIS systems as shown in the following illustration.

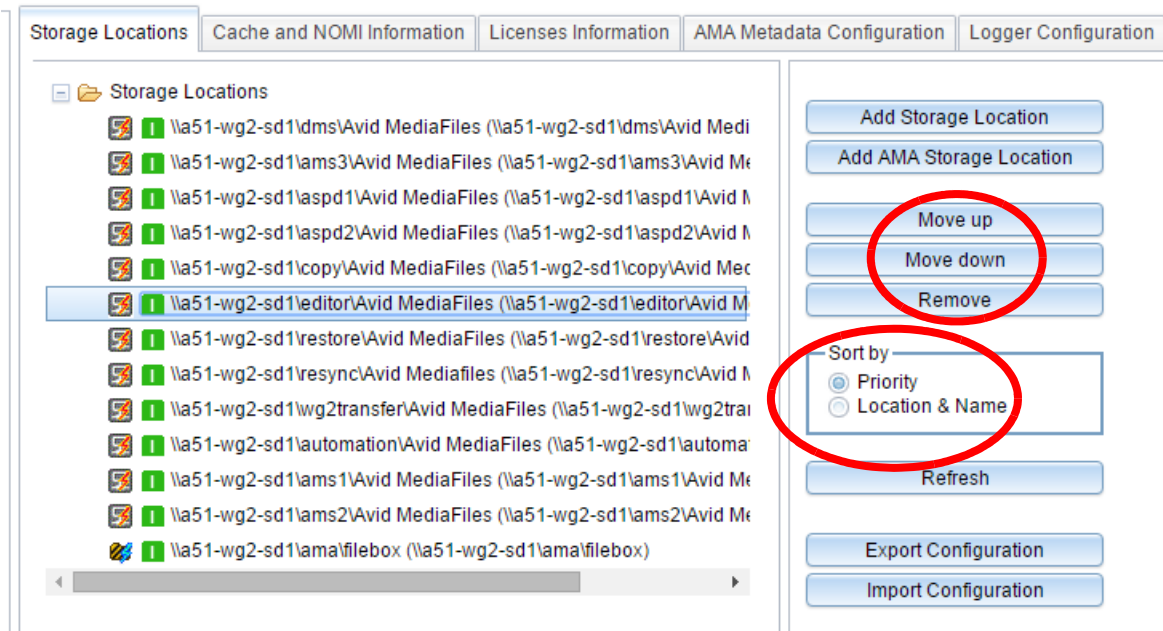


- Configure the leader of the Network of Media Indexers to see all of the storage locations that you want to access on both ISIS systems. For example, the following illustration shows connections to the workspaces on two ISIS systems. The two System Directors are named A51-wg1-sd1 and BSI-Multi-ISIS.



Managing the Order of Storage Locations

You can use the Configuration tab on the web interface to change the order of storage locations. The following illustration shows the related interface elements.



- Move up and Move down buttons: Allow you to move the storage locations within the list. When Sorting by Priority is enabled, you can manage the priority by moving the storage locations up and down in the list. The highest priority storage location is at the top of the list.
- Sort by buttons: Allow you to switch between the default priority order and alphabetical order.



In the case of a local Interplay Media Indexer, the local results are always returned before the results from the Network of Media Indexers.

Viewing the Size of Indexed Workspaces and Folders

You can view the total size of the files indexed by a storage location and by a particular folder. The following illustration shows the total size of each storage location in the Statistics tab.

The screenshot shows the 'Statistics' tab in the Interplay Media Indexer interface. The 'Storage Statistics' section displays the following data:

Storage Workspaces	
Workspaces:	15
Workspace \\a51-wg1-sd1lams1lavid mediafiles	
Folders:	13
Files:	3120
Total size:	2.179 TB
Unc Path:	\\a51-wg1-sd1lams1lavid
Runtime Id:	1
Flags:	1
Recording Type:	Unknown
Workspace \\a51-wg1-sd1lams2lavid mediafiles	
Folders:	3
Files:	232
Total size:	255.885 GB
Unc Path:	\\a51-wg1-sd1lams2lavid
Runtime Id:	2
Flags:	1
Recording Type:	Unknown



The Interplay Media Indexer only calculates the size of files that are indexed. If there are non-indexed files in the folder they are not reported in this value.

The following illustration shows the size of folders in the Storage Browser tab. The Deep value is the size of the currently selected folder and all subfolders under this folder. The value on the right is the size of the currently selected folder. In this example the top folder is empty but the subfolders contain indexed media.

The screenshot shows the 'Storage Browser' tab in the Interplay Media Indexer interface. The 'Folder Path' field shows the selected path: \\a51-wg1-sd1lams1lavid mediafiles\mxf. The 'Deep Size' is 2.179 TB, and the 'Size' is 0 bytes. The 'Metadata Found' is 0. The folder structure is displayed in the left pane, showing the selected folder and its subfolders.

Troubleshooting Interplay | Media Indexer

This section describes what to do if you have media that is showing as offline. It usually means that the Interplay Media Indexer does not see the media; there are several things to check.

Before you begin troubleshooting, verify the following:

- Shared storage volumes (ISIS workspaces) are mounted and the user that mounted the workspace has Read/Write privileges.
- On Avid ISIS, Interplay Media Indexers that monitor shared storage must use letterless drive mapping. In general other Interplay clients, such as Avid editing applications, must use Lettered drive mappings. For more information on mounting drives, see the Avid ISIS Client Manager Help.
- The editing application is logged in to the Interplay Engine with a user that has sufficient privileges.
- The Windows firewall is disabled.
- It is possible to ping the machine that runs the shared storage Interplay Media Indexer by IP address and by name from the editing system and vice versa.

Quarantined Files

MI v3.8 and later defaults to a new approach for handling corrupted files or files that the MI is unable to work with. MI no longer moves files into quarantine folders. Now the MI internally marks the files as to-be-ignored.

Ignored files will be reconsidered for processing when the file modification time changes. This new approach maintains separate to-be-ignored lists for each indexing approach; one for classic and one for format independence. Now a file that is not supported by one indexing approach may still be supported by the other approach.

Heartbeat Information

You can view heartbeat information for each workspace being monitored by the Interplay Media Indexer. Heartbeat refers to the network connectivity between the Interplay Media Indexer and the shared storage workspace.

The system displays the Heartbeat information for each workspace in the Local Adapters view as shown in the following illustration.

Local Adapters	
Adapters:	12
Adapter:	\\a51-wg2-sd1\ams1\Avid MediaFiles
Health Status Message:	Normal
Monitor Root:	\\a51-wg2-sd1\ams1\Avid MediaFiles
Runtime Id:	15
Flags:	1
Internal file addition notifications:	24
Internal file update notifications:	35
Internal file removal notifications:	0
Internal folder addition notifications:	0
Internal folder update notifications:	133
Internal folder removal notifications:	0
Created file last notification:	1/6/16 1:51 PM Eastern Standard Time
Deleted file last notification:	None yet
Full resyncs count:	6
ISIS drive type:	ISIS_5000 (REALTIME)
Time since last heartbeat notification triggered (in s):	584
Time since last heartbeat notification received (in s):	577
Last heartbeat notification delay (in ms):	6851

The “Last heartbeat notification delay” is the value of interest. Check the value periodically and record the value. Over time you will get an idea of the normal delay value for that workspace. If you notice a spike in the value it may indicate that there is a problem with the connectivity to shared storage. Check the ISIS System Director Administrator user interface to see if there are any problems or additional loads being placed on the shared storage system.

In most cases a delay of up to 30 seconds is acceptable. If it takes several minutes, it is a sign that there may be a problem with the connection between Interplay Media Indexer and ISIS shared storage.

Each received heartbeat notification is also logged to the storage-notification log in the following directory:

%Program Files%\Avid\AvidMI\state\log

If a problem occurs you can check the log to determine when the problem started and correlate it with any other occurrences in your Interplay environment that happened around the same time. For example, it can be useful to check the heartbeat values if there is a problem with new media not coming online fast enough. In that case, compare the values on the other Interplay Media Indexers in the Network of Media Indexers to see if there is a problem with one of them or to rule them out as part of the problem.

The default for the heartbeat is 600 seconds (10 minutes). You can change the heartbeat interval value by editing the following line in the jini.config file:

```
static notiactionsHeartbeatInterval = new Integer( 600 );
```

The jini.config file is located in the following directory on the Interplay Media Indexer server:

%Program Files%\Avid\AvidMI\state\config\jini

Log File and Cache File Locations

Log and cache files are located in the following folder on Windows 7, Windows 8, Windows 10, Windows Server 2008, and Windows Server 2012:

%ProgramData%\Avid\AvidMI

This usually translates to the following location:

C:\Program Data\Avid\AvidMI

On Windows Server 2003, log and cache files are in one of the following locations:

- log files: %Program Files%\Avid\AvidMI\state\log
- cache files: %Program Files%\Avid\AvidMI\state

On a Macintosh editor, the log and cache files are in the following location:

- log files: /Applications/Avid Media Indexer/state/log
- cache files: /Applications/Avid Media Indexer/state

Archived Log Files

Log files are now automatically archived in the following folder:

%ProgramData%\Avid\AvidMI\log\archive

This usually translates to the following location:

C:\Program Data\Avid\AvidMI\log\archive

One zip file per day is stored in the folder with the following limitations:

- If there is less than 10 GB free space on the drive that stores the log and cache files, the Interplay Media Indexer starts removing zip files, oldest first, until more than 10 GB is free. Note that the current day's archive is not removed.
- Zipped log archives are removed automatically by size and age restrictions. The details of the restrictions are configurable in the jini.config file at the following location on a Windows system:

%Program Files%\Avid\AvidMI\state\config\jini

On a Macintosh editor, the log archives and jini config files are in the following locations:

log archive files: /Applications/Avid Media Indexer/state/log/archive

jini config file: /Applications/Avid Media Indexer/state/config/jini

Default jini config file values

The default values for configuring the saving and purging of archive files are as follows:

```
// Period for scheduling task which purges log archives (in milliseconds).
static archivesPurgerPeriod = new Long( 300000 );

// Retained archives count for task which purges log archives.
static archivesCount = new Integer( 60 );

// Retained archives size for task which purges log archives (in bytes).
static archivesSize = new Long( 104857600 );
```

Media Indexer Cache Files Format

This section describes the MI cache files. Note that cache files are binary files and are not intended to be human-readable.

adapters.cache

The adapters.cache file contains information about which workspaces or file locations are configured for indexing. In the case of a local Interplay Media Indexer, it also identifies the Interplay Media Indexer server in the Network of Media Indexers that the local Interplay Media Indexer is connected to. For those familiar with the Interplay Media Indexer cache files, the information that used to be contained in the ummds.cache files is now in adapters.cache.

quality.cache

The quality.cache file contains information about the quality or resolution of indexed files. For those familiar with Interplay Media Indexer cache files, this information used to be in the ummds.cache file.

metadata_xx_y.cache

The metadata cache files are the persistent storage for the actual essence metadata that the Interplay Media Indexer is monitoring. While the adapters.cache file has the information about which locations on the file system to index, the metadata_xx_y.cache file contains the result of indexing those locations. On a local Interplay Media Indexer, the results will only contain data that is indexed locally, while on a server Interplay Media Indexer, the results will usually contain results from indexing shared storage.

For those familiar with Interplay Media Indexer cache files, the information that used to be contained in the channelxxx.cache is now in the metadata_xx_yy.cache files. The value for xx is always 20, and the values for y range from 1 to 20. Data will be spread across these files.

adapters.txt

Adapters.txt is a readable file that contains the configured workspaces. Interplay Media Indexer keeps that in sync with the caches but does not need this for loading data from the cache. The file is informational only and is overwritten each time the cache is flushed or the storage locations are changed. The file is basically a readable version of adapters.cache. Interplay Media Indexer does not read any information from the file.

The following table lists the MI database files including the cache files.

Type	Naming Scheme	File Type	Available since MI version
MXF Media MetaData	metadata_xx_yy.cache	MXF-DB-File	NA
MXF MetaData Indexes	mdLinks.cache	MXF-DB-File	NA
MXF Plugin-ID-to-Name-Mapping	plugins.cache	MXF-DB-File	v3.0.x
MXF Quality Descriptor	quality.cache	MXF-DB-File	NA
MXF Quality Descriptor Index	quality_idx.cache	MXF-DB-File	NA
MXF Index Locations	adapters.cache	MXF-DB-File	NA
AMA Media MetaData	amaMetadata_xx_yy.cache	AMA-DB-File	3.0.x
AMA MetaData Indexes	amaMdPathLinkCache.cache	AMA-DB-File	3.0.x
AMA Plugin-ID-to-Name-Mapping	amaPlugins.cache	AMA-DB-File	3.0.x
AMA Quality Descriptors	amaQuality.cache	AMA-DB-File	3.0.x
AMA Quality Descriptor Index	amaQuality_idx.cache	AMA-DB-File	3.0.x
AMA Essence Locations	amaAdapters.cache	AMA-DB-File	3.0.x
Storage Licenses	licenses.cache	Internal	v3.1.x
Password Store	mi.keystore	Internal	v3.2.x
User Store	userCache.cache	Internal	v3.2.x

Cache File Behavior for Local Media Indexers

The local Media Indexer does not cache data that is available on the Interplay Media Indexer server. Refer to the server cache files for that information.

Media Indexer utest_amq Application

The Media Indexer utest application is not supported for MI v3.5 and later.

The new replacement for utest is utest_amq which provides all the features that were present in utest along with several new features. Note that utest_amq is a command line interface and does not provide a user interface mode or a topology print. The following illustration shows the utest_amq command and its options.

```

Administrator: C:\Windows\System32\cmd.exe
C:\Program Files\Avid\AvidMI\bin>utest_amq.bat
C:\Program Files\Avid\AvidMI\bin>rem Copyright 2013 by Avid Technology, Inc.
"..\.jre\bin\java.exe" -server -Xverify:none -Xms128m -Djava.security.manager -Djava.securit
edia-server.jar" com.avid.mi.ummd.test.Utest

utest_amq usage:
  utest_amq [parameters of connection] [activetest | passivetest | poisontest | largemessag
EventListenerTest [test parameters]

    parameters of connection  host1 ! protocol://host2,host3:portNumber failover() will be
                                and instead of missing parameters will use default parameter
                                url::protocol://host:port ! url::failover:(protocol://host:

    activetest                Does lookups on a regular basis to stress
                                Parameters: frequency(per second) duration
    passivetest               Registers extra listeners (for channels a
                                Parameters: number of listeners, duration
    poisontest                 Sends poison message on a regular basis to
                                Parameters: name of queue, requests per s
    largemessagetest           Sends large message on a regular basis to
                                Parameters: name of queue, path to file,
    expmessagetest             Sends message with expiration time on a r
                                Parameters: name of unique queue, expirat
    heapdump                   Create heap dump file of Media Indexer
                                Parameters: Hostname, port, path to file
    beanreport                  Create text report with findings beans
                                Parameters: Hostname, port, path to file
    UnresponsiveWorkspaceEventListenerTest
                                UnresponsiveWorkspaceEventListenerTest All
                                Parameters: (without parameters)
    UnresponsiveChannelEventListenerTest
                                UnresponsiveChannelEventListenerTest Alloc
                                Parameters: sourceId slotId
    metadata-extractor [path-to-file] Print extracted md in json format

Examples:
utest_amq url::nio://localhost:61616 passivetest 10 20
utest_amq url::failover:(tcp://localhost:61616) activetest 100 20
utest_amq tcp://localhost poisontest mi.MediaBrowseService.v.3.1.listMediaByPathPaged 10 2
utest_amq localhost largemessagetest mi.MediaLookup.v.3.1.findPathByFileMobId pathIo/msg.
utest_amq localhost expmessagetest QueueExpiredMessagesTest 10 10 20
utest_amq localhost 24444 heapdump pathIo/heapDump
utest_amq localhost 24444 beanreport pathIo/report org.apache.activemq:*
utest_amq localhost UnresponsiveWorkspaceEventListenerTest
utest_amq localhost UnresponsiveChannelEventListenerTest 060a2b340101010101010f0013-000000
utest_amq metadata-extractor D:\testmedia\AB1c106f.4A01.4F2054F205376.mxf
  
```

The command resides in the following directory:

Windows:

%Program Files%\Avid\AvidMI\bin --> usually C:\ProgramFiles\Avid\AvidMI\bin

Mac OS X:

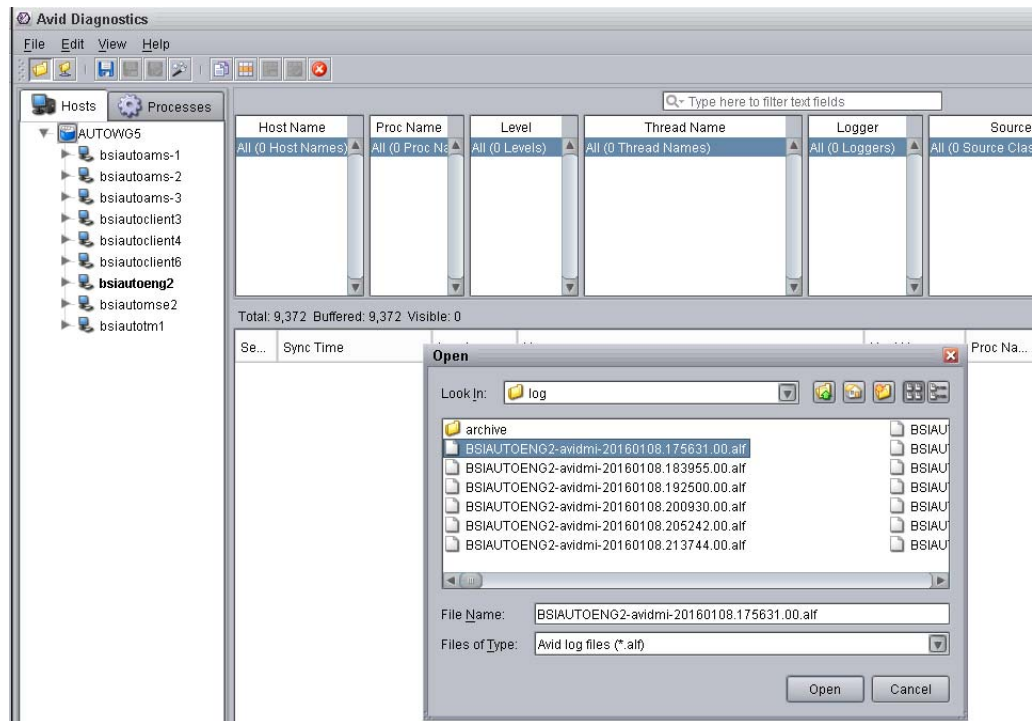
/Applications/Avid/Media Indexer/bin

Type “utest_amq.bat” on Win or “./utest_amq.sh” on Mac for a list of the available options.

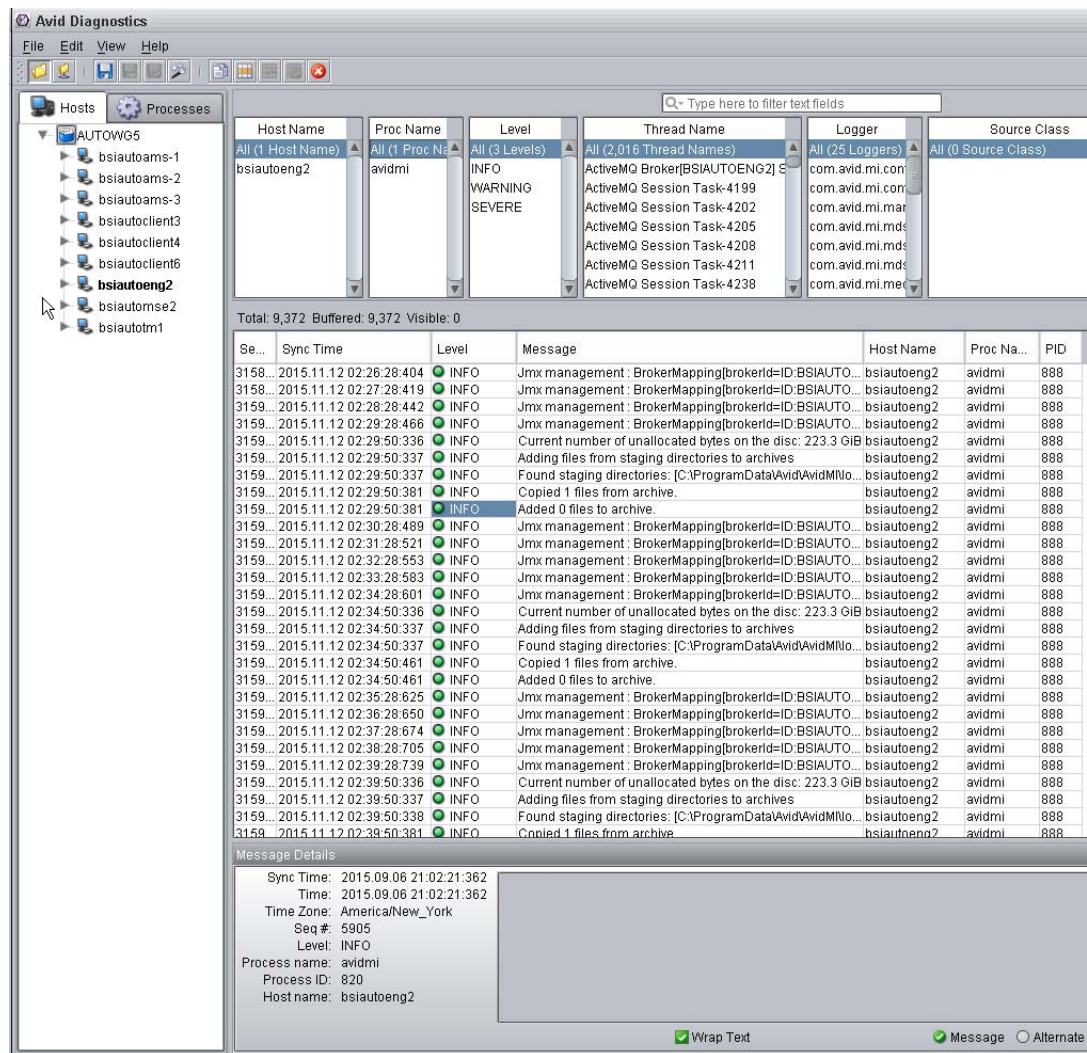
Media Indexer and the ASF Diagnostics Tool

The ASF Diagnostics tool is supported for Media Indexer v3.5 and greater. However, the Live Mode is no longer supported because the MI no longer appears in the ASF process tree. You can still open individual .alf log files, a folder, or a zip with .alf files in the Diagnostics tool.

To open individual .alf log files in the Diagnostics tool, copy the log files to a system running ASF and use File > Open or File > Folder Open as shown in the following illustration.



The following illustration shows the .alf file open in the Diagnostics tool.



You can filter on the following values to obtain diagnostics information.

- Filter on the logger name “com.avid.ummd” to filter all messages issued by the Interplay Media Indexer. For example, this can provide additional information about quarantined files.
- Filter on logger name “com.avid.workgroup” to filter messages issued by the Service Framework. For example, this can provide information regarding Interplay Media Indexer election issues.
- Filter on logger name “UMMDTableCacheLoader” to determine problems with loading the cache file.
- Filter on “INDEXING ERROR” to view indexing errors as they happen.
- Filter on “Quarantined file” to view information on bad media files as they are quarantined. Note that if you see a lot of Quarantined file messages about OMFI media files, it can be a sign of the presence of legacy OMF files, or still unsupported OMF precompute media files.
- Filter on the Message “Commit” to see when the Interplay Media Indexer saved the cache.

Audit Trail Logging

The MI audit trail logging mechanism allows you to collect all Interplay Media Indexer events that could affect file count. It uses a different logger name than the default Java logger so that it can be configured separately.

To enable and configure the audit trail logging capability:

1. Open the following file in an application such as Notepad:
`.\AvidMI\state\config\logging.properties`
2. Locate the following line:
`#com.avid.workgroup.diagnostics.logging.AvidFileHandler.count = 100`
3. Remove the # sign to uncomment the line, and change the value to 500 as follows:
`com.avid.workgroup.diagnostics.logging.AvidFileHandler.count = 500`
4. Add the following line after the modified line:
`com.avid.ummd.util.AuditTrail.level = FINE`
5. Save the file and restart the Interplay Media Indexer.
 This enables the new audit trail mechanism to track file system operations. This level of logging can be increased to FINER or FINEST if more detailed information is required.



Only use the audit trail for troubleshooting. The feature uses Interplay Media Indexer resources and can affect system performance. After you finish troubleshooting, comment out the modified lines and restart the Interplay Media Indexer. For example:

```
#com.avid.workgroup.diagnostics.logging.AvidFileHandler.count = 500
#com.avid.ummd.util.AuditTrail.level = FINE
```

Storage Notification Logging

Media Indexer provides a logging feature that collects information on notifications from shared storage. The information is stored in the following log file:

`.\AvidMI\state\config\storage-notifications-x.alf`

where *x* is the index of the file.

By default, the Interplay Media Indexer logs warnings in the file when folders are found with more than 5,500 (warning) and 10,000 (severe warning) files.

In addition, the number of log files saved and the file size limit for each log file are configurable.

To modify the number of saved log files and the size of saved log files:

1. Open the following file in an application such as Notepad:
`.\AvidMI\state\config\logging.properties`

2. Locate the following lines:

```
# the size limit for each log file in bytes.
# Default value is 2 * 1024 * 1024 = 2097152 = 2 MB
com.avid.ummd.util.StorageNotificationsHandler.limit = 2097152
# the log files count. Default value is 10
com.avid.ummd.util.StorageNotificationsHandler.count = 10
```

3. Modify either the limit (file size) or count (number of log files) value as necessary.
4. Save the file and restart the Interplay Media Indexer.

For additional information on using the logging properties file, see [“Audit Trail Logging” on page 253](#).

Improving Interplay | Media Indexer Performance by Limiting the File Count in Media File Folders

Avid strongly recommends that you keep no more than 5,000 files in an ISIS folder that is being indexed by a Interplay Media Indexer. The number of files per folder affects the speed at which the Interplay Media Indexer can traverse the folder. Interplay Media Indexer must traverse a folder when one of the following occurs:

- When the Interplay Media Indexer starts up and must rescan the folders
- When the Interplay Media Indexer starts up with an empty cache
- During a daily resync
- When you delete, move, or rename a file or folder

If you have more than 5,000 files in a folder you can use one of the following techniques to redistribute the files:

- Use the Production Services Move Provider to move the media files associated with master clips
- Move the files manually as described in the following procedure

To manually move media files to a different media files folder:

1. Identify the folder that contains too many files and determine which shared storage workspace it resides on.
2. Open the Configuration tab on the Media Indexer Web interface and remove the workspace from the Storage Locations section.
3. Create a new folder on the same shared storage workspace and move enough media files to get the file count down below 5,000 files. For example, sort by creation date and move all the files created after a certain date to the new folder.
4. Add the workspace back into the Interplay Media Indexer storage configuration.

The files will still be online in the Avid editing application but they will be offline in Interplay Access.

To bring the files back online in Interplay Access, do one of the following:

- ▶ In Interplay Access, select all the files that you moved, right-click the selection, and choose Update Status from Interplay Media Indexer. For example, search by creation date, select all the files, and then perform the Update Status from Interplay Media Indexer.
- ▶ Load the files into the Avid editing application monitor and then check the files back into the Interplay database.

Limiting the Number of Files Captured by the Editor or by Interplay | Production

The Avid editing applications keep track of the number of media files in a folder and automatically create a new folder when the total exceeds 5,000 files. This includes files created via the editor's Capture command as well as files created from other methods in the editing application such as rendering and consolidating. The new folder has a .x extension such as *FolderName.2*. For earlier versions of the editing applications you need to keep track of the number of media files in a folder. For performance reasons you should split up the files in the folders after they reach 5,000 files as described in this section.

In general, you should create a workflow that avoids capturing too many files to a single folder. If you know that you will have recurring ingests, you should manage the folders for those ingests. For example, you could name folders for the days of the week or for specific types of ingest. Then you could regularly empty those folders as part of your delete workflow. For additional information, see the chapter on Delete Guidelines in the *Interplay Best Practices Guide*.

Interplay | Media Indexer Restart Behavior

You have the following options when you restart a Interplay Media Indexer that has existing cache files:

- **REINDEX_CHANGED:** This is the default option. The Interplay Media Indexer indexes media files that have been added or changed within the last two hours since the cache was last saved. The amount of time is configurable. The Interplay Media Indexer does not reindex all known files to validate the cache.
- **REINDEX_ALWAYS:** This is identical to the default behavior of previous releases as follow:
 - Index any media files added to the storage locations since the cache was last saved.
 - Reindex all known files to validate the cache information.
- **REINDEX_CHANGED_LOCAL:** Reindex new or changed files on local storages and reindex all known files on shared storage locations.
- **REINDEX_CHANGED_SHARED:** Reindex new or changed media files on shared storage locations and reindex all media files on local storage locations
- **REINDEX_NEVER:** Do not check if any media files have been added or changed since the cache was written and do not validate the cache. This gives the fastest performance at startup but should only be used temporarily on the recommendation of Avid support.

You can set these options in the `jini.config` file located at the following location:

`%Program Files%\Avid\AvidMI\state\config\jini\jini.config`

The following shows the relevant portion of the file:

```
// Tolerance on reindexing of known files
// Media files modified up to this specified time, before the last time
// the MI cache was saved will NOT need to be re-validated.
// (depends on reindexingMode)
static timeStampOptimizationErrorMargin = new Long( 7200000 ); //2 hours

// Used to set reindexing mode for known files
// Reindexing of changed files adjusted by
// timeStampOptimizationErrorMargin parameter
// Possible values:
```

```
// REINDEX_ALWAYS - Always reindex known files
// REINDEX_NEVER - Never reindex known files
// REINDEX_CHANGED - Reindex known files on all storages if they were
// changed
// REINDEX_CHANGED_LOCAL - Reindex known files on local storages
// if they were changed and all known files on shared storages
// REINDEX_CHANGED_SHARED - Reindex known files on shared storages if
// they were changed and all known files on local storages
// Default value: REINDEX_CHANGED
static reindexingMode = ReindexingMode.REINDEX_CHANGED;
```

The time range value is in milliseconds and is set to 2 hours by default. After you edit and save the file, restart the Interplay Media Indexer service as described in [“Media Indexer Web UI Shutdown” on page 241](#).

Changing the Interplay | Media Indexer Full Resync Time

The full Resync time is configurable. By default it is once a day at 1:00 AM.

To change the full resync time:

1. Edit the following file:
`%Program Files%\Avid\AvidMI\state\config\jini\jini.config`
2. Change the resync time. The following shows the relevant portion of the file:

```
// Default time for workspaces with scheduled time resync policy
static scheduledResyncTime = "01:00:00 AM";
```
3. Save your changes and restart the Interplay Media Indexer service as described in [“Media Indexer Web UI Shutdown” on page 241](#).
 See also, [“Viewing the Scan or Reindex Progress” on page 233](#).

Indexing Files Starting with a Period

Starting at Interplay Media Indexer v3.3.1 and v3.1.4, files starting with a period (“.”) are not indexed by default. This change was made in response to problems indexing some resource fork files generated by Mac OS X systems. Resource fork files start with a period and underscore (._).

If you wish to change the default behavior, you can edit the jini.config file on the MI server. The procedure in this section shows how to instruct MI to ignore files beginning with “._” and index other files start with a period.

To instruct MI to index files that start with a period:

1. Locate the following file on the MI server:
`%Program Files%\Avid\AvidMI\state\config\jini\jini.config`
2. Edit the file in an application such as Notepad and search for the block of text beginning with “static excludePatterns” as shown below. Line numbers are added for reference. Note that the lines may be numbered differently in different versions of the file.

```
113 static excludePatterns = new Pattern[] {
114 Pattern.compile( ".*\\\\\\Creating((.*\\\\\\.*)|\\\\\\.*)?", flags ),
115 Pattern.compile( ".*\\/Creating(\\/.*)?", flags ),
116 Pattern.compile( ".*\\/Creating((.*\\/.*)|\\\\\\.*)?", flags ),
117 Pattern.compile( ".*\\\\\\+Quarantined Files(\\\\\\.*)?", flags ),
```

[illegible]

You will edit the last three lines of this block of text.

3. Edit lines 123 and 124 as shown the example below. The following examples show the text before and after the edits.

Before:

```
Pattern.compile( ".*\\.mdb", flags ),
Pattern.compile(".*\\\\\\\\\\\\\\\\.*(\\\\\\\\\\\\.*)?", flags),
Pattern.compile(".*\\\\\\\\.*(/.*)?", flags));
```

After:

```
Pattern.compile( ".*\\.mdb", flags ),
Pattern.compile(".*\\\\\\\\\\\\\\\\.\\.(\\\\\\\\.\\.)?", flags),
Pattern.compile(".*\\/\\.\\.(\\/\\.\\.)?", flags));
```

4. Save your changes and restart the Media Indexer service.

After you restart the MI, it automatically begins ignoring files that start with “_” and indexes files that start with a just a period. No manual rescan or re-index is required.

5. Repeat these steps on all of your Media Indexer v3.1.4, v3.3.1, or v3.4 servers.



If you want the client MI to index files on its local drive that start with a period then you should modify the client MI. If the files only reside on your local drive then you do not have to modify the server MI.

Estimating the Indexing Time

This section describes how to estimate how long it will take for a Media Indexer server to index or re-index all the files it is currently monitoring. This can be helpful when determining how long it might take to upgrade the Media Indexer servers to a new version.



You can minimize the downtime at a site by pre-building the MI cache as described in “Prebuilding a Media Indexer Cache During an Upgrade” on page 259.

To estimate the time to index the files monitored by an MI server:

1. Open the Statistics tab on the Media Indexer web interface.
2. Calculate the overall number of files that are monitored by the MI (both MXF and SPAAF files) as follows:
 - a. Note the Files value on the “Storage Statistics” page.
 - b. Note the Files value on the “AMA Metadata (SPAAF) Workspaces” view.
 - c. Add the two values together. This is a rough estimate of the number of files being monitored by the MI server.
3. Determine the two average extraction time values. Check the “Average Worker Thread Execution Time” values in the following Statistics views:

- “Indexing Workers Thread Pool” view
- “FI Indexing Workers Thread” Pool

These two indexing operations (in milliseconds) occur at roughly the same speed but run independently and may drift apart in performance. You will use these values to compute the estimated time to perform indexing for both operations. For MI v3.5 and earlier, just use the “Indexing Workers Thread Pool” view.

The most accurate average indexing time values are the ones taken after the indexing process begins. However, you can use the current values to get an estimate of how long the process may take for an upcoming indexing session.

4. Determine number of threads in the pool as follows:

- The number of threads per CPU per indexing pool is 2 for MI 3.5 and earlier but not larger than 48 per pool (there is only one indexing pool in MI-3.5 and earlier).
- The number of threads per CPU per indexing pool is 1 for MI 3.6 and later but not larger than 24 per pool.
- The CPU value includes hyper-threaded cores

Example

This example uses the following values:

- Filecount = 1 million
- Average extraction time = 500 ms
- Threads in the pool (CPU cores) = 24

Calculation

Indexing time = 1 million * 500 ms / 24
 = 500,000 seconds/24
 = 20833 seconds
 = 5.79 hours

Perform two calculations, using the two average extraction times listed in the procedure. The two indexing operations (in milliseconds) occur at roughly the same speed but run independently and may drift apart in performance. During the actual indexing process, monitor both values as described in the next section.

Monitoring the Indexing Process

During the indexing operation, monitor the two average extraction times. If they vary from the original estimate you can calculate a new estimation time.

To monitor the indexing process, open the four “Thread Pool Statistics” views on the MI Statistics tab. There are two for the classic MI database. The two FI (format independence) views are for the MongoDB database.

When all the queue length values are to 0, then the MI is done indexing. Note that the queue length values will go up and down during indexing as more subfolders with files are discovered.



The indexing worker sections are for indexing files and the scanning workers sections are for discovering sub folders.

Prebuilding a Media Indexer Cache During an Upgrade

Note that the information in this section applies to all Media Indexer v3.x upgrades. That is why the term HAG is still used.

A Media Indexer Service might take a long time to fully re-index all shared storage locations, depending on the size and the structural organization of media files on those storage locations.

The Media Indexer database format often changes between MI releases. When this happens, the existing cache files from the previous version cannot be loaded. As a result, the Media Indexer must re-index the files on shared storage from scratch after an upgrade. During the re-index period, any media that has not been indexed yet is essentially offline to the Interplay system.

This re-indexation can be time consuming at a large site. Because there is often a need to keep the upgrade maintenance window as short as possible, the following procedure can be used to help minimize the re-indexation effort.

The main idea is to have a separate MI that is not part of the HAG run the new software version and index the same shared storage locations that are indexed by the HAG MI nodes. One way to do this is to temporarily remove one of the MI nodes from the HAG for the duration of the cache prebuilding.



This requires that you run the new MI software on one node to index your storage while the HAG MI nodes are still running the older MI software version.

Also note that it is best to have the MI machine that prebuilds the cache files set to run at the same time zone as the actual HAG MI nodes that the cache files get copied to.

To prebuild cache files during an upgrade:

1. Either setup a new MI or remove one of the existing MIs from the HAG.
2. Install the new MI software on the separate MI and have it index all of the storage locations that are monitored by the HAG.
3. Once the separate MI finishes indexing all the storage locations, copy all the cache files (%ProgramData%/Avid/AvidMI/*.cache) to a safe location.
4. Do not add this MI node back into the HAG until all MI nodes in the HAG are updated.

To load the prebuilt cache files on the HAG MIs:

1. On the MI in the HAG to be upgraded, turn off the Service Starter service. This will prevent the MI from automatically starting up while you are copying over the cache files.
2. Perform the MI upgrade. The MI will restart after the upgrade.
3. Shut down the MI service after it restarts from the upgrade.
4. Once the MI service is down, make sure that AvidMI.exe is no longer showing as a running process in the Task Manager.
5. Copy all the *.cache files from the previously safe location over to replace all existing *.cache files on the machine that the MI version was just updated on.
6. After the copy is complete, start the MI service and restart the Service Starter Service.

Now the updated and newly started MI service will be able to load all the data from the copied cache files (as they are the same cache schema version). Then it will only scan the folders to find and index new or changed files that have changed or got added since that other special MI has indexed them.

7. If there are multiple MI nodes in the HAG, apply the same procedure to each of them separately.
8. If you removed an MI from the HAG to perform the cache pre-building, put it back into the HAG after the other MI nodes are updated.
9. As soon as you remove the last MI node to be upgraded from the HAG, you can add the already-migrated MIs to the HAG. At that time the MI clients can start using the newly upgraded HAG.

As mentioned above, the updated and newly started MI service will be able to load all the data from the copied cache files (as they are the same cache schema version). Then it will only scan the folders to find and index new or changed files that have changed or got added since that other special MI has indexed them. The less files that have changed since then, the quicker the startup sequence of the Media Indexer. Best case (no indexing required), this takes only a few minutes per 1 million files to index. Typical performance should be roughly two minutes per 1 million files.

A Interplay | Production Maintenance Recommendations

This chapter provides a list of maintenance activities that should be performed regularly in an Interplay Production environment.

- [Interplay Weekly Maintenance Check List](#)
- [Interplay Monthly Maintenance Check List](#)
- [Avid ISIS Recommended Maintenance](#)
- [Complete Server Room Shutdown](#)

The Interplay Administrator includes a set of maintenance tools for specific problems. You can also run these regularly, depending on your workflow.

Interplay Weekly Maintenance Check List

Interplay Weekly Maintenance Check List

Task	For more information
Avid ISIS	
☐ See “Avid ISIS Recommended Maintenance” on page 265	
Interplay Engine and Interplay Archive Engine	
☐ Ensure that individual ingest folders do not contain more than 5,000 objects each.	See “Folder Limitations” on page 42 and “Improving Interplay Media Indexer Performance by Limiting the File Count in Media File Folders” on page 254.
☐ Delete older AvDeletes log files from the following folder: <code>\\InterplayEngineName\C\$\Program Files\Avid\Avid Interplay Engine\Logs\Machines\InterplayEngineName</code>	
Interplay Transfer Server	
☐ Starting in Interplay v3.0, Interplay Transfer no longer requires a weekly reboot. A monthly reboot is sufficient. This is because Interplay Transfer v3.x creates a separate process for each playback, ingest, or DET job. If there is a problem with one job it only affects that particular job and does not affect the Interplay Transfer Engine.	
Interplay Production Services	
☐ Purge all jobs. To purge jobs, use the Interplay Production Services and Transfer Status window.	<i>Interplay Production Services Setup and User's Guide</i>
 <i>Avid recommends that you turn on Auto Purge so you don't have to manually purge jobs.</i>	
 <i>Starting in Interplay v3.x you do not need to restart the Interplay Production servers every week. Once a month is sufficient.</i>	
Interplay Media Indexer	
☐ Rebalance the Interplay Media Indexer configuration and/or storages to make sure sufficient indexing memory is available for the Interplay Media Indexer.	Working with Interplay Media Indexers v3.5 and Later
Use the Interplay Media Indexer Web interface to check and modify the Interplay Media Indexer configuration.	

Interplay Monthly Maintenance Check List

Interplay Monthly Maintenance Check List

Task	For more information
Interplay Engine	
☐ Check the ratio between the number of database pages and the number of objects in the database. This value is automatically calculated and displayed in the Interplay Administrator tool. Note that the Interplay Engine does not need to be restarted except for the following reasons: • To verify system health before an upgrade. See the section “Best Practices for Performing an Engine Upgrade” in the Interplay Production Readme. • Testing the failover capability on a cluster system during regular company maintenance windows (for example, twice a year).	See “ Determining Interplay Database Scalability ” on page 139.
Interplay Transfer Server	
☐ Inspect the Transfer Engine internal disc drive for minimal normal free space. If the discs are reporting higher than normal use, inspect for the presence of large logging or other error reporting files with recent creation dates. Report any unusual findings to the site management for follow on activities.	
☐ Reboot Interplay Transfer server CPUs. To do this, stop the Interplay Transfer Engine application and then reboot the CPU.	Interplay Transfer Setup and User’s Guide
☐ Restart Interplay Transfer Engine application from the desktop shortcut.	
Interplay Production Services	
☐ Restart the Production Services server and Provider systems, including Transcode Provider, and Archive Provider.	<i>Interplay Production Services Setup and User’s Guide</i>
 <i>Do not shut down the Production Services server while jobs are in progress. It is fine to shut down the server while jobs are pending. They will restart after the server is restarted.</i>	

Interplay Monthly Maintenance Check List(Continued)

Task	For more information
Interplay Streaming Server (not supported in Interplay Production v3.5 and later)	
☐ Reboot the server. and clean up the log files. In the log folder, keep the following files (these are the current ones) C:\Program Files\Darwin Streaming Server\Logs • StreamingServer.log • Error.log • server_status and delete the following categories of log files: • Error.xxyyzzaaaa.log • mp3_access.xxyyzzaaa.log • StreamingServer.xxyyzzaaa.log	
Avid ISIS	
☐ See “Avid ISIS Recommended Maintenance” on page 265	
Avid Service Framework	
☐ Use the Health Monitor to check memory and CPU usage of server-side Framework services (Lookup, System Configuration, Time Sync Master, Email). Check to make sure that none have memory that is increasing at an unusual rate and that none have persistently high CPU usage.	

Avid ISIS Recommended Maintenance

Typically, the Avid ISIS does not need to be power cycled. All components of the Avid ISIS stack can be individually replaced or restarted without interfering with the production of Avid ISIS stack.



Power cycling the entire stack (all the components at the same time) could risk the stability of the Avid ISIS stack.

For a detailed description of the maintenance procedures for Avid ISIS, see the *Avid ISIS 7000 Setup Guide*, *Avid ISIS 5000 Setup Guide*, or the *Avid ISIS 2000 Setup Guide*.

Complete Server Room Shutdown

There is no requirement to power cycle the entire Avid infrastructure but, if the need arises to turn off *all* the equipment (such as a relocating the server room), turn off the components in the following order. When turning on the components, use the reverse order.

To shut down the entire Interplay Production workgroup (server room):

1. Shut down all Avid editing systems and attached media I/O equipment.
2. Shut down all capture and playout servers such as AirSpeed Multi Streams.
3. Shut down Interplay Capture servers.
4. Shut down Interplay Transfer Server.
5. Shut down Interplay Production Services and Providers
6. Shut down Interplay Engine and Interplay Archive Engine.
7. Shut down Interplay Media Indexers — Don't stop Interplay Media Indexer while it is indexing storage.
8. Shut down systems running the Service Framework Lookup Service.
9. Shut down Backup Avid ISIS System Director.
10. Shut down Primary Avid ISIS System Director.
11. Shut down Avid ISIS Drive Chassis.
12. Shut down network switches.



Turn on the entire rack of equipment in reverse order as above and verify that all clients have mounted the necessary Avid ISIS Workspaces.

B Archive Migration (Workgroup 4)

Starting at Interplay v1.1.4, you can migrate MediaManager v4.5.x (also known as Workgroup 4) archived assets to an Interplay Archive environment. See your Avid representative for details on migrating.



Conversion of OMF to AAF metadata does not happen during the migration process. The OMF to AAF conversion occurs when the metadata is checked in and out of an Interplay Window using a separate Interplay API. Media files remain as OMF files until you create new media, for example, using transcode or render to create AAF media files.

Searching for Migrated Workgroup 4 Archive Items

After importing the Workgroup 4 assets into an Interplay Archive system database, you can use Interplay Access to filter the assets you want to restore from the imported Workgroup 4 assets.

There are two special attributes “WG4 Archived Date” and “WG4 Tape Group” that apply only to Workgroup 4 clips. These two items are available as Interplay Archive search values.

The following illustration shows the Workgroup 4 attributes and the icons used to identify Workgroup 4 assets before the restore process. The Workgroup 4 icons are similar to the masterclip, subclip, and sequence icons used with Interplay assets, except they include a 4. After the restore process the 4 is removed from the Workgroup 4 icons to display the type of asset, such as clip, subclip, sequence, and so on.

	Name	WG4 Archived Date	WG4 Tape Group
	NTSC Nov OMF Clip.17	3/28/07 4:37 PM	sgl-node3-1
	NTSC November1Clip.05.Sub.01	3/28/07 7:39 PM	sgl-node3-1
	NTSC November1Clip.02.Sub.01	3/28/07 7:34 PM	sgl-node3-1
	NTSC November1Clip.Sub.01	3/28/07 7:43 PM	sgl-node3-1
	NTSC Nov OMF Clip.16	3/28/07 3:19 PM	sgl-node3-1
	NTSC Nov OMF Clip.15	3/28/07 3:15 PM	sgl-node3-1
	NTSC Nov OMF Clip.14	3/28/07 3:05 PM	sgl-node3-1
	Mon OMF Bin Clip.38	3/28/07 3:31 PM	sgl-node3-1
	AAFPALmpeg50	3/27/07 4:48 PM	sgl-node3-1
	911_press.Transcoded.01	3/27/07 2:48 PM	sgl-node3-1
	AAFPALmpeg51	3/27/07 5:11 PM	sgl-node3-1
	AAFPALdv25	3/27/07 5:55 PM	sgl-node3-1

Workgroup 4 asset icon after import process but before restore process.

To search for Workgroup 4 assets in Interplay Access:

- ▶ Perform a normal search using the type of asset you want to restore.
The values allowed are masterclip, subclip, and sequence.

Restoring Migrated Assets

After you migrate the Workgroup 4 archive metadata, you can restore the migrated clips at your convenience. However, to take advantage of some Interplay Production restore features, such as partial restore and MultiRez, it is also necessary re-archive your Workgroup 4 media into Interplay Archive. For more information, see [“Working with OMF Media Files After an Archive Migration” on page 272](#).

For information on restoring different versions of MediaManager clips, see [“Restoring Different Versions of a MediaManager Archive Clip” on page 268](#).

Be aware of the following when restoring migrated clips:

- Avid recommends that you schedule time to restore all of your archived Workgroup 4 media and then re-archive it in Interplay Archive. See [“Working with OMF Media Files After an Archive Migration” on page 272](#).
- Avid recommends that restored Workgroup 4 clips be rearchived to a separate partition to keep the restored Workgroup 4 assets separate from the Workgroup 4 tapes. This makes it easier for deletions; see [“Working with OMF Media Files After an Archive Migration” on page 272](#).
- After the Workgroup 4 assets are restored in Interplay Archive database, Avid recommends that you change the OMF metadata to AAF metadata. From an Avid editing application, check in and then check out the assets to change the OMF metadata to AAF metadata.
- You don’t need to manually convert the AAF/MXF media track length information because the Media Duration Update tool is automatically run when performing a restore.
- Partial Restore is not supported with migrated OMF media files. If you want to use Partial Restore with these media files, you can transcode the assets to create MXF media files and rearchive, and then subclip or Mark IN/OUT into a sequence to partial restore.

To restore the migrated clips:

- ▶ Perform the operation as a normal Restore. See *Interplay Production Services Setup and User’s Guide*.

In Interplay Access, the media status shows the media as online. However, an Avid editing application might show the media as offline until you update the status.

You might need to click in the Interplay Access window and then click the File Locations tab to see the updated file path.

Restoring Different Versions of a MediaManager Archive Clip

The Migration tool imports the latest version of MediaManager (WG4) clips into Interplay Production. If the latest version is not the version you want, follow the steps in [“Locating the Correct Version to Restore” on page 268](#) to locate and restore the correct version.

After performing the archive migration, the FlashNet server database contains entries for different versions of the metadata for Avid assets such as sequences and master clips. These entries are stored in an SQL table named “avid.” You can perform a search in the SQL database to locate the version that you want to restore. After determining the correct clip versions, you can use Interplay Access to restore the desired version.



Before you begin, make sure you have a recent backup version of your Interplay database.

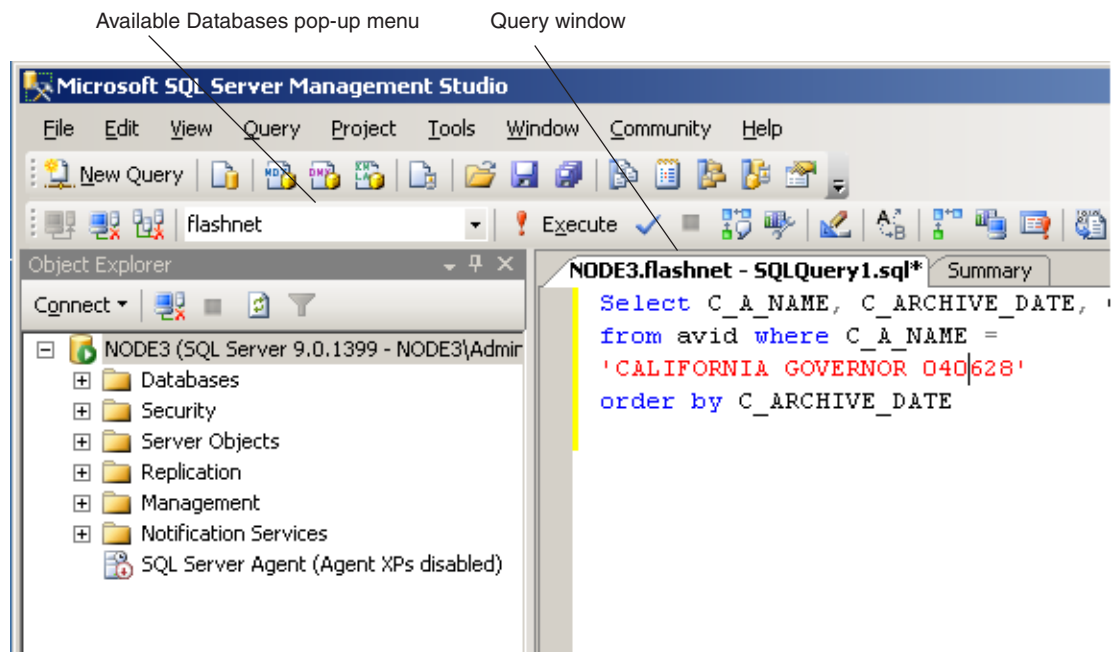
Locating the Correct Version to Restore

To use SQL Management Studio to locate the correct version:

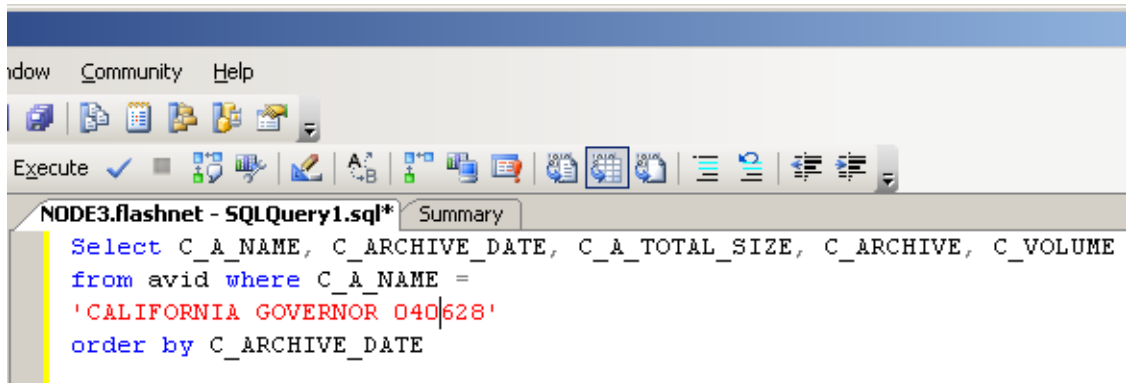
1. On the FlashNet server, select Start > Programs > Microsoft SQL Server 2005 > SQL Server Management Studio.
2. Select the appropriate database and select New Query.
3. Type the following into the Query window:

```
Select C_A_NAME, C_ARCHIVE_DATE, C_A_TOTAL_SIZE, C_ARCHIVE, C_VOLUME
from avid where C_A_NAME =
'clip name'
order by C_ARCHIVE_DATE
```

where *clip name* is the name of the clip you want to search for. The following illustration shows an example using a clip named CALIFORNIA GOVERNOR 040628.



The following illustration shows the Query window in more detail.



The following illustration shows the result panel for this search.

	C_A_NAME	C_ARCHIVE_DATE	C_A_TOTAL_SIZE	C_ARCHIVE	C_VOLUME
1	CALIFORNIA GOVERNOR 040628	2004-07-02 19:30:52.000	508732416	123	RNC154
2	CALIFORNIA GOVERNOR 040628	2004-07-06 23:29:03.000	508733440	123	RNC164
3	CALIFORNIA GOVERNOR 040628	2004-07-08 23:02:30.000	148480	109	RNC168
4	CALIFORNIA GOVERNOR 040628	2004-07-10 17:38:19.000	148480	62	RNC172
5	CALIFORNIA GOVERNOR 040628	2004-07-14 03:44:31.000	148480	8	RNC178
6	CALIFORNIA GOVERNOR 040628	2004-07-16 22:12:23.000	148480	40	RNC180
7	CALIFORNIA GOVERNOR 040628	2004-07-25 18:04:26.000	148480	19	RNC192
8	CALIFORNIA GOVERNOR 040628	2004-07-27 01:09:32.000	140288	50	RNC189

The migration tool imports only the latest archived asset. In the preceding illustration, this is the version archived at 2004-07-27. In this example, the version archived at 2004-07-06 is a more desired version. The clip archive number is 123 and the Volume name is RNC164. The file size is also larger than the 2004-07-27 version.

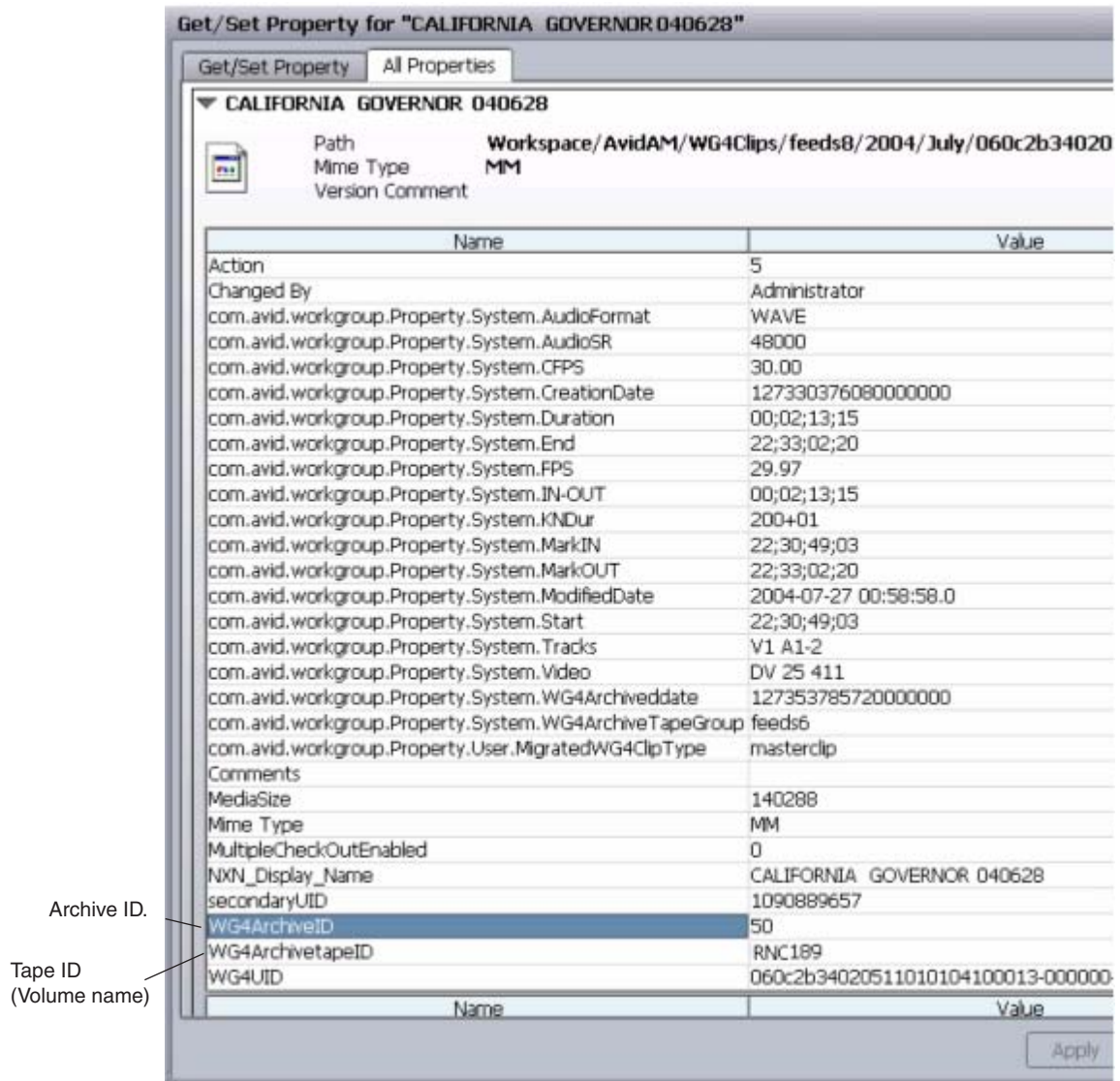
Setting the Correct Version Information for the Clip in Interplay | Access

You use Interplay Access to change the clip's archive number and volume name in the Interplay database.

To use Interplay Access to change the version to restore:

1. Log in to Interplay Access as an Administrator and search for the clip.
2. Right-click the clip and select Advanced.
3. Click Get/Set Property and click the All properties tab.
4. Scroll down the Name list and locate WGArchive ID and WGArchiveTapeID.

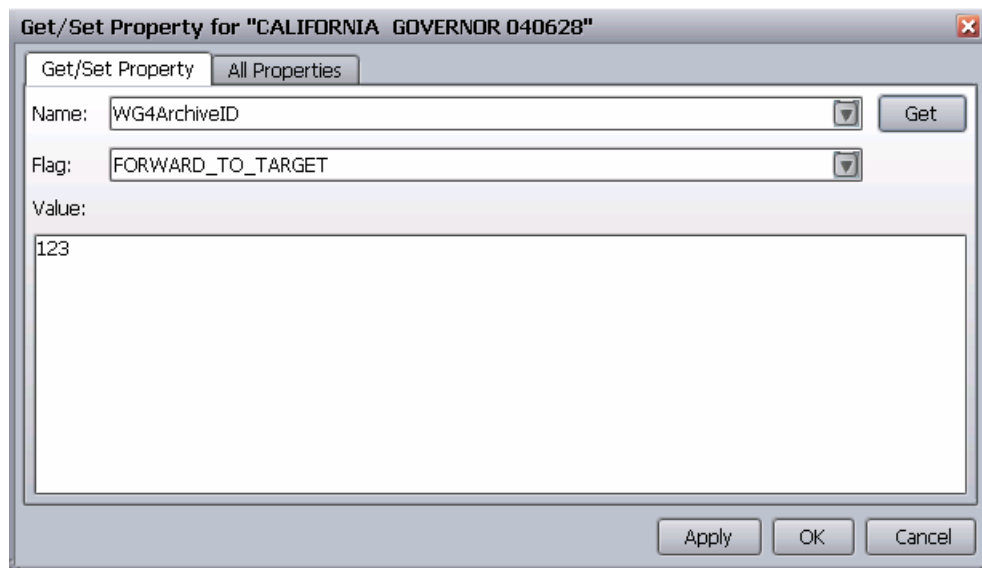
The following illustration shows the selections for this example.



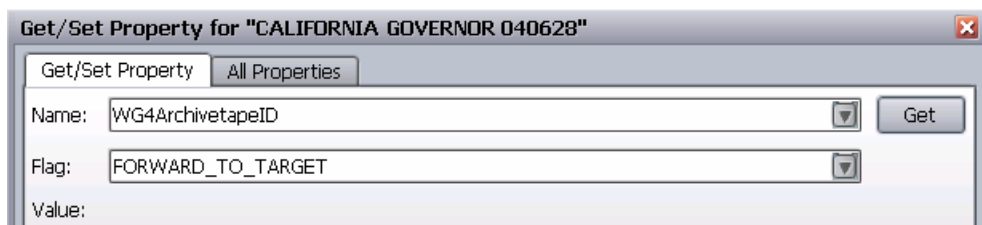
The All Properties tab shows the current WG4Archive ID and Archive Tape ID values. In this example, we want to change the WGArchive ID to 123 and change the WG4ArchiveTapeID to RNC164.

5. Click the Get/Set Property tab and select WG4ArchiveID from the Name list.
6. Click the Get button. The system displays the current value in the Value window.
7. Type the new value in the Value window and click the Apply button.

The following illustration shows the new value for this example.



8. Select the WG4ArchivetapeID from the Name menu and click Get. The system displays the current value in the Value window.



9. Change the value, and click the Apply button. In this example we change the value to RNC164.
10. Click the All Properties tab and verify that the new values are correct.
11. Click OK to close the dialog box.

Restoring the Clip from the MediaManager Database

To restore the correct version of the clip:

1. Locate the clip that you want to restore in Interplay Access.
2. Delete the media for the clip.



If you have already restored the clip and want to restore a different version, delete the Interplay Production version of the media before you perform the restore.

3. Right-click the clip in Interplay Access and select Restore.

Working with OMF Media Files After an Archive Migration

You can troubleshoot problems that might occur with OMF media that is restored after an archive migration. The conversion of OMF to AAF metadata does not happen during the restore process. Media files remain as OMF files until you create new media. For example, using transcode or render to create MXF media files. For specific steps on performing the archive migration, see your Avid representative.

To create MXF media files after migrating a Workgroup 4 archive:

1. After you migrate the Workgroup 4 archive database to Interplay Production, restore the media to a specific workspace, for example, one named WG4.
2. If you check out a reference clip from the sequence and notice that some of the media is offline, use the Avid editing application's Reveal File command to verify whether the online media comes from the "WG4" workspace. If it does, you are working with OMF media. For example, if you have an OMF sequence containing alpha, the video fill is online but the alpha media is offline.
3. Do the following in the Avid editing application:
 - a. Change the Dynamic Relink settings by doing one of the following:
 - Select If no match is found > Keep Existing Media.
 - Turn off Dynamic Relink.
 - b. Check out the sequence again.
At this point the media should be online.
4. Transcode the sequence to an MXF resolution by doing the following:
 - a. Open the Consolidate/Transcode dialog box.
 - b. Select Transcode.
 - c. Select Convert Video.
 - d. (Option) Select Create New Sequence, if you want to create a new sequence for the MXF media that is separate from the OMF media.
 - e. Click Transcode.

New MXF media is created for the entire sequence. Turn on Dynamic Relink to link the sequence to the new MXF media.

5. (Option) Rename the OMF sequence so you can identify it as being linked to OMF media. This makes it easier to delete the OMF version later. See ["Deleting the Online OMF Media" on page 273](#).
6. (Option) Rearchive the sequence that is linked to the MXF media. This lets you take advantage of partial restore the next time you restore the sequence.

Deleting the Online OMF Media

Use the following methods to delete the online OMF media files:

- If you used Create New Sequence to create the MXF media, you have two sequences: one linking to OMF media and another linking to MXF media. After you perform the transcode, rename the OMF version so you can identify it for deletion. Then you can use Interplay Access to search for sequences with that text in the name. Delete the sequence and the media.
- If you did not select Create New Sequence in the Consolidate/Transcode dialog box before you transcoded the sequence, the sequence's clips link to both the OMF media and the new MXF media. In this case there is no way to delete just the OMF media from Interplay Access. If you want to free up disk space, you must manually remove the OMF media files from your Interplay environment.

To manually delete online OMF media files:

1. Use one of the following methods to identify the media files:
 - ▶ Select a clip from the sequence in Interplay Access and select the File Locations tab in the Object Inspector, then identify the .omf file in the File Path column.
 - ▶ Use Reveal File in the Avid editing application to identify the OMF files associated with the clip.
2. Use Windows Explorer to delete the associated .omf files.



This procedure should only be performed by an Administrator to avoid deleting the wrong media files. In general, you should only use the Interplay Production applications to delete media files.

Retained Workgroup 4 Clip Attributes

The following table list the Workgroup 4 attributes that carry over to the Interplay Archive database.



These attributes are not available in Interplay Access until after you perform a restore of the Workgroup 4 assets.

Workgroup 4 Clip Attributes Retained in Interplay Environment			
Attribute	Master clip	Subclip	Sequence
Archive date	X	X	X
Tape group	X	X	X
File location	X		X
Start	X	X	X
Duration	X	X	X
End	X	X	X
Mark_In	X	X	X
Mark_Out	X	X	X
In_Out	X	X	X
FPS	X	X	X
KN_dur	X	X	X
Video	X	X	X
Tracks	X	X	X
Audio_Format	X	X	X
Audio_SR	X	X	X
CFPS	X	X	X
Created_by	X	X	X
Tape_ID	X	X	X
Creation_Date	X	X	X
Modified_Date	X	X	X
Media_Size	X	X	X
Comment	X	X	X

Index

Symbols

.spaaf files
described [85](#)

Numerics

1080p/23.976 proxy support [170](#)
1080p/23.976 support [167](#)
1080p/24 support [167](#)
1080p/24 support proxy [170](#)
720p/23.976 proxy support [170](#)
720p/23.976 support [167](#)

A

Access control [24](#), [26](#), [34](#)
 Instinct and Interplay Assist User settings [37](#)
AFD (Active Format Description)
 support for [206](#)
AirSpeed Multi Stream
 configuring for H.264 workflow [154](#)
AirSpeed Muti Stream operating points [191](#)
AirSpeed server
 Frame Chase editing with [200](#)
AMA (Avid Media Access) material
 .spaaf files [85](#)
 browsing in Access [97](#)
 configuring media files location [89](#)
 configuring metadata location [88](#)
 deleting in Access [101](#)
 license for third-party storage [93](#)
 Media Indexer [85](#)
 resyncing [102](#)
 support for [82](#)
 workflow [82](#)
Ancillary data
 ingest and playout [205](#)
 support for AFD [206](#)
Apple ProRes support [171](#)
Archive
 restoring a clip [133](#)
Archive migration [266](#)
Asset manager
 Administrator client [15](#)
 description [14](#)
Assets [14](#)

Avid
 online support [10](#)
 training services [12](#)
Avid assets [14](#)
 deleting [128](#)
Avid Instinct
 support for stereoscopic 3D [181](#)
Avid Logging application [20](#)
Avid Service Framework [23](#)

C

Capture
 creating folders [37](#)
 editing during [198](#)
 Frame Chase Capture [198](#)
 Frame Chase editing [198](#)
Custom properties
 example of search [77](#)

D

Deleting
 Avid assets and media [128](#)
 best practices guidelines [39](#)
 determining how much space will be freed up [68](#)
 introduction [38](#)
 media [24](#)
 media and metadata [39](#)
Digital Rights Management [20](#)
DNxHD 100 support [184](#)
DNxHR support [187](#)
Dual-ingest configuration
 Frame Chase editing [199](#)
Dynamic Relink
 described [13](#)
Dynamic relink
 working with a single resolution [207](#)

E

Edit while capture
 See Frame Chase editing
Editing
 during capture [198](#)
Editor Database settings [34](#)

F

File assets [14](#)
Film option [167](#)
Folders
 defining a database [26](#)
 for capturing media [37](#)
Frame Chase editing
 described [198](#)
 dual-ingest configuration [199](#)
 limitations [203](#)
 overview [198](#)
 workflow [200](#)

H

H.264 media
 AirSpeed Multi Stream encoder card [154](#)
 archiving, restoring, deleting [166](#)
 editing [166](#)
 supported resolutions [149](#)
H.264 workflow [149](#)
 acquiring media [162](#)
 archiving, restoring, deleting [166](#)
 configuring AirSpeed Multi Stream [154](#)
 configuring Interplay Capture [158](#)
 dual-ingest [162](#)
 folders and shared-storage workspaces [154](#)
 installation and configuration [151](#)
 Production Services profiles [159](#)
 transcoding [165](#)
 with AirSpeed Multistream [151](#)
 with Interplay Transcode [151](#)

I

Icons
 new for Interplay 2.3 [24](#)
Incoming Media folder
 and performance [37](#)
In-progress clips
 editing during capture [198](#)
 limitations [203](#)
Instinct [19](#)
Instinct and Interplay Assist User Settings [37](#)
Interplay Administrator [15](#)
Interplay Assist [20](#)
 support for stereoscopic 3D [181](#)
Interplay Capture [23](#)
 configuring for H.264 [158](#)
Interplay Engine [14](#)
Interplay Media Indexer [23](#)
 AMA material [85](#)
 description [218](#)
Interplay Portal
 viewing [10](#)
Interplay Production Services [19](#)

Interplay Transfer [18](#)
Interplay Window
 support for Stereoscopic 3D [181](#)

J

JPEG 2000 support [174](#)

L

Logging application [20](#)
Long GOP [106](#)
 Capture workflow [109](#)
 Output workflow [114](#)

M

Manage Database Roles [29](#), [32](#)
Media deletion [24](#)
Media files
 deleting [128](#)
Media Indexer [23](#)
 see Avid Interplay Media Indexer [218](#)
Migration from MediaManager [22](#)
MPEG-2 [106](#)
MultiRez
 described [13](#)
 partially online clips, restoring from an archive [120](#)
MXF OP-1A [108](#)
MXF OP-ATOM [108](#)

O

OMF files
 working with [272](#)
Omneon
 operating points [191](#)
Online support [10](#)
OP-1A file format [108](#)
OP-ATOM file format [108](#)
Operating points [189](#)
 AirSpeed Multi Stream [191](#)
 Omneon [191](#)
Operating points for Long GOP workflow [106](#), [108](#)

P

Partially online clips
 restoring from an archive [120](#)
Permissions
 assigning [26](#)
Profiles
 Production Services for H.264 workflow [159](#)
Projects
 assigning permission to users [31](#)

Properties
 stereoscopic 3D [181](#)
 ProRes
 support [171](#)

R

Restoring a clip [133](#)
 Restricted flag [20](#)

S

Searches
 custom properties,example [77](#)
 text media attribute [74](#)
 Server Hostname Settings [37](#)
 Sony XAVC Long GOP support [195](#)
 Stereoscopic 3D support
 in Interplay Access [178](#)
 in Interplay Transcode [180](#)
 in Interplay Window, Interplay Assist, Avid Instinct
 [181](#)
 media and clips [176](#)
 overview [175](#)
 properties [181](#)
 STP Encode service
 workflow [116](#)
 System maintenance [261](#)

T

Training services [12](#)
 TransferManager [18](#)
 Troubleshooting [10](#)

U

UHDTV1 support [187](#)
 Unique identifier
 custom property for [77](#)
 User database
 defining [26](#)

W

Workflow
 configuring Interplay Media Indexer [218](#)
 delete workflows [52](#)
 frame chase editing [198](#)
 H.264 MultiRez [149](#)
 setting up user database [26](#)
 system maintenance [261](#)
 Workgroups
 migration from MediaManager [22](#)

X

XAVC Long GOP support [195](#)
 XAVC-Intra support [192](#)
 XDCAM HD and XDCAM EX [106](#)
 XDCAM HD and XDCAM EX file formats [108](#)

